

Mensuration

2-D: Under 2 dimensions we will study

- Triangles
- Quadrilateral
- Polygons
- Circle

3-D: Under 3 dimensions we will study

- Cube
- Cuboids
- Box
- Cylinder
- Prism
- Cone
- Pyramid
- Frustum
- Sphere
- Hemisphere
- Tetrahedral

Note: For 2D figures, area and perimeter are calculated and for 3D figures, volume and surface area are calculated.

TWO DIMENSIONAL FIGURES

TRIANGLE

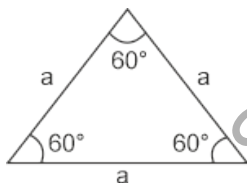
For a triangle with height, h, and base, b;

$$\text{Area} = \frac{1}{2} \times b \times h$$

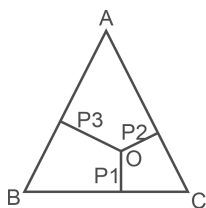
Perimeter = Sum of all sides

Special cases:

1) Equilateral Triangle - All sides are equal and each angle is 60° .



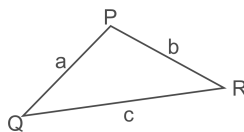
- Area = $\frac{\sqrt{3}}{4} a^2$
- Height = $\frac{\sqrt{3}}{2} a$
- Perimeter = $3a$
- Inradius(r) = $\frac{a}{2\sqrt{3}}$
- Circumradius(R) = $\frac{a}{\sqrt{3}}$



If P1, P2 and P3 are perpendicular to a side from a point inside the triangle, then

- Height of triangle = $P1 + P2 + P3$

2) Scalene Triangle - All sides will be unequal lengths.



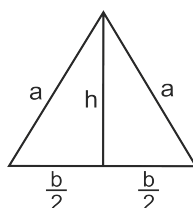
Area = $\sqrt{s(s-a)(s-b)(s-c)}$,
This formula is called Heron's formula.

Where, Semi-perimeter,

$$s = \frac{a+b+c}{2}$$

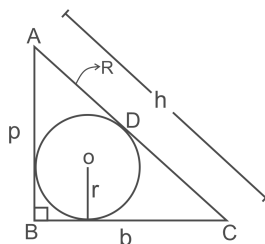
- Perimeter = $a + b + c$

3) Isosceles Triangle :- two sides and two angles are equal. Altitude bisects the base.



- Area = $\frac{b}{4} \sqrt{4a^2 - b^2}$
- Height = $\sqrt{a^2 - \left(\frac{b}{2}\right)^2}$
 $= \frac{1}{2} \sqrt{4a^2 - b^2}$
- Perimeter = $a + a + b$
 $= 2a + b$

4) Right angled Triangle :- One of the angles is 90° .



Here, p = perpendicular, b = base and h = hypotenuse

- Area = $\frac{1}{2} \times b \times p$
- Perimeter = $p + b + h$
- Pythagoras Theorem:
 $h^2 = p^2 + b^2$
- Inradius(r) = $\frac{p + b - h}{2}$
- Circumradius(R) = $\frac{h}{2}$

Note: Common Pythagoras triplets:

(1, 1, $\sqrt{2}$); (1, 2, $\sqrt{5}$); (3, 4, 5); (5, 12, 13);
(7, 24, 25); (20, 21, 29); (9, 40, 41);
(8, 15, 17); (12, 35, 37); (60, 11, 61);
(65, 72, 97); (96, 110, 146).

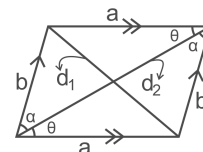
Try remembering them.

QUADRILATERAL

A figure enclosed by four sides is called a quadrilateral. A quadrilateral has four angles and sum of these angles is equal to 360° .

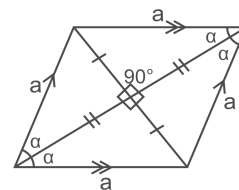
Special Cases:

1) Parallelogram - It is a quadrilateral with opposite sides parallel and equal.



- Area = base $\times$ height
- Perimeter = $2(a + b)$
- $d_1^2 + d_2^2 = 2(a^2 + b^2)$
- In a parallelogram, opposite sides are equal, opposite angles are equal and diagonals bisect each other.

2) Rhombus - It is a parallelogram with all four sides equal. The opposite angles in a rhombus are equal.

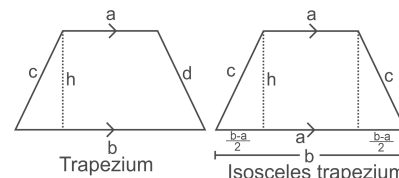


Here, a = side; d_1 and d_2 are diagonals.

- Area = $\frac{1}{2} \times d_1 \times d_2$
- Side (a) = $\frac{1}{2} \sqrt{d_1^2 + d_2^2}$
- Perimeter = $4a$
- $4a^2 = d_1^2 + d_2^2$

Diagonals bisect each other at right angles.

3) Trapezium - It is a quadrilateral with one pair of opposite sides parallel.



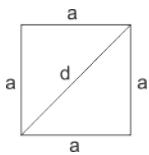
Here, a and b are parallel sides and h is the height or perpendicular distance between a and b.

- Area = $\frac{1}{2} \times \text{height} \times (\text{sum of parallel sides})$
 $= \frac{1}{2} \times h \times (a + b)$

4) Rectangle – It is a parallelogram with equal opposite sides and each angle is 90°

- Area = Length $\times$ Breadth = $L \times B$
- Perimeter = $2(L + B)$
- Diagonal (d) = $\sqrt{L^2 + B^2}$
- Area of the walls of a rectangular room
 $= 2 \times (L + B) \times H$

5) Square – It is a parallelogram with all four sides equal and each angle is equal to 90° .



- Area = $(side)^2 = a^2 = \frac{1}{2}d^2$
- Perimeter = $4 \times side = 4a$
- Diagonal (d) = $a\sqrt{2}$

NOTE: Important points about Quadrilaterals

1. The diagonals of a parallelogram bisect each other.
2. Diagonal of a parallelogram divides it into two triangles of equal area.
3. The diagonals of a rectangle are of equal lengths and bisect each other.
4. The diagonals of a square are equal and bisect each other at right angles.
5. A rhombus has unequal diagonals and they bisect each other at right angles.
6. A parallelogram and a rectangle have equal areas if they are on the same base and between the same parallel lines.

REGULAR POLYGON

In a regular polygon all sides and all interior angles are equal. A polygon is called a pentagon, hexagon, heptagon, octagon, nonagon and a decagon as they have 5, 6, 7, 8, 9, 10 sides, respectively.

If each side of a regular polygon of 'n' sides is equal to 'a' then:

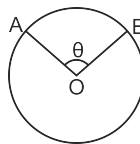
- Area of regular pentagon
 $= \frac{6.9}{4} \times a^2$
- Area of regular hexagon
 $= 6 \times \frac{\sqrt{3}}{4} a^2$
- Each exterior angle = $\frac{360^\circ}{n}$
- Each interior angle

= 180° – Exterior angle

- Number of diagonals
 $= \frac{n(n-3)}{2}$
- Sum of all interior angles
 $= (n-2) \times 180$
- Sum of all exterior angle = 360°

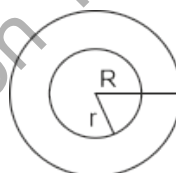
CIRCLE

It is a plane figure enclosed by a line on which every point is equally distant from a fixed point (centre) inside the circle.



- Area = πr^2
- Circumference (perimeter)
 $= 2\pi r$
- Diameter = $2r$
- Length of Arc (AB)
 $= 2\pi r \times \frac{\theta}{360^\circ}$
- Area of sector AOB
 $= \pi r^2 \times \frac{\theta}{360^\circ}$
- Length of Arc (AB) = $angle \times radius$, (where angle in radian)

CIRCULAR RING



Here, R = radius of bigger ring,
 r = radius of smaller ring

- Area = $\pi(R^2 - r^2)$
- Difference in circumference of both the rings = $2\pi R - 2\pi r$

Short Cut methods/Tricks

1. If the length and breadth of a rectangle are increased by a% and b%, the area of the rectangle will be increased by

$$(a + b + \frac{ab}{100})\%$$

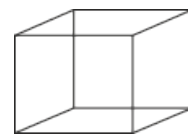
2. If any of the two sides of the rectangle is decreased then use '-ve' values for that side.
3. All the sides of any two dimensional figure changed by a%, then its area will change by $(2a + \frac{a^2}{100})\%$
 Whenever there is a decrease, use negative value for 'a'
4. If all the sides of any two dimensional

figure has changed (increased or decreased) by a% then its perimeter also changes by a%. In the case of a circle such changes take place because of the change in radius (or diameter).

5. If the area of a square is 'a' square unit. Then the area of the circle formed with the same perimeter is given by square units.
6. Area of a square inscribed in a circle of radius 'r' is equal to $2r^2$.
7. The area of the largest triangle inscribed in a semicircle of radius r is equal to r^2 .
8. If a pathway of width x is made inside or outside a rectangular plot of length L and breadth B, then area of the pathway is
 (i) $2x(L + B + 2x)$, if path is made outside the plot
 (ii) $2x(L + B - 2x)$, if a path is made inside the plot.
9. If two paths, each of width x are made parallel to length (L) and breadth (B) of the rectangular plot in the middle of the plot crossing each other, then,
 Area of the path = $x(L + B - x)$

THREE DIMENSIONAL FIGURES

CUBE



All sides are equal. It has six faces and 12 edges.

- Volume = a^3
- Total surface area = $6a^2$
- Diagonal = $a\sqrt{3}$
- Sum of all edges = $12a$

Here, a = length of the side

CUBOID

A rectangular body having 3D rectangular shape, is called a cuboid.

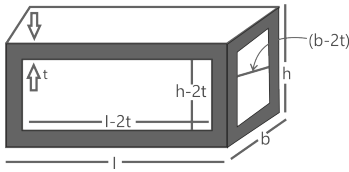


- Volume = $l \times b \times h$
- Total surface area =
 $2(lb + bh + lh)$
- Diagonal = $\sqrt{l^2 + b^2 + h^2}$

BOX

A box has its shape like a cube or cuboid. The amount that a box can hold or

contain is called the capacity of the box.
Capacity means internal volume.



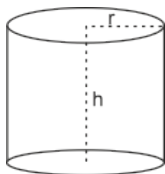
- Surface area of an open box
 $= 2(\text{length} + \text{breadth}) \times \text{height} + \text{length} \times \text{breadth}$
 $= 2(l + b) \times h + l \times b$
- Capacity of box
 $= (l - 2t)(b - 2t)(h - 2t);$
 where, t = thickness of box
- Volume of the material of the box = External volume – Internal volume(or capacity) =
 $lbh - (l - 2t)(b - 2t)(h - 2t)$

ROOM

A rectangular room has four walls (surfaces) and opposite walls have equal area.

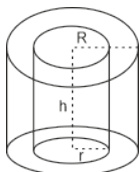
- Total Area of walls
 $= 2(l + b) \times h$
- Total volume of the room
 $= l \times b \times h$
- Area of floor or roof = $l \times b$

CYLINDER



- Volume of cylinder = area of base $\times$ height = $\pi r^2 h$
- Curved surface area = Perimeter of base $\times$ height = $2\pi r h$
- Total surface Area = curved surface area + area of both the circles
 $= 2\pi r h + 2\pi r^2 = 2\pi(r + h)r$
- When the rectangular sheet is folded along its length, then the length becomes the circumference of the base of the cylinder and breadth becomes the height of the cylinder.

HOLLOW CYLINDER



- Volume of hollow cylinder
 $= \pi(R^2 - r^2)h$
- Curved surface area
 $= 2\pi(R + r)h$
- Total surface area
 $= 2\pi(R + r)h + 2\pi(R^2 - r^2)$
 $= 2\pi(R + r)\{h + R - r\}$

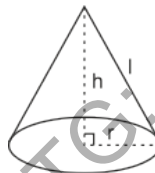
Where, R = External radius of cylinder, r = internal radius of cylinder, h = height

PRISM

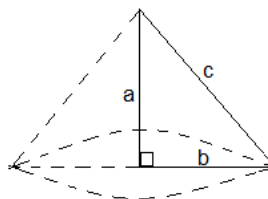
- Volume of prism
 $= \text{area of base} \times \text{height}$
- Lateral surface area = Perimeter of base $\times$ height
- Total surface Area = Lateral surface area + area of base and top surface

CONE

A solid and round body with a round base and pointed peak.



- Volume = $\frac{1}{3} \times \text{base area} \times \text{height} = \frac{1}{3} \pi r^2 h$
- Slant height (l) = $\sqrt{r^2 + h^2}$
- Curved surface area = $\pi r l$
 $= \pi r \sqrt{r^2 + h^2}$
- Total surface area = $\pi r l + \pi r^2$
 $= \pi r(l + r)$
- Cone formed by rotating right angled triangle about its height:



- Volume of cone so formed
 $= \frac{1}{3} \pi b^2 a$
- Similarly, Cone formed by rotating right angled triangle about its base:
 Volume of cone so formed
 $= \frac{1}{3} \pi a^2 b$

- Similarly, Cone formed by rotating right angled triangle about its hypotenuse :
 Volume of cone so formed
 $= \frac{1}{3} \pi r^2 c$, (where r is the altitude on hypotenuse and $r = \frac{a \times b}{c}$)

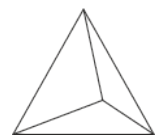
Note: If the base is not round, it will be called a pyramid. A pyramid can have various shapes of the base example: square, rectangular, triangular etc.

PYRAMID

- Pyramid means a structure with regular polygon as its base and sloping sides that meet in a point at the top.
- In Pyramid, with n sided regular polygon at its base, total number of vertices = $n + 1$
- Volume = $\frac{1}{3} \times \text{base area} \times \text{height}$
- Slant height (l) = $\sqrt{r^2 + h^2}$
- Lateral surface area
 $= \frac{\text{Perimeter} \times \text{slant height}}{2}$
- Total Surface area = Lateral surface area + Area of base

TETRAHEDRON

It is a 3D figure made by joining four equilateral triangles.

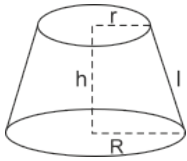


- Volume (V)
 $= \frac{1}{3} \times \text{base area} \times \text{height}$
 $\Rightarrow V = \frac{1}{3} \times \frac{\sqrt{3}}{4} a^2 \times \left(\frac{\sqrt{6}}{12} a + \frac{\sqrt{6}}{4} a \right)$
 $\Rightarrow V = \frac{\sqrt{2}}{12} a^3$ (Remember this formula)
- Lateral surface area = $3 \times \frac{\sqrt{3}}{4} a^2$
- Total surface area
 $= 4 \times \frac{\sqrt{3}}{4} a^2$
- Height of Tetrahedron = $\frac{\sqrt{6}}{3} a$

FRUSTUM OF CONE

If a cone is cut by a plane parallel to its base, so as to divide the cone into two

parts: upper part and lower part, then the lower part is called frustum.



- Slant height (l)

$$= \sqrt{h^2 + (R - r)^2}$$
- Curved Surface Area

$$= \pi(R + r)l$$
- Total surface area

$$= \pi(R + r)l + \pi R^2 + \pi r^2$$

$$= \pi\{(R + r)l + R^2 + r^2\}$$
- Volume

$$= \frac{1}{3}\pi h(r^2 + R^2 + rR)$$

SPHERE

- Volume of sphere

$$= \frac{4}{3}\pi r^3$$
- Curved Surface area

$$= \text{Total surface area} = 4\pi r^2$$

HOLLOW SPHERE OR SPHERICAL SHELL:

- Volume of hollow sphere

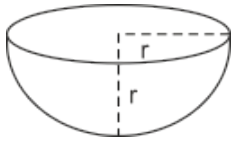
$$= \frac{4}{3}\pi(R^3 - r^3)$$
- Internal surface area

$$= 4\pi r^2$$
- External surface area

$$= 4\pi R^2$$

Here R = external radius and
 r = internal radius

HEMISPHERE



- Volume of the hemisphere

$$= \frac{2}{3}\pi r^3$$
 - Total surface area

$$= 3\pi r^2$$
 - Curved surface area

$$= 2\pi r^2$$
- Where, r = radius

IMP. UNIT CONVERSION :

- $1m^3 = 1000$ litres and
- $1 \text{ litre} = 1000 \text{ cm}^3$
- $1 \text{ meter} = 10 \text{ decimeter}$
 $= 100 \text{ cm} = 1000 \text{ millimeter}$
- $1 \text{ meter} = 10^{-1} \text{ decameter}$
 $= 10^{-2} \text{ hectometer}$
 $= 10^{-3} \text{ kilometer}$

Variety Questions

Q.1. At what angle is the arc length half the perimeter of the circle?

SSC CGL 17/09/2024 (1st Shift)

- (a) 120° (b) 180° (c) 90° (d) 270°

Q.2. What is the central angle of a sector of a circle whose area and perimeter are, respectively, equal to 209 cm^2 and 63 cm if it is given that its radius is a non-negative integer (roundup to one decimal place) ?

SSC CGL 10/09/2024 (3rd Shift)

- (a) 55.5° (b) 53.5° (c) 49.5° (d) 51.5°

Q.3. The area of a square is $16x^2 + 40x + 25$ square units. Find the perimeter of the square.

SSC CGL 10/09/2024 (3rd Shift)

- (a) $3(10x + 8)$ (b) $3(8x + 10)$
 (c) $2(10x + 8)$ (d) $2(8x + 10)$

Q.4. A right triangle with sides 3 cm, 4 cm and 5 cm is rotated about the sides of 3 cm to form a cone. The volume of the cone so formed is:

SSC CGL 10/09/2024 (2nd Shift)

- (a) $20\pi \text{ cm}^3$ (b) $16\pi \text{ cm}^3$
 (c) $25\pi \text{ cm}^3$ (d) $28\pi \text{ cm}^3$

Q.5. Find the area (in cm^2) of the sector whose perimeter is $\frac{64}{3} \text{ cm}$ and central angle is 60° . (Use $\pi = \frac{22}{7}$)

SSC CGL 09/09/2024 (3rd Shift)

- (a) $\frac{47}{3}$ (b) $\frac{77}{3}$ (c) $\frac{68}{3}$ (d) $\frac{85}{3}$

Q.6. A cuboid with dimensions 20cm, 12cm, and 10 cm is cut into 8 identical pieces by 3 cuts. What will be the total surface area of all the pieces?

SSC CHSL 10/07/2024 (3rd Shift)

- (a) 2240 cm^2 (b) 1220 cm^2
 (c) 3360 cm^2 (d) 1680 cm^2

Q.7. A rectangle with the largest area is cut out of a circle with radius 5 units. What is the area of the remaining portion?

SSC CHSL 10/07/2024 (1st Shift)

- (a) $25(2\pi - 3)$ (b) $25(2\pi - 5)$
 (c) $25(\pi - 3)$ (d) $25(\pi - 2)$

Q.8. A cylinder has a curved surface area equal to 350% of the curved surface area of another cylinder. If their radii are in the ratio 3 : 1, then the volume of the smaller cylinder is approximately ____% of the larger cylinder (the smaller cylinder has its radius and height smaller when compared to the larger).

SSC CHSL 09/07/2024 (1st Shift)

- (a) 10.16 (b) 8.86 (c) 7.28 (d) 9.52

Q.9. The inner and outer radii of a hemispherical wooden bowl are 6 cm and 8 cm, respectively. Its entire surface has to be polished and the cost of polishing $\pi \text{ cm}^2$ is Rs.50. How much will it cost to polish the bowl?

SSC CHSL 08/07/2024 (4th Shift)

- (a) Rs.11,600 (b) Rs.11,400
 (c) Rs.10,000 (d) Rs.12,000

Q.10. Which of the following statements is INCORRECT about the circle?

SSC CHSL 08/07/2024 (4th Shift)

- (a) The area of a sector = $\frac{1}{2}$ (arc length $\times R$)
 (b) The radius of an incircle of an equilateral triangle of side 's' is $\frac{s}{2\sqrt{3}}$
 (c) The length of an arc = $\frac{\text{central angle made by the arc}}{360^\circ} \times \pi r^2$
 (d) Circles are congruent if they have same radii while similar circles may have different radii.

Q.11. The average of the areas of 2 similar triangles is 706.5 m^2 whose perimeters are in the ratio of 6 : 11. What is 20% of the difference (in m^2) in areas of both triangles ?

SSC CHSL 08/07/2024 (1st Shift)

- (a) 164 (b) 149 (c) 153 (d) 157

Q.12. Two equal circles are drawn on a square in such a way that opposite side of the square forms the diameter of each circle. If the remaining area of the square is 42 cm^2 , what is the measurement (in cm) of the diameter of each circle?

SSC CHSL 02/07/2024 (2nd Shift)

- (a) 14 (b) 2.5 (c) 7 (d) 3.5

Q.13. The ratio of the diameters of two spheres is given as 1 : 4. The larger sphere is melted and 125 identical spheres are made out of the molten material. The smaller sphere is melted and 27 identical spheres are made out of the molten material. If the ratio of the volume of each of the 125 identical spheres to the volume of each of the 27 identical spheres is given as 1 : m, what is the value of m ?

SSC CHSL 01/07/2024 (1st Shift)

- (a) $\frac{1728}{125}$ (b) $\frac{125}{1728}$ (c) $\frac{27}{8000}$ (d) $\frac{64}{3375}$

Q.14. Two rectangular sheets of paper, each $60 \text{ cm} \times 36 \text{ cm}$, are made into two right circular cylinders, one by rolling the paper along its length and the other along the breadth. The ratio of the volumes of the two cylinders, thus formed, is:

SSC CPO 29/06/2024 (3rd Shift)

- (a) 5 : 6 (b) 8 : 3 (c) 7 : 4 (d) 5 : 3

Q.15. Three circles each of radius 5 cm touch one another. The area (in cm^2) subtended between them is:

SSC CPO 28/06/2024 (3rd Shift)

- (a) $50(\sqrt{3} - \frac{\pi}{2})$ (b) $25(\sqrt{3} + \frac{\pi}{2})$
(c) $25(2\sqrt{3} - \frac{\pi}{2})$ (d) $25(\sqrt{3} - \frac{\pi}{2})$

Q.16. A gear 15 cm in diameter is turning a gear 21 cm in diameter. When the smaller gear has 105 revolutions, how many has the larger one made?

SSC CPO 28/06/2024 (3rd Shift)

- (a) 75 (b) 100 (c) 85 (d) 90

Q.17. A solid metal cylinder of 12 cm height and 7 cm radius is melted and recast into two cones in the proportion of 1 : 2 (volume), keeping the height 12 cm. What would be the percentage change in the flat surface area before and after [use $\pi = \frac{22}{7}$]

SSC CPO 28/06/2024 (3rd Shift)

- (a) 35% (b) 20% (c) 45% (d) 50%

Q.18. Three solid iron cubes of edges 8 cm, 10 cm and 12 cm are melted together to make a new cube. It is observed that 496 cm^3 of the melted material is lost due to improper handling. The area (in cm^2) of the whole surface of the newly formed cube is :

Matriculation Level 26/06/2024 (Shift - 1)

- (a) 1331 (b) 1176 (c) 2197 (d) 3375

Q.19. If the semi-perimeter and area of a rectangular field whose length and breadth are 'x' and 'y' is 12 cm and 28 cm^2 , respectively, then find the value of $x^4 + x^2y^2 + y^4$.

Graduate Level 20/06/2024 (Shift - 3)

- (a) 6690 (b) 6960 (c) 6609 (d) 6906

Q.20. The base of a right prism is an equilateral triangle with each side measuring 4 cm. If the lateral surface area is 120 cm^2 , find the volume (in cm^3) of the prism.

SSC CHSL Tier II (10/01/2024)

- (a) $30\sqrt{3}$ (b) $40\sqrt{3}$ (c) $10\sqrt{3}$ (d) $20\sqrt{3}$

Q.21. Some ice pieces, spherical in shape, of diameter 6 cm are dropped in a cylindrical container containing some juice and are fully submerged. If the diameter of the container is 18 cm and level of juice rises by 40 cm, then how many ice pieces are dropped in the container?

SSC CPO 05/10/2023 (3rd Shift)

- (a) 90 (b) 80 (c) 85 (d) 95

Q.22. Some medicine in liquid form is prepared in a hemispherical container of

diameter 36 cm. When the container is full of medicine, the medicine is transferred to small cylindrical bottles of diameter 6 cm and height 6 cm. How many bottles are required to empty the container?

SSC CPO 04/10/2023 (2nd Shift)

- (a) 70 (b) 75 (c) 72 (d) 76

Q.23. The length and the breadth of the floor of a rectangular hall are 126 feet and 90 feet, respectively. What will be the area (in square feet) of each of the largest identical square tiles that can be used to tile this floor in a way that no part of the floor remains uncovered?

SSC CPO 03/10/2023 (3rd Shift)

- (a) 196 (b) 256 (c) 324 (d) 484

Q.24. The radii of the ends of a frustum of a solid right-circular cone 45 cm high are 28 cm and 7 cm. If this frustum is melted and reconstructed into a solid right circular cylinder whose radius of base and height are in the ratio 3 : 5, find the curved surface area (in cm^2) of this cylinder. [Use $\pi = \frac{22}{7}$]

SSC CPO 03/10/2023 (1st Shift)

- (a) 4610 (b) 4620 (c) 4580 (d) 4640

Q.25. A square and a rhombus have the same base and the rhombus is inclined at 45° , then what will be the ratio of the area of the square to the area of the rhombus?

SSC MTS 14/09/2023 (3rd Shift)

- (a) $\sqrt{3} : 1$ (b) $1 : \sqrt{3}$ (c) $\sqrt{2} : \sqrt{3}$ (d) $\sqrt{2} : 1$

Q.26. A closed wooden box measures externally 10 cm long, 8 cm broad and 6 cm high. If the thickness of the wood is 0.5 cm, then the volume of wood required is

SSC MTS 14/09/2023 (3rd Shift)

- (a) 165 cm^3 (b) 300 cm^3
(c) 230 cm^3 (d) 150 cm^3

Q.27. The weight of a cube varies directly as the product of its volume and its density. The ratio of densities of the materials of the first cube and second cube is 27:16. If the weight of the first cube is 4 times the weight of the second cube, then what is the ratio of the edges of the first cube and second cube?

SSC MTS 14/09/2023 (2nd Shift)

- (a) 2 : 3 (b) 3 : 2 (c) 3 : 4 (d) 4 : 3

Q.28. The weight of a circular disc varies directly as the product of the square of the radius and its thickness. Two identical/similar discs have their thickness in the ratio of 16 : 9. What is the ratio of their radii if the weight of the first is four times that of the second?

SSC MTS 12/09/2023 (3rd Shift)

- (a) 3 : 2 (b) 2 : 3 (c) 1 : 2 (d) 2 : 1

Q.29. A hemispherical dome of a building needs to be painted. If the circumference of the base of the dome is 154 cm, then find the cost of painting it if the cost of painting is ₹4 per 100 cm^2 (use $\pi = \frac{22}{7}$).

SSC CHSL 10/08/2023 (4th Shift)

- (a) ₹150.92 (b) ₹150.66
(c) ₹105.29 (d) ₹105.66

Q.30. The area of a sector of a circle is 616 cm^2 with a central angle of 10° . The radius of the circle is _____. (use $\pi = \frac{22}{7}$)

SSC CHSL 08/08/2023 (2nd Shift)

- (a) 84 cm (b) 21 cm (c) 48 cm (d) 28 cm

Q.31. Find the volume of the largest right circular cone that can be cut out from a cube whose edge is $3\frac{1}{2} \text{ cm}$, correct to two places of decimals (use $\pi = \frac{22}{7}$).

SSC CHSL 07/08/2023 (4th Shift)

- (a) 13.21 cm^3 (b) 21.31 cm^3
(c) 11.23 cm^3 (d) 12.13 cm^3

Q.32. The sum of the radius of the base and the height of a cylinder is 42 m. If the total surface area of the cylinder is 6336 m^2 , find the curved surface area of the cylinder correct to two places of decimals (use $\pi = \frac{22}{7}$)

SSC CHSL 02/08/2023 (4th Shift)

- (a) 2157.43 m^2 (b) 2571.43 m^2
(c) 2715.43 m^2 (d) 2517.43 m^2

Q.33. Three cubes with sides in the ratio of 3 : 4 : 5 are melted to form a single cube whose diagonal is $18\sqrt{3} \text{ cm}$. The sides of the three cubes are:

SSC CGL 20/07/2023 (1st shift)

- (a) 21 cm, 28 cm and 35 cm
(b) 9 cm, 12 cm and 15 cm
(c) 18 cm, 24 cm and 30 cm
(d) 12 cm, 16 cm and 20 cm

Q.34. The perimeter of equilateral triangle is $3 \times (\sqrt[4]{\frac{16}{3}})$ units. Determine the area of the triangle.

Matriculation Level 30/06/2023 (Shift - 4)

- (a) 4 unit^2 (b) 1 unit^2 (c) 3 unit^2 (d) 2 unit^2

Q.35. A spherical ball of diameter 8 cm is cut into two equal parts. The curved area of one such part has to be painted with green colour, while the other part has to be painted with red colour. The bases of both the hemispheres are to be painted with blue colour. The cost of painting with blue is ₹2/ cm^2 , while the

cost of painting the curved area is ₹3/cm². What will be the cost (in ₹) of painting the hemispheres? Take $\pi = 3.14$
Matriculation Level 28/06/2023 (Shift - 4)
(a) ₹451.92 (b) ₹492.92
(c) ₹803.84 (d) ₹401.92

Q.36. A cube of side 5 cm, has been assembled using 125 cubes each of 1 cm edge. One cube is removed from the middle of each of the faces of the cube. What is the surface area (in cm²) of the remaining solid?
SSC CHSL Tier II (26/06/2023)
(a) 155 (b) 144 (c) 150 (d) 174

Q.37. A prism and a pyramid have the same base and the same height. Find the ratio of the volumes of the prism and the pyramid.
SSC CGL Tier II (07/03/2023)
(a) 2 : 3 (b) 3 : 1 (c) 1 : 3 (d) 3 : 2

Q.38. How many metres of 2 m wide cloth will be required to make a conical tent with the diameter of the base as 14 m and slant height as 9 m to ignore wastage?
SSC CGL 12/12/2022 (1st Shift)
(a) 66 m (b) 88 m (c) 99 m (d) 77 m

Q.39. The minute hand of a clock is 20 cm long. Find the area on the face of the clock swept by the minute hand between 8 a.m. and 8:45 a.m.
SSC CGL 12/12/2022 (1st Shift)
(a) $\frac{6600}{7}$ cm² (b) $\frac{6600}{9}$ cm²
(c) $\frac{6600}{18}$ cm² (d) $\frac{6600}{14}$ cm²

Q.40. The external diameter of an iron pipe is 20 cm and its length is 12 cm. If the thickness of the pipe is 1 cm, find the surface area of the pipe (take $\pi = \frac{22}{7}$) correct to two places of decimal.
SSC CGL 06/12/2022 (2nd Shift)
(a) 1,662.67 cm² (b) 1,552.57 cm²
(c) 1,442.48 cm² (d) 1,772.76 cm²

Q.41. If the length of a certain rectangle is decreased by 4 cm and breadth is increased by 2 cm, it would result in a square of the same area. What is the perimeter of the original rectangle?
SSC CGL 06/12/2022 (1st Shift)
(a) 15 cm (b) 24 cm (c) 20 cm (d) 10 cm

Q.42. Two identical circles each of radius 30 cm intersect each other such that the circumference of each one passes through the center of the other. What is the area of the intersecting region?
SSC CGL 03/12/2022 (4th Shift)
(a) $400\pi - 250\sqrt{3}$ cm²
(b) $300\pi - 150\sqrt{3}$ cm²

(c) $500\pi - 350\sqrt{3}$ cm²
(d) $600\pi - 450\sqrt{3}$ cm²

Q.43. If the sum of the diagonals of a rhombus is L and the perimeter is 4P, find the area of the rhombus?
SSC CPO 11/11/2022 (Morning)

(a) $\frac{1}{4}(L^2 - P^2)$ (b) $\frac{1}{4}(L^2 - 4P^2)$
(c) $\frac{1}{2}(L^2 - 4P^2)$ (d) $\frac{1}{4}(L^2 + 3P^2)$

Q.44. Water is flowing through a cylindrical canal with an internal diameter of 7 m at the speed of 18 km/h. Find the volume of water flowing through the canal in 30 minutes. (take $\pi = \frac{22}{7}$)

SSC CPO 09/11/2022 (Afternoon)
(a) 3,76,500 m³ (b) 3,56,500 m³
(c) 3,66,500 m³ (d) 3,46,500 m³

Q.45. ABCDEF is a regular hexagon. Side of the hexagon is 36 cm. What is the area of the triangle ABC?
SSC CGL Tier II (08/08/2022)

(a) $324\sqrt{3}$ cm² (b) $360\sqrt{3}$ cm²
(c) $240\sqrt{3}$ cm² (d) $192\sqrt{3}$ cm²

Q.46. The frustum of a right circular cone has the radius of the base as 5 cm, radius of the top as 3 cm, and height as 6 cm. What is its volume?
SSC CHSL 27/05/2022 (Afternoon)
(a) 98π cm³ (b) 100π cm³
(c) 96π cm³ (d) 90π cm³

Q.47. Three sides of a triangle are $\sqrt{a^2 + b^2}$, $\sqrt{(2a)^2 + b^2}$ and $\sqrt{a^2 + (2b)^2}$ units. What is the area (in unit squares) of a triangle?
SSC CGL Tier II (03/02/2022)
(a) $\frac{5}{2}ab$ (b) $3ab$ (c) $4ab$ (d) $\frac{3}{2}ab$

Q.48. A square park has been divided into two rectangles of equal area. If the perimeter of each of these rectangles is 39 m, then what will be the perimeter of the square park?
SSC CHSL 09/08/2021 (Afternoon)
(a) 104 m (b) 39 m (c) 78 m (d) 52 m

Q.49. The circumference of a circle is 'a π ' units and the area of the circle is 'b π ' square units. If a:b is equal to 4:5, then the radius of the circle is:
SSC CHSL 06/08/2021 (Evening)
(a) 3 cm (b) 2.5 cm (c) 5 cm (d) 2 cm

Q.50. The area of a triangular field whose sides are 65m, 72m and 97m is equal to the area of a rectangular park whose sides are in the ratio of 5 : 13. What is the perimeter (in m) of the rectangular park?

SSC CHSL 19/04/2021 (Morning)
(a) 108 (b) 180 (c) 216 (d) 144

Q.51. A bicycle wheel has a radius of 42 cm. It makes 40 revolutions in 25 seconds. What is its speed (in kmph, up to one decimal place)?
SSC CHSL 16/04/2021 (Morning)
(a) 3.5 (b) 11.6 (c) 15.2 (d) 9.5

Q.52. A wire is bent to form a square of area 169 cm². If the same wire is bent to form a circle, then what is its area in cm² (to the nearest whole number)?
SSC CHSL 13/04/2021 (Afternoon)
(a) 215 (b) 227 (c) 532 (d) 531

Q.53. A sector of radius 10.5 cm with the central angle 120° is folded to form a cone by joining the two bounding radii of the sector. What is the volume (in cm³) of the cone so formed?
SSC CGL Tier II (13/09/2019)

(a) $\frac{343\sqrt{2}}{6}\pi$ (b) $\frac{343\sqrt{3}}{6}\pi$
(c) $\frac{343\sqrt{3}}{12}\pi$ (d) $\frac{343\sqrt{2}}{12}\pi$

Q.54. The base of a right prism is a triangle with sides 20 cm, 21 cm and 29 cm. If its volume is 7560 cm³, then its lateral surface area (in cm²) is:
SSC CGL Tier II (12/09/2019)
(a) 2484 (b) 2556 (c) 2520 (d) 2448

Q.55. If the radius of a sphere is increased by 4 cm, its surface area is increased by 464 π cm². What is the volume (in cm³) of the original sphere?
SSC CGL Tier II (11/09/2019)
(a) $\frac{15625}{6}\pi$ (b) $\frac{35937}{8}\pi$
(c) $\frac{11979}{2}\pi$ (d) $\frac{15625}{8}\pi$

Q.56. The base of right prism is a trapezium whose parallel sides are 11 cm and 15cm and the distance between them is 9 cm. If the volume of the prism is 1731.6 cm³, then the height (in cm) of the prism will be:
SSC CGL Tier II (11/09/2019)
(a) 15.6 (b) 15.2 (c) 14.8 (d) 14.2

Q.57. A ribbon of equal width and length l is wrapped around the curved surface of a right circular cylinder to cover it completely. If the base circumference of this cylinder is c, then how many times the ribbon was wrapped around the cylinder?
SSC MTS 05/08/2019 (Evening)
(a) $\frac{l}{4c}$ (b) $\frac{l}{c}$ (c) $\frac{l}{2c}$ (d) $\frac{2l}{c}$

Q.58. ABCD is a rhombus with each side being equal to 8 cm. If $BD = 10$ cm, $AC = 2\sqrt{x}$ cm, what is the value of $\sqrt{x + 10}$?
SSC CHSL 11/07/2019 (Afternoon)
(a) $2\sqrt{3}$ (b) $3\sqrt{2}$ (c) 7 (d) 5

Q.59. The two diagonals of a rhombus are respectively, 14 cm and 48 cm. The perimeter of the rhombus is equal to :
SSC CHSL 11/07/2019 (Morning)
(a) 120cm (b) 160cm (c) 80cm (d) 100cm

Q.60. The area of a sector of a circle with central angle 60° is A. The circumference of the circle is C. Then A is equal to :
SSC CHSL 10/07/2019 (Evening)

(a) $\frac{c^2}{6\pi}$ (b) $\frac{c^2}{18\pi}$ (c) $\frac{c^2}{24\pi}$ (d) $\frac{c^2}{4\pi}$

Q.61. Equilateral triangles are drawn on the hypotenuse and one of the perpendicular sides of a right-angled isosceles triangle. Their areas are H and A respectively. $\frac{A}{H}$ is equal to :

SSC CHSL 10/07/2019 (Afternoon)
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{\sqrt{2}}$

Q.62. Twelve sticks, each of length one unit, are used to form an equilateral triangle. The area of the triangle is :
SSC CHSL 10/07/2019 (Morning)
(a) $3\sqrt{3}$ (b) $2\sqrt{3}$ (c) $4\sqrt{3}$ (d) $8\sqrt{3}$

Q.63. If the length of a rectangle is decreased by 11% and the breadth is increased by 11%, its area will undergo :
SSC CHSL 09/07/2019 (Afternoon)
(a) 13.13% increase (b) 1.21% increase
(c) 1.21% decrease (d) 13.13% decrease

Q.64. In triangle ABC, the length of BC is less than twice the length of AB by 2 cm. The length of AC exceeds the length of AB by 10 cm. The perimeter is 32cm. The length (in cm) of the smallest side of the triangle is :
SSC CHSL 05/07/2019 (Afternoon)
(a) 4 (b) 10 (c) 8 (d) 6

Q.65. The parallel sides of a trapezium are 20 cm and 10 cm and its non-parallel sides are equal to each other. If its area is 180 cm^2 , then what is the length (in cm) of each non parallel side?
SSC CHSL 04/07/2019 (Morning)
(a) 11 (b) 13 (c) 12 (d) 15

Q.66. The length of a rectangular park is 20m more than its breadth. If the cost of fencing the park at Rs. 53 per metre is Rs. 21,200, then what is the area (In square metres) of the park ?
SSC CHSL 03/07/2019 (Evening)
(a) 9504 (b) 8925 (c) 9240 (d) 9900

Q.67. A wire is in the shape of a rectangle whose sides are in the ratio 7 : 4. It was initially in the shape of a circle of radius , very nearly equal to 31.5 cm. The length of smaller side of the rectangle is : (Take $\pi = \frac{22}{7}$)

SSC CHSL 03/07/2019 (Afternoon)
(a) 44 cm (b) 36 cm (c) 40 cm (d) 32cm

Q.68. A circle circumscribes a rectangle whose sides are in the ratio 4 : 3. If the perimeter of the rectangle is 56 cm, then what is the area (In square cm) of the circle ?
SSC CHSL 03/07/2019 (Morning)
(a) 70π (b) 96π (c) 90π (d) 100π

Q.69. The curved surface area and volume of a cylindrical pole are 132 square metres and 528 cubic metres, respectively. What is the height (In m) of the pole ? (Take $\pi = \frac{22}{7}$)

SSC CHSL 02/07/2019 (Afternoon)
(a) $2\frac{1}{2}$ (b) $3\frac{5}{8}$ (c) $3\frac{1}{2}$ (d) $2\frac{5}{8}$

Q.70. The area of a field in the shape of a hexagon is $2400\sqrt{3}$ square meters. What will be the cost(Rs.) of fencing it at Rs. 18.50 per meter?
SSC CHSL 01/07/2019 (Evening)
(a) 4440 (b) 5920 (c) 5550 (d) 5180

Q.71. Six cubes, each of edge 2 cm, are joined end to end. What is the total surface area of the resulting cuboid
SSC CGL 13/06/2019 (Morning)
(a) 96cm^2 (b) 144cm^2
(c) 104cm^2 (d) 128cm^2

Q.72. The radius of a sphere is reduced by 40%. By what percent, will its volume decrease?
SSC CGL 10/06/2019 (Afternoon)
(a) 60% (b) 64% (c) 72.5% (d) 78.4%

Q.73. The area of a field in the shape of a triangle with each side x meter is equal to the area of another triangular field having sides 50m, 70m and 80m. The value of x is closest to :
SSC CGL 07/06/2019 (Afternoon)
(a) 65.5 (b) 63.2 (c) 62.4 (d) 61.8

Q.74. A sector is cut out from a circle of diameter 42 cm. If the angle of the sector is 150° , then its area (in square cm) is :
(Take $\pi = \frac{22}{7}$)
SSC CGL 07/06/2019 (Morning)
(a) 564 (b) 574 (c) 580.6 (d) 577.5

Q.75. The radii of three concentric circles are in the ratio of 4:5:7. What is the ratio

of the area between the two inner circles to that between the two outer circles ?
SSC CGL 06/06/2019 (Evening)
(a) 4 : 7 (b) 5 : 9 (c) 4 : 5 (d) 3 : 8

Q.76. The length of the metallic pipe is 7.56 m. Its external and internal radii are 2.5 cm and 1.5 cm respectively. If 1 cubic cm of the metal weigh 7.5 g, then the weight(kg) of the pipe is
SSC CGL 06/06/2019 (Afternoon)
(a) 72.82 (b) 70.14 (c) 71.28 (d) 69.68

Q.77. The volume of a metallic cylindrical pipe is 7480 cm^3 . If its length is 1.4 m and its external radius is 9 cm, then its thickness (given $\pi = \frac{22}{7}$) is:

SSC CGL 06/06/2019 (Morning)
(a) 1 cm (b) 0.8 cm (c) 0.9 cm (d) 1.2cm

Q.78. The areas of the three adjacent faces of a cuboid are 32cm^2 , 24cm^2 and 48cm^2 . What is the volume (in cm^3) of the cuboid ?
SSC CGL 04/06/2019 (Evening)
(a) 192 (b) 256 (c) 288 (d) 128

Q.79. The area of a triangle is 15 sq cm and the radius of its incircle is 3 cm. Its perimeter is equal to :
SSC CGL 04/06/2019 (Evening)
(a) 12 cm (b) 20 cm (c) 5 cm (d) 10 cm

Q.80. How much iron sheet (in m^2) will be needed to construct a rectangular tank measuring $10 \text{ m} \times 8 \text{ m} \times 6 \text{ m}$, if a circular opening of radius one meter is to be left at the top of the tank? (correct to one decimal place)
SSC CGL 04/06/2019 (Afternoon)
(a) 371.6 (b) 370.4 (c) 372.9 (d) 370.8

Q.81. A solid cube of volume 13824 cm^3 is cut into 8 cubes of equal volumes. The ratio of the surface area of the original cube to the sum of the surface areas of three of the smaller cubes is :
SSC CGL 04/06/2019 (Morning)
(a) 2 : 3 (b) 4 : 3 (c) 8 : 3 (d) 2 : 1

Q.82. The unequal side of an isosceles triangle is 2cm. The medians drawn to the equal sides are perpendicular. The area of the triangle is :
SSC CPO 16/03/2019 (Evening)
(a) 2cm^2 (b) 3cm^2 (c) 5cm^2 (d) 1cm^2

Q.83. A steel vessel has a base of length 60 cm and breadth 30 cm. Water is poured in the vessel. A cubical steel box having an edge of 30 cm is immersed completely in the vessel. By how much will the water rise?
SSC CPO 16/03/2019 (Evening)
(a) 12 cm (b) 15 cm (c) 10 cm (d) 9 cm

Q.84. Original breadth of a rectangular box is 20cm. The box was then remade in such a way that its length increased by 30% but the breadth decreased by 20% and the area increased by 100 cm^2 . What is the new area(in cm^2) of the box ?
SSC CPO 16/03/2019 (Evening)
(a) 2500 (b) 2200 (c) 2600 (d) 2400

Q.85. Find the cost of carpeting a room which is 11m long and 6m broad by a carpet which is 60cm broad at the rate of rs 112.50 per meter.
SSC CPO 16/03/2019 (Afternoon)
(a) 12,375 (b) 13,280
(c) 11,695 (d) 12,040

Q.86. A field is $119\text{m} \times 18\text{m}$ in dimension. A tank $17\text{m} \times 6\text{m} \times 3\text{m}$ is dug out in the middle and the soil removed is evenly spread over the remaining part of the field. The increase in the level on the remaining part of the field is: (in cm)
SSC CPO 16/03/2019 (Afternoon)
(a) 14 cm (b) 13 cm (c) 15 cm (d) 12 cm

Q.87. A swimming pool is 40 m in length, 30 m in breadth and 2.2 m in depth. The cost of cementing its floor and the four sides at Rs. $25/\text{m}^2$ is:
SSC CPO 16/03/2019 (Afternoon)
(a) Rs.43,980 (b) Rs.37,540
(c) Rs.34,260 (d) Rs.37,700

Q.88. A square cardboard with side 3 m is folded through one of its diagonal to make a triangle, the height(m) of the triangle is :
SSC CPO 16/03/2019 (Morning)
(a) $\frac{3}{\sqrt{2}}$ (b) $2\sqrt{3}$ (c) $3\sqrt{2}$ (d) $\frac{2}{\sqrt{3}}$

Q.89. If the area of a regular hexagon is $108\sqrt{3}\text{cm}^2$, its perimeter (in cm) is:
SSC CPO 15/03/2019 (Evening)
(a) $36\sqrt{2}$ (b) $42\sqrt{3}$ (c) $28\sqrt{3}$ (d) 24

Q.90. A square piece of cardboard with side 12 cm has a small square of 2 cm cut out from each of the corners. The resulting flaps are turned up to make a box 2 cm deep. The volume (in cm^3) of the box is:
SSC CPO 15/03/2019 (Evening)
(a) 128 (b) 94 (c) 102 (d) 11

Q.91. The area (in cm^2) of a right angled triangle having base 24 cm and hypotenuse 25 cm is :
SSC CPO 15/03/2019 (Evening)
(a) 72 (b) 108 (c) 92 (d) 84

Q.92. The length of the longest pole that can be placed in a room 16 m long, 8 m

wide and 11 m high is:
SSC CPO 15/03/2019 (Evening)
(a) 21 m (b) 20 m (c) 18 m (d) 19 m

Q.93. If the height of an equilateral triangle is $10\sqrt{3}\text{ cm}$, the area (in cm^2) is:
SSC CPO 15/03/2019 (Morning)
(a) $124\sqrt{3}$ (b) $75\sqrt{3}$ (c) $80\sqrt{3}$ (d) $100\sqrt{3}$

Q.94. The dimensions of a swimming pool are $66\text{m} \times 35\text{m} \times 3\text{m}$. How many hours will it take to fill the pool by a pipe of diameter 35 cm with water flowing at speed 8 m/s ?
SSC CPO 15/03/2019 (Morning)
(a) 2.75 h (b) 3.5 h (c) 2.5 h (d) 3.2 h

Q.95. Which of the following has the maximum number of vertices ?
SSC CPO 14/03/2019 (Evening)
(a) Cuboid
(b) Triangular Prism
(c) Hexagonal pyramid
(d) Tetrahedron

Q.96. A diagonal of a quadrilateral is 40 cm. The length of the perpendicular to the opposite ends is 7.5 cm and 8.6 cm. What is the area (in cm^2) of quadrilateral?
SSC CPO 14/03/2019 (Evening)
(a) 434 (b) 322 (c) 368 (d) 28

Q.97. The area of each square of a chessboard having 64 squares is 4 cm^2 . If there is a border on all the sides of the chessboard of 2 cm, then the perimeter of the chessboard is:
SSC CPO 14/03/2019 (Morning)
(a) 128cm (b) 80cm (c) 70cm (d) 256cm

Q.98. The sides of a triangle are 24cm, 45cm and 51cm. At each of its vertices, circles of radius 10.5cm are drawn. What is the area of the triangle, excluding the portion covered by sectors of the circles?
SSC CPO 13/03/2019 (Evening)
(a) 244.75cm^2 (b) 366.75cm^2
(c) 464.75cm^2 (d) 327.75cm^2

Q.99. The radius of a cylinder is increased by 150 % and its height is decreased by 20 %. What is the percentage increase in its volume?
SSC CPO 12/03/2019 (Morning)
(a) 400% (b) 600% (c) 500% (d) 80%

Practice Questions

SSC CHSL 2024 Tier - 2

Q.100. The surface area of a cube is 1728 cm^2 . Find its volume.
SSC CHSL Tier II (18/11/2024)
(a) $6453\sqrt{2}\text{ cm}^3$ (b) $5436\sqrt{2}\text{ cm}^3$

(c) $3456\sqrt{2}\text{ cm}^3$ (d) $4356\sqrt{2}\text{ cm}^3$

Q.101. A cylinder has a radius $\frac{8}{\pi}\text{ cm}$ and its volume is 448 cm^3 . What is the height of the cylinder (taken $\pi = \frac{22}{7}$) ?
SSC CHSL Tier II (18/11/2024)
(a) 24 cm (b) 21 cm (c) 23 cm (d) 22 cm

Q.102. The radii of the two spheres are in the ratio 4 : 5. Their volume will be in the ratio:
SSC CHSL Tier II (18/11/2024)
(a) 64 : 125 (b) 125 : 64
(c) 16 : 25 (d) 4 : 5

Q.103. For making a toy, a hemisphere is attached at one end of a cylinder and a cone is attached at the other end of the cylinder. The cylinder, the cone and the hemisphere have a common radius of 4.2 cm. The height of the cylinder and that of the cone is 7 cm. Find the volume (in cm^3) of the toy. (Use $\pi = \frac{1}{22}$)
SSC CHSL Tier II (18/11/2024)
(a) 358.8 (b) 762.255
(c) 672.672 (d) 863.25

SSC MTS 2024 Tier - 1

Q.104. The area of a regular hexagon is $2048\sqrt{3}\text{ cm}^2$. What is the length (in cm) of each side of the hexagon ?
SSC MTS 30/09/2024 (Morning)
(a) $32\sqrt{3}$ (b) $\frac{64\sqrt{3}}{3}$ (c) $\frac{32\sqrt{3}}{3}$ (d) $64\sqrt{3}$

Q.105. A right circular cone of height h and radius R is cut by a plane parallel to the base at a distance $\frac{h}{4}$ from the base. The ratio of the curved surface areas of the resulting cone to the frustum is:
SSC MTS 30/09/2024 (Afternoon)
(a) 7 : 16 (b) 16 : 9 (c) 16 : 7 (d) 9 : 7

Q.106. What is the number of gold coins, each having a radius of 0.875 cm and a thickness of 2 mm, that need to be melted to create a cuboid with dimensions $5.5\text{ cm} \times 5\text{ cm} \times 3.5\text{ cm}$?
(Use $\pi = \frac{22}{7}$)
SSC MTS 30/09/2024 (Evening)
(a) 250 (b) 175 (c) 200 (d) 225

Q.107. A cylindrical tank of height 10 m and radius 5 m has a small circular cut of 0.5 m radius on its top surface. What would be the total surface area of the tank ?
SSC MTS 01/10/2024 (Morning)
(a) $\frac{599\pi}{4}\text{ m}^2$ (b) $156\pi\text{ m}^2$

(c) $\frac{299\pi}{2} \text{ m}^2$ (d) $150 \pi \text{ m}^2$

Q.108. A solid cylinder with a height of 8 units and a radius of 3.5 units is to be covered with a paper sheet. If the cost of the paper sheet is ₹40 per 5 square units, the cost (in ₹) of covering the cylinder is _____ (Take $\pi = \frac{22}{7}$)

SSC MTS 01/10/2024 (Afternoon)

(a) 2268 (b) 1664 (c) 2024 (d) 1012

Q.109. A hemispherical bowl is 264 cm round at the brim. Suppose that the bowl is half-full of beverage. If the beverage is served in hemispherical glasses 6 cm in the diameter at the top, how many people would be served?

SSC MTS 07/10/2024 (Morning)

(a) 1170 (b) 1372 (c) 1270 (d) 1470

Q.110. What is the area of the remaining portion of a circle of radius 14 cm when a square with maximum possible area inscribed in it is removed from it? (Use $\pi = \frac{22}{7}$)

SSC MTS 07/10/2024 (Morning)

(a) 518 cm^2 (b) 520 cm^2
(c) 224 cm^2 (d) 616 cm^2

Q.111. The volume of the hemisphere is $486\pi \text{ cm}^3$. If the radius is tripled, then what will be its curved surface area (in cm^2)?

SSC MTS 07/10/2024 (Afternoon)

(a) 1458 (b) 1025π (c) 1458π (d) 1546

Q.112. A hemispherical bowl of internal radius 15 cm contains a liquid. The liquid is to be filled into cylindrical shaped bottles of diameter 5 cm and height 6 cm. The number of bottles required to empty the bowl is:

SSC MTS 08/10/2024 (Morning)

(a) 40 (b) 30 (c) 60 (d) 50

Q.113. A solid cone of base radius 7 cm and height 18 cm is melted and recast into a hollow metal hemisphere of internal and external radii 20 cm and R cm, respectively. What is the value of R (in cm)?

SSC MTS 08/10/2024 (Afternoon)

(a) $\sqrt[3]{8144}$ (b) $\sqrt[3]{8441}$
(c) $\sqrt[3]{8448}$ (d) $\sqrt[3]{8568}$

Q.114. A solid spherical item is melted into the shape of a cylinder mounted by a cone having same radius. The diameters of the cylinder and sphere are 21 cm each. What is the height (in cm) of the cone if the height of the cylinder is 7 cm?

(Take $\pi = \frac{22}{7}$)

SSC MTS 08/10/2024 (Evening)

(a) 21 (b) 10.5 (c) 14 (d) 7

Q.115. ABCDEF is a regular hexagon, and $m(\overline{AD}) = 12 \text{ cm}$. What is the area (in cm^2) of the given hexagon?

SSC MTS 09/10/2024 (Morning)

(a) $216\sqrt{3}$ (b) $54\sqrt{3}$ (c) $108\sqrt{3}$ (d) $162\sqrt{3}$

Q.116. A right circular cone of height h and radius R is cut by a parallel to the base at a distance $\frac{h}{5}$ from the base. The

ratio of the curved surface area of the frustum and total surface area of the resulting cone is:

SSC MTS 09/10/2024 (Afternoon)

(a) $16(R+L) : 9L$ (b) $9(R+L) : 16L$
(c) $16L : 9(R+L)$ (d) $9L : 16(R+L)$

Q.117. The radius of the base and height of a right circular cone are in the ratio 8 : 15. Find the slant height of the cone if its volume is $1005\frac{5}{7} \text{ cm}^3$

SSC MTS 09/10/2024 (Afternoon)

(a) 12 cm (b) 17 cm (c) 21 cm (d) 15 cm

Q.118. A rectangular sheet of length 28 cm and breadth 16 cm is folded to meet the edges so that it is made into a cylindrical pipe in two possible ways – one along the length and the other along the breadth. The larger volume is how much percentage of smaller volume?

SSC MTS 09/10/2024 (Evening)

(a) 165% (b) 180% (c) 185% (d) 175%

Q.119. A conical tent is to accommodate 12 persons. Each person must have 5 m^2 of the space on the ground and 20 m^3 of air to breathe. The height of the tent should be:

SSC MTS 09/10/2024 (Evening)

(a) 12 m (b) 13 m (c) 12.5 m (d) 22 m

Q.120. A medicine capsule is shaped like a cylinder with two hemispheres attached to each of its ends. The length of the entire capsule is 15 mm, and its diameter is 6 mm. Calculate the surface area (in mm^2 , rounded to the nearest integer) of the capsule. (Use $\pi = \frac{22}{7}$)

SSC MTS 14/10/2024 (Afternoon)

(a) 286 (b) 283 (c) 289 (d) 280

Q.121. A hemispherical container of total surface area 7392 cm^2 is filled with oil.

However, $\frac{1}{4}$ of the container is empty.

This oil is to be transferred in small hemispherical bottles having radius of 6.3 cm. How many bottles will be required (rounded off to the nearest whole number)? (Take $\pi = \frac{22}{7}$.)

SSC MTS 14/10/2024 (Evening)

(a) 351 (b) 66 (c) 75 (d) 156

Q.122. A cone of radius 16 cm and height 12 cm is cut, parallel to the base, at some height closer to the top. If the ratio of the heights of the solids is 1 : 3, then the ratio of the total surface areas of the two solids formed is approximately given by:

SSC MTS 17/10/2024 (Morning)

(a) 9 : 143 (b) 1 : 8 (c) 2 : 15 (d) 7 : 180

Q.123. A conical cavity is drilled in a solid circular cylinder of 7 cm height and 48 cm base diameter. The height and base diameter of the cone are the same as those of the cylinder. What is the total surface area of the remaining solid?

SSC MTS 17/10/2024 (Afternoon)

(a) $1512\pi \text{ cm}^2$ (b) $1076\pi \text{ cm}^2$
(c) $1420\pi \text{ cm}^2$ (d) $1654\pi \text{ cm}^2$

Q.124. The side of a square exceeds the side of another square by 3 cm and the sum of the areas of the two squares is 225 cm^2 . Find the sum of the perimeter (in cm) of both squares.

SSC MTS 17/10/2024 (Evening)

(a) 84 (b) 74 (c) 64 (d) 94

Q.125. The external curved surface area of a metallic hemispherical bowl is 308 cm^2 . If the density of metal is 8.7 gm per cm^3 , then the mass of the bowl is equal to: (Take $\pi = \frac{22}{7}$)

SSC MTS 18/10/2024 (Morning)

(a) 6254.2 gm (b) 6252.4 gm
(c) 5262.4 gm (d) 5264.2 gm

Q.126. From a square with area a^2 another square made on mid points of the sides of the square is removed. What is the area left in the square?

SSC MTS 18/10/2024 (Morning)

(a) $\frac{a^2}{2}$ (b) $\frac{2a^2}{3}$ (c) $\frac{a^2}{4}$ (d) $\frac{3a^2}{4}$

Q.127. In a lead pipe 4.2 m long, the external diameter is 5.4 cm and the thickness of the pipe is 3 mm. Find the weight of the pipe, if 1 cm^3 of lead mass is 25 grams. (Use $\pi = \frac{22}{7}$)

SSC MTS 18/10/2024 (Afternoon)

(a) 54.90 kg (b) 45.09 kg
(c) 59.40 kg (d) 50.49 kg

Q.128. A solid sphere of iron having radius 15 cm is to be melted and recast in the shape of a cylindrical pipe having internal and external radii 4 cm and 6 cm, respectively. Find the length (in m) of the pipe thus formed.

SSC MTS 18/10/2024 (Evening)

(a) 2.15 (b) 2.25 (c) 2.45 (d) 2.35

Q.129. The sum of the radius and slant height of a cone is square of the height of the cone, while the difference between the radius and slant height is 1 cm only. What is the total surface area (in cm^2) of the cone having height of 9 cm ? (Take $\pi = \frac{22}{7}$) (Rounded off to 2 decimal places)

SSC MTS 21/10/2024 (Morning)

- (a) 10,437.43 (b) 11,282.85
(c) 10,182.86 (d) 10,128.85

Q.130. A cylinder of radius 10 cm and height 30 cm is cut, parallel to the base, into two pieces of different sizes. The total surface area of the larger piece formed is 70% of that of the given cylinder. The smaller piece is melted and made into a sphere. The radius (in cm) of the sphere is:

SSC MTS 21/10/2024 (Morning)

- (a) $\sqrt[3]{821}$ (b) $\sqrt[3]{900}$ (c) $\sqrt[3]{986}$ (d) $\sqrt[3]{844}$

Q.131. Find the volume of the largest right circular cone which is inscribed in a hollow cube of length 6 units and is standing on its base.

SSC MTS 21/10/2024 (Afternoon)

- (a) $18\pi \text{ units}^3$ (b) $9\pi \text{ units}^3$
(c) $21\pi \text{ units}^3$ (d) $10\pi \text{ units}^3$

Q.132. A cylindrical vessel of height 14 cm has outer and inner radii 14 cm and 10 cm, respectively. It is to be filled by transferring water in it by using a small cylinder of height 7 cm and outer and inner radii 7 cm and 5 cm, respectively. What will be the unfilled height of the big vessel after transferring water 6 times by using the smaller cylinder? (Use $\pi = \frac{22}{7}$)

SSC MTS 21/10/2024 (Evening)

- (a) 2.2 cm (b) 3.5 cm
(c) 2.9 cm (d) 3.8 cm

Q.133. A cylindrical tube, which is open at both ends, is crafted from a metal. The internal diameter of the tube is 12 cm, and its length is 35 cm. The metal throughout has a thickness of 2 cm. Determine the total surface area (in cm^2) of the metal tube. (Take $\pi = \frac{22}{7}$)

SSC MTS 22/10/2024 (Morning)

- (a) 3696 (b) 3256 (c) 3854 (d) 3145

Q.134. The perimeter of a rhombus is 160 cm and one of its diagonals has a length 50 cm. Find the area (in cm^2) of the rhombus.

SSC MTS 22/10/2024 (Morning)

- (a) $280\sqrt{39}$ (b) $300\sqrt{39}$
(c) $250\sqrt{39}$ (d) $175\sqrt{39}$

Q.135. The diagonals of two squares are in the ratio of 5 : 6. What is the ratio of

their areas ?

SSC MTS 22/10/2024 (Afternoon)

- (a) 18 : 20 (b) 25 : 36 (c) 23 : 18 (d) 2 : 3

Q.136. An ice-cream cone with a lid has a total surface area equal to the area of another circle of certain radius. If the radius of the circle is equal to average of the slant height and the radius of the cone, then the ratio of the slant height of the cone to the radius of the circle is:

SSC MTS 23/10/2024 (Morning)

- (a) 3 : 1 (b) 2 : 3 (c) 3 : 2 (d) 1 : 3

Q.137. The diameter of a hollow hemisphere is 30 cm. Water is filled in it up to its brim. When spherical lead shots each of radius 0.75 cm are dropped into the hemisphere, two-fifth of the water flows out. How many lead shots were dropped into the hemisphere ?

SSC MTS 29/10/2024 (Evening)

- (a) 1575 (b) 2250 (c) 1600 (d) 1380

Q.138. In a conical tent of diameter 13.2 m and vertical height 14 m, how many full bags of wheat can be emptied, if the space required for the wheat in each bag is 29.04 m^3 ? [Use $\pi = \frac{22}{7}$]

SSC MTS 30/10/2024 (Afternoon)

- (a) 25 (b) 28 (c) 22 (d) 21

Q.139. What is the radius of the largest right circular cone that can be cut out from a cube of edge 8.4 cm ?

SSC MTS 12/11/2024 (Morning)

- (a) 4.1 cm (b) 4.2 cm
(c) 2.2 cm (d) 2.1 cm

Q.140. A right circular solid metallic cone is 7.2 cm high and radius of its base 3.5 cm. It is melted and recast into two right circular solid metallic cones equal in all respects, each with base radius 2.1 cm and height 'h'. What is the value of 'h' (in cm) ?

SSC MTS 13/11/2024 (Morning)

- (a) 14 (b) 10 (c) 21 (d) 12

Q.141. The radius of a circle is equal to the side of a square. The perimeters of the square and circle are in the ratio 7 : 11, then the ratio of the areas will be :

(Use $\pi = \frac{22}{7}$)

SSC MTS 13/11/2024 (Afternoon)

- (a) 14 : 22 (b) 22 : 14 (c) 22 : 7 (d) 7 : 22

Q.142. The internal and external diameters of a bowl in the shape of a hemisphere are 26.6 cm and 28 cm, respectively. What is the cost (in ₹, to the nearest rupee) of painting it, all over, at the rate of ₹3.75/ cm^2 ? (Take $\pi = \frac{22}{7}$).

SSC MTS 13/11/2024 (Afternoon)

- (a) 9,380 (b) 9,015 (c) 8,950 (d) 9,000

Q.143. If the radii of two right circular cylinders are in the ratio of 2 : 3 and their heights are in the ratio of 5 : 4, then find the ratio of their curved surface areas ?

SSC MTS 14/11/2024 (Morning)

- (a) 6 : 5 (b) 5 : 6 (c) 6 : 10 (d) 10 : 6

Q.144. A hollow hemispherical steel bowl can contain $\frac{686}{3}\pi$ cubic unit of liquid.

The outer radius is 9 units. How much steel is used in making the bowl ?

SSC MTS 14/11/2024 (Afternoon)

- (a) $\frac{772}{3}\pi$ cubic units

- (b) $\frac{433}{3}\pi$ cubic units

- (c) $\frac{727}{3}\pi$ cubic units

- (d) $\frac{343}{3}\pi$ cubic units

Q.145. Air is leaking from a spherical-shaped advertising balloon at the rate of 24 cubic feet per minute. If the radius of the balloon is 8 feet, how long would it take for the balloon to empty? Round your answer to the nearest minute. Use the approximate value of π , that is 3.14.

SSC MTS 14/11/2024 (Afternoon)

- (a) 89 minutes (b) 80 minutes
(c) 65 minutes (d) 94 minutes

Q.146. A solid hemisphere of radius 12 cm is melted into N identical cones, each of radius 6 cm and height 8 cm. What is the value of N ?

SSC MTS 14/11/2024 (Evening)

- (a) 12 (b) 8 (c) 10 (d) 15

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Q.147. The radii of the two cones are in the ratio of 2 : 5 and their volumes are in the ratio of 3 : 5. What is the ratio of their heights?

SSC CGL 09/09/2024 (1st Shift)

- (a) 15 : 4 (b) 11 : 15 (c) 13 : 11 (d) 4 : 11

Q.148. What is the volume (in cm^3) of a cylinder if the radius of the cylinder is 8 cm and the height is 14 cm?

(Take $\pi = \frac{22}{7}$)

SSC CGL 09/09/2024 (2nd Shift)

- (a) 2686 (b) 2816 (c) 2784 (d) 2456

Q.149. From a circle with the radius of 15.75 cm, a sector with the arc length of 11 cm is cut off. Find the area (in cm^2) of this sector.

SSC CGL 09/09/2024 (2nd Shift)

- (a) 86.525 (b) 86.625

(c) 86.875 (d) 86.125

Q.150. A right-angled isosceles triangle has an area of 50 square units. Its hypotenuse is (in units):

SSC CGL 09/09/2024 (2nd Shift)

(a) $10\sqrt{3}$ (b) $5\sqrt{5}$ (c) $10\sqrt{2}$ (d) $5\sqrt{2}$

Q.151. What is the perimeter (in cm, rounded off to one decimal place) of the sector of a circle with a radius of 10 cm and of angle 30° ? (Use $\pi = 3.14$)

SSC CGL 09/09/2024 (3rd Shift)

(a) 20.4 (b) 35.7 (c) 82.8 (d) 25.2

Q.152. If a right circular cone of height 24 cm has a volume of 1232 cm^3 , then the total surface area of the cone is

(use $\pi = \frac{22}{7}$):

SSC CGL 10/09/2024 (1st Shift)

(a) 806 cm^2 (b) 704 cm^2 (c) 904 cm^2 (d) 608 cm^2

Q.153. A rectangular piece of paper is 50 cm long and 22 cm wide. If a cylinder is formed by rolling the paper along its breadth, then the volume of the cylinder is:

SSC CGL 10/09/2024 (3rd Shift)

(a) $2,125 \text{ cm}^3$ (b) $1,925 \text{ cm}^3$ (c) $2,025 \text{ cm}^3$ (d) $1,825 \text{ cm}^3$

Q.154. A rectangular courtyard is 4m 95cm long and 16m 65cm broad. It is to be paved with the square tiles of the same size. Find the least number of such square tiles required to pave the rectangular courtyard.

SSC CGL 11/09/2024 (1st Shift)

(a) 877 (b) 944 (c) 388 (d) 407

Q.155. The inner and outer radii of a circular track are 28 m and 25 m, respectively. The cost (in ₹) of levelling

the track at ₹28 per m^2 is: (Use $\pi = \frac{22}{7}$)

SSC CGL 11/09/2024 (1st Shift)

(a) 15,421 (b) 13,992

(c) 11,229 (d) 17,355

Q.156. If the perimeter of an equilateral triangle is 36 units, then its area is (in square units):

SSC CGL 11/09/2024 (1st Shift)

(a) $144\sqrt{2}$ (b) $36\sqrt{2}$ (c) $144\sqrt{3}$ (d) $36\sqrt{3}$

Q.157. If the ratio of the areas of two similar triangles is 25 : 144, the ratio of their corresponding sides is:

SSC CGL 11/09/2024 (2nd Shift)

(a) 5 : 12 (b) 5 : 144 (c) 25 : 12 (d) 12 : 5

Q.158. A cuboid of dimensions 50 cm, 150 cm, 175 cm can be divided into how many identical largest cubes?

SSC CGL 11/09/2024 (3rd Shift)

(a) 75 (b) 84 (c) 85 (d) 90

Q.159. The height of a cylinder is 20 cm. The lateral surface area is 1760 cm^2 . Its volume is:

SSC CGL 12/09/2024 (1st Shift)

(a) 12032 cm^3 (b) 12302 cm^3 (c) 12203 cm^3 (d) 12320 cm^3

Q.160. Two cylinders have the radii of their bases in the ratio 4 : 5 and their heights are in the ratio 5 : 7. Find the ratio of their curved surface areas.

SSC CGL 12/09/2024 (2nd Shift)

(a) 5 : 4 (b) 4 : 7 (c) 1 : 2 (d) 4 : 5

Q.161. The circumference of the base of a conical tent of height 8 m is $32 \pi \text{ m}$. Find its curved surface area.

SSC CGL 12/09/2024 (3rd Shift)

(a) $164\sqrt{5} \pi \text{ m}^2$ (b) $162\sqrt{3} \pi \text{ m}^2$ (c) $132\sqrt{3} \pi \text{ m}^2$ (d) $128\sqrt{5} \pi \text{ m}^2$

Q.162. A circular arc whose radius is 12 cm makes an angle of 30° at the centre. Find the perimeter (in cm) of the sector formed. (Use $\pi = 3.14$)

SSC CGL 12/09/2024 (3rd Shift)

(a) 32.38 (b) 30.28 (c) 28.64 (d) 26.24

Q.163. One-quarter of a circular pizza of radius 21 cm was removed from the whole pizza. What is the perimeter (in cm) of the remaining pizza? (Use $\pi = \frac{22}{7}$)

SSC CGL 13/09/2024 (1st Shift)

(a) 99 (b) 141 (c) 128 (d) 131

Q.164. What is the height (in m) of a conical water tank having radius of 70 m and volume of $46,200 \text{ m}^3$? (Take $\pi = \frac{22}{7}$)

SSC CGL 13/09/2024 (1st Shift)

(a) 27 (b) 9 (c) 18 (d) 54

Q.165. A and B are centres of two circles with radii 2 cm and 1 cm respectively, where $AB = 5 \text{ cm}$. C is the centre of another circle of radius r cm, which touches each of the above two circles externally. If $\angle ACB = 90^\circ$, then the value of r is:

SSC CGL 13/09/2024 (1st Shift)

(a) 5 cm (b) 2 cm (c) 3 cm (d) 4 cm

Q.166. From a circle of radius 7 units, an arc length is cut by a chord of length 7 units. What is the arc length of the smaller portion (in units)?

SSC CGL 13/09/2024 (1st Shift)

(a) $\frac{8}{3} \pi$ (b) $\frac{7}{6} \pi$ (c) $\frac{7}{3} \pi$ (d) $\frac{7}{4} \pi$

Q.167. A chord of the larger among two concentric circles is of length 20 cm and it is tangent to the smaller circle. What is the area (in cm^2) of the annular portion

between the two circles?

SSC CGL 13/09/2024 (2nd Shift)

(a) 100π (b) 164π (c) 122π (d) 175π

Q.168. What is the volume in cubic units of a cylinder with a height equal to the diameter and a radius is 4 units?

SSC CGL 13/09/2024 (2nd Shift)

(a) 128π (b) $\frac{64\pi}{3}$ (c) 32π (d) $\frac{32\pi}{3}$

Q.169. The angle turned by minutes hand in 25 minutes in a clock is:

SSC CGL 13/09/2024 (3rd Shift)

(a) π (b) $\frac{\pi}{2}$ (c) $\frac{5\pi}{3}$ (d) $\frac{5\pi}{6}$

Q.170. Find the measure of the central angle of a sector if its area is 16π and the radius is 8.

SSC CGL 17/09/2024 (1st Shift)

(a) 75° (b) 60° (c) 108° (d) 90°

Q.171. The circumference of the base of a cylindrical vessel is 44 cm and its height is 25 cm. How many liters of water can it hold? (Take $\pi = \frac{22}{7}$)

SSC CGL 17/09/2024 (1st Shift)

(a) 38.5 (b) 3.58 (c) 3.85 (d) 83.5

Q.172. M and N are, respectively, the centres of two circles of radii 12 cm and 8 cm. QR is the common tangent to the circles. If $MN = 16 \text{ cm}$, then what will be the length of QR?

SSC CGL 17/09/2024 (2nd Shift)

(a) $5\sqrt{17} \text{ cm}$ (b) $4\sqrt{15} \text{ cm}$ (c) $3\sqrt{19} \text{ cm}$ (d) $6\sqrt{8} \text{ cm}$

Q.173. AB and CD are two parallel chords drawn in a circle with centre O. The distance between the two chords is 21 cm. If the lengths of AB and CD are 24 cm and 18 cm, respectively, then the radius of the circle is equal to:

SSC CGL 17/09/2024 (2nd Shift)

(a) 15 cm (b) 20 cm (c) 18 cm (d) 24 cm

Q.174. Find the length of the longest stick that can be fitted in a cubical vessel of edge 70 cm.

SSC CGL 17/09/2024 (2nd Shift)

(a) $35\sqrt{3} \text{ cm}$ (b) $70\sqrt{3} \text{ cm}$ (c) $10\sqrt{3} \text{ cm}$ (d) $70\sqrt{2} \text{ cm}$

Q.175. In a rhombus PQRS, O is any interior point such that $OP = OR$. What is the degree measure of $\angle SOQ$?

SSC CGL 17/09/2024 (2nd Shift)

(a) 120° (b) 180° (c) 240° (d) 200°

Q.176. If the height and diameter of a cylinder are 12 cm and 28 cm, respectively, then find the ratio of the total surface area to the curved surface

area.

SSC CGL 17/09/2024 (2nd Shift)

(a) 13 : 5 (b) 10 : 3 (c) 13 : 6 (d) 11 : 6

Q.177. A circular arc makes an angle of 15° at the centre. If the radius of the circular arc is 14 cm, what is the perimeter (correct up to two decimal places) of the sector so formed? ($\pi = 3.14$)?

SSC CGL 17/09/2024 (3rd Shift)

(a) 17.66 cm (b) 21.32 cm
(c) 35.32 cm (d) 31.66 cm

Q.178. The perimeter of a rectangular garden is 48 m. If the length is 6 m more than the breadth, the area (in m^2) of the garden is:

SSC CGL 18/09/2024 (1st Shift)

(a) 135 (b) 96 (c) 84 (d) 112

Q.179. If three cubes of volume 512 cm^3 each are joined end to end, find the surface area of the resulting cuboid.

SSC CGL 18/09/2024 (1st Shift)

(a) 896 cm^2 (b) 986 cm^2
(c) 895 cm^2 (d) 869 cm^2

Q.180. The radius of the base of a right circular cone is 9 cm and its slant height is 49 cm. Find the curved surface area of the cone.

SSC CGL 18/09/2024 (2nd Shift)

(a) 1386 cm^2 (b) 1265 cm^2
(c) 1262 cm^2 (d) 1490 cm^2

Q.181. The area of a parallelogram with base 26 cm and height 18 cm is:

SSC CGL 19/09/2024 (1st Shift)

(a) 234 cm^2 (b) 936 cm^2
(c) 932 cm^2 (d) 468 cm^2

Q.182. The base radius of a circular cylinder is 10.5 cm. If the area of its curved surface is 792 cm^2 , then what is the volume of the cylinder?

SSC CGL 19/09/2024 (2nd Shift)

(a) 4185 cm^3 (b) 4518 cm^3
(c) 4152 cm^3 (d) 4158 cm^3

Q.183. In a rectangular field, the sum of the length and the breadth is 9 cm and area is 18 cm^2 . If the length is 'a' and breadth is 'b', then find the value of $(a^2 - ab + b^2)$.

SSC CGL 19/09/2024 (3rd Shift)

(a) 36 (b) 24 (c) 18 (d) 27

Q.184. If the area of a sector of a circle is 462 cm^2 and the central angle is 120° , then its radius is _____.

SSC CGL 19/09/2024 (3rd Shift)

(a) 18 cm (b) 17 cm (c) 21 cm (d) 14 cm

Q.185. The total surface area of a closed cube is given as 1152 cm^2 . What is the length (in cm) of each side of the cube?

SSC CGL 19/09/2024 (3rd Shift)

(a) $9\sqrt{2}$ (b) $4\sqrt{13}$ (c) $8\sqrt{3}$ (d) $8\sqrt{2}$

Q.186. The length, breadth and height of a room is 15 m, 9 m, and 5 m, respectively. From each can of paint 40 square metre of area is painted. How many cans of paint will be needed to paint only the walls of the room?

SSC CGL 23/09/2024 (2nd Shift)

(a) 12 (b) 6 (c) 4 (d) 8

Q.187. The volume of a right circular cylindrical box of radius 30cm is 600 cm^3 . Find the height of the box.

SSC CGL 24/09/2024 (1st Shift)

(a) 0.61cm (b) 0.31cm
(c) 0.51cm (d) 0.21cm

Q.188. The radii of two circles are 8 cm and 15 cm. The perimeter of the circle having area equal to the sum of the areas of the two circles (in cm) is:

SSC CGL 24/09/2024 (2nd Shift)

(a) 32π (b) 30π (c) 28π (d) 34π

Q.189. The ratio of the length, the breadth, and the height of a solid cuboid is given as 3 : 2 : 1, and the breadth of this cuboid is given as 12 cm. Find the volume (in cm^3) of the cuboid.

SSC CGL 24/09/2024 (3rd Shift)

(a) 1250 (b) 1296 (c) 1200 (d) 1500

Q.190. The circumference of the base of the cylindrical vessel is 154 cm and its height is 49 mm. How many litres of water can it hold? (correct to three

places of decimals, use $\pi = \frac{22}{7}$)

SSC CGL 25/09/2024 (1st Shift)

(a) 9.244 (b) 2.439 (c) 3.924 (d) 4.329

Q.191. A cylindrical tank of radius 14 cm is full of water. If 616 litres of water are drawn out, then the water level in the tank is dropped by ____ m. (Take $\pi = \frac{22}{7}$)

SSC CGL 25/09/2024 (2nd Shift)

(a) 1000 (b) 100 (c) 1 (d) 10

Q.192. Find the diameter (in cm) of a circle whose circumference is equal to the sum of the circumferences of two circles of radii 12 cm and 16 cm.

SSC CGL 25/09/2024 (2nd Shift)

(a) 32 (b) 56 (c) 64 (d) 28

Q.193. A well with 14 cm radius is dug 23 cm deep. Find the volume of the earth taken out of it.

SSC CGL 25/09/2024 (3rd Shift)

(a) $14,178 \text{ cm}^3$ (b) $14,108 \text{ cm}^3$
(c) $14,168 \text{ cm}^3$ (d) $15,168 \text{ cm}^3$

Q.194. What is the total surface area of a cone whose curved surface area is 550 cm^2 and radius is 7 cm?

(Take $\pi = \frac{22}{7}$)

SSC CGL 26/09/2024 (1st Shift)

(a) 764.71 cm^2 (b) 704.00 cm^2
(c) 682.60 cm^2 (d) 714.05 cm^2

Q.195. From a point A, which is at a distance of 17 cm from the centre C of a circle with radius 8 cm, the pair of tangents AB and AD to the circle are drawn. The area of the quadrilateral ABCD is _____ cm^2 .

SSC CGL 26/09/2024 (1st Shift)

(a) 192 (b) 120 (c) 60 (d) 360

Q.196. The central angle of a sector is 80° and whose length is 96π . What is the radius of the circle?

SSC CGL 26/09/2024 (2nd Shift)

(a) 196 units (b) 204 units
(c) 116 units (d) 216 units

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Q.197. A largest possible sphere is carved from a cube of side 14 cm. What is its volume in cm^3 ?

SSC CHSL 01/07/2024 (1st Shift)

(a) $205\frac{1}{3}$ (b) $1707\frac{1}{3}$
(c) $1437\frac{1}{3}$ (d) $1600\frac{1}{3}$

Q.198. The respective ratio between numerical values of the curved surface area and the volume of a right circular cylinder is 2 : 3. If the respective ratio between the radius and the height of the cylinder is 3 : 7, what is the total surface area of the cylinder?

SSC CHSL 01/07/2024 (2nd Shift)

(a) $62\pi \text{ cm}^2$ (b) $45\pi \text{ cm}^2$
(c) $65\pi \text{ cm}^2$ (d) $60\pi \text{ cm}^2$

Q.199. The surface areas of two spheres are in the ratio 25 : 36. What is the ratio of their volumes?

SSC CHSL 01/07/2024 (2nd Shift)

(a) $\frac{36}{115}$ (b) $\frac{125}{216}$ (c) $\frac{36}{125}$ (d) $\frac{25}{36}$

Q.200. The sides of a cuboid are in the ratio 3 : 4 : 5 and its surface area is equal to the surface area of the cube with side 6 units. What is the approximate volume of the Cuboid?

SSC CHSL 01/07/2024 (2nd Shift)

(a) 240 (b) 210 (c) 150 (d) 180

Q.201. The radius of a roller is 49 cm and its length is 200 cm. It takes 700 complete revolutions to move once over to level a playground. Find the area of the playground (Use $\pi = \frac{22}{7}$)

SSC CHSL 01/07/2024 (3rd Shift)

(a) $4312 \times 10^4 \text{ cm}^2$ (b) $4312 \times 10^2 \text{ cm}^2$
(c) $4322 \times 10^4 \text{ cm}^2$ (d) $4312 \times 10^3 \text{ cm}^2$

Q.202. The curved surface area of a cone is 2200 cm^2 and its radius is 28 cm , what is the slant height (in cm) of the cone? (Use $\pi = \frac{22}{7}$)

SSC CHSL 01/07/2024 (3rd Shift)

- (a) 22 (b) 23 (c) 24 (d) 25

Q.203. If the side of a cube is 3 , which of the following options is equal to the ratio between the surface area and the volume of the cube?

SSC CHSL 01/07/2024 (4th Shift)

- (a) $2:3$ (b) $2:1$ (c) $2:5$ (d) $2:7$

Q.204. The surface area of a sphere is 2464 cm^2 . Find its volume.

SSC CHSL 01/07/2024 (4th Shift)

- (a) $13121 \frac{1}{3} \text{ cm}^3$ (b) $11498 \frac{2}{3} \text{ cm}^3$
(c) $11892 \frac{2}{3} \text{ cm}^3$ (d) $11621 \frac{1}{3} \text{ cm}^3$

Q.205. Which of the following are the sides of a right - angled triangle?

SSC CHSL 01/07/2024 (4th Shift)

- (a) 16 cm , 31 cm , 34 cm
(b) 12 cm , 16 cm , 20 cm
(c) 6 cm , 7 cm , 10 cm
(d) 5 cm , 11 cm , 13 cm

Q.206. Find the diameter of a spherical ball whose volume is 38.808 cm^3 .

(Use $\pi = \frac{22}{7}$)

SSC CHSL 01/07/2024 (4th Shift)

- (a) 4.8 cm (b) 4.4 cm
(c) 4.6 cm (d) 4.2 cm

Q.207. The perimeter of a minor sector of a circle of radius 4 units subtending an angle of 45° is:

SSC CHSL 02/07/2024 (1st Shift)

- (a) $4 - \pi$ units (b) $8 + \pi$ units
(c) $8 - \pi$ units (d) $4 + \pi$ units

Q.208. If the diameter of a hemisphere is 21 cm , then the curved surface area (in cm^2) of the hemisphere is: Take $\pi = \frac{22}{7}$

SSC CHSL 02/07/2024 (1st Shift)

- (a) 639 (b) 939 (c) 693 (d) 963

Q.209. The capacity of a cylinder tank is 8316 cm^3 . If the radius of its base is 21 cm , then find the depth of the tank.

SSC CHSL 02/07/2024 (1st Shift)

- (a) 6 cm (b) 8 cm (c) 4 cm (d) 10 cm

Q.210. The volume of a cube is reduced to 72.9% . By how much is the side of the cube reduced?

SSC CHSL 02/07/2024 (2nd Shift)

- (a) 9.5% (b) 10% (c) 9% (d) 8.5%

Q.211. A hemisphere has a radius of 6.2 cm . Find its total surface area (use $\pi = 3.14$) (round off to two decimal places).

SSC CHSL 02/07/2024 (2nd Shift)

- (a) 276.35 cm^2 (b) 282.25 cm^2
(c) 358.52 cm^2 (d) 362.10 cm^2

Q.212. If the side of a square is increased by 5% , then the percentage increase in its area will be:

SSC CHSL 02/07/2024 (2nd Shift)

- (a) 11% (b) 10.25% (c) 10% (d) 10.5%

Q.213. If the brick size is $25 \text{ cm} \times 12 \text{ cm} \times 9 \text{ cm}$, then how many bricks are required to construct a wall of length 10 m , breadth 22.5 cm and height 6 m ?

SSC CHSL 02/07/2024 (2nd Shift)

- (a) 4500 (b) 6000 (c) 8000 (d) 5000

Q.214. An iron box with external dimensions 60 cm , 40 cm , and 20 cm is made of 1 cm thick sheet. If 1 cm^3 of iron weighs 50 gm , the weight of the empty box is:

SSC CHSL 02/07/2024 (3rd Shift)

- (a) 416.40 kg (b) 240 kg
(c) 214.05 kg (d) 400 kg

Q.215. If the diameter of a hemisphere is 42 cm , then the volume of hemisphere (in cm^3) is: Take $\pi = \frac{22}{7}$

SSC CHSL 02/07/2024 (3rd Shift)

- (a) 19154 (b) 19254 (c) 19444 (d) 19404

Q.216. Find the difference between the curved surface area and total surface area of a right circular cylinder having a length of 68 cm and base diameter of 28 cm .

SSC CHSL 02/07/2024 (3rd Shift)

- (a) 1432 cm^2 (b) 1332 cm^2
(c) 1132 cm^2 (d) 1232 cm^2

Q.217. The perimeter of a triangle is 36 units. Its area cannot be more than _____ square units.

SSC CHSL 02/07/2024 (3rd Shift)

- (a) $12\sqrt{7}$ (b) $36\sqrt{2}$ (c) $36\sqrt{3}$ (d) $18\sqrt{5}$

Q.218. The outer surface of a sphere having diameter 10 m is painted at the rate of $\text{₹} \frac{80}{\pi}$ per m^2 . What is the cost of painting?

SSC CHSL 02/07/2024 (4th Shift)

- (a) $\text{₹}8,000$ (b) $\text{₹}9,000$
(c) $\text{₹}4,000$ (d) $\text{₹}6,000$

Q.219. The triangle has sides 3 cm , 4 cm and 5 cm . What is the length of the perpendicular from the opposite vertex to the side whose length is 5 cm ?

SSC CHSL 02/07/2024 (4th Shift)

- (a) 3.5 cm (b) 2.4 cm
(c) 2.2 cm (d) 1.4 cm

Q.220. The radius and height of a cylinder are in the ratio $6:7$ and its

volume is 792 cm^3 . Calculate its curved surface area in cm^2

SSC CHSL 02/07/2024 (4th Shift)

- (a) 262 (b) 490 (c) 264 (d) 226

Q.221. The internal length, breadth and height of a cardboard box are 6 cm , 10 cm and 12 cm , respectively. How many boxes are needed in which cubes whose volume is 2160 cm^3 can be packed?

SSC CHSL 03/07/2024 (1st Shift)

- (a) 6 (b) 5 (c) 3 (d) 4

Q.222. The heights of two cones are in the ratio of $5:3$ and their radii are in the ratio $6:5$. Find the ratio of their volumes.

SSC CHSL 03/07/2024 (1st Shift)

- (a) $10:5$ (b) $13:5$ (c) $12:5$ (d) $11:5$

Q.223. If the radius of a sphere is doubled, then the volume of the sphere becomes _____ times the previous volume.

SSC CHSL 03/07/2024 (1st Shift)

- (a) two (b) one (c) eight (d) four

Q.224. The curved surface area of a right circular cylinder of height 14 cm , is 88 cm^2 . Find the total surface area of the cylinder.

SSC CHSL 03/07/2024 (2nd Shift)

- (a) $\frac{360}{7} \text{ cm}^2$ (b) $\frac{660}{7} \text{ cm}^2$
(c) $\frac{560}{7} \text{ cm}^2$ (d) $\frac{460}{7} \text{ cm}^2$

Q.225. The area of a rectangular field is 3600 m^2 . If the field is 45 m long, what is its perimeter?

SSC CHSL 03/07/2024 (2nd Shift)

- (a) 255 m (b) 260 m (c) 250 m (d) 265 m

Q.226. If the surface area of cuboid is 108 m^2 , then find the height of the cuboid, whose length is 6 m and breadth is 3 m .

SSC CHSL 03/07/2024 (2nd Shift)

- (a) 5.5 m (b) 5 m (c) 4.5 m (d) 4 m

Q.227. For a given circle of radius 4 cm , the angle of its sector is 45° . Find the area (in cm^2) of the sector. (Use $\pi = 3.14$.)

SSC CHSL 03/07/2024 (3rd Shift)

- (a) 7.28 (b) 6.28 (c) 6.18 (d) 7.18

Q.228. If the height of a right circular cone is 24 cm and radius of its base is 10.5 cm , then the volume of right circular

cone (in cm^3) is: Take $\pi = \frac{22}{7}$

SSC CHSL 03/07/2024 (3rd Shift)

- (a) 2727 (b) 2772 (c) 2277 (d) 2237

Q.229. If the radius of a sphere is increased by 12% , then the volume will be increased by approximately:

SSC CHSL 03/07/2024 (3rd Shift)

- (a) 41.49% (b) 45.49%

- (c) 40.49 percent (d) 42.5 percent

Q.230. A cylindrical iron rod whose height is twelve times its radius is melted and cast into spherical balls, the radius of each of which is one third of the radius of the cylinder. Find the number of spherical balls.

SSC CHSL 03/07/2024 (4th Shift)

- (a) 243 (b) 198 (c) 224 (d) 212

Q.231. If the area of a triangle whose base measures 6 cm is 18 cm^2 , then its height is:

SSC CHSL 03/07/2024 (4th Shift)

- (a) 6 cm (b) 3.1 cm (c) 9 cm (d) 2.2 cm

Q.232. The edges of a rectangular box are in the ratio of 3 : 5 : 8, and its surface area is 632 cm^2 .

What will be the volume of the box ?

SSC CHSL 04/07/2024 (1st Shift)

- (a) 801 cm^3 (b) 960 cm^3
(c) 990 cm^3 (d) 510 cm^3

Q.233. Three solid cubes of metal have sides 6 cm, 8 cm and 10 cm, respectively. They are melted to form a new cube. What is the surface area of this new cube?

SSC CHSL 04/07/2024 (1st Shift)

- (a) 812 cm^2 (b) 950 cm^2
(c) 915 cm^2 (d) 864 cm^2

Q.234. If the diameter of the base and curved surfaced area of a cylinder are 14cm and 352 cm^2 , respectively, then the total surface area of the cylinder is equal to:

SSC CHSL 04/07/2024 (1st Shift)

- (a) 660 cm^2 (b) 654 cm^2
(c) 760 cm^2 (d) 754 cm^2

Q.235. The radius of a sphere is increased by 25%. What is the increase percent in surface area?

SSC CHSL 04/07/2024 (2nd Shift)

- (a) 15.55% (b) 36.45%
(c) 56.25% (d) 25.25%

Q.236. What is the curved surface area of cylinder whose radius is 3cm and height is 14cm ? (Take $\pi = \frac{22}{7}$)

SSC CHSL 04/07/2024 (2nd Shift)

- (a) 262 cm^2 (b) 264 cm^2
(c) 263 cm^2 (d) 261 cm^2

Q.237. The total surface area of a solid hemisphere of diameter 28 cm is:

SSC CHSL 04/07/2024 (3rd Shift)

- (a) 1848 cm^2 (b) 1784 cm^2
(c) 1488 cm^2 (d) 1648 cm^2

Q.238. Find the capacity of an overhead cylindrical water tank, in litres, whose radius is 2.1 m and height is 6.3 m.

SSC CHSL 04/07/2024 (3rd Shift)

- (a) 61802 (b) 611726
(c) 87318 (d) 65622

Q.239. The diameter of a cylinder is 14 cm and its height is 9 cm. Find the total surface area of the cylinder.

SSC CHSL 04/07/2024 (3rd Shift)

- (a) 704 cm^2 (b) 714 cm^2
(c) 708 cm^2 (d) 700 cm^2

Q.240. From a circle, a sector with a central angle of 30° is cut off. If the length of the arc of this sector is 5.5 cm, then find the radius (in cm) of the circle.

[Use $\pi = \frac{22}{7}$]

SSC CHSL 04/07/2024 (4th Shift)

- (a) 12 (b) 10 (c) 11 (d) 10.5

Q.241. If the height of a solid right circular cylinder is 27 cm and diameter of its base is 30 cm, the total surface area (in cm^2) of the right circular cylinder is:

(Take $\pi = \frac{22}{7}$)

SSC CHSL 04/07/2024 (4th Shift)

- (a) 3978 (b) 3960 (c) 3976 (d) 3964

Q.242. 3 solid spheres of diameters 6 cm, 8 cm and 10 cm are melted together to form a single solid sphere. Find its radius.

SSC CHSL 04/07/2024 (4th Shift)

- (a) 3 cm (b) 12 cm (c) 9 cm (d) 6 cm

Q.243. The ratio of the lateral surface area to the total surface area of a cylinder with base diameter 60 cm and height 10 cm is:

SSC CHSL 04/07/2024 (4th Shift)

- (a) 1 : 9 (b) 1 : 3 (c) 1 : 7 (d) 1 : 4

Q.244. The length, breadth and height of a cuboid are in ratio of 3 : 4 : 5. If the volume of the cuboid is 1620 m^3 , find the total surface area of the cuboid.

SSC CHSL 05/07/2024 (1st Shift)

- (a) 646 m^2 (b) 846 m^2
(c) 423 m^2 (d) 712 m^2

Q.245. A cylindrical oil tank has a radius of 5 m and a total surface area of 628 m^2 , what would be its curved surface area? (round off to the second decimal place)

SSC CHSL 05/07/2024 (1st Shift)

- (a) 401.43 m^2 (b) 371.83 m^2
(c) 470.8 m^2 (d) 481.43 m^2

Q.246. Rajesh is playing with a football having diameter of 28 cm. He wants to get a photo of his favourite player painted on the entire ball. The painter charges ₹2/cm². What will be the cost (in ₹) of painting the ball (take $\pi = \frac{22}{7}$) ?

SSC CHSL 05/07/2024 (1st Shift)

- (a) 4,928 (b) 2,414 (c) 4,828 (d) 2,464

Q.247. A solid copper sphere of radius 9 cm is drawn into a wire of radius 4 mm. Find the length of the wire

SSC CHSL 05/07/2024 (2nd Shift)

- (a) 70.25 m (b) 60.75 m
(c) 20.75 m (d) 65.25 m

Q.248. Find the height of the triangle whose base is two-thirds of its height and whose area is 363 cm^2

SSC CHSL 05/07/2024 (2nd Shift)

- (a) 33 cm (b) 27 cm (c) 29 cm (d) 31 cm

Q.249. Find the percentage increase in surface area of a cube by triple each side.

SSC CHSL 05/07/2024 (3rd Shift)

- (a) 700% (b) 800% (c) 650% (d) 500%

Q.250. If the length, breadth and total surface area of a cuboid are 30 cm, 20 cm and 2600 cm^2 , respectively, then find its height.

SSC CHSL 05/07/2024 (3rd Shift)

- (a) 50 cm (b) 30 cm (c) 14 cm (d) 15 cm

Q.251. Two similar triangles have area 490 sq. cm. and 640 sq. cm. If the length of the one side of the first triangle is 21 cm, then what is the length of the corresponding side of the second triangle?

SSC CHSL 05/07/2024 (4th Shift)

- (a) 16 cm (b) 14 cm (c) 24 cm (d) 21 cm

Q.252. The diameter of a roller is 35 cm and its length is 100 cm. It takes 200 complete revolutions to level a playground. Find the area of the playground in m^2

SSC CHSL 05/07/2024 (4th Shift)

- (a) 220 (b) 2,20,00,000
(c) 1,10,000 (d) 110

Q.253. The perimeter of a rectangle is 70 cm and its diagonal is 25 cm. The area of the rectangle is :

SSC CHSL 08/07/2024 (1st Shift)

- (a) 425 cm^2 (b) 420 cm^2
(c) 300 cm^2 (d) 625 cm^2

Q.254. A solid sphere of radius 8 cm is melted and then recast into small spherical balls each of diameter 4 cm. Find the number of balls thus obtained

(used $\pi = \frac{22}{7}$).

SSC CHSL 08/07/2024 (1st Shift)

- (a) 64 (b) 48 (c) 32 (d) 24

Q.255. If the radius of a sphere is increased by 3.5 cm, its surface area increases by 550 cm^2 . What was the original radius of the sphere ?

(use $\pi = \frac{22}{7}$)

SSC CHSL 08/07/2024 (2nd Shift)

(a) 4 cm (b) 4.5 cm (c) 6 cm (d) 3.5 cm

Q.256. The perimeter of a square whose area is the same as the area of a semi-circle of radius $2\sqrt{2}$ cm is:

SSC CHSL 08/07/2024 (2nd Shift)

(a) $8\sqrt{\pi}$ cm (b) $8\sqrt{2\pi}$ cm
(c) $16\sqrt{\pi}$ cm (d) $4\sqrt{\pi}$ cm

Q.257. A metallic solid cuboid of dimensions 36 cm × 22 cm × 12 cm is melted and recast in the form of cubes of side 6 cm. Find the number of the cubes so formed.

SSC CHSL 08/07/2024 (2nd Shift)

(a) 45 (b) 46 (c) 43 (d) 44

Q.258. The area of an isosceles right angled triangle is 16 m^2 . Its hypotenuse is _____.

SSC CHSL 08/07/2024 (3rd Shift)

(a) 0.8 m (b) 8 m (c) 4 m (d) 8 cm

Q.259. Find the cost of painting a ball that is in the shape of a sphere with a radius of 14 cm. The painting cost of the ball is ₹5 per square centimetre

(take $\pi = \frac{22}{7}$).

SSC CHSL 08/07/2024 (4th Shift)

(a) ₹12,230 (b) ₹12,320
(c) ₹13,022 (d) ₹13,220

Q.260. Two cubes, each of edge 5 cm, are joined face to face. Find the surface area of the cuboid thus formed.

SSC CHSL 08/07/2024 (4th Shift)

(a) 250 cm^2 (b) 300 cm^2
(c) 125 cm^2 (d) 200 cm^2

Q.261. The length of a rectangular shaped

park exceeds its breadth by 15 m. If the perimeter of the park is 110 m, find the length of the park (in m).

SSC CHSL 09/07/2024 (1st Shift)

(a) 35 (b) 40 (c) 25 (d) 20

Q.262. If the lateral surface area of a cube is 144 cm^2 , then find its volume.

SSC CHSL 09/07/2024 (1st Shift)

(a) 864 cm^3 (b) 1728 cm^3
(c) 117.58 cm^3 (d) 216 cm^3

Q.263. If the areas of three adjacent faces of a cuboid are 5 cm^2 , 15 cm^2 , 27 cm^2 , respectively, the surface area of the cuboid is:

SSC CHSL 09/07/2024 (1st Shift)

(a) 75 cm^2 (b) 47 cm^2
(c) 45 cm^2 (d) 94 cm^2

Q.264. A closed cylindrical tank of radius

1.5 m and height 3 m is made from a sheet of metal, how much sheet of metal is required to do it?

SSC CHSL 09/07/2024 (2nd Shift)

(a) 42.24 square metre
(b) 24.24 square metre
(c) 42.42 square metre
(d) 24.42 square metre

Q.265. Two cubes have their volumes in the ratio 1 : 125. Find the ratio of their surface areas.

SSC CHSL 09/07/2024 (2nd Shift)

(a) 1 : 125 (b) 1 : 25 (c) 1 : 5 (d) 25 : 1

Q.266. If the radius of the base and slant height of the cone is 6 cm and 10 cm, respectively, then the volume of the cone is equal to:

SSC CHSL 09/07/2024 (3rd Shift)

(a) $108 \pi \text{ cm}^3$ (b) $98 \pi \text{ cm}^3$
(c) $106 \pi \text{ cm}^3$ (d) $96 \pi \text{ cm}^3$

Q.267. Find the number of paper balls, each 2 cm in diameter, that can be made from a sphere of diameter 18 cm.

SSC CHSL 09/07/2024 (4th Shift)

(a) 1020 (b) 484 (c) 625 (d) 729

Q.268. A cuboidal water purifier can carry 10 litres of water. Its breadth is $\frac{1}{5}$

of its height and length is $\frac{1}{3}$ of the sum of the breadth and height. The height of the tank is:

SSC CHSL 10/07/2024 (1st Shift)

(a) 20 cm (b) 10 cm (c) 80 cm (d) 50 cm

Q.269. A cube of edge 4 cm is cut into cubes each of edge of 2 cm. The ratio of the total surface area of one of the small cubes to that of the large cube is equal to:

SSC CHSL 10/07/2024 (2nd Shift)

(a) 1 : 4 (b) 1 : 64 (c) 1 : 8 (d) 1 : 16

Q.270. The curved surface area of a right circular cone is $20\pi \text{ cm}^2$. If its radius is 4 cm, then find the volume of cone.

SSC CHSL 10/07/2024 (2nd Shift)

(a) $21 \pi \text{ cm}^3$ (b) $20 \pi \text{ cm}^3$
(c) $16 \pi \text{ cm}^3$ (d) $18 \pi \text{ cm}^3$

Q.271. The ratio of the outer and the inner circumference of a circular path is 7 : 6. If the path is 45 m wide, then what is the radius (in m) of the inner circle?

SSC CHSL 10/07/2024 (2nd Shift)

(a) 240 (b) 270 (c) 150 (d) 120

Q.272. A solid metallic cube having surface area of 54 cm^2 is melted to form smaller cubes of surface area $\frac{8}{12} \text{ cm}^2$ each. Find the number of smaller cubes.

SSC CHSL 10/07/2024 (4th Shift)

(a) 512 (b) 343 (c) 216 (d) 729

Q.273. Find the surface area (in cm^2) of a cardboard box in the shape of a cuboid whose length is 60 cm, breadth is half its length and height is one-third of its breadth.

SSC CHSL 11/07/2024 (1st Shift)

(a) 2700 (b) 10800 (c) 8100 (d) 5400

Q.274. From a circle of radius r units, the largest equilateral triangle is cut out. What is the length (in units) of the side of the triangle?

SSC CHSL 11/07/2024 (1st Shift)

(a) $\frac{r}{\sqrt{2}}$ (b) $\sqrt{2}r$ (c) $\sqrt{3}r$ (d) $\frac{r}{\sqrt{3}}$

Q.275. If the area of a rhombus is 70 m^2 and its height is 5 m, its perimeter is ____.

SSC CHSL 11/07/2024 (2nd Shift)

(a) 64 m (b) 35 m (c) 56 m (d) 25 m

Q.276. The measure of each side of an equilateral triangle is 16 cm. What is the ratio of its perimeter to the sum of its medians?

SSC CHSL 11/07/2024 (2nd Shift)

(a) $\sqrt{3} : 2$ (b) $2\sqrt{3} : 1$ (c) $2 : \sqrt{3}$ (d) $1 : 2\sqrt{3}$

Q.277. A hemispherical bowl is made of silver and has an inner diameter of 4 cm if the thickness of silver is 0.5 cm. Find the volume of silver used in making the bowl correct to two places of decimals (use $\pi = 3.14$)

SSC CHSL 11/07/2024 (2nd Shift)

(a) 15.69 cm^3 (b) 51.96 cm^3
(c) 15.96 cm^3 (d) 51.69 cm^3

Q.278. The volume of the rectangular block is 12288 m^3 . Its dimensions are in the ratio of 4 : 3 : 2. If the entire surface is polished at the rate 2 paise per m^2 , then find the total cost of polishing:

SSC CHSL 11/07/2024 (4th Shift)

(a) ₹44.42 (b) ₹11.14
(c) ₹66.56 (d) ₹33.28

SSC CPO 2024 Tier - 1

Q.279. A hemispherical bowl is 88 cm round the brim. Assuming it to be full, how many persons may be served from it in hemispherical glasses, 7 cm in diameter at the top? (Where $\pi = \frac{22}{7}$)

SSC CPO 27/06/2024 (1st Shift)

(a) 74 (b) 68 (c) 64 (d) 70

Q.280. The chord length of a chord made on an arc of a circle is equal to the radius of the circle. The length of the arc is _____ (in units), if the radius of the circle is 21 units. (Take $\pi = \frac{22}{7}$)

SSC CPO 27/06/2024 (1st Shift)

(a) 20 (b) 22 (c) 24 (d) 21

Q.281. The base of a parallelogram is twice its height. If the area of the parallelogram is 338 cm^2 , then find its height (in cm).

SSC CPO 27/06/2024 (1st Shift)

(a) 13 (b) 11 (c) 14 (d) 12

Q.282. What is the total surface area of a cone whose radius is $\frac{r}{4}$ and slant height is $4l$?

SSC CPO 27/06/2024 (1st Shift)

(a) $8\pi r(l+r)$ (b) $\pi r(l + \frac{r}{16})$ (c) $\pi r(l + \frac{r}{4})$ (d) $4\pi r(l+r)$

Q.283. How many metres of cloth will be required to make a conical tent, the radius of whose base is 21 metres and height is 28 metres. The width of the cloth is 5 metres. (Where $\pi = \frac{22}{7}$)

SSC CPO 27/06/2024 (1st Shift)

(a) 470 (b) 462 (c) 456 (d) 478

Q.284. The volume of a cube is four times the volume of a cuboid. If the sides of the cuboid are 32 cm, 8 cm and 4 cm, then find the ratio of the total surface area of the cube to that of the cuboid.

SSC CPO 27/06/2024 (2nd Shift)

(a) 23 : 13 (b) 24 : 13

(c) 21 : 13 (d) 22 : 13

Q.285. A copper sphere of diameter 12 cm is drawn into a wire of diameter 4 mm. What is the length (in cm) of the wire? (Where $\pi = \frac{22}{7}$)

SSC CPO 27/06/2024 (2nd Shift)

(a) 7200 (b) 7823 (c) 8342 (d) 9000

Q.286. The length of the largest possible rod that can be placed in a cubical room is $42\sqrt{3}$ m. The surface area (in m^2) of the largest possible sphere that fit within the cubical room is: [Use $\pi = \frac{22}{7}$]

SSC CPO 27/06/2024 (2nd Shift)

(a) 3590 (b) 4589 (c) 2564 (d) 5544

Q.287. Find the area of a minor sector of a circle whose circumference is 88 cm and the length of its minor arc is 22 cm (use $\pi = \frac{22}{7}$).

SSC CPO 27/06/2024 (2nd Shift)

(a) 154 cm^2 (b) 451 cm^2 (c) 415 cm^2 (d) 145 cm^2

Q.288. A rectangular sheet of metal is 60 cm by 25 cm. Equal squares of side 5 cm are cut off at the corners and the remainder is folded up to form an open

rectangular box. Find the volume of the box.

SSC CPO 27/06/2024 (3rd Shift)

(a) 2750 cm^3 (b) 5750 cm^3 (c) 3750 cm^3 (d) 4750 cm^3

Q.289. A drainage tile is a cylindrical shell 42 cm long. The inside and outside diameters are 8 cm and 14 cm, respectively. What is the volume (in cm^3) of clay required for the tile? ((Use $\pi = \frac{22}{7}$)

SSC CPO 27/06/2024 (3rd Shift)

(a) 5241 (b) 4356 (c) 4881 (d) 4125

Q.290. One laddoo with a radius of 5 cm belongs to a shopkeeper. How many laddoos of radius 2.5 cm may be created from the same one laddoo?

SSC CPO 28/06/2024 (1st Shift)

(a) 2 (b) 6 (c) 4 (d) 8

Q.291. The cost of painting a spherical vessel of diameter 14 cm is ₹21,560. What is the cost of painting (in ₹) per square centimetre? (Use $\pi = \frac{22}{7}$)

SSC CPO 28/06/2024 (1st Shift)

(a) 30 (b) 32 (c) 28 (d) 35

Q.292. The radius of the base and height of a cone are 5 cm and 6 cm, respectively, whereas the radius of the base and height of a cylinder are 2.5 cm and 3 cm, respectively. The ratio of the volume of the cone to that of the cylinder is: (Where $\pi = \frac{22}{7}$)

SSC CPO 28/06/2024 (1st Shift)

(a) 8 : 5 (b) 9 : 4 (c) 8 : 3 (d) 3 : 5

Q.293. A cylinder of radius 7 cm has a curved surface area of 2200 cm^2 . Find its total surface area. (Use $\pi = \frac{22}{7}$)

SSC CPO 28/06/2024 (2nd Shift)

(a) 2508 cm^2 (b) 2260 cm^2 (c) 2550 cm^2 (d) 1580 cm^2

Q.294. A hollow sphere of external and internal diameters of 10 cm and 6 cm, respectively, is melted and made into another solid in the shape of a right circular cone of base diameter 10 cm. Find the height of the cone.

SSC CPO 28/06/2024 (2nd Shift)

(a) 13.68 cm (b) 14.68 cm

(c) 16.68 cm (d) 15.68 cm

Q.295. A spherical box has a surface area 528 cm^2 . Find the radius of the spherical box.

SSC CPO 28/06/2024 (2nd Shift)

(a) $\sqrt{37}$ cm (b) $\sqrt{42}$ cm(c) $\sqrt{29}$ cm (d) $\sqrt{35}$ cm

Q.296. The distance between the parallel sides of a trapezium is 18 cm. If the area of the trapezium is 1188 cm^2 , then what is the sum of the lengths of the parallel sides?

SSC CPO 28/06/2024 (3rd Shift)

(a) 150 cm (b) 115 cm

(c) 126 cm (d) 132 cm

Q.297. The area of the canvas cloth needed to erect a right conical tent of height 20ft. and circular base of circumference 30π ft. is:

SSC CPO 28/06/2024 (3rd Shift)

(a) 455π sq. ft. (b) 375π sq. ft.(c) 589π sq. ft. (d) 470π sq. ft.

Q.298. The volume of a right circular cone is $150\pi \text{ cm}^3$ and its height is 18 cm. Find its slant height (in cm) (correct to 2 decimal places).

SSC CPO 28/06/2024 (3rd Shift)

(a) 18.68 (b) 25.35 (c) 28.68 (d) 15.25

Q.299. A school has two rectangular playgrounds, which are similar in shape. If the length and the breadth of the first playground are given as 224 m and 160 m, respectively, then which of the following options CANNOT be the dimensions of the second playground?

SSC CPO 29/06/2024 (1st Shift)

(a) Length = 273 m; Breadth = 195 m

(b) Length = 168 m; Breadth = 120 m

(c) Length = 210 m; Breadth = 150 m

(d) Length = 343 m; Breadth = 255 m

Q.300. The height and radius of a cone-shaped storage tank are 9 ft. and 7 ft., respectively. Find the volume of water the tank can hold? (Use $\pi = \frac{22}{7}$)

SSC CPO 29/06/2024 (1st Shift)

(a) 462 cubic ft. (b) 527 cubic ft.

(c) 1386 cubic ft. (d) 1423 cubic ft.

Q.301. The breadth of a room is twice its height and half its length. The volume of the room is 1728 m^3 . The length of the room is:

SSC CPO 29/06/2024 (1st Shift)

(a) 20 m (b) 24 m (c) 34 m (d) 28 m

Q.302. A woman working in a company was supplied a rectangular solid (cuboid shape) of wax with dimensions 70 cm, 44 cm and 20 cm to prepare cylindrical candles, each 14 cm in diameter and 20 cm of height. Find the number of candles.

SSC CPO 29/06/2024 (1st Shift)

(a) 15 (b) 20 (c) 10 (d) 30

Q.303. A cone of slant height 4.2 units has a lateral surface area 13.2 units^2 . The radius of the base is:

SSC CPO 29/06/2024 (1st Shift)

- (a) 1 unit (b) 1.5 units
(c) 2.5 units (d) 2 units

Q.304. The in-radius of a triangle is 8 cm, and the sum of the lengths of its sides is 125 cm. The area of the triangle (in cm^2) is

SSC CPO 29/06/2024 (2nd Shift)

- (a) 480 (b) 1000 (c) 250 (d) 500

Q.305. Melting a metal ball of radius 3 units, 27 identical dice and a ball of radius 1 units are made. What is the side of dice?

SSC CPO 29/06/2024 (2nd Shift)

- (a) 1.7 units (b) 1.5 units
(c) 1.6 units (d) 1.4 units

Q.306. The ratio of the slant height and the height of a cone is 4 : 3. If the curved surface area of the cone is $4\sqrt{7}\pi$ square units, then the radius of the cone is _____ units.

SSC CPO 29/06/2024 (3rd Shift)

- (a) $7\sqrt{7}$ (b) 7 (c) $\sqrt{7}$ (d) $\frac{\sqrt{7}}{7}$

Q.307. The length of a side of an equilateral triangle is 18 cm. The area (in cm^2) of the region lying between the circumcircle and the incircle of the triangle is [use $\pi = \frac{22}{7}$]

SSC CPO 29/06/2024 (3rd Shift)

- (a) $254\frac{3}{7}$ (b) $254\frac{1}{7}$
(c) $254\frac{2}{7}$ (d) $254\frac{4}{7}$

Q.308. The area of a circle is 256π square cm. The length of its longest chord (in cm) is:

SSC CPO 29/06/2024 (3rd Shift)

- (a) 34 (b) 36 (c) 6 (d) 32

SSC Selection Post (Phase - XII)

Q.309. The perimeter of one face of a cube is 40 cm. Its volume will be:

Matriculation Level 20/06/2024 (Shift - 1)

- (a) 10 cm^3 (b) 100 cm^3
(c) 1000 cm^3 (d) 10000 cm^3

Q.310. If the length of each of the two equal sides of an isosceles triangle is 15 cm and the adjacent angle is 30° , then the area of the triangle is:

Matriculation Level 20/06/2024 (Shift - 1)

- (a) 66.25 cm^2 (b) 36.25 cm^2
(c) 56.25 cm^2 (d) 26.25 cm^2

Q.311. The lateral surface area of a cone is 550 cm^2 . If the diameter of the cone is 14 cm, then the height of the cone is:

Matriculation Level 20/06/2024 (Shift - 1)

- (a) 24 cm (b) 12.5 cm
(c) 12 cm (d) 25 cm

Q.312. The radius of a hemisphere is 9 cm. Its volume will be:

Matriculation Level 20/06/2024 (Shift - 4)

- (a) $712\pi\text{ cm}^3$ (b) $243\pi\text{ cm}^3$
(c) $486\pi\text{ cm}^3$ (d) $972\pi\text{ cm}^3$

Q.313. If the cylinder height is 9 cm and the base area is $25\pi\text{ cm}^2$, then find the total surface area.

Matriculation Level 20/06/2024 (Shift - 4)

- (a) 240 cm^2 (b) 140 cm^2
(c) 440 cm^2 (d) 340 cm^2

Q.314. The height of one cone is 3 times the height of another cone, while its radius is half of the radius of the other cone. If their total volume is 100 unit^3 , then the difference in the volumes of the cones is _____ unit^3 .

Matriculation Level 21/06/2024 (Shift - 1)

- (a) 12.5 (b) 14.3 (c) 13.4 (d) 15.5

Q.315. A cube is having surface area of 5400 square units. Its volume would increase approximately _____ if the side is increased by 5 units.

Matriculation Level 21/06/2024 (Shift - 1)

- (a) 59% (b) 45% (c) 55% (d) 50%

Q.316. If the radius of a sphere is increased by 2 cm, its surface area increases by 704 cm^2 . What was the radius of the sphere before the increase?

(Use $\pi = \frac{22}{7}$)

Matriculation Level 21/06/2024 (Shift - 1)

- (a) 14 cm (b) 11 cm (c) 12 cm (d) 13 cm

Q.317. What must be the total percentage change in the volume of a cuboid if its length and breadth are decreased by 10% and 20%, respectively, while its height is increased by 40%?

Matriculation Level 21/06/2024 (Shift - 4)

- (a) 0.8% increase (b) 0.8% decrease
(c) 1.8% increase (d) 1.8% decrease

Q.318. There is a large cone with a diameter of 14 m and height of 12 m. It is being filled with water at a rate of 2 m^3 in every 20 seconds. How long will it take to fill the cone?

Matriculation Level 21/06/2024 (Shift - 4)

- (a) 313 sec (b) 6160 sec
(c) 616 sec (d) 12320 sec

Q.319. The volume of a cuboid is twice the volume of a cube. If the dimensions of the cuboid are 12 cm, 16 cm and 18 cm, then the total surface area of the cube is:

Matriculation Level 21/06/2024 (Shift - 4)

- (a) 625 cm^2 (b) 864 cm^2
(c) 484 cm^2 (d) 764 cm^2

Q.320. The height of a solid cylinder is 4 times its radius. Its curved surface area is _____ times its base area.

Matriculation Level 21/06/2024 (Shift - 4)

- (a) 2 (b) 6 (c) 8 (d) 4

Q.321. The lateral surface area of a cylinder is 3862.2 cm^2 and its height is 15 cm. Find its volume (take $\pi = 3.14$) correct to one decimal place.

Matriculation Level 24/06/2024 (Shift - 1)

- (a) 79175.1 cm^3 (b) 88275.2 cm^3
(c) 78275.2 cm^3 (d) 89175.1 cm^3

Q.322. The length of cold storage of cuboidal shape is double its breadth and its height is half of its breadth. If its volume is 216 m^3 , then what are the dimensions of the cold storage?

Matriculation Level 24/06/2024 (Shift - 1)

- (a) $12\text{ m} \times 6\text{ m} \times 3\text{ m}$ (b) $12\text{ m} \times 9\text{ m} \times 5\text{ m}$
(c) $14\text{ m} \times 7\text{ m} \times 5\text{ m}$ (d) $8\text{ m} \times 4\text{ m} \times 2\text{ m}$

Q.323. The volume of a right circular cylinder of base radius r is obtained by multiplying its curved surface area by :

Matriculation Level 24/06/2024 (Shift - 4)

- (a) $\frac{2}{r}$ (b) $\frac{1}{2r}$ (c) $2r$ (d) $\frac{r}{2}$

Q.324. How many cubes of length 50 cm can be accommodated in a container of 5 m in length, 2 m in width and 5 m in height ?

Matriculation Level 25/06/2024 (Shift - 1)

- (a) 400 (b) 500 (c) 450 (d) 250

Q.325. Find the area of a right angle whose base is 15 cm and hypotenuse is 17 cm.

Matriculation Level 26/06/2024 (Shift - 1)

- (a) 40 cm^2 (b) 55 cm^2
(c) 60 cm^2 (d) 75 cm^2

Q.326. The dimensions of a rectangular field when extended by 6 metres are in the ratio of 4 : 3 and when decreased by 6 metres are in the ratio of 2 : 1. Find the dimensions of the field.

Higher Secondary 20/06/2024 (Shift - 2)

- (a) 16 metres, 12 metres
(b) 10 metres, 6 metres
(c) 18 metres, 12 metres
(d) 14 metres, 10 metres

Q.327. The radius of a circle is increased in such a way that its circumference increases by 7%. As a result, the area of the circle also increases by:

Higher Secondary 20/06/2024 (Shift - 2)

- (a) 18.69% (b) 10.28%
(c) 14.49% (d) 12.20%

Q.328. The radii of the two concentric circles are 27 cm and 18 cm. The area (in cm^2) of the closed figure bounded by the boundaries of the circle is:

Higher Secondary 20/06/2024 (Shift - 2)
(a) 505π (b) 405π (c) 288π (d) 504π

Q.329. A hemispherical bowl has a 21 cm radius. It is to be painted inside as well as outside. The cost of painting it at the rate of ₹0.05 per cm^2 and assuming that the thickness of the bowl is negligible, is: (Use $\pi = \frac{22}{7}$)

Higher Secondary 24/06/2024 (Shift - 2)
(a) ₹410.10 (b) ₹188.30
(c) ₹277.20 (d) ₹388.20

Q.330. If the area of a regular pentagon is $3920\sqrt{3}\text{ cm}^2$, then how long is its each side?

Higher Secondary 25/06/2024 (Shift - 2)
(a) 38 cm (b) 56 cm (c) 58 cm (d) 46 cm

Q.331. The base of a parallelogram is thrice its height. If the area of the parallelogram is 1020 cm^2 , then what is its height?

Higher Secondary 25/06/2024 (Shift - 2)
(a) $2\sqrt{85}\text{ cm}$ (b) $\sqrt{89}\text{ cm}$
(c) $\sqrt{82}\text{ cm}$ (d) $2\sqrt{84}\text{ cm}$

Q.332. A sphere and a cube have equal surface areas. The ratio of the volume of the sphere to that of the cube is:

Higher Secondary 25/06/2024 (Shift - 4)
(a) $\sqrt{6} : \sqrt{\pi}$ (b) $\sqrt{\pi} : \sqrt{8}$
(c) $\sqrt{3} : \sqrt{\pi}$ (d) $\sqrt{2} : \sqrt{\pi}$

Q.333. The surface area of a sphere is 2464 cm^2 . Calculate its volume.

(Use $\pi = \frac{22}{7}$)

Higher Secondary 26/06/2024 (Shift - 2)
(a) 11598.67 cm^3 (b) 11478.67 cm^3
(c) 11488.67 cm^3 (d) 11498.67 cm^3

Q.334. The area of the rhombus (in cm^2) having each side equal to 13 cm and one of its diagonals equal to 24 cm is:

Graduate Level 20/06/2024 (Shift - 3)
(a) 110 (b) 120 (c) 60 (d) 130

Q.335. Find the area (in cm^2) of a circle with a maximum radius that can be inscribed in a rectangle of length 18 cm and breadth 12 cm.

Graduate Level 20/06/2024 (Shift - 3)
(a) 72π (b) 28π (c) 136π (d) 36π

Q.336. The difference between the total surface area and lateral surface area of a cube, if the side of the cube is 11 cm (in square cm), is _____.

Graduate Level 26/06/2024 (Shift - 3)
(a) 244 (b) 243 (c) 242 (d) 241

Q.337. If the radius of the right circular cone is increased by 20% and its height is decreased by 25%, then the volume of

the right circular cone will be increased by:

Graduate Level 26/06/2024 (Shift - 4)
(a) 15% (b) 12% (c) 10% (d) 8%

Q.338. Half the perimeter of a rectangular garden, whose length is 8 cm more than its width, is 42 cm. Find the area of the rectangular garden.

Graduate Level 26/06/2024 (Shift - 4)
(a) 524 cm^2 (b) 425 cm^2
(c) 254 cm^2 (d) 542 cm^2

Q.339. The volume of a cube is 64 cm^3 . The total surface area of the cube is:

Graduate Level 26/06/2024 (Shift - 4)
(a) 16 cm^2 (b) 64 cm^2
(c) 96 cm^2 (d) 128 cm^2

SSC CHSL 2023 Tier - 2

Q.340. The diameter of a sphere is increased by 30%. What is the percentage increase in its volume?

SSC CHSL Tier II (02/11/2023)
(a) 69% (b) 119.7% (c) 41.7% (d) 90.5%

Q.341. The diameter of a hemispherical bowl is 21 cm. The volume of the bowl is equal to: (use $\pi = \frac{22}{7}$)

SSC CHSL Tier II (02/11/2023)
(a) $2,624.5\text{ cm}^3$ (b) $2,425.5\text{ cm}^3$
(c) $2,725.5\text{ cm}^3$ (d) $2,524.5\text{ cm}^3$

Q.342. The radius and height of a right circular cone is increased by 25%. Find the increase in percentage of the volume of the cone (rounded off to 2 decimal places).

SSC CHSL Tier II (02/11/2023)
(a) 94.31% (b) 95.31%
(c) 94.51% (d) 93.51%

Q.343. A hollow cube is made of paper to have a volume of 512 cubic units. How much paper in square units will be required to make the cube?

SSC CHSL Tier II (10/01/2024)
(a) 328 (b) 348 (c) 288 (d) 384

Q.344. A right rectangular box's measurements are 1.6 m long, 90 cm wide, 60 cm in height. Soap cakes of measurement $6\text{ cm} \times 5\text{ cm} \times 40\text{ mm}$ are to be packed in the box, so that no space is left. How many cakes can be packed in the box?

SSC CHSL Tier II (10/01/2024)
(a) 6000 (b) 5600 (c) 6500 (d) 7200

SSC CGL 2023 Tier - 2

Q.345. The ratio of the length, width and height of a closed cuboid is given as 6 : 3 : 2. The total surface area of this cuboid is given as 1800 cm^2 . Find the volume (in cm^3) of this cuboid.

SSC CGL Tier II (26/10/2023)
(a) 4650 (b) 4800 (c) 4500 (d) 4200

Q.346. A solid cube, whose each edge is of length 48 cm, is melted. Identical solid cubes, each of volume 64 cm^3 , are made out of this molten cube, without any wastage. How many such small cubes are obtained?

SSC CGL Tier II (26/10/2023)
(a) 1728 (b) 1718 (c) 1748 (d) 1738

SSC CPO 2023 Tier - 1

Q.347. The cost of painting the total surface area of a 30 m high solid right circular cylinder at the rate of ₹25 per m^2 is ₹18,425. What is the volume (in m^3) of

this cylinder [use $\pi = \frac{22}{7}$]?

SSC CPO 03/10/2023 (2nd Shift)
(a) 1210 (b) 1155 (c) 1145 (d) 1122

Q.348. What will be the difference between the total surface area and the curved surface area of a hemisphere having 4 cm diameter?

SSC CPO 03/10/2023 (2nd Shift)
(a) $4.4\pi\text{ cm}^2$ (b) $5\pi\text{ cm}^2$
(c) $4\pi\text{ cm}^2$ (d) $8\pi\text{ cm}^2$

Q.349. A conical shape vessel has a radius of 21 cm and has a slant height of 25 cm. If the curved part of the vessel is to be painted white, find the cost (in ₹) of painting at the rate of ₹1.5 per cm^2 .

SSC CPO 03/10/2023 (2nd Shift)
(a) 2475 (b) 825 (c) 1650 (d) 1250

Q.350. A solid sphere made of wax of radius 12 cm is melted and cast into solid hemispheres of radius 4 cm each. Find the number of such solid hemispheres.

SSC CPO 03/10/2023 (3rd Shift)
(a) 14 (b) 54 (c) 28 (d) 27

Q.351. Find the area of a rhombus if the perimeter of the rhombus is 52 cm, and one of its diagonals is 10 cm long.

SSC CPO 03/10/2023 (3rd Shift)
(a) 164 cm^2 (b) 144 cm^2
(c) 160 cm^2 (d) 120 cm^2

Q.352. The total surface area of a square-based right pyramid is 1536 m^2 , of which 37.5% is the area of the base of the pyramid. What is the volume (in m^3) of this pyramid?

SSC CPO 04/10/2023 (1st Shift)
(a) 3108 (b) 3048 (c) 3072 (d) 3144

Q.353. If the diameter of a sphere is reduced to its half, then the volume would be:

SSC CPO 04/10/2023 (2nd Shift)

- (a) reduced by $\frac{1}{8}$ of the former volume
 (b) increased by $\frac{1}{4}$ of the former volume
 (c) reduced by $\frac{1}{4}$ of the former volume
 (d) increased by $\frac{1}{8}$ of the former volume

Q.354. The heights of two cones are in the ratio 7 : 5 and their diameters are in the ratio 10 : 21. What is the ratio of their volumes ? (Where $\pi = \frac{22}{7}$)

SSC CPO 04/10/2023 (2nd Shift)

- (a) 17 : 21 (b) 14 : 19
 (c) 20 : 63 (d) 26 : 47

Q.355. The diameter of a hemisphere is equal to the diagonal of a rectangle of length 4 cm and breadth 3 cm. Find the total surface area (in cm^2) of the hemisphere.

SSC CPO 04/10/2023 (3rd Shift)

- (a) 25π (b) $\frac{50\pi}{4}$ (c) $\frac{25\pi}{4}$ (d) $\frac{75\pi}{4}$

Q.356. The height and the radius of the base of a right circular cone are in the ratio of 12 : 5. If its volume is 314 cm^3 , then what is the slant height of the cone ? (Use $\pi = 3.14$)

SSC CPO 05/10/2023 (1st Shift)

- (a) 11 cm (b) 14 cm (c) 12 cm (d) 13 cm

Q.357. A right square pyramid having lateral surface area is 624 cm^2 . If the length of the diagonal of the square is $24\sqrt{2}$, then the volume of the pyramid is:

SSC CPO 05/10/2023 (1st Shift)

- (a) 780 cm^3 (b) 1150 cm^3
 (c) 1083 cm^3 (d) 960 cm^3

Q.358. A cylinder has some water in it at a height of 16 cm. If a sphere of radius 9 cm is put into it, then find the rise in the height of the water if the radius of the cylinder is 12 cm.

SSC CPO 05/10/2023 (1st Shift)

- (a) 1.6 cm (b) 8.75 cm
 (c) 8 cm (d) 6.75 cm

Q.359. What will be the area of a plot of quadrilateral shape, one of whose diagonals is 20 m and lengths of the perpendiculars from the opposite vertices on it are 12 m and 18 m, respectively ?

SSC CPO 05/10/2023 (2nd Shift)

- (a) 250 m^2 (b) 400 m^2
 (c) 200 m^2 (d) 300 m^2

Q.360. Two metallic balls P and Q are such that the diameter of P is four times the diameter of Q. What is the ratio between the volumes of P and Q?

SSC CPO 05/10/2023 (2nd Shift)

- (a) 8 : 1 (b) 16 : 1 (c) 64 : 1 (d) 32 : 1

Q.361. If the slant height of a cone is 29 cm and its height is 20 cm, find the ratio between the magnitudes of total surface area and the volume.

SSC CPO 05/10/2023 (3rd Shift)

- (a) 3 : 7 (b) 3 : 14 (c) 5 : 14 (d) 7 : 15

Q.362. The radius of a metallic spherical ball is 3 cm. If the metallic ball is melted and recast into x number of hemispheres of radius equal to half the radius of the metallic spherical ball, then find the value of x.

SSC CPO 05/10/2023 (3rd Shift)

- (a) 14 (b) 16 (c) 13 (d) 15

Q.363. A hollow cylindrical tube made of plastic is 3 cm thick. If the external diameter is 20 cm and length of the tube is 49 cm, find the volume of the plastic.

SSC CPO 05/10/2023 (3rd Shift)

- (a) 7644 cm^3 (b) 7238 cm^3
 (c) 7854 cm^3 (d) 7546 cm^3

SSC MTS 2023 Tier - 1

Q.364. The diameter of a metallic sphere is 6 cm. It is melted and drawn into a wire having a diameter of the cross section as 0.4 cm. Find the length of the wire (in meters).

SSC MTS 01/09/2023 (1st Shift)

- (a) 18 (b) 9 (c) 12 (d) 10

Q.365. From a cuboid of dimension of $8 \times 6 \times 6$ units, small cubes of dimension $0.5 \times 0.5 \times 0.5$ unit are cut. How many small cubes can be cut from the big cuboid ?

SSC MTS 01/09/2023 (3rd Shift)

- (a) 576 (b) 1152 (c) 2304 (d) 1552

Q.366. The radius and the height of a right circular cone are in the ratio 3 : 4. If its volume is 301.44 m^3 , find the slant height (use $\pi = 3.14$).

SSC MTS 04/09/2023 (1st Shift)

- (a) 5 cm (b) 20 cm (c) 10 cm (d) 15 cm

Q.367. Water is pouring into a cuboidal reservoir at the rate of 50 litres per minute. The volume of the reservoir is 120 m^3 . How many hours will it take to fill the reservoir ?

SSC MTS 04/09/2023 (3rd Shift)

- (a) 35 (b) 45 (c) 40 (d) 30

Q.368. Using the middle points of sides of a square with an area of 36 square units, another square is made. What is the area of a new square made in square units?

SSC MTS 05/09/2023 (2nd Shift)

- (a) 24 (b) 27 (c) 18 (d) 21

Q.369. The length of the median of an

equilateral triangle is $6\sqrt{3}$ cm. The area of the triangle is :

SSC MTS 05/09/2023 (3rd Shift)

- (a) $36\sqrt{3} \text{ cm}^2$ (b) 36 cm^2
 (c) 72 cm^2 (d) $72\sqrt{3} \text{ cm}^2$

Q.370. The areas of three adjacent faces of a cuboid are p , q and $r \text{ unit}^2$, respectively. What is its volume ?

SSC MTS 06/09/2023 (3rd Shift)

- (a) $p^3 q^3 r^3 \text{ unit}^3$ (b) $\sqrt{pqr} \text{ unit}^3$
 (c) $pqr \text{ unit}^3$ (d) $\sqrt[3]{pqr} \text{ unit}^3$

Q.371. A cylinder has a total surface area of 64π square units, where the height of the cylinder is equal to its radius. If the height is doubled and the radius is halved, then the total surface area of the new cylinder is _____ square units.

SSC MTS 08/09/2023 (3rd Shift)

- (a) 54π (b) 40π (c) 64π (d) 48π

Q.372. A hollow metal sphere of internal and external radii 5 cm and 6 cm, respectively, is melted into a solid cone of base radius 5.2 cm. What is the height (in cm, rounded off to 1 decimal place) of the cone ?

SSC MTS 11/09/2023 (1st Shift)

- (a) 11.8 (b) 12.8 (c) 13.5 (d) 14.5

Q.373. Find the area (in cm^2) of the rhombus, lengths of whose diagonals are, respectively, 7.5 cm and 5.6 cm.

SSC MTS 11/09/2023 (3rd Shift)

- (a) 21 (b) 28 (c) 27 (d) 30

Q.374. If a hollow sphere of an external and internal diameters of 6 cm and 4 cm, respectively, is melted into a cone of base diameter 8 cm, then the height of the cone is:

SSC MTS 12/09/2023 (1st Shift)

- (a) $4\frac{3}{4} \text{ cm}$ (b) $3\frac{1}{4} \text{ cm}$
 (c) $5\frac{1}{4} \text{ cm}$ (d) $2\frac{3}{4} \text{ cm}$

Q.375. A conical tent is set to accommodate 15 persons. Each person must have 5 square metres of space on the ground and 30 cubic metres of air to breathe. What is the height of the cone?

SSC MTS 12/09/2023 (2nd Shift)

- (a) 15 m (b) 22 m (c) 13 m (d) 18 m

Q.376. A solid metallic cuboid of length 11 cm, breadth 8 cm and height 6 cm is melted and reshaped to form a cone of height 14 cm. What is the radius of the cone thus formed ?

SSC MTS 13/09/2023 (3rd Shift)

- (a) 6 cm (b) 9 cm (c) 7 cm (d) 8 cm

Q.377. The length, breadth and height of

a cuboid are in the ratio of 6 : 4 : 3. If the total surface area of the cuboid is 972 cm², what is the length (in cm) of the diagonal of the cuboid ?

SSC MTS 14/09/2023 (1st Shift)

- (a) $5\sqrt{65}$ (b) $3\sqrt{61}$ (c) $3\sqrt{65}$ (d) $5\sqrt{61}$

Q.378. A triangle has its sides 5 cm, 12 cm and 13 cm. What will be the area of the triangle formed by joining the mid point of the three sides ?

SSC MTS 14/09/2023 (2nd Shift)

- (a) $7\frac{1}{2}$ cm² (b) $19\frac{1}{2}$ cm²
(c) $13\frac{1}{2}$ cm² (d) 15 cm²

Q.379. The height of a right circular cone is 20 cm, and its base diameter is 30 cm. What is the curved surface area and the total surface area of the right circular cone, respectively ?

SSC MTS 14/09/2023 (2nd Shift)

- (a) 335π cm² and 525π cm²
(b) 390 cm² and 475π cm²
(c) 450 cm² and 625π cm²
(d) 375π cm² and 600π cm²

SSC CHSL 2023 Tier - 1

Q.380. Aman wants to get his spherical overhead water tank having a diameter of 7 m, painted. If the cost of painting is ₹13 per square metre, find the cost of painting Aman's spherical overhead water tank. (Use $\pi = \frac{22}{7}$)

SSC CHSL 02/08/2023 (2nd Shift)

- (a) ₹1,872 (b) ₹1,984
(c) ₹2,002 (d) ₹1,926

Q.381. If the height and slant height of a cone are 21 cm and 29 cm, respectively.

Find the volume. (use $\pi = \frac{22}{7}$)

SSC CHSL 02/08/2023 (3rd Shift)

- (a) 8800 cm³ (b) 8780 cm³
(c) 8440 cm³ (d) 8080 cm³

Q.382. The perimeter of a parallelogram is 48 cm. If the height of the parallelogram is 6 cm and the length of the adjacent side is 8 cm. find its area.

SSC CHSL 03/08/2023 (1st Shift)

- (a) 90 cm² (b) 80 cm²
(c) 84 cm² (d) 96 cm²

Q.383. In an isosceles triangle, if the unequal side is 8 cm and equal side is 5 cm, then the area of the triangle is:

SSC CHSL 03/08/2023 (1st Shift)

- (a) 12 cm² (b) 25 cm²
(c) 6 cm² (d) 11 cm²

Q.384. Find the volume of the largest possible sphere circumscribed by a cube

of edge 7 cm.

SSC CHSL 03/08/2023 (2nd Shift)

- (a) 179.67 cm³ (b) 169.67 cm³
(c) 189.67 cm³ (d) 199.67 cm³

Q.385. If a map of a city was drawn on a scale of 0.2 cm to a km, what area would represent a country if it has an area of 22500 km²?

SSC CHSL 04/08/2023 (2nd Shift)

- (a) 2500 cm² (b) 2250 cm²
(c) 5625 cm² (d) 900 cm²

Q.386. The sides of the three cubes of metal are 15 cm, 18 cm and 25 cm, respectively. Find the side (in cm) of the new cube formed by melting these cubes together

SSC CHSL 04/08/2023 (2nd Shift)

- (a) $9\sqrt[3]{388}$ (b) $6\sqrt[3]{388}$ (c) $7\sqrt[3]{388}$ (d) $4\sqrt[3]{388}$

Q.387. A shopkeeper usually makes one big besan laddoo (spherical in shape) with a radius of 9 cm. With the same laddoo, How many besan laddoos of radius 3 cm can be made? (Use $\pi = \frac{22}{7}$)

SSC CHSL 04/08/2023 (3rd Shift)

- (a) 9 (b) 18 (c) 36 (d) 27

Q.388. The area of a triangle is half the area of a square. The perimeter of the square is 172 cm. What is the area of the triangle (in cm²) ?

SSC CHSL 04/08/2023 (4th Shift)

- (a) 924.5 (b) 1984 (c) 1849 (d) 992

Q.389. A cuboidal water tank is 12 m long, 9 m wide and 7.5 m deep. How many litres of water can it hold?

SSC CHSL 08/08/2023 (1st Shift)

- (a) 840000 (b) 805000
(c) 780000 (d) 810000

Q.390. What is the area of a triangular flower bed of a garden whose perimeter is 20 metres and two sides are 5 metres and 6 metres.

SSC CHSL 08/08/2023 (4th Shift)

- (a) $10\sqrt{2}$ m² (b) $2\sqrt{10}$ m²
(c) $5\sqrt{2}$ m² (d) $4\sqrt{5}$ m²

Q.391. The perimeter of a sector of a circle of radius 7.4 cm is 20.6 cm. The area of this sector is:

SSC CHSL 10/08/2023 (3rd Shift)

- (a) 42.92 cm² (b) 5.8 cm²
(c) 10.73 cm² (d) 21.46 cm²

Q.392. The perimeter of a minor sector of a circle of radius 5 cm is 15.5 cm. Find the area of the minor sector (use $\pi = \frac{22}{7}$).

SSC CHSL 11/08/2023 (1st Shift)

- (a) 17.53 cm² (b) 13.75 cm²

- (c) 13.57 cm² (d) 17.35 cm²

Q.393. A copper wire is bent in the form of a square and it encloses an area of 30.25 cm². If the same wire is bent to form a circle, then find the area of the circle. (use $\pi = \frac{22}{7}$).

SSC CHSL 11/08/2023 (2nd Shift)

- (a) 38.50 cm² (b) 42.25 cm²
(c) 35 cm² (d) 30.25 cm²

Q.394. A hemispherical bowl is made of 1 cm thick steel. The inside radius of the bowl is 4 cm. The volume of the steel (in cm³) used in making the bowl is:

SSC CHSL 17/08/2023 (2nd Shift)

- (a) $40\frac{2}{3}\pi$ (b) $27\frac{1}{3}\pi$
(c) $41\frac{1}{3}\pi$ (d) $35\frac{1}{3}\pi$

Q.395. A rod of 5 cm radius and 52 cm length is converted into a wire of 100 cm length of uniform thickness. The radius of the wire (in cm) is:

SSC CHSL 17/08/2023 (4th Shift)

- (a) $\sqrt{13}$ (b) $\sqrt{15}$ (c) $\sqrt{23}$ (d) $\sqrt{17}$

SSC CGL 2023 Tier - 1

Q.396. Ranu carries water to school in a cylindrical flask with diameter 12 cm and height 21 cm. Determine the amount of water that she can carry in the flask.

(Use $\pi = \frac{22}{7}$)

SSC CGL 14/07/2023 (4th shift)

- (a) 2372 cm³ (b) 2370 cm³
(c) 2376 cm³ (d) 2374 cm³

Q.397. If the area of a circle is 616 cm² and a chord XY = 10 cm, then find the perpendicular distance from the center of the circle to the chord XY.

SSC CGL 17/07/2023 (1st shift)

- (a) $\sqrt{171}$ cm (b) $\sqrt{161}$ cm
(c) $\sqrt{117}$ cm (d) $\sqrt{181}$ cm

Q.398. The area of two triangles is in the ratio 5 : 3 and their heights are in the ratio 5 : 7. Find the ratio of their bases.

SSC CGL 17/07/2023 (1st shift)

- (a) 3 : 5 (b) 7 : 3 (c) 7 : 5 (d) 2 : 3

Q.399. If the side of an equilateral triangle is increased by 34%, then by what percentage will its area increase?

SSC CGL 17/07/2023 (2nd shift)

- (a) 70.65% (b) 79.56%
(c) 68.25% (d) 75.15%

Q.400. An arc of length 23.1 cm subtends an 18° angle at the centre. What is the area of the circle ?

[Use $\pi = \frac{22}{7}$]

SSC CGL 17/07/2023 (2nd shift)

- (a) 16978.50 cm² (b) 16988.50 cm²
(c) 16878.50 cm² (d) 16798.50 cm²

Q.401. What is the volume of a cube if the perimeter of one face of the cube is 40 cm ?

SSC CGL 17/07/2023 (3rd shift)

- (a) 1000 cm³ (b) 820 cm³
(c) 600 cm³ (d) 800 cm³

Q.402. A spherical ball of lead, 3 cm in diameter, is melted and recast into three spherical balls. The diameters of two of these balls are $\frac{3}{2}$ cm and 2 cm, respectively. Find the diameter of the third ball.

SSC CGL 17/07/2023 (4th shift)

- (a) 2.1 cm (b) 3.3 cm (c) 3 cm (d) 2.5 cm

Q.403. The area of the sector of a circle of radius 12 cm is 32π cm². Find the length of the corresponding arc of the sector.

SSC CGL 17/07/2023 (4th shift)

- (a) $\frac{16}{3}\pi$ cm (b) $\frac{13}{3}\pi$ cm
(c) $\frac{10}{3}\pi$ cm (d) $\frac{8}{3}\pi$ cm

Q.404. The ratio of the length of each equal side and the third side of an isosceles triangle is 3 : 5. If the area of the triangle is $30\sqrt{11}$ cm², then the length of the third side (in cm) is:

SSC CGL 18/07/2023 (2nd shift)

- (a) $10\sqrt{6}$ (b) $5\sqrt{6}$ (c) $13\sqrt{6}$ (d) $11\sqrt{6}$

Q.405. A conical tent has a radius of 7 m and a vertical height of 24 m. How many full bags of rice can be emptied in it if the space occupied by the rice in each bag is 2 m³ ?

SSC CGL 19/07/2023 (1st shift)

- (a) 566 (b) 616 (c) 716 (d) 650

Q.406. To have a surface area of 9π square units of a ball, what should be the diameter (in units) of the ball?

SSC CGL 19/07/2023 (3rd shift)

- (a) 2.5 (b) 2 (c) 1.5 (d) 3

Q.407. A solid cone with curved surface area twice its base area has slant height of $6\sqrt{3}$ cm. Its height is :

SSC CGL 20/07/2023 (2nd shift)

- (a) $6\sqrt{2}$ cm (b) 9 cm (c) 6 cm (d) $3\sqrt{6}$ cm

Q.408. The width of a rectangle is 2 m less than its length. If the perimeter of the rectangle is 68 m, then what is the length (in metres) of the rectangle?

SSC CGL 20/07/2023 (4th shift)

- (a) 20 (b) 18 (c) 16 (d) 17

Q.409. The length, breadth and height of

a cuboid are in the ratio 1 : 2 : 3. The length, breadth and height of the cuboid are increased by 200%, 300% and 300%, respectively. Then compared to the original volume, the increase in the volume of the cuboid will be:

SSC CGL 20/07/2023 (4th shift)

- (a) 47 times (b) 26 times
(c) 17 times (d) 15 times

Q.410. The volume of a cuboid is twice that of a cube. If the dimensions of the cuboid are (8 m × 8 m × 16 m), the total surface area of the cube is:

SSC CGL 21/07/2023 (1st shift)

- (a) 316 m² (b) 288 m²
(c) 324 m² (d) 384 m²

Q.411. The perimeter of an isosceles triangle is 100 cm. If the base is 36 cm, then find its semi perimeter (in centimetres).

SSC CGL 21/07/2023 (2nd shift)

- (a) 64 (b) 60 (c) 45 (d) 50

Q.412. The area of the sector of a circle of radius 28 cm is 112 cm². Find the length of the corresponding arc of the sector.

SSC CGL 21/07/2023 (3rd shift)

- (a) 4 cm (b) 8 cm (c) 6 cm (d) 5 cm

Q.413. If the volume of one brick is 0.0014 m³, then how many bricks will be required to construct a wall of length 14 m, breadth 0.125 m and height 5 m?

SSC CGL 21/07/2023 (3rd shift)

- (a) 5250 (b) 3250 (c) 6250 (d) 4250

Q.414. If the inner radius of a hemispherical bowl is 5 cm and its thickness is 0.25 cm, then find the volume of the material required in making the bowl. (use $\pi = \frac{22}{7}$) (Rounded

up to two places of decimals).

SSC CGL 21/07/2023 (4th shift)

- (a) 34 cm³ (b) 44 cm³
(c) 45.34 cm³ (d) 41.28 cm³

Q.415. A one metre pipe is made with inner diameter equal to outer radius. How much material (in cubic units) is required to make the pipe, if it can hold $\frac{88}{7}$ cubic metres water in it ?

SSC CGL 24/07/2023 (1st shift)

- (a) 37.7 (b) 35.5 (c) 33.3 (d) 36.6

Q.416. The ratio of the length to width of a certain rectangle is 3 : 2 and the area is 150 cm². The perimeter of the rectangle (in cm) is :

SSC CGL 24/07/2023 (1st shift)

- (a) 20 (b) 30 (c) 50 (d) 40

Q.417. The length of each side of a

triangle is 12 cm. What is the length of the circumradius of the triangle?

SSC CGL 24/07/2023 (1st shift)

- (a) $8\sqrt{3}$ cm (b) $2\sqrt{3}$ cm
(c) $6\sqrt{3}$ cm (d) $4\sqrt{3}$ cm

Q.418. A ball is to be made with inner radius of 2 units and outside radius of 3 units. How much material is required to make the ball ?

SSC CGL 24/07/2023 (4th shift)

- (a) $\frac{19}{3}\pi$ (b) 19π (c) $\frac{76}{3}\pi$ (d) π

Q.419. A metallic cube has each of its side of length 12 cm. It is melted and recast into three small cubes. Out of these cubes, two have their sides as 6 cm and 8 cm, respectively. The length of each side of the third cube is:

SSC CGL 25/07/2023 (1st shift)

- (a) 7 cm (b) 9 cm (c) 10 cm (d) 5 cm

Q.420. The field is 150 m long and 80 m wide. How many times (rounded off to 1 decimal place) is its perimeter to the length of its diagonal?

SSC CGL 25/07/2023 (1st shift)

- (a) 2.3 (b) 2.5 (c) 2.7 (d) 3.1

Q.421. The length, breadth and height of a hall are 10 m, 20 m and 15m respectively. Find the cost of whitewashing the walls of the inside of the hall and ceiling at the rate of ₹10.20/m²

SSC CGL 25/07/2023 (2nd shift)

- (a) ₹13,394 (b) ₹11,220
(c) ₹15,320 (d) ₹16,542

Q.422. A cuboid with sides 4, 6 and 8 units is covered with paper. The paper is removed and a square is made from it. What is the side (in units) of the square?

SSC CGL 25/07/2023 (4th shift)

- (a) 12.24 (b) 16.62 (c) 14.42 (d) 12.62

Q.423. What is the surface area (in cm²) of a spherical sculpture whose radius is 35 cm ? (Take $\pi = \frac{22}{7}$)

SSC CGL 26/07/2023 (2nd shift)

- (a) 16540 (b) 15700
(c) 15400 (d) 14500

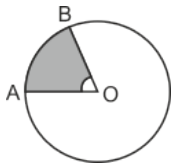
Q.424. A petrol tank is of length 50 m, breadth 25 m, and depth 10 m. How many litres of petrol can be accommodated in the petrol tank when it is completely filled?

SSC CGL 26/07/2023 (2nd shift)

- (a) 1,25,00,000 (b) 12,50,000
(c) 1,25,000 (d) 12500

Q.425. In the following figure, the length of arc AB is equal to five times of the radius r of the circle. Find the area of

sector AOB ?



SSC CGL 27/07/2023 (1st shift)

- (a) $2.5r^2$ (b) $\frac{1}{2}r^2$ (c) $3r^2$ (d) $2r^2$

Q.426. A hemispherical bowl has a radius of 7 cm. It is to be painted from inside as well as outside. Find the cost of painting it at the rate of Rs.40 per 20 cm²? (Take the thickness of the bowl as negligible).

SSC CGL 27/07/2023 (1st shift)

- (a) Rs.1,232 (b) Rs.1,125
(c) Rs.1,432 (d) Rs.1,550

Q.427. A circular arc whose radius is 4 cm makes an angle 45° at the centre. Find the perimeter of the sector formed.

(Take $\pi = \frac{22}{7}$)

SSC CGL 27/07/2023 (3rd shift)

- (a) $\frac{78}{7}$ cm (b) $\frac{72}{7}$ cm
(c) $\frac{74}{7}$ cm (d) $\frac{76}{7}$ cm

SSC Selection Post (Phase - XI)

Q.428. If the volume of a cube is given as 12167 cm³, then the surface area of the cube will be:

Graduate Level 27/06/2023 (Shift - 4)

- (a) 3475 cm² (b) 4574 cm²
(c) 3174 cm² (d) 5413 cm²

Q.429. What is the side of a cube (in metres) in which an iron rod of maximum length 10 metres can be put?

Graduate Level 28/06/2023 (Shift - 3)

- (a) 5.77 (b) 7.57 (c) 5.57 (d) 7.75

Q.430. The volume of a cylinder is 1650 m³, whereas the surface area of its base is $78\frac{4}{7}$ m². Find the height of the cylinder.

Graduate Level 30/06/2023 (Shift - 1)

- (a) 7.5 m (b) 14 m (c) 21 m (d) 2.1 m

Q.431. Find the length of a diagonal of a cuboid whose dimensions are 12 m, 10 m and 8 m. (up to one decimal point)

Matriculation Level 27/06/2023 (Shift - 1)

- (a) 18m (b) 16.5m (c) 17m (d) 17.54m

Q.432. The internal and external diameters of a hollow hemispherical bowl are 6 cm and 10 cm, respectively. If it is melted and recast into a solid cylinder of diameter 14 cm, what is the height of the cylinder in cm?

Matriculation Level 27/06/2023 (Shift - 1)

- (a) 1.1 (b) 1.22 (c) 1.33 (d) 1

Q.433. The perimeter of an isosceles right - angled triangle having an area of 200 cm² is:

Matriculation Level 28/06/2023 (Shift - 1)

- (a) 68.3 cm (b) 78.2 cm
(c) 70.6 cm (d) 58.6 cm

Q.434. A solid sphere has a surface area of 616 cm². This sphere is now cut into two hemispheres. What is the total surface area of one of the hemispheres?

Matriculation Level 28/06/2023 (Shift - 1)

- (a) 440 cm² (b) 462 cm²
(c) 452 cm² (d) 390 cm²

Q.435. The curved surface area of a cylindrical pillar is 528 m² and its volume is 1848 m³. The ratio between its radius and height is _____.

Matriculation Level 30/06/2023 (Shift - 3)

- (a) 9 : 10 (b) 7 : 12 (c) 7 : 3 (d) 9 : 5

Q.436. To pack a set of books, Surbhi made a carton that was 48 inches long, 27 inches wide and 30 inches tall. What will be the total surface area (in square feet) of this carton, once it is closed from all sides? [Use 1 foot = 12 inches]

Matriculation Level 30/06/2023 (Shift - 4)

- (a) 49.25 (b) 48.25 (c) 48.75 (d) 49.75

SSC CHSL 2022 Tier - 2

Q.437. The length of a rectangular plot is thrice its breadth. If the area of the rectangular plot is 2700 m², then what is the breadth of the rectangular plot?

SSC CHSL Tier II (26/06/2023)

- (a) 20 m (b) 12 m (c) 30 m (d) 28 m

SSC MTS 2022 Tier - 1

Q.438. When a wire is bent in the form of a square, then the area enclosed by it is 5929 cm². If wire is bent into the form of a circle, then what will be the area enclosed by the wire?

SSC MTS 02/05/2023 (Afternoon)

- (a) 7546 cm² (b) 8542 cm²
(c) 7632 cm² (d) 6954 cm²

Q.439. What will be the area of the semi-circle of perimeter 3.6 meters?

SSC MTS 02/05/2023 (Afternoon)

- (a) 1.54 m² (b) 1.76 m²
(c) 1.10 m² (d) 0.77 m²

Q.440. The ratio of the outer and the inner circumference of a circular path is 17 : 15. If path is 40 metres wide, then what is the radius of the inner circle?

SSC MTS 02/05/2023 (Evening)

- (a) 280 metres (b) 340 metres

- (c) 360 metres (d) 300 metres

Q.441. If the area of a square is 529 cm², then what is the length of its diagonal?

SSC MTS 03/05/2023 (Morning)

- (a) 23 cm (b) $26\sqrt{3}$ (c) $23\sqrt{2}$ (d) 46 cm

Q.442. The length of a cuboid is six times of its breadth and the height of a cuboid is four times of its breadth. If the breadth of the cuboid is 3 cm, then what is the total surface area of the cuboid?

SSC MTS 03/05/2023 (Evening)

- (a) 484 cm² (b) 612 cm²
(c) 576 cm² (d) 676 cm²

Q.443. The ratio of radii of a cylinder to a cone is 3 : 1. If their heights are equal, What is the ratio of their volumes?

SSC MTS 03/05/2023 (Evening)

- (a) 1 : 3 (b) 27 : 1 (c) 9 : 1 (d) 1 : 9

Q.444. The ratio of the sides of a triangle is 11 : 11 : 4. If the area of triangle is $2\sqrt{117}$ cm², then what is the length of the equal sides?

SSC MTS 04/05/2023 (Morning)

- (a) 3 cm (b) 13 cm (c) 11cm (d) 9 cm

Q.445. The two adjacent sides of a parallelogram are 12 cm and 5 cm respectively. If one of the diagonals is 13 cm long, then what is the area of the parallelogram?

SSC MTS 04/05/2023 (Afternoon)

- (a) 60 cm² (b) 30 cm²
(c) 75 cm² (d) 25 cm²

Q.446. A cuboid whose sides are 8 cm, 27 cm and 64 cm is melted to form a new cube. What is the respective ratio of the total surface area of the cuboid and cube?

SSC MTS 08/05/2023 (Morning)

- (a) 307: 216 (b) 291: 203
(c) 349: 248 (d) 329: 237

Q.447. Find $\frac{1}{21}$ -th of the volume of a hemisphere whose radius is 4.2 m

SSC MTS 08/05/2023 (Afternoon)

- (a) 6.6 m³ (b) 7.3 m³ (c) 8.2 m³ (d) 7.1 m³

Q.448. Two cubes each of edge 9 cm are joined to form a single cuboid. What is the total surface area of the cuboid formed?

SSC MTS 08/05/2023 (Afternoon)

- (a) 775 cm² (b) 810 cm²
(c) 990 cm² (d) 1024 cm²

Q.449. The radius of the base of a conical tent is 8 m and its height is 15 m, what is the cost of the material needed to make it if it costs Rs.54 per π m²?

SSC MTS 08/05/2023 (Evening)

- (a) Rs.6454 (b) Rs.7344

(c) Rs.8678

(d) Rs.7454

Q.450. The largest possible sphere is carved out of a cube of side 6 cm. What is the volume of the sphere?

SSC MTS 09/05/2023 (Morning)

(a) $\frac{792}{7}$ cm³ (b) $\frac{144}{7}$ cm³(c) $\frac{133}{3}$ cm³ (d) $\frac{444}{7}$ cm³

Q.451. Find the area of the rhombus whose diagonals are 16 cm and 20 cm.

SSC MTS 09/05/2023 (Morning)

(a) 110 cm² (b) 150 cm²(c) 160 cm² (d) 120 cm²

Q.452. Two cubes of iron whose edges are 7 cm and 11 cm are melted and formed into a single cube. Find the volume (in cm³) of the new cube formed.

SSC MTS 09/05/2023 (Afternoon)

(a) 1881 (b) 1674 (c) 1444 (d) 1221

Q.453. Length of a rectangle is 58 meters, while its breadth is 30 meters. What is the total cost of covering it with grass bed if cost is Rs. 20 per m²?

SSC MTS 09/05/2023 (Evening)

(a) Rs. 34800 (b) Rs. 32400

(c) Rs. 28500 (d) Rs. 17400

Q.454. Find the weight of a cylindrical iron container with base radius 12.6 cm and height is 4 m, where the weight of 1 cm³ of iron is 2 grams?

SSC MTS 10/05/2023 (Morning)

(a) 299.284 kg (b) 199.584 kg

(c) 399.168 kg (d) 499.482 kg

Q.455. Find the area of the triangle whose height is 56 cm and the corresponding base is $\frac{1}{4}$ th of the height.

SSC MTS 10/05/2023 (Evening)

(a) 544 cm² (b) 440 cm²(c) 392 cm² (d) 480 cm²

Q.456. The outer radius of a spherical shell is 13 cm and the thickness of the shell is 3 cm. What is the volume of the metal used for the shell? ($\pi = \frac{22}{7}$)

SSC MTS 11/05/2023 (Morning)

(a) 1197 cm³ (b) 2197 cm³(c) 3024 cm³ (d) 5016 cm³

Q.457. What will be the approximately volume of sphere with radius 65 cm?

SSC MTS 11/05/2023 (Afternoon)

(a) 4.5×10^6 cm³ (b) 6×10^6 cm³(c) 1.15×10^6 cm³ (d) 2×10^6 cm³

Q.458. The lateral area is thrice as big as the base area of a cone. If the diameter of the cone is 1 cm, then what is the total surface area of the cone?

SSC MTS 11/05/2023 (Evening)

(a) π cm² (b) 3π cm² (c) 2π cm² (d) 4π cm²

Q.459. If the radius of a sphere is $\frac{3}{4}$ of the radius of a hemisphere, then what will be the ratio of the volumes of sphere and hemisphere?

SSC MTS 12/05/2023 (Morning)

(a) 9:16 (b) 51:64 (c) 27:32 (d) 18:64

Q.460. The area of the base of a rectangular tank is 720 m² and the volume of water contained in it is 28800 m³. What is the depth of the water?

SSC MTS 12/05/2023 (Morning)

(a) 30 m (b) 40 m (c) 50 m (d) 60 m

Q.461. Sides of a triangle are 6 cm, 8 cm and 10 cm. What is the area of the triangle?

SSC MTS 12/05/2023 (Afternoon)

(a) 20 cm² (b) 28 cm²(c) 24 cm² (d) 16 cm²

Q.462. What is the area of the lateral surface of a right circular cylinder, If the circumference of the base is 22 cm and its height is four times of its radius?

SSC MTS 12/05/2023 (Evening)

(a) 388 cm² (b) 308 cm²(c) 408 cm² (d) 288 cm²

Q.463. If the heights of two cylinders are in the ratio of 7:9 and their diameters are in the ratio of 6:7, then what is the ratio of the volumes of the cylinders?

SSC MTS 15/05/2023 (Afternoon)

(a) 6:7 (b) 7:9 (c) 5:9 (d) 4:7

Q.464. If the curved surface area of a solid cylinder is two-fifth of its total surface area, then what is the ratio of its diameter to its height?

SSC MTS 15/05/2023 (Evening)

(a) 2:5 (b) 1:2 (c) 2:3 (d) 3:1

Q.465. Difference between the circumferences of two circles is 4.4 cm and radius of the smaller circle is 1.4 cm. The radius of the larger circle is:

SSC MTS 16/05/2023 (Morning)

(a) 4.1 cm (b) 0.7 cm

(c) 2.1 cm (d) 3.5 cm

Q.466. Nine times the area of a circle is same as the three times the area of a square. What is the ratio of the diameter of the circle and diagonal of the square?

SSC MTS 16/05/2023 (Afternoon)

(a) $\sqrt{2} : \sqrt{3}\pi$ (b) $2 : \sqrt{3}\pi$ (c) $2 : 3\pi$ (d) $\sqrt{5} : \sqrt{7}\pi$

Q.467. How many litre of water can a hemispherical bowl of diameter 21 m hold? (1m³ - 1000 litre)

SSC MTS 16/05/2023 (Afternoon)

(a) 2100000 litre (b) 2425500 litre

(c) 2322500 litre (d) 2250500 litre

Q.468. A solid cylinder having radius of base as 28 cm and height as 24 cm is bisected from its height to get two identical cylinders. What will be the percentage increase in the total surface area?

SSC MTS 17/05/2023 (Morning)

(a) 61.72 percent (b) 41.92 percent

(c) 53.85 percent (d) 48.64 percent

Q.469. The height and curved surface area of the cylinder are 15 cm and 330 cm². Find the diameter of the cylinder.

SSC MTS 17/05/2023 (Morning)

(a) 2 cm (b) 5 cm (c) 7 cm (d) 6 cm

Q.470. The length, breadth, height of a room are 6 m 40 cm, 4 m 80 cm and 3 m 20 cm respectively. Find the longest tape which can measure the dimensions of the room exactly.

SSC MTS 17/05/2023 (Morning)

(a) 1 m 80 cm (b) 1 m 20 cm

(c) 1 m 60 cm (d) 1 m 40 cm

Q.471. A solid hemisphere of radius 8 cm is melted into N identical cone whose radius is 4 cm and height is 2 cm. Then, what is the value of N?

SSC MTS 17/05/2023 (Afternoon)

(a) 26 (b) 10 (c) 32 (d) 25

Q.472. The ratio of the outer and the inner circumferences of a circular path is 11:7. If path is 20 metres wide, then what is the radius of the inner circle?

SSC MTS 17/05/2023 (Evening)

(a) 20 metres (b) 55 metres

(c) 65 metres (d) 35 metres

Q.473. The longest side of the obtuse triangle is 7 cm and the other two sides of the triangle are 4 cm and 5 cm. Find the area of the triangle.

SSC MTS 18/05/2023 (Morning)

(a) $1\sqrt{3}$ cm² (b) $6\sqrt{3}$ cm²(c) $3\sqrt{2}$ cm² (d) $4\sqrt{6}$ cm²

Q.474. If the diagonal of a square is 18.4 cm, then the area of that square is: (in cm²)

SSC MTS 18/05/2023 (Evening)

(a) 84.64 (b) 338.56 (c) 324.42 (d) 169.28

Q.475. A cone and a cylinder have same height and the radius of the cone is twice of the radius of the cylinder. What is the ratio of the volume of the cone to that of the cylinder?

SSC MTS 19/05/2023 (Afternoon)

(a) 2:5 (b) 4:5 (c) 3:2 (d) 4:3

Q.476. The difference between the length and breadth of a rectangle is 40 metres. If its perimeter is 96 metres, then what will be the area of rectangle?

SSC MTS 19/05/2023 (Evening)

- (a) 254 metre² (b) 110 metre²
(c) 176 metre² (d) 152 metre²

Q.477. A farmer's land is in the shape of a trapezium which has its parallel sides measuring 7.58 yards and 6.42 yards and the distance between the parallel sides is 6.50 yards. The cost of ploughing the land is Rs. 2000 per square yard. What amount have to be spent in order to plough the entire land ?

SSC MTS 13/06/2023 (Morning)

- (a) Rs. 85000 (b) Rs. 91000
(c) Rs. 72000 (d) Rs. 102000

Q.478. If the radius of a cylinder is decreased by 12 percent, then by how much percent its height must be increased so that the volume of the cylinder remains same ?

SSC MTS 14/06/2023 (Morning)

- (a) 29.13 percent (b) 21.78 percent
(c) 42.56 percent (d) 34.27 percent

Q.479. The radius of a right circular cylinder is 7 cm. Its height is thrice its radius. What is the curved surface area of the cylinder ?

SSC MTS 14/06/2023 (Afternoon)

- (a) 724 cm² (b) 666 cm²
(c) 924 cm² (d) 264 cm²

Q.480. The ratio of the sides of a triangle is 3 : 3 : 4. If the area of triangle is $32\sqrt{5}$ cm², then what is the length of the equal sides ?

SSC MTS 14/06/2023 (Evening)

- (a) 12 cm (b) 16 cm (c) 10 cm (d) 15 cm

Q.481. The length of a cuboid is 4 cm. If the breadth of the cuboid is four times of its length and height of the cuboid is twice of its length, then what is the lateral surface area of the cuboid?

SSC MTS 15/06/2023 (Afternoon)

- (a) 380 cm² (b) 440 cm²
(c) 260 cm² (d) 320 cm²

Q.482. One of the diagonals of a rhombus is 70 percent of the other diagonal. What is the ratio of the area of rhombus to the square of the length of the larger diagonal?

SSC MTS 15/06/2023 (Evening)

- (a) 5 : 17 (b) 7 : 20 (c) 6 : 19 (d) 20 : 7

Q.483. If the radius of a cylinder is decreased by 16 percent, then by how much percent its height must be increased, so that the volume of the cylinder remains the same ?

SSC MTS 15/06/2023 (Evening)

- (a) 32.96% (b) 41.72%
(c) 45.28% (d) 36.43%

Q.484. The sum of the length and breadth of a cuboid is 16 cm. If the height of the cuboid is $\frac{1}{4}$ of the sum of

its length and breadth, then what is the lateral surface area of the cuboid ?

SSC MTS 16/06/2023 (Afternoon)

- (a) 128 cm² (b) 196 cm²
(c) 96 cm² (d) 156 cm²

Q.485. The length of a cuboid is 10 cm. If the breadth of the cuboid is half of its length and height of the cuboid is twice of its length, then what is the lateral surface area of the cuboid?

SSC MTS 16/06/2023 (Evening)

- (a) 600 cm² (b) 500 cm²
(c) 720 cm² (d) 560 cm²

Q.486. The perimeter of the top of a rectangular table is 56 meter and its area is 192 meter². What is the length of its diagonal ?

SSC MTS 20/06/2023 (Morning)

- (a) 22 metre (b) 20 metre
(c) 16 metre (d) 18 metre

Q.487. If the area of non-spherical portion of a solid hemisphere is 1560 m², then the total surface area of the hemisphere is:

SSC MTS 20/06/2023 (Afternoon)

- (a) 6240 m² (b) 4680 m²
(c) 5140 m² (d) 3120 m²

Q.488. The sum of the length and breadth of a cuboid is 40 cm. If the height of the cuboid is $\frac{1}{5}$ of the sum of

its length and breadth, then what is the lateral surface area of the cuboid?

SSC MTS 20/06/2023 (Evening)

- (a) 580 cm² (b) 320 cm²
(c) 640 cm² (d) 720 cm²

SSC CHSL 2022 Tier - 1

Q.489. If the radius of the sphere is decreased by 10 percent, then volume of the sphere will be decreased by what percentage ?

SSC CHSL 10/03/2023 (1st Shift)

- (a) 32.5 percent (b) 30.6 percent
(c) 25.6 percent (d) 27.1 percent

Q.490. Three circles of radius 7 cm are placed in such a way that each circle touches the other two. What will be the area of the portion enclosed by these three circles ?

SSC CHSL 10/03/2023 (3rd Shift)

- (a) $49\sqrt{3} - 77$ cm² (b) $40\sqrt{3} - 66$ cm²
(c) $50\sqrt{3} - 66$ cm² (d) $55\sqrt{3} - 77$ cm²

Q.491. There is a rectangular garden of 240 metres × 80 metres. A path of width

4 metres is build outside the garden along its four sides. What is the area of the path ?

SSC CHSL 10/03/2023 (4th Shift)

- (a) 2826 metre² (b) 2542 metre²
(c) 2916 metre² (d) 2624 metre²

Q.492. The sum of the length, breadth, and height of a cuboid box is 20 cm and the total surface area of cuboid is 256 sq. cm. What is the maximum length (in approximate) of a stick that can be placed inside the cuboid box ?

SSC CHSL 13/03/2023 (2nd Shift)

- (a) 32 cm (b) 24 cm (c) 16 cm (d) 12 cm

Q.493. X, Y and Z are three equilateral triangles. The sum of the areas of X and Y is equal to the area of Z. If the side lengths of X and Y are 6 cm and 8 cm respectively, then what is the side length of Z ?

SSC CHSL 13/03/2023 (2nd Shift)

- (a) 10.5 cm (b) 10 cm (c) 9.5 cm (d) 9 cm

Q.494. An incircle with center O is drawn inside an equilateral triangle ABC and the length of the side is 12 cm. Find the area (in sq. cm) of the triangle which is not in the incircle.

SSC CHSL 13/03/2023 (2nd Shift)

- (a) 21.48 (b) 30.64 (c) 37.71 (d) 24.64

Q.495. What is the area of a rhombus whose diagonals are of 8 cm and 13 cm?

SSC CHSL 13/03/2023 (4th Shift)

- (a) 52 cm² (b) 96 cm²
(c) 44 cm² (d) 84 cm²

Q.496. If the radius of a sphere is thrice than the radius of a hemisphere, then what will be the respective ratio of their volumes ?

SSC CHSL 14/03/2023 (1st Shift)

- (a) 27 : 1 (b) 16 : 1 (c) 54 : 1 (d) 18 : 1

Q.497. The ratio of the volume of the first and second cylinder is 32 : 9 and the ratio of their heights is 8 : 9. If the area of the base of the second cylinder is 616 cm², then what will be the radius of the first cylinder ?

SSC CHSL 14/03/2023 (2nd Shift)

- (a) 24 cm (b) 20 cm (c) 28 cm (d) 36 cm

Q.498. Half of the perimeter of a rectangle is 30 cm. If the length of the rectangle is 6 cm more than its breadth, then what is the area of the rectangle?

SSC CHSL 14/03/2023 (4th Shift)

- (a) 216 cm² (b) 250 cm²
(c) 185 cm² (d) 195 cm²

Q.499. If the square of product of three sides of the triangle is 1024 units and the area of its circumcircle is 16π , what is

the area of the given triangle ?

SSC CHSL 15/03/2023 (1st Shift)

- (a) 2 sq. units (b) 3 sq. units
(c) 2.5 sq. units (d) 4 sq. units

Q.500. The breadth of a rectangular floor is $\frac{3}{5}$ of its length. If the area of the floor

is 60 metre^2 , then what is the difference between the length and breadth of the floor ?

SSC CHSL 15/03/2023 (2nd Shift)

- (a) 6 m (b) 7.5 m (c) 5 m (d) 4 m

Q.501. A solid cylinder having radius of base as 7 cm and height as 5 cm is bisected from its height to get two identical cylinders. What will be the percentage increase in the total surface area?

SSC CHSL 16/03/2023 (2nd Shift)

- (a) 58.33 percent (b) 64.56 percent
(c) 69.44 percent (d) 52.56 percent

Q.502. If in triangle ABC, MN is parallel to BC, and M and N are points on AB and AC respectively. The area of quadrilateral MBCN = 130 cm^2 . If AN: NC = 4:5, then the area of Triangle MAN is:

SSC CHSL 17/03/2023 (1st Shift)

- (a) 40 cm^2 (b) 65 cm^2
(c) 32 cm^2 (d) 45 cm^2

Q.503. The circumference of the two circles is 264 m and 308 m respectively. What is the difference between the area of the larger circle and the smaller circle?

SSC CHSL 17/03/2023 (1st Shift)

- (a) 2002 m^2 (b) 2010 m^2
(c) 1890 m^2 (d) 1990 m^2

Q.504. If the heights of two cylinders are in the ratio of 2 : 3 and their radius are in the ratio of 6 : 5, then what is the ratio of the volumes of the cylinders ?

SSC CHSL 17/03/2023 (4th Shift)

- (a) 24 : 29 (b) 21 : 31
(c) 24 : 25 (d) 28 : 31

Q.505. The sum of the length and breadth of a cuboid is 4 cm. Length is three times of breadth. If the height of the cuboid is half of the sum of its length and breadth, then what is the total surface area of the cuboid ?

SSC CHSL 20/03/2023 (2nd Shift)

- (a) 64 cm^2 (b) 22 cm^2
(c) 32 cm^2 (d) 30 cm^2

Q.506. The area of a square is 1156 cm^2 . What is the respective ratio between the length and the breadth of a rectangle whose length is twice the side of the square and whose breadth is 14 cm less than the side of the square?

SSC CHSL 20/03/2023 (4th Shift)

- (a) 16 : 7 (b) 19 : 5 (c) 17 : 5 (d) 21 : 19

Q.507. In triangle ABC, AB=20 cm, BC=16 cm and AC = 12 cm and the radius of circle inscribed in it is 4 cm. Find the area of triangle ABC.

SSC CHSL 21/03/2023 (1st Shift)

- (a) 48 cm^2 (b) 72 cm^2
(c) 96 cm^2 (d) 120 cm^2

Q.508. A trapezium plate having two parallel sides of length 15 cm and 9 cm. and distance between them is 6 cm. Copper plating is to be done on the plate at a rate of Rs.3 per square cm. What will be the total cost of copper plating on the upper side of the plate?

SSC CHSL 21/03/2023 (1st Shift)

- (a) Rs.432 (b) Rs.216 (c) Rs.72 (d) Rs.108

Q.509. ΔABC is similar to ΔPQR , AB = 4 cm, BC = 8 cm and AC = 10 cm. If QR = 16 cm, then what is the area of ΔPQR ?

SSC CHSL 21/03/2023 (2nd Shift)

- (a) $2\sqrt{231} \text{ cm}^2$ (b) $4\sqrt{231} \text{ cm}^2$
(c) $4\sqrt{462} \text{ cm}^2$ (d) $8\sqrt{462} \text{ cm}^2$

SSC CGL 2022 Tier - 2

Q.510. What is the volume of the largest sphere that can be carved out of a wooden cube of sides 21 cm? ($\pi = \frac{22}{7}$)

SSC CGL Tier II (02/03/2023)

- (a) 3851 cm^3 (b) 6858 cm^3
(c) 4851 cm^3 (d) 5821 cm^3

Q.511. A rectangular park is 120 m long and 104 m wide. A 1 - m wide path runs along the boundary of the park, remaining completely inside the park area. Thus, the outside edges of the path run along the boundary wall of the park. The inside edges of the path are marked with a white line of negligible thickness. If it costs ₹2.50 to mark each metre with the white line, then how much would it cost (in ₹) to fully mark the inside edges of the path ?

SSC CGL Tier II (02/03/2023)

- (a) 1090 (b) 1080 (c) 1120 (d) 1100

Q.512. What is the total surface area of a pyramid whose base is a square with side 8 cm and height of the pyramid is 3 cm ?

SSC CGL Tier II (03/03/2023)

- (a) 169 cm^2 (b) 121 cm^2
(c) 144 cm^2 (d) 184 cm^2

Q.513. The breadth 'b' of a room is twice its height and half of its length. Find the length of the longest diagonal of the room.

SSC CGL Tier II (03/03/2023)

- (a) $\frac{\sqrt{20}b}{2}$ (b) $\frac{b}{2}$ (c) $\frac{\sqrt{21}b}{2}$ (d) $\frac{\sqrt{19}b}{2}$

Q.514. A pyramid has an equilateral triangle as its base, of which each side is 8 cm. Its slant edge is 24 cm. The whole surface area of the pyramid (in cm^2) is:

SSC CGL Tier II (06/03/2023)

- (a) $(16\sqrt{3} + 24\sqrt{35})$ (b) $(12\sqrt{3} + 24\sqrt{35})$
(c) $(24\sqrt{3} + 36\sqrt{35})$ (d) $(16\sqrt{3} + 48\sqrt{35})$

Q.515. The difference between the length of two parallel sides of a trapezium is 12 cm. The perpendicular distance between these two parallel sides is 60 cm. If the area of the trapezium is 1380 cm^2 , then find the length of each of the parallel sides (in cm).

SSC CGL Tier II (06/03/2023)

- (a) 27, 15 (b) 31, 19 (c) 29, 17 (d) 24, 12

SSC CGL 2022 Tier - 1

Q.516. How many spherical lead shots each of diameter 8.4 cm can be obtained from a rectangular solid of lead with dimension 88 cm, 63 cm, 42 cm.

(take $\pi = \frac{22}{7}$) ?

SSC CGL 01/12/2022 (1st Shift)

- (a) 920 (b) 750 (c) 650 (d) 860

Q.517. A solid metallic sphere of radius 13 cm is melted and recast into a cone having diameter of the base as 13 cm. What is the height (cm) of the cone ?

SSC CGL 01/12/2022 (3rd Shift)

- (a) 246 (b) 152 (c) 174 (d) 208

Q.518. Which of the following two will provide more volume ?

I) A cuboid of edges 6 cm, 7 cm and 8 cm

II) A cube of edge 7 cm

SSC CGL 02/12/2022 (1st Shift)

- (a) Cube will have more volume
(b) Both will have equal volume
(c) Cuboid will have more volume
(d) Cannot be determined

Q.519. A hemispherical bowl made of iron has an inner diameter of 84 cm. Find the cost of tin plating it on the inside at the rate of ₹21 per 100 cm^2 (take $\pi = \frac{22}{7}$)

) correct to two places of decimal.

SSC CGL 02/12/2022 (1st Shift)

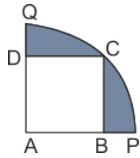
- (a) ₹2,328.48 (b) ₹2,425.48
(c) ₹2,425.60 (d) ₹2,355.48

Q.520. The radius of a right circular cylinder is five times of its height. If the height of the cylinder is 3.5 cm, then what is the volume of the cylinder?

SSC CGL 02/12/2022 (3rd Shift)

- (a) 3368.75 cm³ (b) 3872.75 cm³
(c) 3146.75 cm³ (d) 3524.25 cm³

Q.521. In the given figure, a square ABCD is inscribed in a quadrant APCQ. If AB = 16 cm, find the area of the shaded region (take $\pi = 3.14$) correct to two places of decimal.



SSC CGL 02/12/2022 (4th Shift)

- (a) 155.98cm² (b) 179.68cm²
(c) 163.85cm² (d) 145.92cm²

Q.522. If the volume of a sphere is 24,416.64 cm³, find its surface area (take $\pi=3.14$) correct to two places of decimal.
SSC CGL 03/12/2022 (1st Shift)

- (a) 3069.55 cm² (b) 4069.44 cm²
(c) 5069.66 cm² (d) 6069.67 cm²

Q.523. The radius of a right circular cylinder is thrice its height. If the height of the cylinder is 2.1 cm, then what is the volume of the cylinder ?

SSC CGL 03/12/2022 (3rd Shift)

- (a) 224.65 cm³ (b) 194.72 cm³
(c) 324.86 cm³ (d) 261.95 cm³

Q.524. A solid copper sphere of radius 9 cm is hammered and moulded into a wire of radius 2 cm. What is the length of this wire ?

SSC CGL 03/12/2022 (3rd Shift)

- (a) 224 cm (b) 183 cm
(c) 198 cm (d) 243 cm

Q.525. The length of the side of a cube is 5.6 cm. What is the volume of the largest sphere that can be taken out of the cube ?

SSC CGL 05/12/2022 (1st Shift)

- (a) 91.98 cm³ (b) 99.96 cm³
(c) 96.98 cm³ (d) 90.69 cm³

Q.526. A hollow sphere has an outer radius of 4 cm and inner radius of 1 cm. What is the volume of this hollow sphere?

SSC CGL 05/12/2022 (3rd Shift)

- (a) 84 π cm³ (b) 68 π cm³
(c) 48 π cm³ (d) 92 π cm³

Q.527. Three cubes of equal volume are joined end to end. Find the surface area of the resulting cuboid if the diagonal of the cube is $6\sqrt{3}$ cm.

SSC CGL 05/12/2022 (4th Shift)

- (a) 509 cm² (b) 504 cm²
(c) 516 cm² (d) 512 cm²

Q.528. A solid hemisphere has radius 21

cm. It is melted to form a cylinder such that the ratio of its curved surface area to total surface area is 2 : 5. What is the radius (in cm) of its base (take $\pi = \frac{22}{7}$) ?

SSC CGL 06/12/2022 (1st Shift)

- (a) 23 (b) 21 (c) 17 (d) 19

Q.529. Calculate the area of a sector of a circle with radius 10 metres and angle of 90 degrees at the centre.

SSC CGL 06/12/2022 (3rd Shift)

- (a) 79.3 m² (b) 79.5 m²
(c) 78.5 m² (d) 75 m²

Q.530. The ratio of the outer and the inner circumference of a circular path is 5 : 4. If the path is 50 metres wide, then what is the radius of the inner circle?

SSC CGL 06/12/2022 (3rd Shift)

- (a) 250 metres (b) 300 metres
(c) 200 metres (d) 210 metres

Q.531. The diagonal of the square is $8\sqrt{2}$ cm. Find the diagonal of another square whose area is triple that of the first square.

SSC CGL 06/12/2022 (4th Shift)

- (a) $8\sqrt{5}$ cm (b) $8\sqrt{3}$ cm
(c) $8\sqrt{2}$ cm (d) $8\sqrt{6}$ cm

Q.532. Steel is used to make a hemispherical bowl that is 0.37 cm thick. The bowl's inner radius is 6 cm. Find the bowl's outer curved surface area (take $\pi = \frac{22}{7}$).

SSC CGL 07/12/2022 (1st Shift)

- (a) 532 cm² (b) 255.0548 cm²
(c) 484 cm² (d) 523.4107 cm²

Q.533. An iron rod with diameter 4 cm and length 12 cm is drawn into a wire of length 12 m of uniform thickness. Determine the thickness of the wire.

SSC CGL 08/12/2022 (2nd Shift)

- (a) 0.6 cm (b) 0.7 cm
(c) 0.3 cm (d) 0.4 cm

Q.534. A solid cone of height 42 cm with diameter of its base 42 cm is cut out from a wooden solid sphere of radius 24 cm. Find the percentage of wood wasted correct to two places of decimal.

SSC CGL 09/12/2022 (1st Shift)

- (a) 75.56% (b) 56.65%
(c) 66.50% (d) 67.50%

Q.535. The length of the side of a cube is 2.8 cm. What is the volume of the largest sphere that can be taken out of the cube ?

SSC CGL 09/12/2022 (2nd Shift)

- (a) 11.50 cm³ (b) 1.15 cm³
(c) 11.55 cm³ (d) 115 cm³

Q.536. The curved surface area of a right

circular cone of base radius 21 cm is 594 sq cm. What is the slant height of the cone ?

SSC CGL 09/12/2022 (3rd Shift)

- (a) 15 cm (b) 11 cm (c) 9 cm (d) 6 cm

Q.537. The number of solid spheres, each of diameter 6 cm, that could be moulded to form a solid metal cylinder of height 90 cm and diameter 4 cm is:

SSC CGL 09/12/2022 (4th Shift)

- (a) 10 (b) 14 (c) 16 (d) 13

Q.538. The radii of the ends of a frustum of a cone 7 cm high are 5 cm and 3 cm. Find its volume correct to one decimal place. (Use $\pi = \frac{22}{7}$)

SSC CGL 12/12/2022 (2nd Shift)

- (a) 345.6 cm³ (b) 359.3 cm³
(c) 379.3 cm³ (d) 369.3 cm³

Q.539. Area of the floor of a cubical room is 64 m². The length of the longest rod that can be kept in the room is:

SSC CGL 12/12/2022 (3rd Shift)

- (a) $16\sqrt{3}$ m (b) $4\sqrt{3}$ m
(c) $12\sqrt{3}$ m (d) $8\sqrt{3}$ m

Q.540. A conical vessel, whose internal radius is 20 cm and height is 27 cm, is full of water. If this water is poured into a cylindrical vessel with internal radius 15 cm, what will be the height to which the water rises in it ?

SSC CGL 12/12/2022 (4th Shift)

- (a) 16 cm (b) 10 cm
(c) 12 cm (d) 14 cm

Q.541. If the numerical value of twice the curved surface area of a right circular cylinder is equal to the numerical value of its volume, then what is the numerical value of the radius of the base of the cylinder ?

SSC CGL 13/12/2022 (1st Shift)

- (a) 2 (b) 3 (c) 5 (d) 4

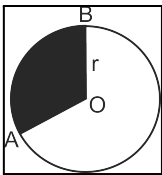
SSC CPO 2022 Tier - 1

Q.542. A field is in the shape of a rectangle of length 90 m and breadth 75 m. In one corner of the field, a pit, which is 18 m long, 15 m broad and 6 m deep, has been dug out. The Earth taken out of it is evenly spread over the remaining part of the field. Find the rise in the level of the field ?

SSC CPO 09/11/2022 (Morning)

- (a) 27 cm (b) 25 cm (c) 28 cm (d) 24 cm

Q.543. In the given figure, the length of arc AB is equal to twice the length of radius r of the circle. Find the area of sector OAB in terms of the radius r.



SSC CPO 09/11/2022 (Morning)

- (a) $3r$ (b) $2r$ (c) πr^2 (d) r^2

Q.544. Determine the total surface area of a hemisphere closed at bottom.

Radius of the hemisphere is $\sqrt{\frac{25}{\pi}}$ unit.

SSC CPO 09/11/2022 (Morning)

- (a) 75 unit^2 (b) 70 unit^2
(c) 60 unit^2 (d) 50 unit^2

Q.545. Find the perimeter of major sector of a circle of radius 12 metres, whose minor sector subtends an angle of 75° at the centre.

SSC CPO 09/11/2022 (Morning)

- (a) $24 + 5\pi$ metres (b) $24 + 19\pi$ metres
(c) $24 - 5\pi$ metres (d) $24 - 19\pi$ metres

Q.546. A frustum of a right circular cone, whose height is 18 cm, large base radius is 25 cm, and small base radius is 20 cm, is melted to form a small sphere of diameter 2 cm. How many spheres will be formed?

SSC CPO 09/11/2022 (Afternoon)

- (a) 6287 (b) 6862 (c) 4575 (d) 5857

Q.547. The radius of a hollow sphere is 21 cm. It is cut into two equal halves by a plane passing through its centre. What is 75% of the curved surface area (in cm^2 , rounded off to the nearest tens) of the

hemisphere? (Take $\pi = \frac{22}{7}$)

SSC CPO 09/11/2022 (Evening)

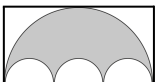
- (a) 4160 (b) 3470 (c) 2080 (d) 2770

Q.548. The total surface area of a solid metallic hemisphere is 462 cm^2 . This is melted and moulded into a right circular cone. If the radius of the base of the cone is the same as that of the hemisphere, then its height is: (use $\pi = \frac{22}{7}$)

SSC CPO 09/11/2022 (Evening)

- (a) 14 cm (b) 7 cm (c) 21 cm (d) 28 cm

Q.549. As shown in the given figure, inside the large semicircle, three semicircles (with equal radii) are drawn so that their diameters all sit on the large semicircle's diameter. What is the ratio between the unshaded and shaded areas?



SSC CPO 09/11/2022 (Evening)

- (a) 4 : 3 (b) 1 : 2 (c) 2 : 1 (d) 3 : 4

Q.550. What is the length of the longest pole that can fit itself in a hall 60 feet long, 30 feet broad and 20 feet high?

SSC CPO 10/11/2022 (Morning)

- (a) 50 ft. (b) 70 ft. (c) 30 ft. (d) 20 ft.

Q.551. The volume of a wall, six times as high as its breadth and half as long as its height, is 23958 cm^3 . What is the breadth of the wall?

SSC CPO 10/11/2022 (Morning)

- (a) 21 cm (b) 15 cm (c) 11 cm (d) 18 cm

Q.552. The curved surface area and circumference at the base of a solid right circular cylinder are 2200 cm^2 and 110 cm, respectively. Find the height of the cylinder.

SSC CPO 10/11/2022 (Morning)

- (a) 24 cm (b) 22 cm (c) 20 cm (d) 18 cm

Q.553. The length of a cold storage is 5 times its breadth, and its height is 8 m. The area of its four walls (including the doors) is 720 m^2 . Find the total surface area of cold storage.

SSC CPO 10/11/2022 (Afternoon)

- (a) 1290.6 m^2 (b) 1275.3 m^2
(c) 1265.25 m^2 (d) 1282.5 m^2

Q.554. A cylinder of height 8 cm and radius 6 cm is melted and converted into 3 cones of the same radius and height that of cylinder. Determine the total curved surface area of cones.

SSC CPO 10/11/2022 (Evening)

- (a) $180\pi \text{ cm}^2$ (b) $60\pi \text{ cm}^2$
(c) $144\pi \text{ cm}^2$ (d) $120\pi \text{ cm}^2$

Q.555. The heights of two right circular cones are in the ratio 1 : 5 and the perimeter of their bases are in the ratio 5: 3. Find the ratio of their volumes.

SSC CPO 10/11/2022 (Evening)

- (a) 8 : 11 (b) 7 : 6 (c) 5 : 9 (d) 3 : 4

Q.556. The area of the sector of a circle is 128 cm^2 . If the length of the arc of that sector is 64 cm, then find the radius of the circle.

SSC CPO 10/11/2022 (Evening)

- (a) 4 cm (b) 8 cm (c) 2 cm (d) 16 cm

Q.557. A cone-shaped storage tank's height and radius are 9 and 7 feet, respectively. Determine how much liquid the tank can contain. (Take $\pi = 3.14$)

SSC CPO 11/11/2022 (Morning)

- (a) 471 cubic feet
(b) 450 cubic feet
(c) 435.32 cubic feet
(d) 461.58 cubic feet

Q.558. A hall of 10 m length, 5 m width and 6 m height is to be painted (one side only). Determine the total surface area to

be painted. (neglect the area of windows and doors) & (floor area is NOT to be painted)

SSC CPO 11/11/2022 (Morning)

- (a) 300 m^2 (b) 280 m^2
(c) 330 m^2 (d) 230 m^2

Q.559. A hemispherical bowl of internal radius 6 cm contains a liquid. This liquid is to be filled into cylindrical shaped small bottles of diameter 2 cm and height 4 cm. How many bottles will be needed to empty the bowl?

SSC CPO 11/11/2022 (Afternoon)

- (a) 32 (b) 37 (c) 38 (d) 36

Q.560. The difference between the semi-perimeter and the sides of ΔPQR are 18 cm, 17 cm and 25 cm, respectively. Find the area of the triangle

SSC CPO 11/11/2022 (Evening)

- (a) $330\sqrt{510} \text{ cm}^2$ (b) $230\sqrt{510} \text{ cm}^2$
(c) $80\sqrt{510} \text{ cm}^2$ (d) $130\sqrt{510} \text{ cm}^2$

SSC CGL 2021 Tier - 2

Q.561. The sum of the curved surface area and the total surface area of a solid cylinder is 2068 cm^2 . If the radius of its base is 7 cm, then what is the volume of this cylinder? (use $\pi = \frac{22}{7}$)

SSC CGL Tier II (08/08/2022)

- (a) 2060 cm^3 (b) 2480 cm^3
(c) 3080 cm^3 (d) 2760 cm^3

Q.562. The slant height of a cone is 20 cm. If area of its base is 616 cm^2 , then what is the curved surface area of this cone?

SSC CGL Tier II (08/08/2022)

- (a) 960 cm^2 (b) 440 cm^2
(c) 1760 cm^2 (d) 880 cm^2

Q.563. The height of a cylinder is 6 cm more than the radius of its base. If its radius is 14 cm, then what will be volume of this cylinder? (use $\pi = \frac{22}{7}$)

SSC CGL Tier II (08/08/2022)

- (a) 13560 cm^3 (b) 14340 cm^3
(c) 10440 cm^3 (d) 12320 cm^3

Q.564. ABCD is a quadrilateral. The length of diagonal AC is 24 cm. The sum of perpendiculars drawn from vertex B and D on the diagonal AC is 10 cm. What is the area of the quadrilateral?

SSC CGL Tier II (08/08/2022)

- (a) 240 cm^2 (b) 180 cm^2
(c) 120 cm^2 (d) 90 cm^2

Q.565. The difference between the areas

of two concentric circles is 264 cm^2 . What is the difference between the square of their radius? (use $\pi = \frac{22}{7}$)

SSC CGL Tier II (08/08/2022)

- (a) 70 cm^2 (b) 140 cm^2
(c) 84 cm^2 (d) 64 cm^2

Q.566. Volume of a cone whose radius of a base and height are r and h respectively, is 400 cm^3 . What will be the volume of a cone whose radius of base and height are $2r \text{ cm}$ and $h \text{ cm}$ respectively?

SSC CGL Tier II (08/08/2022)

- (a) 100 cm^3 (b) 1600 cm^3
(c) 1200 cm^3 (d) 800 cm^3

SSC Selection Post (Phase - X)

Q.567. A conical vessel has radius 4 cm and its curved surface area is $20\pi \text{ cm}^2$. The volume of conical vessel is:

Graduate Level 02/08/2022 (Shift - 3)

- (a) $18\pi \text{ cm}^3$ (b) $12\pi \text{ cm}^3$
(c) $26\pi \text{ cm}^3$ (d) $16\pi \text{ cm}^3$

Q.568. A student wants to know the area of a rhombus, one of whose diagonals is 6 cm . He measures another diagonal as 8 cm . What is the area of the rhombus?

Graduate Level 05/08/2022 (Shift - 3)

- (a) 25 cm^2 (b) 20 cm^2
(c) 24 cm^2 (d) 48 cm^2

Q.569. The volumes of two cubes are in the ratio of $8 : 27$. If the volume of the larger cube is 513 cm^3 more than that of the smaller cube, find the length of each side of the smaller cube.

Higher Secondary 01/08/2022 (Shift - 3)

- (a) 27 cm (b) 7 cm (c) 6 cm (d) 16 cm

Q.570. A chord of a circle of radius 21 cm makes a right angle at the center. Find the area of major segment. (take $\pi = \frac{22}{7}$) closest to nearest integer.

Higher Secondary 03/08/2022 (Shift - 4)

- (a) 1360 cm^2 (b) 1436 cm^2
(c) 1260 cm^2 (d) 1160 cm^2

SSC MTS 2021 Tier - 1

Q.571. A goat is tied with a 10.5 m -long rope to one corner of a plot which is in the shape of an equilateral triangle each side of which is 16 m long. Find the area that the goat can graze on.

SSC MTS 05/07/2022 (Morning)

- (a) 62.25 m^2 (b) 53.5 m^2
(c) 57.75 m^2 (d) 51.75 m^2

Q.572. If the height and the radius of the base of a cone are both doubled, the volume of the cone becomes _____ of its previous volume.

SSC MTS 05/07/2022 (Morning)

- (a) 4 times (b) 3 times
(c) 6 times (d) 8 times

Q.573. A smaller circle touches a bigger circle internally and also passes through the centre 'O' of the bigger circle. If the area of the smaller circle is 192 cm^2 , the area of the bigger circle (in cm^2) is:

SSC MTS 05/07/2022 (Evening)

- (a) 768 (b) 384 (c) 1024 (d) 720

Q.574. The perimeter of a square is equal to the perimeter of a rectangle, and the latter has a width of 12 cm . The area of the square is 400 cm^2 . Find the area of the rectangle.

SSC MTS 06/07/2022 (Morning)

- (a) 412 cm^2 (b) 394 cm^2
(c) 324 cm^2 (d) 336 cm^2

Q.575. If the total surface area of a sphere of radius $r \text{ cm}$ is equal to the total surface area of a cube of side $s \text{ cm}$, which of the following is true?

SSC MTS 06/07/2022 (Morning)

- (a) $r > s$ (b) $r \geq s$ (c) $r = s$ (d) $r < s$

Q.576. The volume of a wall whose height is six times its breadth and length is half its height, is 23958 cm^3 . What is the breadth of the wall?

SSC MTS 06/07/2022 (Afternoon)

- (a) 21 cm (b) 15 cm (c) 11 cm (d) 18 cm

Q.577. A goldsmith uses 110 cm^3 of gold to make a wire 1 mm in diameter. He will use this wire to make gold chains of equal lengths. If each chain requires 56 cm length of wire, then how many gold chains can he make? [use $\pi = \frac{22}{7}$]

SSC MTS 07/07/2022 (Morning)

- (a) 320 (b) 250 (c) 160 (d) 210

Q.578. The measures of the outer dimensions of a closed metal box are given as $15 \text{ cm} \times 18 \text{ cm} \times 8 \text{ cm}$. The thickness of the metal is 1.5 cm . What will be the volume of the metal required to make the box?

SSC MTS 07/07/2022 (Evening)

- (a) 1680 cm^3 (b) 950 cm^3
(c) 1440 cm^3 (d) 1260 cm^3

Q.579. The length of a side of an equilateral triangle is 120 cm less than its perimeter. What is the area of the equilateral triangle?

SSC MTS 07/07/2022 (Evening)

- (a) $1800\sqrt{3} \text{ cm}^2$ (b) $900\sqrt{3} \text{ cm}^2$
(c) $450\sqrt{3} \text{ cm}^2$ (d) $1200\sqrt{3} \text{ cm}^2$

Q.580. The volume of a cuboid is 3300 cm^3 . The sum of the length and the breadth of the cuboid is 37 cm . If the difference between the length and the breadth of the cuboid is 13 cm , then what is the height of the cuboid?

SSC MTS 08/07/2022 (Morning)

- (a) 18 cm (b) 11 cm (c) 14 cm (d) 16 cm

Q.581. The length and the breadth of a rectangular cardboard are 40 cm and 35 cm , respectively. If the biggest possible square is cut out from this cardboard, what is the area of the remaining part of the cardboard?

SSC MTS 08/07/2022 (Afternoon)

- (a) 175 cm^2 (b) 275 cm^2
(c) 425 cm^2 (d) 325 cm^2

Q.582. The diagonal of a rectangle is 12 cm long and it is twice as long as one of the sides of the rectangle. Find area of the rectangle?

SSC MTS 08/07/2022 (Evening)

- (a) $25\sqrt{3} \text{ cm}^2$ (b) $30\sqrt{3} \text{ cm}^2$
(c) $36\sqrt{3} \text{ cm}^2$ (d) $42\sqrt{3} \text{ cm}^2$

Q.583. A solid sphere whose radius is 21 cm is melted and recast into a regular pyramid of square base. If the height of the pyramid is 66 cm , then what is the length of each side of the base of the pyramid? [use $\pi = \frac{22}{7}$].

SSC MTS 08/07/2022 (Evening)

- (a) 68 cm (b) 85 cm (c) 42 cm (d) 48 cm

Q.584. the total surface area of a closed cube is 128 cm^2 more than the area of a square. If the length of each side of the cube is 8 cm , then what is the length of each side of the square?

SSC MTS 11/07/2022 (Morning)

- (a) 16 cm (b) 12 cm (c) 18 cm (d) 9 cm

Q.585. In a camp, there are tents of the same shape and size. Each tent is cylindrical up to a height of 4 m and conical above it. The diameters of the bases of the cylinder and the cone are both 10.5 m and the slant height of the conical part is 10 m . If a total of 3861 m^2 canvas is used in making all the tents, then how many tents are there in the

camp? [use $\pi = \frac{22}{7}$]

SSC MTS 11/07/2022 (Evening)

- (a) 11 (b) 7 (c) 19 (d) 13

Q.586. A rectangular LCD has an area of 480 cm^2 and a perimeter of 92 cm. If the size of the LCD is defined in term of length of the diagonal, then what is the size of the LCD?

SSC MTS 12/07/2022 (Evening)

(a) 34 cm (b) 39 cm (c) 31 cm (d) 35 cm

Q.587. If the circumference of a circle is equal to the perimeter of a square and the radius of the given circle is positive, then which of the following options is correct?

SSC MTS 12/07/2022 (Evening)

(a) Area of the circle > Area of the square
(b) Area of the circle $\geq$ Area of the square
(c) Area of the circle < Area of the square
(d) Area of the circle = Area of the square

Q.588. If the length of one side of a rectangle is twice that of the other side, and its perimeter is 54 cm, then what is the length of the longer side of this rectangle?

SSC MTS 13/07/2022 (Evening)

(a) 14 cm (b) 9 cm (c) 27 cm (d) 18 cm

Q.589. A person uses a rectangular metallic sheet of $45 \text{ m} \times 30 \text{ m}$ for making an open box. From each corner of the sheet he cut offs a square. If the length of one side of each of the squares is 10 m, then what will be the volume of the box (in m^3)?

SSC MTS 13/07/2022 (Evening)

(a) 5000 (b) 4500 (c) 7000 (d) 2500

Q.590. The length of the longest diagonal that can be placed in a closed cube is $50\sqrt{3} \text{ cm}$. What is the total surface area of this cube?

SSC MTS 14/07/2022 (Evening)

(a) 15000 cm^2 (b) 10000 cm^2
(c) 12000 cm^2 (d) 6000 cm^2

Q.591. The larger diagonal of a rhombus is 150% of its smaller diagonal, and its area is 432 cm^2 . Find the length (in cm) of the side of the rhombus?

SSC MTS 14/07/2022 (Evening)

(a) $8\sqrt{13}$ (b) $4\sqrt{13}$ (c) $6\sqrt{13}$ (d) $2\sqrt{13}$

Q.592. The length of each side of a rhombus is 10 cm. If the length of one of its diagonals is 16 cm, then what is the area of the rhombus?

SSC MTS 15/07/2022 (Evening)

(a) 108 cm^2 (b) 112 cm^2
(c) 128 cm^2 (d) 96 cm^2

Q.593. Two cubes, each having a total surface area of 150 cm^2 , are joined together to form a cuboid. Find the total surface area of the resulting cuboid.

SSC MTS 18/07/2022 (Afternoon)

(a) 225 (b) 250 (c) 275 (d) 300

Q.594. The circumference of a circle is equal to the perimeter of an equilateral triangle. If the radius of the circle is 84 cm, then what will be the length of each side of the equilateral triangle?

[use $\pi = \frac{22}{7}$]

SSC MTS 19/07/2022 (Afternoon)

(a) 192 cm (b) 176 cm
(c) 172 cm (d) 184 cm

Q.595. The height of a cylinder is 3 cm more than the radius of its base. If the height is 10 cm, then what is the volume of this cylinder? [use $\pi = \frac{22}{7}$]

SSC MTS 20/07/2022 (Morning)

(a) 1540 cm^3 (b) 1860 cm^3
(c) 2160 cm^3 (d) 1240 cm^3

Q.596. What is the radius of a circular field whose area is equal to the sum of the area of three smaller circular fields of radii 8 m, 9 m and 12 m respectively?

SSC MTS 20/07/2022 (Morning)

(a) 20 m (b) 21 m (c) 29 m (d) 17 m

Q.597. A toy is formed by placing a solid circular cone on a solid hemisphere and the radius of the common base of the cone and the hemisphere is 15 cm. If the volume of the toy is $2850\pi \text{ cm}^3$, then find the total surface area of that toy.

SSC MTS 20/07/2022 (Morning)

(a) $815\pi \text{ cm}^2$ (b) $755\pi \text{ cm}^2$
(c) $705\pi \text{ cm}^2$ (d) $685\pi \text{ cm}^2$

Q.598. A large solid cube is melted and cast into 'N' small solid spheres, each of radius 3 cm, and 'N+2' small solid cuboids, Each of dimensions $4 \text{ cm} \times 4 \text{ cm} \times 6.5 \text{ cm}$. If the length of each side of the large solid cube is 12 cm, then find the value of 'N' [use $\pi = \frac{22}{7}$]

SSC MTS 20/07/2022 (Afternoon)

(a) 8 (b) 5 (c) 7 (d) 6

Q.599. The height of a trapezium is 68 cm and the sum of its parallel sides is $75 \frac{6}{17}$ cm. If the area of the trapezium is $\frac{6}{17}$ times of the area of a square, then the length of the diagonal of the square is:

(take $\sqrt{2} = 1.41$)

SSC MTS 21/07/2022 (Morning)

(a) 127.39 cm (b) 183.49 cm
(c) 119.85 cm (d) 102.39 cm

Q.600. Find the total surface area of a closed hemisphere whose diameter is 7 cm. [Use $\pi = \frac{22}{7}$]

SSC MTS 21/07/2022 (Evening)

(a) 77 cm^2 (b) 154 cm^2
(c) 110 cm^2 (d) 115.5 cm^2

Q.601. The ratio of the length to the width of a rectangle is 3:2. If the length of this rectangle is increased by 25% and its width is kept constant, then the area of the rectangle increases by 24 m^2 . What is the width of this rectangle?

SSC MTS 22/07/2022 (Morning)

(a) 12 m (b) 10 m (c) 8 m (d) 15 m

Q.602. If the ratio of an external angle and an internal angle of a regular polygon is 1 : 5, then what is the number of sides in the polygon?

SSC MTS 22/07/2022 (Afternoon)

(a) 24 (b) 22 (c) 12 (d) 30

Q.603. The area of a square is 1296 cm^2 and the radius of a circle is $\frac{7}{6}$ of the length of a side of the square. What is the ratio of the perimeter of the square and the circumference of the circle?

(use $\pi = \frac{22}{7}$)

SSC MTS 25/07/2022 (Morning)

(a) 13:11 (b) 8:11 (c) 6:11 (d) 3:7

Q.604. The length, the breadths, and the volumes of two cuboids are in the ratios of 4 : 5, 3 : 4 and 2 : 3 respectively. What is the ratio of their heights?

SSC MTS 25/07/2022 (Afternoon)

(a) 3 : 5 (b) 10 : 9 (c) 8 : 15 (d) 2 : 3

Q.605. The breadth of a rectangle is $\frac{4}{5}$ of the radius of a circle. The radius of the circle is $\frac{1}{5}$ of the side of a square, whose area is 625 cm^2 . What is the area of the rectangle if the length of rectangle is 20 cm?

SSC MTS 25/07/2022 (Afternoon)

(a) 150 cm^2 (b) 60 cm^2
(c) 100 cm^2 (d) 80 cm^2

Q.606. The diameter of a circle is $\frac{7}{4}$ times of the base of a triangle, and the height of the triangle is 14 cm. If the area of the triangle is 56 cm^2 , then what is the circumference (in m) of the circle - (use $\pi = \frac{22}{7}$)

SSC MTS 26/07/2022 (Morning)

(a) 0.44 m (b) 2.46 m
(c) 0.48 m (d) 1.74 m

Q.607. A square of side p is taken. A rectangle is cut out from this square

such that the length of one side of the rectangle is equal to half of the length of one side of the square and the length of another side of the rectangle is equal to $\frac{1}{3}$ of the length of the first side of the rectangle. What is the area of the portion of the square that remained after the rectangle is cut out?

SSC MTS 26/07/2022 (Afternoon)

- (a) $\frac{7}{8}p^2$ (b) $\frac{3}{4}p^2$ (c) $\frac{11}{12}p^2$ (d) $\frac{15}{16}p^2$

Q.608. The length of a rope, by which a calf is tied, is decreased from 9 m to 5 m. Find the decrease in the area that can be grazed by the calf. [use $\pi = \frac{22}{7}$]

SSC MTS 26/07/2022 (Evening)

- (a) $112 m^2$ (b) $154 m^2$
(c) $122 m^2$ (d) $176 m^2$

SSC CHSL 2021 Tier - 1

Q.609. A pillar 11 m in radius is 21 m high. How much material (in m^3) was used to construct it? (Use $\pi = \frac{22}{7}$)

SSC CHSL 24/05/2022 (Morning)

- (a) 7996 (b) 7886 (c) 7989 (d) 7986

Q.610. A sphere passes through the eight corners of a cube of side 14 cm. Find the volume (in cm^3) of the sphere.

($\pi = \frac{22}{7}$)

SSC CHSL 25/05/2022 (Morning)

- (a) $4312\sqrt{3}$ (b) $4012\sqrt{3}$
(c) $4312\sqrt{2}$ (d) $4012\sqrt{2}$

Q.611. A rectangular water tank is 1.5 m high, 3 m long and 3.5 m wide. How much water (in litres) can it hold?

SSC CHSL 25/05/2022 (Evening)

- (a) 13520 (b) 14510 (c) 15750 (d) 11780

Q.612. What is the volume of the largest right circular cone that can be cut out from a cube whose edge is 10 cm?

SSC CHSL 26/05/2022 (Morning)

- (a) $\frac{250\pi}{3}$ (b) $\frac{351\pi}{3}$ (c) $\frac{145\pi}{3}$ (d) $\frac{150\pi}{3}$

Q.613. 15 identical solid spherical balls of radius 5 cm are melted to form a single sphere. In this process, 35% of the mass is wasted. What is the radius (in cm) of the single sphere that is now formed? [Use $\pi = \frac{22}{7}$, and give your answer correct to two decimal places.]

SSC CHSL 26/05/2022 (Evening)

- (a) 12.68 (b) 16.68 (c) 10.68 (d) 34.68

Q.614. As air is pumped into a spherical balloon, its radius increases from 7 cm to

14 cm. Find the ratio of volume of the balloon's in both circumstances.

SSC CHSL 27/05/2022 (Morning)

- (a) 2 : 5 (b) 1 : 9 (c) 1 : 8 (d) 3 : 7

Q.615. A solid right-circular cylinder, whose radius of the base is 15 cm and height is 12 cm, is melted and moulded into the solid right - circular cone, whose radius of the base is 24 cm. What will be the height of this cone?

SSC CHSL 27/05/2022 (Evening)

- (a) 14.0625 cm (b) 14.0675 cm
(c) 14.6025 cm (d) 14.0525 cm

Q.616. A solid toy is in the shape which is a combination of a cylinder, cone and a hemispherical bowl. The cylinder contributes to 50% of the total volume of the toy, the cone contributes to 20% of the volume. Find the ratio of the contribution (in terms of volume) of the cone, cylinder and hemisphere.

SSC CHSL 30/05/2022 (Afternoon)

- (a) 2 : 3 : 5 (b) 5 : 2 : 3
(c) 2 : 5 : 3 (d) 4 : 5 : 3

Q.617. What is the volume (cm^3) of a right circular cylinder whose curved surface area is $12\pi cm^2$ and the circumference of its base is $4\pi cm$?

SSC CHSL 30/05/2022 (Evening)

- (a) 48π (b) 36π (c) 12π (d) 24π

Q.618. The sum of the length, breadth and depth of a cuboid is 23 cm, and its diagonal is $5\sqrt{7}$ cm. Its total surface area (cm^2) is:

SSC CHSL 31/05/2022 (Morning)

- (a) 144 (b) 354 (c) 188 (d) 111

Q.619. A cone-shaped temple top has slant height of 10 m and radius of base is 11 m. Find the total cost for painting its curved surface area at the rate of ₹7/ m^2 .

SSC CHSL 31/05/2022 (Afternoon)

- (a) ₹2440 (b) ₹2040
(c) ₹2420 (d) ₹2240

Q.620. The area of a triangle is $96 cm^2$ and the length of one of its sides is 24 cm. What is the length of the perpendicular drawn to this side of the triangle from the opposite vertex?

SSC CHSL 01/06/2022 (Morning)

- (a) 16 cm (b) 8 cm (c) 4 cm (d) 12 cm

Q.621. Find the volume (in cm^3) of the cuboid, if its length is twice more than its breadth and its breadth is 8 cm, which is twice its height.

SSC CHSL 01/06/2022 (Afternoon)

- (a) 768 (b) 769 (c) 767 (d) 766

Q.622. If the length of a cuboid is 20 m,

breadth is 5 m and height is 4 m, then find the length of diagonal of the cuboid.

SSC CHSL 01/06/2022 (Evening)

- (a) 12 m (b) 21 m (c) 18 m (d) 24 m

Q.623. The radius of a hemispherical bowl is given as 7 cm. What is the capacity (cm^3) of the bowl? [$\pi = \frac{22}{7}$]

SSC CHSL 02/06/2022 (Afternoon)

- (a) $718\frac{2}{3}$ (b) $812\frac{1}{3}$ (c) $618\frac{1}{3}$ (d) $518\frac{2}{3}$

Q.624. The length of each side of a regular hexagon is 10 cm. What is the area (cm^2) of the hexagon?

SSC CHSL 02/06/2022 (Evening)

- (a) $156\sqrt{3}$ (b) $144\sqrt{3}$ (c) $148\sqrt{3}$ (d) $150\sqrt{3}$

Q.625. The area of a square is $9m^2 + 12m + 4$. Find the measure of the side of the square.

SSC CHSL 02/06/2022 (Evening)

- (a) $3m + 2$ (b) $3m + 4$
(c) $3m^2 + 2$ (d) $3m^2 + 2m$

Q.626. The area of a square is $4x^2 - 12x + 9$, which of the following will be the side of the square?

SSC CHSL 03/06/2022 (Afternoon)

- (a) $(2x + 3)$ (b) $(2x - 3)$
(c) $(3x - 2)$ (d) $(3x + 2)$

Q.627. If the areas of three adjacent faces of a cuboidal box are $729 cm^2$, $529 cm^2$ and $289 cm^2$, respectively, then find the volume of the box.

SSC CHSL 03/06/2022 (Evening)

- (a) $10557 cm^3$ (b) $10560 cm^3$
(c) $10555 cm^3$ (d) $10551 cm^3$

Q.628. The diameter of the base and the slant height of a right circular cone are 10 cm and 16 cm, respectively. Find its

total surface area (cm^2). [Use $\pi = \frac{22}{7}$]

SSC CHSL 06/06/2022 (Morning)

- (a) $\frac{5720}{7}$ (b) $\frac{720}{7}$ (c) 330 (d) $\frac{1760}{7}$

Q.629. The diagonal of a rhombus is 8 cm and its area is $48 cm^2$, then the length of each side of the rhombus is:

SSC CHSL 06/06/2022 (Afternoon)

- (a) $\sqrt{13}$ (b) $2\sqrt{13}$ (c) $6\sqrt{13}$ (d) $5\sqrt{13}$

Q.630. The total surface area of a solid cylinder whose radius is r cm and height is $\frac{r}{2}$ cm (in cm^2) is:

SSC CHSL 07/06/2022 (Morning)

- (a) $3\pi \frac{r^2}{2}$ (b) $4\pi r^2$ (c) $3\pi r^2$ (d) $\pi \frac{r^3}{2}$

Q.631. The base of a hemispherical bowl

has a circumference of 132 cm. Find the capacity of the bowl [Use $\pi = \frac{22}{7}$]

SSC CHSL 07/06/2022 (Afternoon)

- (a) 18240 cm^3 (b) 19404 cm^3
(c) 19708 cm^3 (d) 16540 cm^3

Q.632. Find the volume of the cylinder circumscribing a sphere with a diameter of 14 cm. [Use $\pi = \frac{22}{7}$]

SSC CHSL 07/06/2022 (Evening)

- (a) 2156 cm^3 (b) 2158 cm^3
(c) 2155 cm^3 (d) 2157 cm^3

Q.633. The surface area of a spherical volleyball is $100\pi \text{ cm}^2$. Find the volume of air enclosed in the ball.

SSC CHSL 08/06/2022 (Morning)

- (a) $\frac{400}{3} \pi$ (b) $\frac{350}{3} \pi$
(c) $\frac{500}{3} \pi$ (d) $\frac{200}{3} \pi$

Q.634. Find the length of diagonal (m) of a cuboid 13 m long, 7 m broad and 3 m high.

SSC CHSL 08/06/2022 (Evening)

- (a) $\sqrt{207}$ (b) $\sqrt{232}$ (c) $\sqrt{327}$ (d) $\sqrt{227}$

Q.635. The radius of a roller is 14 cm and its length 20cm. It takes 235 complete revolutions to move once over to level a playground. Find the area of the playground. (Use $\pi = \frac{22}{7}$)

SSC CHSL 09/06/2022 (Afternoon)

- (a) 4136 cm^2 (b) $4136 \times 10^3 \text{ cm}^2$
(c) 41360 cm^2 (d) $4136 \times 10^2 \text{ cm}^2$

Q.636. The length of the hypotenuse of a right-angled triangle is 37 cm and the length of one of its other two sides is 12 cm. What is the area of the triangle in cm^2 ?

SSC CHSL 09/06/2022 (Evening)

- (a) 222 (b) 204 (c) 216 (d) 210

Q.637. The area of quadrilateral is 336 m^2 and the perpendiculars drawn to one diagonal from the opposite vertices are 16 m and 12 m long. Find the length of this diagonal.

SSC CHSL 10/06/2022 (Morning)

- (a) 28 cm (b) 26 cm (c) 21 cm (d) 24 cm

Q.638. How many spherical balls of radius 5 cm can be made by melting a spherical clay ball having a radius of 15 cm?

SSC CHSL 10/06/2022 (Afternoon)

- (a) 18 (b) 3 (c) 9 (d) 27

Q.639. What is the height of a solid right circular cylinder whose radius is 3 cm

and total surface area is $60\pi \text{ cm}^2$?

SSC CHSL 10/06/2022 (Evening)

- (a) 3 cm (b) 5 cm (c) 7 cm (d) 9 cm

SSC CGL 2021 Tier - 1

Q.640. A solid cube of side 8 cm is dropped into a rectangular container of length 16cm, breadth 8 cm and height 15 cm which is partly filled with water. If the cube is completely submerged, then the rise of water level (in cm) is :

SSC CGL 11/04/2022 (Morning)

- (a) 2 (b) 6 (c) 5 (d) 4

Q.641. A solid cube of side 8 cm is dropped into a rectangular container of length 16cm, breadth 8 cm and height 15cm which is partly filled with water. If the cube is completely submerged, then the rise of water level (in cm) is :

SSC CGL 11/04/2022 (Morning)

- (a) 2 (b) 6 (c) 5 (d) 4

Q.642. What is the difference in the volume (in cm^3) of a sphere of radius 7 cm and that of a cone of a radius 7 cm and height of 10 cm? [Use $\pi = \frac{22}{7}$]

SSC CGL 12/04/2022 (Morning)

- (a) 205 (b) 704 (c) 924 (d) 1078

Q.643. If the length of a diagonal of a square is (a + b), then the area of the square is:

SSC CGL 13/04/2022 (Morning)

- (a) $a^2 + b^2$ (b) $\frac{1}{2}(a^2 + b^2) + ab$
(c) $a^2 + b^2 + 2ab$ (d) $\frac{1}{2}(a^2 + b^2)$

Q.644. Let $x \text{ cm}^2$ be the surface area and $y \text{ cm}^3$ be the volume of a sphere such that $y = 14x$. What is the radius (in cm) of the sphere?

SSC CGL 13/04/2022 (Evening)

- (a) 102 (b) 42 (c) 51 (d) 68

Q.645. A hemispherical depression of diameter 4 cm is cut out from each face of a cubical block of sides 10 cm. Find the surface area of the remaining solid

(in cm^2) (Use $\pi = \frac{22}{7}$)

SSC CGL 18/04/2022 (Morning)

- (a) $900\frac{4}{7}$ (b) $112\frac{4}{7}$
(c) $675\frac{3}{7}$ (d) $713\frac{1}{7}$

Q.646. The area of a cardboard (in cm^2) needed to make a closed box of size 20 cm $\times$ 10 cm $\times$ 8 cm will be:

SSC CGL 18/04/2022 (Evening)

- (a) 880 (b) 690 (c) 750 (d) 960

Q.647. From the body of a solid cube of edge 7 cm, a solid sphere is removed. The volume of the remaining solid was found to be $163\frac{1}{3} \text{ cm}^3$. What is the diameter (in cm) of the sphere?

(Use $\pi = \frac{22}{7}$)

SSC CGL 19/04/2022 (Morning)

- (a) 10 (b) 7 (c) 5 (d) 8

Q.648. The length of a wire (in cm) of 0.1mm radius that can be drawn from melting a solid copper sphere of a diameter 6 cm is:

SSC CGL 19/04/2022 (Evening)

- (a) 250000 (b) 360000
(c) 440000 (d) 810000

Q.649. The inner circumference of a circular path enclosed between two concentric circles is 264m. The uniform width of the circular path is 3m. What is the area (in m^2 , to the nearest whole

number) of the path? (Use $\pi = \frac{22}{7}$)

SSC CGL 20/04/2022 (Morning)

- (a) 696 (b) 756 (c) 820 (d) 948

Q.650. The total surface area of a right pyramid, with base as a square of side 8 cm, is 208 cm^2 . What is the slant height (in cm) of the pyramid?

SSC CGL 20/04/2022 (Afternoon)

- (a) 7 (b) 10 (c) 9 (d) 8

Q.651. A 22.5 m high tent is the shape of a frustum of a cone surmounted by a hemisphere. If the diameter of the upper and the lower circular end of the frustum are 21 m and 39 m respectively, then find the areas of the cloth (in m^2) used to make the tent (ignoring the wastage).

(Use $\pi = \frac{22}{7}$)

SSC CGL 21/04/2022 (Evening)

- (a) $787\frac{2}{7}$ (b) $2107\frac{2}{7}$
(c) $1635\frac{6}{7}$ (d) $2800\frac{2}{7}$

SSC CGL 2020 Tier - 2

Q.652. A solid metallic cuboid of dimensions 12 cm $\times$ 54 cm $\times$ 72 cm is melted and converted into 8 cubes of the same size. What is the sum of the lateral surface areas (in cm^2) of 2 such cubes?

SSC CGL Tier II (29/01/2022)

- (a) 2592 (b) 2268 (c) 3888 (d) 1944

Q.653. The radius of a spherical balloon is inflated from 3.5 cm to 4.9 cm by pushing air into it. What is the percentage increase in the volume of the

original balloon ?

SSC CGL Tier II (29/01/2022)

(a) 74.4% (b) 174.4% (c) 73.6% (d) 173.6%

Q.654. The radius of the base of a cylindrical tank is 4 m. If three times the sum of the areas of its two circular faces is twice the area of its curved surface, then the capacity (in kilolitres) of the tank is:

SSC CGL Tier II (29/01/2022)

(a) 54π (b) 108π (c) 96π (d) 144π

Q.655. The slant height and radius of a right circular cone are in the ratio 29 : 20.

If its volume is $4838.4 \pi \text{ cm}^3$, then its radius is:

SSC CGL Tier II (29/01/2022)

(a) 20 cm (b) 28 cm (c) 24 cm (d) 25 cm

Q.656. The areas of three adjacent faces of a cuboidal solid block of wax are 216 cm^2 , 96 cm^2 and 144 cm^2 . It is melted and 8 cubes of the same size are formed from it. What is the lateral surface area (in cm^2) of 3 such cubes?

SSC CGL Tier II (03/02/2022)

(a) 648 (b) 432 (c) 576 (d) 288

Q.657. A well with inner radius 3 m, is dug 6 m deep. The soil taken out of it has been spread evenly all around it to a width of 2 m to form an embankment. The height (in m) of the embankment is:

SSC CGL Tier II (03/02/2022)

(a) $4\frac{1}{2}$ (b) $4\frac{1}{4}$ (c) $3\frac{1}{4}$ (d) $3\frac{3}{8}$

Q.658. A solid metallic sphere of radius 4 cm is melted and recast into spheres of 2 cm each. What is the ratio of the surface area of the original sphere to the sum of surface areas of the spheres, so formed?

SSC CGL Tier II (03/02/2022)

(a) 2 : 1 (b) 2 : 3 (c) 1 : 2 (d) 1 : 4

Q.659. A cylindrical tube, open at both ends, is made of a metal sheet which is 0.5 cm thick. Its outer radius is 4 cm and length is 2 m. How much metal (in cm^3) has been used in making the tube ?

SSC CGL Tier II (03/02/2022)

(a) 800π (b) 450π (c) 750π (d) 550π

Q.660. The base of a right prism is a triangle whose sides are 8 cm, 15 cm and 17 cm, and its lateral surface area is 480 cm^2 . What is the volume (in cm^3) of the prism?

SSC CGL Tier II (03/02/2022)

(a) 540 (b) 600 (c) 720 (d) 640

Q.661. The curved surface area of a right circular cone is 2310 cm^2 and its radius is 21 cm. If its radius is increased by

100% and height is reduced by 50%, then its capacity (in litres) will be correct to one decimal place) .(Use $\pi = \frac{22}{7}$)

SSC CGL Tier II (03/02/2022)

(a) 27.8 (b) 28.2 (c) 26.7 (d) 25.9

SSC MTS 2020 Tier - 1

Q.662. The area of a square is S and that of the square formed by joining the mid points of the given square is A. Then the value of $\frac{A}{S}$ is equal to:

SSC MTS 05/10/2021 (Morning)

(a) 0.25 (b) 0.5 (c) 0.125 (d) 1.25

Q.663. A container in the shape of a right circular cone, whose radius and depth are equal, gets completely filled by 128000 spherical droplets, each of diameter 2mm. What is the radius (in cm) of the container?

SSC MTS 05/10/2021 (Morning)

(a) 6 (b) 4 (c) 8 (d) 2

Q.664. The total surface area of a solid right circular cylinder of height 13 cm, is 880 cm^2 . Its volume (in cm^3) is $11k$. The value of k is: (Use $\pi = \frac{22}{7}$)

SSC MTS 05/10/2021 (Afternoon)

(a) 208 (b) 182 (c) 104 (d) 91

Q.665. The area of a parallelogram ABCD is 300 cm^2 . The distance between AB and CD is 20cm and the distance between BC and AD is 30cm. What is the perimeter (in cm) of the parallelogram?

SSC MTS 05/10/2021 (Evening)

(a) 60 (b) 50 (c) 40 (d) 100

Q.666. The radii of the circular ends of a frustum of a cone are 20 cm and 13 cm and its height is 12 cm. What is the capacity (in litres) of the frustum (correct to one decimal place)? (Use $\pi = \frac{22}{7}$)

SSC MTS 06/10/ 2021 (Morning)

(a) 10.4 (b) 11.2 (c) 10.8 (d) 11.4

Q.667. If the width of the path around a square field is 4.5 m and the area of the path is 252 m^2 , then the length of the side of the field is:

SSC MTS 06/10/ 2021 (Morning)

(a) 9.5 m (b) 9 m (c) 8 m (d) 8.5 m

Q.668. A solid metal cube of edge 12 cm is immersed completely in water in a cylindrical vessel whose radius is 20 cm and height is 32 cm, and the water in the vessel is up to a height of 15 cm. The height (in cm) by which the water will rise in the vessel is (correct to one decimal

place):

SSC MTS 06/10/2021 (Evening)

(a) 0.9 (b) 1.4 (c) 1.2 (d) 1.1

Q.669. From each of the four corners of a rectangular sheet of dimensions 48 cm $\times$ 27 cm, a square of side 3.5 cm is cut off and a box is made. The volume of the box is:

SSC MTS 07/10/2021 (Morning)

(a) 2880 (b) 2780 (c) 2870 (d) 2860

Q.670. The internal dimensions of a rectangular closed wooden box are 30 cm, 18cm and 23cm,. What space (in cm^3) will it occupy, if the wood used is 1cm thick?

SSC MTS 08/10/2021 (Morning)

(a) 12420 (b) 16000 (c) 14136 (d) 9408

Q.671. A solid metallic cylinder with a base radius of 3 cm and a height of 120 cm is melted and recast into a sphere of radius r cm. What is the value of r if there is a 10% metal loss in the conversion?

SSC MTS 08/10/2021 (Afternoon)

(a) 9 (b) 8 (c) 13.5 (d) 12

Q.672. The difference between the areas of two concentric circles is 88 cm^2 . If the radius of the inner circle is 6 cm then the area (in cm^2) of the larger circle is closest to: (Use $\pi = \frac{22}{7}$)

SSC MTS 08/10/2021 (Afternoon)

(a) 196 (b) 201 (c) 197 (d) 198

Q.673. A rectangular plot is 420 m long and 84 m wide. It is surrounded by semi-circular flower beds along all its sides. The area of the entire plot (in hectares) is x. The value of x (correct to one decimal place) is: (Use $\pi = \frac{22}{7}$)

SSC MTS 08/10/2021 (Evening)

(a) 19.1 (b) 18.7 (c) 17.9 (d) 19.4

Q.674. From a wooden cubical block of side 10 cm, a sphere of radius 4.2 cm is carved out. How much wood is wasted in the process (Use $\pi = \frac{22}{7}$)

SSC MTS 11/10/2021 (Morning)

(a) 476.2 cm^3 (b) 689.536 cm^3
(c) 310.464 cm^3 (d) 523.8 cm^3

Q.675. A typist uses a paper of size 32 cm $\times$ 20 cm. He leaves a margin of 2 cm each on all the sides. If he leaves a margin of 1 cm only on all the sides. then what is the percentage of the increase in the area available for typing (correct to 2 decimal places)?

SSC MTS 11/10/2021 (Morning)

(a) 20.54% (b) 82.69%
(c) 17.04% (d) 79.46%

Q.676. A rectangular water reservoir contains 126 kilolitres of water. The depth of the reservoir is 3.5 m and its length measures 8m. What is the width of the reservoir?

SSC MTS 11/10/2021 (Afternoon)

- (a) 4.5 m (b) 4.2 m (c) 4 m (d) 5 m

Q.677. A hall is 18 m long and 12 m broad. If the sum of the areas of the floor and the ceiling is equal to the sum of the areas of the four walls, the volume (in m^3) of the hall is:

SSC MTS 12/10/2021 (Morning)

- (a) 1175.5 (b) 1876.2
(c) 1555.2 (d) 1376.4

Q.678. A wooden piece in the shape of a cuboid is divided into five equal parts by cutting it with four cuts perpendicular to its length, and it turns out that each piece is a cube of volume $27 cm^3$. What is the total surface area (in cm^2) of the cuboid?

SSC MTS 12/10/2021 (Afternoon)

- (a) 150 (b) 174 (c) 216 (d) 198

Q.679. The material of a sphere of radius r is melted and recast into a hollow cylindrical shell of thickness a and outer radius b . What is its length assuming that no material is lost in recasting?

SSC MTS 12/10/2021 (Afternoon)

- (a) $\frac{4r^3}{3a(2a-b)}$ (b) $\frac{4r^3}{3a(b+a)(b-a)}$
(c) $\frac{4r^3}{3a(b-a)}$ (d) $\frac{4r^3}{3a(2b-a)}$

Q.680. The radius of the base of a cone and also that of a hemisphere each is 14 cm. Also, the volumes of these two solids are equal. The curved surface area of the cone is:

(Take $\pi = \frac{22}{7}$)

SSC MTS 12/10/2021 (Evening)

- (a) $614\sqrt{5}$ (b) $626\sqrt{5}$ (c) $616\sqrt{5}$ (d) $516\sqrt{5}$

Q.681. A boy walked along two adjacent sides of a rectangular field. If he had walked along the diagonal, then he would have saved a distance equal to one-fourth of the larger side. The ratio of the larger to the smaller side is:

SSC MTS 13/10/2021 (Morning)

- (a) 7 : 24 (b) 11 : 18 (c) 24 : 7 (d) 18 : 11

Q.682. The length of a rectangle is three-fifth of the radius of a circle. The radius of the circle is equal to the side of a square whose area is $6400 m^2$. Find the area (in m^2) of the rectangle, if the breadth is 20 m is:

SSC MTS 13/10/2021 (Morning)

- (a) 1200 (b) 480 (c) 960 (d) 1000

Q.683. Kamala has three cylindrical containers, each with radius 20 cm that she wants to fill with grain. If her bag of grain contains 100 kg of grain, and she fills all three containers to a height of 70 cm, how much grain will be left in the bag. If 1 kg grain occupies $3000 cm^3$ space? (Use $\pi = \frac{22}{7}$)

SSC MTS 13/10/2021 (Afternoon)

- (a) 88 kg (b) 50 kg (c) 58 kg (d) 12 kg

Q.684. The radius of a spherical balloon increases from 6 cm to 10 cm when more air is pumped into it. The ratio of the surface area of the original balloon and the inflated balloon is:

SSC MTS 13/10/2021 (Afternoon)

- (a) 4 : 5 (b) 27 : 125 (c) 3 : 5 (d) 9 : 25

Q.685. The radius of a cylinder is reduced to 50% of its actual radius. If its volume remains the same as before, then its height becomes k times the original height. The value of k is:

SSC MTS 14/10/2021 (Evening)

- (a) 2 (b) 16 (c) 8 (d) 4

Q.686. The length and breadth of a rectangle are 35 cm and 12 cm, respectively. What will be the circumference of the circle that inscribes this rectangle? ($\pi = \frac{22}{7}$)

SSC MTS 14/10/2021 (Evening)

- (a) $115\frac{3}{7}$ cm (b) $117\frac{1}{7}$ cm
(c) $114\frac{4}{7}$ cm (d) $116\frac{2}{7}$ cm

Q.687. Four solid cubes, each of volume $1728 cm^3$, are kept in two rows having two cubes in each row. They form a rectangular solid with a square base. The total surface area (in cm^2) of the resulting solid is:

SSC MTS 18/10/2021 (Afternoon)

- (a) 1440 (b) 2304 (c) 576 (d) 1152

Q.688. A right circular cone is inscribed in a cube of side 14 cm occupying the maximum space possible. What is the ratio of the volume of the cube to the volume of the cone? ($\pi = \frac{22}{7}$)

SSC MTS 18/10/2021 (Afternoon)

- (a) 81 : 22 (b) 11 : 42
(c) 42 : 11 (d) 22 : 81

Q.689. Three circles, each having radius equal to 4 cm, are drawn with the vertices of an equilateral triangle as the centres. If the length of each side of the triangle is equal to 8 cm, then what is the area (in cm^2) of the portion of the triangle

that is NOT covered by the sectors of the circles?

SSC MTS 18/10/2021 (Evening)

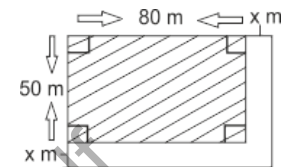
- (a) $16(2\sqrt{3} - \pi)$ (b) $8(2\sqrt{3} - \pi)$
(c) $8(2\sqrt{3} - \frac{\pi}{2})$ (d) $16(2\sqrt{3} - \frac{\pi}{2})$

Q.690. Three cubes each having volume $1728 cm^3$ are placed one above the other. What is the total surface area of the resulting solid?

SSC MTS 20/10/2021 (Morning)

- (a) 2304 (b) 2592 (c) 2160 (d) 2016

Q.691. In the given figure, the area of the unshaded region is 35% of the area of the shaded region. What is the value of x



SSC MTS 20/10/2021 (Morning)

- (a) 20 (b) 15 (c) 10 (d) 5

Q.692. The sum of the height and radius of the base of a solid right circular cylinder is 46 cm. If the total surface area of the solid cylinder is $6072 cm^2$, what is the volume (in cm^3) of the cylinder?

SSC MTS 20/10/2021 (Evening)

- (a) 11025π (b) 3030π
(c) 11020π (d) 3036π

Q.693. A solid right circular cone is cut into two parts by a plane parallel to its base and at a point one-third vertically below its vertex. What is the ratio of the volume of the smaller cone to that of the whole cone?

SSC MTS 20/10/2021 (Evening)

- (a) $\frac{1}{9}$ (b) $\frac{1}{8}$ (c) $\frac{1}{27}$ (d) $\frac{1}{3}$

Q.694. A horse racecourse is in the form of an annular ring whose outer and inner circumferences are 748 m and 396 m, respectively. The width (in m) of the racecourse is: ($\pi = \frac{22}{7}$)

SSC MTS 22/10/2021 (Morning)

- (a) 176 (b) 88 (c) 56 (d) 28

Q.695. A hall is 18 m long and 12 m broad. If the area of the floor is equal to the sum of the areas of the four walls, the volume (in m^3) of the hall is:

SSC MTS 22/10/2021 (Evening)

- (a) 876.2 (b) 777.6 (c) 576.4 (d) 675.5

Q.696. The area of a triangular field with sides 60 m, 175 m and 185 m is equal to that of a rectangular park whose sides are in the ratio of 21 : 10. The perimeter (in m) of the park is:

SSC MTS 26/10/2021 (Afternoon)

(a) 341 (b) 403 (c) 372 (d) 310

Q.697. A circular wire of diameter 56 cm is folded in the shape of a rectangle whose sides are in the ratio 7 : 4. The area enclosed by the rectangle is:

SSC MTS 26/10/2021 (Evening)

(a) 1842 (b) 1792 (c) 1684 (d) 1782

Q.698. A solid metallic toy has a hemispherical base with radius 3.5 cm surmounted by a cone. If the height of the cone is the same as the radius of its base, then the volume of metal used in cm^3

SSC MTS 27/10/2021 (Morning)

(a) $89\frac{2}{3}$ (b) $144\frac{3}{4}$ (c) $134\frac{3}{4}$ (d) $179\frac{2}{3}$

Q.699. A bucket in the shape of the frustum of a cone has its top and bottom radii as 20 cm and 10 cm, respectively. The depth of the bucket is 24 cm. The

capacity of the bucket (in cm^3) is :

$$(\pi = \frac{22}{7})$$

SSC MTS 02/11/2021 (Morning)

(a) 8800 cm^3 (b) 13200 cm^3 (c) 17000 cm^3 (d) 17600 cm^3

Q.700. A rectangle of dimension 10 cm and 5 cm is placed adjacent to another rectangle of the same size to draw an L shape figure. Find the perimeter of the shape so formed.

SSC MTS 02/11/2021 (Morning)

(a) 100 (b) 40 (c) 50 (d) 60

Q.701. The total surface area of a hemisphere is nearly equal to that of an equilateral triangle. The side of the triangle is how many times (approximately) of the radius of the hemisphere?

SSC MTS 02/11/2021 (Evening)

(a) $(\frac{8\pi}{\sqrt{3}})^{0.5}$ (b) $(\frac{4\pi}{3})^{0.5}$ (c) $(2\pi\sqrt{3})^{0.5}$ (d) $(4\pi\sqrt{3})^{0.5}$

Q.702. When a rectangle is divided into three equal parts, each of them turns out to be a square of 16 cm^2 . What is the perimeter (in cm) of the rectangle?

SSC MTS 02/11/2021 (Evening)

(a) 64 (b) 16 (c) 32 (d) 48

Q.703. The area (in cm^2) of a sector of a circle of radius 2 cm is $\frac{4\pi}{5}$. What is the central angle (in degrees) of the sector?

SSC MTS 02/11/2021 (Evening)

(a) 60 (b) 75 (c) 36 (d) 72

SSC CGL 2020 Tier - 1

Q.704. A heap of wheat is in the form of a cone whose base diameter is 8.4 m and height is 1.75 m. The heap is to be covered by canvas. What is the area (in m^2) of the canvas required ? ($\pi = \frac{22}{7}$)

SSC CGL 16/08/2021 (Afternoon)

(a) 60.06 (b) 115.05 (c) 63.6 (d) 115.5

Q.705. The area of the square shaped field is 1764 m^2 . The breadth of a rectangular park is $\frac{1}{3}$ rd the side of the square field and its length is two times its breadth. What is the cost (in Rs.) of leveling the park at Rs.15 per m^2 ?

SSC CGL 16/08/2021 (Evening)

(a) 4200 (b) 4290 (c) 5880 (d) 4320

Q.706. The volume of a wall whose height is 10 times its width and whose length is 8 times its height, is 51.2 m^3 . What is the cost (in Rs.) of painting the wall on one side at the rate of Rs.100/ m^2 ?

SSC CGL 20/08/2021 (Afternoon)

(a) 12750 (b) 12500 (c) 12800 (d) 12250

Q.707. A square has the perimeter equal to the circumference of a circle having radius 7 cm. What is the ratio of the area of the circle to the area of the square?

SSC CGL 23/08/2021 (Afternoon)

(a) 7 : 2 (b) 14 : 11 (c) 7 : 11 (d) 121 : 44

Q.708. The area of the circular path enclosed by two concentric circles is 3080 m^2 . If the difference between the radius of the outer edge and that of the inner edge of the circular path is 10m, what is the sum (in m) of the two radii?

SSC CGL 23/08/2021 (Afternoon)

(a) 70 (b) 112 (c) 98 (d) 84

SSC CHSL 2020 Tier - 1

Q.709. One diagonal of a rhombus is $8\sqrt{3} \text{ cm}$. If the other diagonal is equal to its side, then the area of rhombus is:

SSC CHSL 12/04/2021 (Morning)

(a) $12\sqrt{3}$ (b) $24\sqrt{3}$ (c) $16\sqrt{3}$ (d) $32\sqrt{3}$

Q.710. What is the perimeter of a right-angled triangle whose sides (making 90°) are $20\sqrt{3} \text{ cm}$ and $15\sqrt{3} \text{ cm}$?

SSC CHSL 13/04/2021 (Morning)

(a) $60\sqrt{3}$ (b) $55\sqrt{3}$ (c) $65\sqrt{3}$ (d) $50\sqrt{3}$

Q.711. The perimeter of an isosceles triangle is 91 cm and its base is $1\frac{1}{4}$ times each of its equal sides. What is the length of its base ? (in cm)

SSC CHSL 16/04/2021 (Morning)

(a) 35 (b) 20 (c) 25 (d) 28

Q.712. The lengths of three sides of a triangle are in the ratio 3:4:5. Among three sides, the difference between the largest side and the smallest side of this triangle is 3.6 cm. The area of this triangle is:

SSC CHSL 16/04/2021 (Evening)

(a) 19.44 (b) 32.4 (c) 21.75 (d) 15.64

Q.713. One - quarter of a circular pizza of diameter 28 cm was removed from the whole pizza. What is the perimeter (in cm) of the remaining pizza? ($\pi = \frac{22}{7}$)

SSC CHSL 04/08/2021 (Morning)

(a) 88 (b) 94 (c) 80 (d) 66

Q.714. From a rectangle sheet of dimensions $32 \text{ cm} \times 18 \text{ cm}$, a square of side 3 cm is cut from the four corners of the sheet and a box is made. The volume of the box is:

SSC CHSL 05/08/2021 (Afternoon)

(a) 946 (b) 936 (c) 1305 (d) 1300

Q.715. How many spherical bullets, each bullet being 7 cm in diameter, can be made out of a cube of lead whose edge measures 77 cm? ($\pi = \frac{22}{7}$)

SSC CHSL 05/08/2021 (Evening)

(a) 1452 (b) 4521 (c) 2451 (d) 2541

Q.716. The base of a triangle is equal to the perimeter of a square whose diagonal is $7\sqrt{2} \text{ cm}$ and its height is equal to the side of a square whose area is 169 cm^2 . The area (in cm^2) of the triangle is :

SSC CHSL 06/08/2021 (Afternoon)

(a) 182 (b) 175 (c) 156 (d) 130

Q.717. The perimeter of a rectangle is 86 cm. The measures of its area and breadth are in the ratio of 9 : 1, respectively. The breadth of the rectangle is:

SSC CHSL 11/08/2021 (Evening)

(a) 32 cm (b) 34 cm (c) 36 cm (d) 30 cm

Q.718. The square of the diagonal of a cube is 2175 cm^2 . What is the total surface area (in cm^2) of the cube?

SSC CHSL 12/08/2021 (Afternoon)

(a) 4305 (b) 4272 (c) 4530 (d) 4350

SSC CPO 2020 Tier - 1

Q.719. The sides of a triangle are 24cm, 26cm and 10cm. At each of its vertices, circles of radius 4.2 cm are drawn. What is the area (in cm^2) of the triangle, excluding the portion covered by the sectors of the circle ?

SSC CPO 23/11/2020 (Morning)

(a) 27.72 (b) 120 (c) 105.86 (d) 92.28

Q.720. A 9 cm solid metallic cube and a solid metallic cuboid having dimensions 5 cm, 13cm, 31cm are melted and recast into a single cube. What is the total surface area(in cm^2) of the new cube?
SSC CPO 23/11/2020 (Evening)
(a) 2744 (b) 1362 (c) 865 (d) 1176

Q.721. A rectangular lawn whose length is twice of its breadth is extended by having four semi-circular portions on its sides. What is the total cost (in Rs.) of levelling the entire lawn at the rate of Rs.100 per square metre if the smaller side of the rectangular lawn is 12 m :
SSC CPO 23/11/2020 (Evening)
(a) 85,320 (b) 86,540
(c) 78,650 (d) 97,625

Q.722. A 9 cm solid metallic cube and a solid metallic cuboid having dimensions 5 cm, 13cm, 31cm are melted and recast into a single cube. How much (in Rs.) is the cost to polish the new cube at the rate of Rs.10 per cm^2
SSC CPO 24/11/2020 (Morning)
(a) 13,620 (b) 11,760 (c) 8,650 (d) 27,440

Q.723. A hexagon is $1944\sqrt{3}m^2$. What will be the cost (in Rs) of fencing it at the rate of Rs.11.50 per metre?
SSC CPO 24/11/2020 (Morning)
(a) 2,256 (b) 2,785 (c) 3,200 (d) 2,484

Q.724. The base of a triangle is equal to the perimeter of a square whose diagonal is $6\sqrt{2}cm$, and its height is equal to the side of a square whose area is $144 cm^2$.
The area of the triangle (in m^2)
SSC CPO 24/11/2020 (Evening)
(a) 144 (b) 216 (c) 288 (d) 72

Q.725. The perimeter of a square is half the perimeter of a rectangle. The perimeter of the square is 40m. If its breadth is two-thirds of its length, then what is the area of the rectangle ?
SSC CPO 25/11/2020 (Morning)
(a) 321 (b) 196 (c) 400 (d) 384

Answer Key :-

1.(b)	2.(c)	3.(d)	4.(b)
5.(b)	6.(a)	7.(d)	8.(d)
9.(b)	10.(c)	11.(c)	12.(a)
13.(b)	14.(d)	15.(d)	16.(a)
17.(d)	18.(b)	19.(b)	20.(b)
21.(a)	22.(c)	23.(c)	24.(b)
25.(d)	26.(a)	27.(d)	28.(a)
29.(a)	30.(a)	31.(c)	32.(c)

33.(b)	34.(b)	35.(c)	36.(d)
37.(b)	38.(c)	39.(a)	40.(b)
41.(c)	42.(d)	43.(b)	44.(d)
45.(a)	46.(a)	47.(d)	48.(d)
49.(b)	50.(c)	51.(c)	52.(a)
53.(d)	54.(c)	55.(a)	56.(c)
57.(b)	58.(c)	59.(d)	60.(c)
61.(b)	62.(c)	63.(c)	64.(d)
65.(b)	66.(d)	67.(b)	68.(d)
69.(d)	70.(a)	71.(c)	72.(d)
73.(b)	74.(d)	75.(d)	76.(c)
77.(a)	78.(a)	79.(d)	80.(c)
81.(b)	82.(b)	83.(b)	84.(c)
85.(a)	86.(c)	87.(d)	88.(a)
89.(a)	90.(a)	91.(d)	92.(a)
93.(d)	94.(c)	95.(a)	96.(b)
97.(b)	98.(b)	99.(a)	100.(c)
101.(d)	102.(a)	103.(c)	104.(b)
105.(d)	106.(c)	107.(a)	108.(c)
109.(b)	110.(c)	111.(c)	112.(c)
113.(b)	114.(a)	115.(b)	116.(d)
117.(b)	118.(d)	119.(a)	120.(b)
121.(b)	122.(a)	123.(a)	124.(a)
125.(b)	126.(a)	127.(d)	128.(b)
129.(a)	130.(b)	131.(a)	132.(b)
133.(b)	134.(c)	135.(b)	136.(c)
137.(c)	138.(c)	139.(b)	140.(b)
141.(d)	142.(b)	143.(b)	144.(a)
145.(a)	146.(a)	147.(a)	148.(b)
149.(b)	150.(c)	151.(d)	152.(b)
153.(b)	154.(d)	155.(b)	156.(d)
157.(a)	158.(b)	159.(d)	160.(b)
161.(d)	162.(b)	163.(b)	164.(b)
165.(b)	166.(c)	167.(a)	168.(a)
169.(d)	170.(d)	171.(c)	172.(b)
173.(a)	174.(b)	175.(b)	176.(c)
177.(d)	178.(a)	179.(a)	180.(a)
181.(d)	182.(d)	183.(d)	184.(c)
185.(c)	186.(b)	187.(d)	188.(d)
189.(b)	190.(a)	191.(d)	192.(b)
193.(c)	194.(b)	195.(b)	196.(d)
197.(c)	198.(d)	199.(b)	200.(b)
201.(a)	202.(d)	203.(b)	204.(b)
205.(b)	206.(d)	207.(b)	208.(c)
209.(a)	210.(b)	211.(d)	212.(b)
213.(d)	214.(a)	215.(d)	216.(d)

217.(c)	218.(a)	219.(b)	220.(c)
221.(c)	222.(c)	223.(c)	224.(b)
225.(c)	226.(d)	227.(b)	228.(b)
229.(c)	230.(a)	231.(a)	232.(b)
233.(d)	234.(a)	235.(c)	236.(b)
237.(a)	238.(c)	239.(a)	240.(d)
241.(b)	242.(d)	243.(d)	244.(b)
245.(c)	246.(a)	247.(b)	248.(a)
249.(b)	250.(c)	251.(c)	252.(a)
253.(c)	254.(a)	255.(b)	256.(a)
257.(d)	258.(b)	259.(b)	260.(a)
261.(a)	262.(d)	263.(d)	264.(c)
265.(b)	266.(d)	267.(d)	268.(d)
269.(a)	270.(c)	271.(b)	272.(d)
273.(d)	274.(c)	275.(c)	276.(c)
277.(c)	278.(c)	279.(c)	280.(b)
281.(a)	282.(b)	283.(b)	284.(b)
285.(a)	286.(d)	287.(a)	288.(c)
289.(b)	290.(d)	291.(d)	292.(c)
293.(a)	294.(d)	295.(b)	296.(d)
297.(b)	298.(a)	299.(d)	300.(a)
301.(b)	302.(b)	303.(a)	304.(d)
305.(c)	306.(c)	307.(d)	308.(d)
309.(c)	310.(c)	311.(a)	312.(c)
313.(c)	314.(b)	315.(a)	316.(d)
317.(a)	318.(b)	319.(b)	320.(c)
321.(a)	322.(a)	323.(d)	324.(a)
325.(c)	326.(c)	327.(c)	328.(b)
329.(c)	330.(b)	331.(a)	332.(a)
333.(d)	334.(b)	335.(d)	336.(c)
337.(d)	338.(b)	339.(c)	340.(b)
341.(b)	342.(b)	343.(d)	344.(d)
345.(c)	346.(a)	347.(b)	348.(c)
349.(a)	350.(b)	351.(d)	352.(c)
353.(a)	354.(c)	355.(d)	356.(d)
357.(d)	358.(d)	359.(d)	360.(c)
361.(c)	362.(b)	363.(c)	364.(b)
365.(c)	366.(c)	367.(c)	368.(c)
369.(a)	370.(b)	371.(b)	372.(c)
373.(a)	374.(a)	375.(d)	376.(a)
377.(b)	378.(a)	379.(d)	380.(c)
381.(a)	382.(d)	383.(a)	384.(a)
385.(d)	386.(d)	387.(d)	388.(a)
389.(d)	390.(a)	391.(d)	392.(b)
393.(a)	394.(a)	395.(a)	396.(c)
397.(a)	398.(b)	399.(b)	400.(a)

401.(a)	402.(d)	403.(a)	404.(a)
405.(b)	406.(d)	407.(b)	408.(b)
409.(a)	410.(d)	411.(d)	412.(b)
413.(c)	414.(d)	415.(a)	416.(c)
417.(d)	418.(c)	419.(c)	420.(c)
421.(b)	422.(c)	423.(c)	424.(a)
425.(a)	426.(a)	427.(a)	428.(c)
429.(a)	430.(c)	431.(d)	432.(c)
433.(a)	434.(b)	435.(b)	436.(a)
437.(c)	438.(a)	439.(d)	440.(d)
441.(c)	442.(b)	443.(b)	444.(c)
445.(a)	446.(a)	447.(b)	448.(b)
449.(b)	450.(a)	451.(c)	452.(b)
453.(a)	454.(c)	455.(c)	456.(d)
457.(c)	458.(a)	459.(c)	460.(b)
461.(c)	462.(b)	463.(d)	464.(d)
465.(c)	466.(a)	467.(b)	468.(c)
469.(c)	470.(c)	471.(c)	472.(d)
473.(d)	474.(d)	475.(d)	476.(c)
477.(b)	478.(a)	479.(c)	480.(a)
481.(d)	482.(b)	483.(b)	484.(a)
485.(a)	486.(b)	487.(b)	488.(c)
489.(d)	490.(a)	491.(d)	492.(d)
493.(b)	494.(d)	495.(a)	496.(c)
497.(c)	498.(a)	499.(a)	500.(d)
501.(a)	502.(c)	503.(a)	504.(c)
505.(b)	506.(c)	507.(c)	508.(b)
509.(b)	510.(c)	511.(d)	512.(c)
513.(c)	514.(d)	515.(c)	516.(b)
517.(d)	518.(a)	519.(a)	520.(a)
521.(d)	522.(b)	523.(d)	524.(d)
525.(a)	526.(a)	527.(b)	528.(b)
529.(c)	530.(c)	531.(d)	532.(b)
533.(d)	534.(c)	535.(a)	536.(c)
537.(a)	538.(b)	539.(d)	540.(a)
541.(d)	542.(b)	543.(d)	544.(a)
545.(b)	546.(b)	547.(c)	548.(a)
549.(b)	550.(b)	551.(c)	552.(c)
553.(d)	554.(a)	555.(c)	556.(a)
557.(d)	558.(d)	559.(d)	560.(c)
561.(c)	562.(d)	563.(d)	564.(c)
565.(c)	566.(b)	567.(d)	568.(c)
569.(c)	570.(c)	571.(c)	572.(d)
573.(a)	574.(d)	575.(d)	576.(c)
577.(b)	578.(d)	579.(b)	580.(b)
581.(a)	582.(c)	583.(c)	584.(a)

585.(d)	586.(a)	587.(a)	588.(d)
589.(d)	590.(a)	591.(c)	592.(d)
593.(b)	594.(b)	595.(a)	596.(d)
597.(c)	598.(c)	599.(c)	600.(d)
601.(c)	602.(c)	603.(c)	604.(b)
605.(d)	606.(a)	607.(c)	608.(d)
609.(d)	610.(a)	611.(c)	612.(a)
613.(c)	614.(c)	615.(a)	616.(c)
617.(c)	618.(b)	619.(c)	620.(b)
621.(a)	622.(b)	623.(a)	624.(d)
625.(a)	626.(b)	627.(a)	628.(c)
629.(b)	630.(c)	631.(b)	632.(a)
633.(c)	634.(d)	635.(d)	636.(d)
637.(d)	638.(d)	639.(c)	640.(d)
641.(d)	642.(c)	643.(b)	644.(b)
645.(c)	646.(a)	647.(b)	648.(b)
649.(c)	650.(c)	651.(b)	652.(a)
653.(b)	654.(c)	655.(c)	656.(b)
657.(d)	658.(c)	659.(c)	660.(c)
661.(d)	662.(b)	663.(c)	664.(b)
665.(b)	666.(a)	667.(a)	668.(b)
669.(c)	670.(b)	671.(a)	672.(b)
673.(c)	674.(b)	675.(a)	676.(a)
677.(c)	678.(d)	679.(d)	680.(c)
681.(c)	682.(c)	683.(d)	684.(d)
685.(d)	686.(d)	687.(b)	688.(c)
689.(b)	690.(d)	691.(c)	692.(a)
693.(c)	694.(c)	695.(b)	696.(d)
697.(b)	698.(c)	699.(d)	700.(c)
701.(d)	702.(c)	703.(d)	704.(a)
705.(c)	706.(c)	707.(b)	708.(c)
709.(d)	710.(a)	711.(a)	712.(a)
713.(b)	714.(b)	715.(d)	716.(a)
717.(b)	718.(d)	719.(d)	720.(d)
721.(a)	722.(b)	723.(d)	724.(a)
725.(d)			

Solutions :-**Sol.1.(b)** Let the arc length be l

According to the question,

$$l = \frac{2\pi r}{2} \Rightarrow l = \pi r$$

We know that,

$$\Rightarrow \text{Area of a circle} = \frac{1}{2} \times r \times l$$

$$\Rightarrow \pi r^2 \times \frac{\theta}{360^\circ} = \frac{1}{2} \times r \times \pi r$$

$$\Rightarrow \theta = \frac{360^\circ}{2} = 180^\circ$$

$$\text{Sol.2.(c) Area of circle} = \frac{\theta}{360} \times \pi r^2$$

$$= 209 \Rightarrow \frac{\theta}{360} = \frac{209}{\pi r^2}$$

And perimeter of sector

$$= \frac{\theta}{360} \times 2\pi r + 2r = 63 \Rightarrow \frac{\theta}{360}$$

$$= \frac{63 - 2r}{2\pi r}$$

On equating both:

$$\frac{209}{\pi r^2} = \frac{63 - 2r}{2\pi r} \Rightarrow \frac{209}{r} = \frac{63 - 2r}{2}$$

$$418 = 63r - 2r^2 \Rightarrow 2r^2 - 63r + 418 = 0$$

$$2r^2 - 44r - 19r + 418 = 0$$

$$2r(r - 22r) - 19(r - 22) = 0$$

$$(r - 22)(2r - 19) \Rightarrow r = 22 \text{ or } \frac{19}{2}$$

Now

$$\text{perimeter of sector} = \frac{\theta}{360} \times 2\pi r + 2r$$

$$63 = \frac{\theta}{360} \times 2 \times \frac{22}{7} \times 22 + 44$$

$$\Rightarrow \theta = 49.5^\circ$$

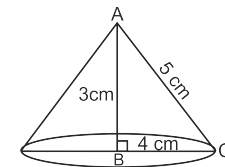
$$\text{Sol.3.(d) Area of square} = (\text{side})^2$$

$$= 16x^2 + 40x + 25 = (4x + 5)^2$$

$$\Rightarrow \text{Side of square} = (4x + 5)$$

$$\Rightarrow \text{Perimeter of square} = 4 \times (4x + 5)$$

$$= 2(8x + 10)$$

Sol.4.(b)

$$\text{Volume of the cone} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \pi \times 4 \times 4 \times 3 = 16\pi \text{ cm}^3$$

Sol.5.(b)

$$\text{Perimeter of sector} = 2r + \frac{\theta}{360} \times 2\pi r$$

$$\Rightarrow 2r + \frac{\theta}{360} \times 2\pi r = \frac{64}{3}$$

$$\Rightarrow 2r + \frac{1}{6} \times 2 \times \frac{22}{7} \times r = \frac{64}{3}$$

$$\Rightarrow 42r + 22r = 64 \times 7$$

$$\Rightarrow 64r = 64 \times 7 \Rightarrow r = 7$$

Now,

$$2r + l = \frac{64}{3} \text{ (given)}$$

$$\Rightarrow 2 \times 7 + l = \frac{64}{3} \Rightarrow l = \frac{64}{3} - 14 = \frac{22}{3}$$

$$\text{Hence, area of sector} = \frac{1}{2}lr$$

$$= \frac{1}{2} \times \frac{22}{3} \times 7 = \frac{77}{3} \text{ cm}^2$$

Sol.6.(a)

$$\text{New length of cuboid} = \frac{20}{2} = 10 \text{ cm}$$

$$\text{New breadth of cuboid} = \frac{12}{2} = 6 \text{ cm}$$

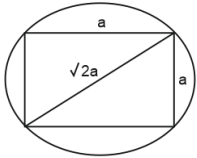
$$\text{New height of cuboid} = \frac{10}{2} = 5 \text{ cm}$$

TSA of all the pieces

$$= 8 \times 2(10 \times 6 + 6 \times 5 + 5 \times 10)$$

$$= 8 \times 2(60 + 30 + 50) = 2240 \text{ cm}^2$$

Sol.7.(d) As we know, Rectangle with the largest area cut out of the circle is square.



$$\text{Area of circle} = \pi \times 5^2 = 25\pi \text{ square units}$$

$$\text{Diameter of circle} = 5 \times 2 = 10 \text{ units}$$

$$\text{Diagonal of square} = \sqrt{2}a = 10$$

$$\Rightarrow a = \frac{10}{\sqrt{2}} = 5\sqrt{2} \text{ unit}$$

$$\text{Area of square} = (5\sqrt{2})^2 = 50 \text{ square units}$$

$$\text{Area of remaining portion} = \text{Area of circle} - \text{Area of square}$$

$$25\pi - 50 = 25(\pi - 2) \text{ square units}$$

Sol.8.(d) According to the question,

$$\text{CSA of the cylinder} = 2\pi rh$$

$$2\pi rh \times \frac{350}{100} = 2\pi RH$$

$$h : H = 6 : 7 \text{ (where radii ratio are, } 3 : 1)$$

$$\text{Volume of the cylinder} = \pi r^2 h$$

$$\text{Small cylinder volume} = \pi(1)^2 \times 6$$

$$\text{Big cylinder volume} = \pi(3)^2 \times 7$$

$$\text{required\%} = \frac{\pi(1)^2 \times 6}{\pi(3)^2 \times 7} \times 100 = 9.52\%$$

Sol.9.(b)

$$\text{Outer area of hemispherical bowl} = 2\pi$$

$$(R)^2 = 2\pi (8)^2 = 128\pi \text{ cm}^2$$

$$\text{Inner area of hemispherical bowl} = 2\pi (r)^2$$

$$= 2\pi (6)^2 = 72\pi \text{ cm}^2$$

$$\text{Ring area of the top} = \pi(R^2 - r^2)$$

$$= \pi(64 - 36) = 28\pi \text{ cm}^2$$

$$\text{Total area to polished}$$

$$= 128\pi + 72\pi + 28\pi = 228\pi \text{ cm}^2$$

$$\text{Hence, cost to polish the bowl} = 228\pi$$

$$= 228 \times 50 = \text{Rs. } 11400$$

Sol.10.(c) Option 'c' is false

$$\text{Length of an arc} = \frac{\theta}{360^\circ} \times 2\pi r$$

Sol.11.(c) Sum of the area of similar triangles = $706.5 \times 2 = 1413 \text{ m}^2$

$$\frac{\text{1st triangle's perimeter}}{\text{2nd triangle's perimeter}}$$

$$= \frac{\text{side of 1st triangle}}{\text{side of 2nd triangle}} = \frac{6}{11}$$

$$\frac{\text{area of 1st triangle}}{\text{area of 2nd triangle}}$$

$$= \left(\frac{\text{side of 1st triangle}}{\text{side of 2nd triangle}}\right)^2 = \left(\frac{6}{11}\right)^2 = \frac{36}{121}$$

$$\text{Sum of both triangles area} = 36 + 121$$

$$= 157 \text{ units} \Rightarrow 157 \text{ units} = 1413 \text{ m}^2$$

$$\text{Area of 1st triangle (36 units)}$$

$$= \frac{1413}{157} \times 36 = 324 \text{ m}^2$$

$$\text{Area of 2nd triangle (121 units)}$$

$$= \frac{1413}{157} \times 121 = 1089 \text{ m}^2$$

$$\text{Required value} = (1089 - 324) \times \frac{1}{5}$$

$$= 153 \text{ m}^2$$

Sol.12.(a) Area of remaining area = area of square - 2 × area of semicircle

According to the question,

$$42 = (2r)^2 - 2 \times \frac{1}{2} \times \pi \times (r)^2$$

$$42 = 4r^2 - \frac{22}{7} \times (r)^2$$

$$42 = r^2 \left(4 - \frac{22}{7}\right) \Rightarrow 42 = r^2 \left(\frac{6}{7}\right)$$

$$r^2 = 42 \times \frac{7}{6} = 49 \Rightarrow r = 7$$

$$\text{So, diameter of each circle (2r)} = 14 \text{ cm}$$

Sol.13.(b) Let the radius of larger and

smaller spheres 2x and $\frac{x}{2}$ respectively.

1st condition :- Volume of larger spheres

$$= 125 \times \text{volume of identical spheres}$$

$$\frac{4}{3} \pi R_1^3 = 125 \times \frac{4}{3} \pi r_1^3$$

$$(2x)^3 = 125 \times r_1^3 \Rightarrow r_1^3 = \frac{(2x)^3}{125}$$

2nd condition :-

Volume of smaller spheres = 27 × volume of identical spheres

$$\frac{4}{3} \pi (R_2)^3 = 27 \times \frac{4}{3} \pi r_2^3$$

$$\frac{4}{3} \pi \left(\frac{x}{2}\right)^3 = 27 \times \frac{4}{3} \pi r_2^3 \Rightarrow r_2^3 = \frac{\left(\frac{x}{2}\right)^3}{27}$$

$$\text{Ratio of the volume} = \frac{(2x)^3}{125} : \frac{\left(\frac{x}{2}\right)^3}{27}$$

$$= \frac{(2x)^3}{125} : \frac{(x)^3}{27 \times 8} = 1728 : 125$$

According to question

$$\frac{1}{m} = \frac{1728}{125}$$

$$\text{Hence the value of } m = \frac{125}{1728}$$

Alternate method :

$$\frac{\frac{\text{volume of larger sphere}}{125}}{\frac{\text{volume of smaller sphere}}{27}} = \frac{1}{m}$$

$$\frac{\frac{\frac{4}{3} \pi \times (2)^3}{125}}{\frac{\frac{4}{3} \pi \times \left(\frac{1}{2}\right)^3}{27}} = \frac{1}{m}$$

$$\Rightarrow \frac{(2)^3 \times 27}{\left(\frac{1}{2}\right)^3 \times 125} = \frac{1}{m}$$

$$\Rightarrow \frac{8 \times 27 \times 8}{125} = \frac{1}{m} \Rightarrow m = \frac{125}{1728}$$

Sol.14.(d)

Case 1: When the sheet of paper is rolled along its length to form a cylinder, then the circumference of the base of the cylinder = length of the rectangular sheet
height of the cylinder = breadth of the rectangular sheet.

$$\text{Let radius} = r \text{ and height} = h = 36 \text{ cm}$$

$$\text{Circumference of base} = 60 \text{ cm}$$

$$2\pi r = 60 \text{ cm} \Rightarrow r = \frac{30}{\pi} \text{ cm}$$

$$\text{Volume of the cylinder formed (V}_1\text{)}$$

$$= \pi r^2 h = \pi \times \frac{30}{\pi} \times \frac{30}{\pi} \times 36 \dots\dots (i)$$

Case 2: When the sheet of paper is rolled along its breadth to form a cylinder, then the circumference of the base of the cylinder = the breadth of the rectangular sheet
height of the cylinder = length of the rectangular sheet.

$$\text{Let radius} = R \text{ and height} = H = 60 \text{ cm}$$

$$\text{Circumference of base} = 36 \text{ cm}$$

$$2\pi R = 36 \text{ cm} \Rightarrow R = \frac{18}{\pi} \text{ cm}$$

$$\text{Volume of the cylinder formed (V}_2\text{)}$$

$$= \pi R^2 H = \pi \times \frac{18}{\pi} \times \frac{18}{\pi} \times 60 \dots\dots (ii)$$

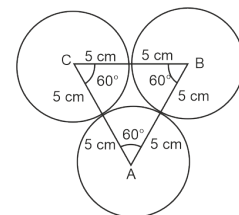
Now,

$$V_1 : V_2 = \pi \times \frac{30}{\pi} \times \frac{30}{\pi} \times 36 : \pi \times \frac{18}{\pi} \times$$

$$\frac{18}{\pi} \times 60 \Rightarrow V_1 : V_2 = 5 : 3$$

Thus, the ratio of the volumes of the two cylinders so formed is 5 : 3.

Sol.15.(d)



Center distance between circle = side of triangle = $2 \times 5 = 10 \text{ cm}$

Since all sides are of 10 cm so, ΔABC is an equilateral triangle

Area subtended between three circle = area of triangle - 3 × area of sector

$$= \frac{\sqrt{3}}{4} \times 10 \times 10 - 3 \times \frac{\theta}{360^\circ} \times \pi r^2$$

$$= 25\sqrt{3} - 3 \times \frac{60^\circ}{360^\circ} \times \pi \times 5 \times 5$$

$$= 25\sqrt{3} - \frac{\pi}{2} \times 25 = 25 \left(\sqrt{3} - \frac{\pi}{2}\right)$$

Sol.16.(a)

$$\text{Circumference of Larger gear (C}_1\text{)} = 2\pi R$$

$$= 2 \times \frac{22}{7} \times \frac{21}{2}$$

$$\text{Circumference of smaller gear (C}_2\text{)} = 2\pi r$$

$$= 2 \times \frac{22}{7} \times \frac{15}{2}$$

$$C_1 : C_2 = 7 : 5$$

$$\text{Ratio} \rightarrow \text{Larger} : \text{Smaller}$$

$$\text{Circumference} \rightarrow 7 : 5$$

Revolution $\rightarrow 5 : 7$

7 unit = 105 revolution

$$5 \text{ unit} = \frac{105}{7} \times 5 = 75$$

Hence, larger gear has 75 revolutions.

Sol.17.(d) Volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 7 \times 7 \times 12 = 1848 \text{ cm}^3$$

Let the two new recast cones volume be x and $2x$

So,

$$\Rightarrow 3x = 1848$$

$$\Rightarrow x = 616 \text{ cm}^3 \text{ and } 2x = 1232 \text{ cm}^3$$

According to question,

$$\Rightarrow \frac{616}{1232} = \frac{\frac{1}{3}\pi r_1^2 h}{\frac{1}{3}\pi r_2^2 h}$$

$$\Rightarrow \frac{616}{1232} = \frac{\frac{1}{3}\pi r_1^2 \times 12}{\frac{1}{3}\pi r_2^2 \times 12}$$

$$\Rightarrow \frac{r_1^2}{r_2^2} = \frac{1}{2} \Rightarrow r_1 : r_2 = 1 : \sqrt{2}$$

$$\text{Now, } \Rightarrow \pi r_1^2 : \pi r_2^2 = 1 : 2$$

$$\text{Required percentage} = \frac{2-1}{2} \times 100$$

$$= 50\%$$

Sol.18.(b) Volume of cubes = $8^3 + 10^3 + 12^3$

$$= 512 + 1000 + 1728 = 3240 \text{ cm}^3$$

Remaining volume of cube

$$= 3240 - 496 = 2744 \text{ cm}^3$$

$$a^3 = 2744 \Rightarrow a = 14 \text{ cm}$$

Area of whole surface of the newly formed cube ($6a^2$)

$$= 6 \times 14 \times 14 = 1176 \text{ cm}^2$$

Sol.19.(b)

$$\text{Semiperimeter} = \frac{2(x+y)}{2} = x+y = 12$$

Area of rectangle = 28

$$xy = 28$$

$$\text{Now, } (x+y)^2 = x^2 + y^2 + (xy+xy)$$

$$(x+y)^2 - xy = x^2 + y^2 + xy$$

$$12^2 - 28 = x^2 + y^2 + xy$$

$$116 = x^2 + y^2 + xy$$

$$\text{Again, } (x+y)^2 = x^2 + y^2 + (3xy - xy)$$

$$(x+y)^2 - 3xy = x^2 + y^2 - xy$$

$$12^2 - 28 \times 3 = x^2 + y^2 - xy$$

$$60 = x^2 + y^2 - xy$$

$$\text{then, } x^4 + x^2y^2 + y^4 = (x^2 + y^2 + xy)(x^2 + y^2 - xy) = 116 \times 60 = 6960$$

Sol.20.(b) Lateral surface area = 120 cm^2

Lateral surface area of prism = perimeter of base $\times$ height

$$120 = 3 \times 4 \times \text{height} \Rightarrow \text{Height} = 10 \text{ cm}$$

Now,

$$\text{volume of prism} = \text{area of base} \times \text{height}$$

$$= \frac{\sqrt{3}}{4} \times 4 \times 4 \times 10 = 40\sqrt{3} \text{ cm}^3$$

Sol.21.(a) According to the question,

Let the number of ice pieces be n

Volume of spherical ice pieces $\times n$

= volume of cylindrical container

$$\frac{4}{3} \times \frac{22}{7} \times 3^3 \times n = \frac{22}{7} \times 9^2 \times 40$$

$$n = 90$$

Sol.22.(c) According to the question,

Volume of hemispherical container

= volume of cylindrical bottles

Let the number of bottles = x

$$\frac{2}{3}\pi r^3 = \pi r^2 h \times x$$

$$= \frac{2}{3} \times 18 \times 18 \times 18 = 3 \times 3 \times 6 \times x$$

$$(\text{Number of bottles}) x = 72$$

Sol.23.(c) H.C.F of 90 and 126 = 18

Largest possible side of the tiles

$$= 18 \text{ feet}$$

$$\text{Area of tiles} = 18 \times 18 = 324 \text{ feet}^2$$

Sol.24.(b)

$$\text{Volume of frustum} = \frac{\pi H}{3}(R^2 + r^2 + Rr)$$

$$= \frac{22 \times 45}{7 \times 3}(28^2 + 7^2 + 28 \times 7)$$

$$= \frac{22 \times 45}{7 \times 3} \times 1029 = 48510$$

Now,

Volume of cylinder = Volume of frustum

$$\frac{22}{7} \times 3x \times 3x \times 5x = 48510$$

$$x^3 = 343 \Rightarrow x = 7$$

So, curved surface area of cylinder

$$= 2\pi rh$$

$$= 2 \times \frac{22}{7} \times 21 \times 35 = 4620 \text{ cm}^2$$

Sol.25.(d) Area of square = Base²

Area of rhombus = Base² $\times \sin \theta$ (θ = angle between the sides)

The ratio of area of the square to the

$$\text{area of rhombus} = \frac{1}{\sin 45} = \sqrt{2} : 1$$

Sol.26.(a) Volume of the wood Required

= (Total volume – inner volume)

$$(10 \times 8 \times 6) - [(10-1) \times (8-1) \times (6-1)]$$

$$= 480 - 315 = 165 \text{ cm}^3$$

Sol.27.(d) According to the question,

Weight $\propto$ (density $\times$ volume)

$$\text{Volume} = k \times \frac{\text{weight}}{\text{density}}$$

Ratio - 1st cube : 2nd cube

$$\text{Density} - 27 : 16$$

$$\text{Weight} - 4 : 1$$

$$\text{Volume} - \frac{4}{27} : \frac{1}{16} \rightarrow 64 : 27$$

$$\text{Side} = \sqrt[3]{64} : \sqrt[3]{27} \rightarrow 4 : 3$$

Sol.28.(a) According to the question,

weight = radii² $\times$ thickness

Ratio of weight = 4 : 1

Ratio of thickness = 16 : 9

$$\begin{aligned} \text{Ratio of their radii} &= \sqrt{\frac{4}{16}} : \sqrt{\frac{1}{9}} \\ &= \frac{2}{4} : \frac{1}{3} = 3 : 2 \end{aligned}$$

Sol.29.(a) Circumference of dome ($2\pi r$)

$$= 154 \text{ cm} \Rightarrow r = 154 \times \frac{7}{44} = 24.5 \text{ cm}$$

Curved surface area of hemispherical

$$\text{dome} = 2 \times \frac{22}{7} \times 24.5 \times 24.5$$

$$\text{Required cost} = \frac{2 \times 22 \times 24.5 \times 24.5}{7 \times 100} \times 4$$

$$= \frac{2 \times 22 \times 3.5 \times 24.5}{25} = 150.92 \text{ ₹}$$

Sol.30.(a) Area of sector = $\frac{\theta}{360^\circ} \times \pi r^2$

$$616 \text{ cm}^2 = \frac{10^\circ}{360^\circ} \times \frac{22}{7} \times r^2$$

$$r^2 = \frac{616 \times 36 \times 7}{22} = 28 \times 36 \times 7$$

$$r = 84 \text{ cm}$$

Sol.31.(c) Volume of cone

$$= \frac{1}{3} \times \frac{22}{7} \times \frac{7}{4} \times \frac{7}{4} \times \frac{7}{2}$$

$$= \frac{11 \times 7 \times 7}{3 \times 16} = \frac{539}{48} = 11.23 \text{ cm}^3$$

Sol.32.(c)

Total surface area of cylinder = $2\pi r(h+r)$

$$6336 = 2 \times \frac{22}{7} \times r \times 42$$

$$r = \frac{6336}{22 \times 12} = 24 \text{ m}$$

Curved surface area of cylinder = $2\pi rh$

$$= 2 \times \frac{22}{7} \times 24 \times 18 = 2715.43 \text{ m}^2$$

Sol.33.(b)

Let the side of cubes is $3x, 4x, 5x$

$$\text{Side of newly formed cube} = \frac{\text{diagonal}}{\sqrt{3}}$$

$$\text{Side of newly formed cube} = \frac{18\sqrt{3}}{\sqrt{3}} = 18$$

Equating the volume of cubes :-

$$(3x)^3 + (4x)^3 + (5x)^3 = 18 \times 18 \times 18$$

$$x^3 = \frac{18 \times 18 \times 18}{216} = 27 \Rightarrow x = 3$$

Sides are :- 9, 12 and 15

Sol.34.(b)

Perimeter of equilateral angle triangle = $3a$

$$\text{Area of equilateral angle triangle} = \frac{\sqrt{3}}{4} a^2$$

$$\text{According to question, } a = \sqrt[4]{\frac{16}{3}}$$

$$\begin{aligned} \text{Therefore, required area} &= \frac{\sqrt{3}}{4} \times \left(\sqrt[4]{\frac{16}{3}}\right)^2 \\ &= \frac{\sqrt{3}}{4} \times \frac{4}{\sqrt{3}} = 1 \text{ unit}^2 \end{aligned}$$

Sol.35.(c)Curved surface area of hemisphere = $2\pi r^2$

Cost for green colour =

$$3 \times (2 \times 3.14 \times 4 \times 4) = 301.44 \text{ Rs.}$$

Cost for red colour

$$= 3 \times (2 \times 3.14 \times 4 \times 4) = 301.44 \text{ Rs.}$$

$$\text{Area of two bases} = \pi r^2 + \pi r^2 = 2\pi r^2$$

$$= (2 \times 3.14 \times 4 \times 4) = 100.48$$

The cost of painting with blue

$$= 2 \times 100.48 = 200.96 \text{ Rs.}$$

$$\text{Total cost} = 301.44 + 301.44 + 200.96$$

$$= 803.84 \text{ Rs.}$$

Sol.36.(d)Total surface area of cube = $(6a^2)$

$$= 6 \times 5^2 = 150 \text{ cm}^2$$

$$\text{Increased area} = 6 \times 5 \text{ cm}^2 - 6 \times 1 \text{ cm}^2$$

$$= 24 \text{ cm}^2$$

$$\text{Total increased surface area} = 150 + 24$$

$$= 174 \text{ cm}^2$$

Sol.37.(b)Volume of Prism = Area of Base $\times$ Height

Volume of Pyramid

$$= \frac{1}{3} \times \text{Area of Base} \times \text{Height}$$

Required ratio

$$\Rightarrow \text{Volume of Prism} : \text{Volume of Pyramid} = 3 : 1$$

Sol.38.(c) ATQ, CSA of cone = total area of rectangular cloth

$$\frac{22}{7} \times \frac{14}{2} \times 9 = 2 \times (\text{length of cloth})$$

$$\Rightarrow 198 = 2 \times l \Rightarrow l = \frac{198}{2} = 99 \text{ m}$$

Sol.39.(a) Angle covered in 45 minutes

$$\text{by minute hand} = \frac{360}{60} \times 45 = 270^\circ$$

$$\text{Required area} = \frac{270^\circ}{360^\circ} \times \frac{22}{7} \times 20 \times 20$$

$$= \frac{6600}{7} \text{ cm}^2$$

Sol.40.(b) External radius (R) = 10 cm

Internal radius (r) = 9 cm and

length (h) = 12 cm

Total surface area of hollow cylinder pipe

$$= 2\pi(R+r)\{h+R-r\}$$

$$= 2 \times \frac{22}{7} \times 19 \times 13$$

$$= \frac{44 \times 247}{7} = 1552.57 \text{ cm}^2$$

Sol.41.(c) Let the length of rectangle is 'l' and breadth of rectangle is 'b'Area of rectangle = length $\times$ breadth = $l \times b$

$$(\text{Area of rectangle}) = (lb) \dots\dots\dots \text{e.q. (1)}$$

According to question,

(Area of resulting square)

$$= (l-4) \times (b+2) \dots\dots\dots \text{e.q. (2)}$$

Side of square are equal, so

$$l-4 = b+2 \Rightarrow l-b = 6 \dots\dots\dots \text{e.q. (3)}$$

Equating e.q. (1) and e.q. (2), we get

$$(lb) = lb + 2l - 4b - 8$$

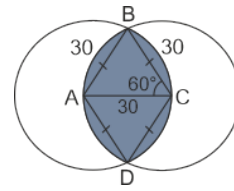
$$l - 2b = 4 \dots\dots\dots \text{e.q. (4)}$$

From e.q. (3) and e.q. (4), we get

$$b = 2, l = 8$$

So, Perimeter of the original rectangle

$$= 2(8+2) = 20 \text{ cm}$$

Sol.42.(d)

$$\text{Area of sector ACB} = \frac{60^\circ}{360^\circ} \times \pi r^2 = \frac{900\pi}{6}$$

$$\text{Area of sector BAC} = \frac{60^\circ}{360^\circ} \times \pi r^2 = \frac{900\pi}{6}$$

Area of equilateral triangle ABC

$$= \frac{\sqrt{3}}{4} r^2 = \frac{900\sqrt{3}}{4}$$

Upper shaded region = area of sector ACB + area of sector BAC - area of equilateral ΔABC

Area of total shaded region

= 2 $\times$ area of upper half shaded region

Required shaded area

$$= 2 \times \left\{ \frac{900\pi}{6} + \frac{900\pi}{6} - \frac{900\sqrt{3}}{4} \right\}$$

$$= 2 \times \left\{ \frac{1800\pi}{6} - \frac{900\sqrt{3}}{4} \right\}$$

$$= (600\pi - 450\sqrt{3}) \text{ cm}^2$$

Short Trick:-

$$\text{Area of shaded region} = r^2 \left(\frac{2}{3} \pi - \frac{\sqrt{3}}{2} \right)$$

$$\text{Area} = 900 \left(\frac{2}{3} \pi - \frac{\sqrt{3}}{2} \right)$$

$$= (600\pi - 450\sqrt{3}) \text{ cm}^2$$

Sol.43.(b) Let side of the rhombus be 'a'Perimeter of rhombus = $4a = 4P \Rightarrow a = P$

As we know that,

$$D_1^2 + D_2^2 = 4a^2, \quad D_1 + D_2 = L$$

$$\text{Then, } D_1 D_2 = \frac{L^2 - 4a^2}{2}$$

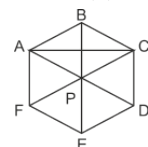
$$\text{Area of Rhombus} = \frac{1}{2} \times D_1 \times D_2$$

$$= \frac{1}{2} \times \frac{L^2 - 4a^2}{2} = \frac{1}{4} \times (L^2 - 4P^2)$$

Sol.44.(d) Volume of water flowing through the canal in 30 min

$$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times (18 \times \frac{5}{18}) \times (30 \times 60)$$

$$= 3,46,500 \text{ m}^3$$

Sol.45.(a) According to the question,

As ABCDEF is a regular hexagon,

$$\text{Area of } \square ABCP = 2 \times \frac{\sqrt{3}}{4} \times (\text{side})^2$$

$$= 2 \times \frac{\sqrt{3}}{4} \times 36^2 = 648\sqrt{3} \text{ cm}^2$$

$$\text{So, area of } \Delta ABC = \frac{1}{2} \times \text{Area of } \square ABCP$$

$$= \frac{1}{2} \times 648\sqrt{3} = 324\sqrt{3} \text{ cm}^2$$

Sol.46.(a)

$$\text{Volume of frustum} = \frac{1}{3} \pi h (R^2 + r^2 + Rr)$$

$$= \frac{1}{3} \pi \times 6 (5^2 + 3^2 + 15)$$

$$= 2\pi (25 + 9 + 15) = 98\pi \text{ cm}^3$$

Sol.47. (d) Let $a = b = 2$ units

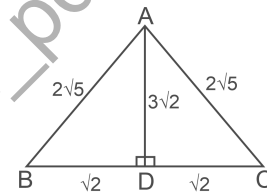
$$BC = \sqrt{a^2 + b^2} = 2\sqrt{2} \text{ units}$$

$$AB = \sqrt{(2a)^2 + b^2} = 2\sqrt{5} \text{ units}$$

$$AC = \sqrt{a^2 + (2b)^2} = 2\sqrt{5} \text{ units}$$

It becomes an isosceles triangle as below.

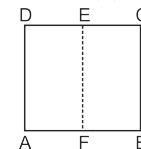
$$AD = \sqrt{(2\sqrt{5})^2 - (\sqrt{2})^2} = 3\sqrt{2} \text{ units}$$



Area of triangle

$$= \frac{1}{2} \times 2\sqrt{2} \times 3\sqrt{2} = 6 \text{ sq units}$$

$$\text{In option (d), } \frac{3}{2} ab = \frac{3}{2} \times 2 \times 2 = 6$$

Sol.48.(d)Let the side of square = a

$$AB = BC = CD = DA = EF = a$$

Perimeter of rectangle ADEF + Perimeter of rectangle BCEF = $39 + 39 = 78$

$$AB + BC + CD + DA + EF + EF = 78$$

$$6a = 78 \text{ cm} \Rightarrow a = 13 \text{ cm}$$

$$\text{Perimeter of Square} = 4a = 52 \text{ cm}$$

Sol.49.(b) Circumference of a circle = $2\pi r = a\pi \Rightarrow 2r = a$

$$\text{Area} = b\pi \Rightarrow \pi r^2 = b\pi \Rightarrow b = r^2$$

$$a : b = 2r : r^2 \text{ or } 2 : r$$

$$4 : 5 = 2 : r \Rightarrow r = 2.5 \text{ cm}$$

Sol.50.(c) 65 m, 72 m and 97 m are triplets of a right-angled triangle.

So area of triangular park

$$= \frac{1}{2} \times 65 \times 72 = 2340$$

Let sides of the rectangle be $5x$ and $13x$.

$$\text{Area of rectangle} = 65x^2$$

$$\text{ATQ, } 65x^2 = 2340 \Rightarrow x = 6$$

$$\text{Perimeter of rectangle} = 2(5x + 13x) = 36x = 216 \text{ m}$$

Sol.51.(c) Radius of bicycle = 0.42 m

Distance travelled in 25 seconds

$$= 2\pi r \times 40 \Rightarrow 2 \times \frac{22}{7} \times 0.42 \times 40$$

$$= 105.6 \text{ m}$$

$$\text{Speed of bicycle (in m/s)} = \frac{105.6}{25}$$

$$= 4.224$$

$$\text{Speed of bicycle (in kmph)} = 4.224 \times \frac{18}{5}$$

$$= 15.2$$

Sol.52.(a) Area of the square = $a^2 = 169$.So, $a = 13$, Perimeter = $4a = 4 \times 13 = 52$ Circumference of the circle = $2\pi r = 52$

$$r = \frac{7}{44} \times 52 = \frac{91}{11}$$

$$\text{Area} = \pi r^2 = \frac{22}{7} \times \frac{91}{11} \times \frac{91}{11}$$

$$= \frac{2366}{11} = 215.09 \approx 215 \text{ cm}^2$$

Sol.53.(d) Slant height of the cone

$$= \text{radius of the sector And } \frac{\theta}{360} \times$$

perimeter of the circle

= perimeter of the base of the cone

Let the radius of cone = r

$$\frac{120}{360} \times 2 \times \pi \times 10.5 = 2 \times \pi \times r$$

$$\Rightarrow r = 3.5$$

$$\text{Height of the cone} = \sqrt{l^2 - r^2}$$

$$= \sqrt{(10.5)^2 - (3.5)^2} = 7\sqrt{2}$$

Volume of the cone

$$= \frac{1}{3} \times \pi \times 3.5 \times 3.5 \times 7\sqrt{2}$$

$$= \frac{343\sqrt{2}}{12} \pi \text{ cm}^3$$

Sol.54.(c) 20, 21 and 29 are triplet pairs, clearly a given triangle is a right angle triangle.

$$\text{Area of the triangle} = \frac{1}{2} \times 20 \times 21$$

$$= 210 \text{ cm}^2$$

Volume of the prism = Base Area $\times$ Height

$$\Rightarrow 7560 = 210 \times \text{Height} \Rightarrow \text{Height} = 36$$

Lateral surface area of the prism

= Base perimeter $\times$ Height

$$= (20 + 21 + 29) \times 36 = 2520 \text{ cm}^2$$

Sol.55.(a) Let the original radius be r .

According to the question

$$4\pi(r + 4)^2 = 4\pi r^2 + 464\pi$$

$$\Rightarrow r^2 + 16 + 8r = r^2 + 116 \Rightarrow r = 12.5$$

$$\text{Volume of the sphere} = \frac{4}{3}\pi r^3$$

$$= \frac{4}{3} \times 12.5 \times 12.5 \times 12.5 \times \pi$$

$$= \frac{15625}{6} \pi \text{ cm}^3$$

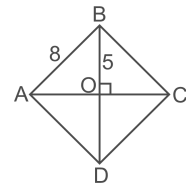
Sol.56.(c)Volume of prism = base area $\times$ height

$$1731.6 = \frac{1}{2} \times (15 + 11) \times 9 \times \text{height}$$

$$\Rightarrow \text{height} = 14.8$$

Sol.57.(b) Let n be the number of turns.Total length of the ribbon = number of turns $\times$ circumference of the base of

$$\text{cylinder } l = n \times c \Rightarrow n = \frac{l}{c}$$

Sol.58.(c)

$$BO = \frac{BD}{2} = \frac{10}{2} = 5$$

$$AO = \frac{AC}{2} = \frac{2\sqrt{x}}{2} = \sqrt{x}$$

$$\text{Now, } AB^2 = AO^2 + BO^2$$

$$8^2 = (\sqrt{x})^2 + 5^2 \Rightarrow x = 39$$

$$\sqrt{x} + 10 \Rightarrow \sqrt{39} + 10 = 7 \text{ cm}$$

Sol.59.(d) Perimeter of rhombus

$$= 2\sqrt{d_1^2 + d_2^2} = 2\sqrt{14^2 + 48^2} = 100 \text{ cm}$$

Sol.60.(c) Circumference of the circle(C)

$$= 2\pi r \Rightarrow r = \frac{C}{2\pi}$$

$$\text{Area of the sector (A)} = \frac{60}{360} \times \pi \times r^2$$

$$= \frac{1}{6} \times \pi \times \left(\frac{C}{2\pi}\right)^2 \Rightarrow \frac{C^2}{24\pi}$$

Sol.61.(b) Let the length of the perpendicular and base of the right angle triangle is a cm. $\Rightarrow$ Hypotenuse of the triangle = $a\sqrt{2}$ Area of Δ formed on hypotenuse(H)

$$= \frac{\sqrt{3}}{4} \times (a\sqrt{2})^2 = \frac{\sqrt{3}a^2}{2}$$

Area of Δ formed on perpendicular(A)

$$= \frac{\sqrt{3}a^2}{4} \Rightarrow \frac{A}{H} = \frac{\frac{\sqrt{3}a^2}{4}}{\frac{\sqrt{3}a^2}{2}} = \frac{1}{2}$$

Sol.62.(c) Since it is an equilateral triangle. Each side will be formed of 4 sticks and its length will be 4 units.

$$\text{Area of the equilateral triangle} = \frac{\sqrt{3}}{4} a^2$$

$$= \frac{\sqrt{3}}{4} \times 4 \times 4 = 4\sqrt{3}$$

Sol.63.(c) % age change in the area

$$= (-11) + 11 + \frac{(-11) \times (11)}{100} = -1.21\%$$

Here, - ve sign indicates that the area is decreasing.

Sol.64.(d) Let $AB = a$ cm

$$AC = a + 10 \Rightarrow BC = 2a - 2$$

According to the question

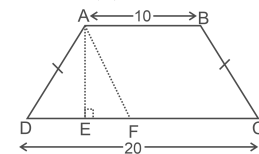
$$a + a + 10 + 2a - 2 = 32 \Rightarrow 4a = 24$$

$$a = AB = 6 \text{ cm}$$

$$AC = 6 + 10 = 16 \text{ cm}$$

$$BC = 2(6) - 2 = 10 \text{ cm}$$

Clearly AB is the smallest side.

Sol.65.(b)

Let ABCD be the given trapezium. And AE be its height.

$$\text{Area of trapezium} = \frac{1}{2} \times (CD + AB) \times AE$$

$$180 = \frac{1}{2} \times (20 + 10) \times AE$$

$$AE = 12 \text{ cm} \quad \text{Draw } AF \parallel BC$$

$$\Rightarrow AB = FC = 10 \text{ cm and } AF = BC$$

$$\Rightarrow DF = DC - FC = 20 - 10 = 10 \text{ cm}$$

$$\text{In } \Delta ADF \Rightarrow AF = AD \quad \dots (AD = BC, \text{ given})$$

$$DE = EF = \frac{DF}{2} = 5 \text{ cm}$$

 $\dots$ (Perpendicular drawn on the base of an isosceles triangle)

$$AF^2 = AE^2 + EF^2, AF = \sqrt{AE^2 + EF^2}$$

$$= \sqrt{12^2 + 5^2} = 13$$

$$\text{Hence } AD = BC = 13 \text{ cm}$$

Sol.66.(d)Let the breadth of the rectangle = b Length of the rectangle (l) = $b + 20$ Perimeter of the rectangle = $2(b + b + 20)$

$$= 4(b + 10)$$

According to the question

$$4(b + 10) \times 53 = 21200$$

$$b + 10 = 100 \Rightarrow b = 90 \text{ m}$$

$$l = 90 + 20 = 110 \text{ m}$$

$$\text{Area of the rectangle} = 90 \times 110 = 9900 \text{ m}^2$$

Sol.67.(b) Let the sides of the rectangle are $7x$ and $4x$.

$$\text{Perimeter of rectangle} = 2(7x + 4x) = 22x$$

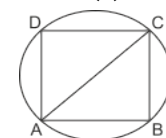
$$\text{Perimeter of circle} = 2\pi r \Rightarrow 2 \times \frac{22}{7} \times 31.5$$

According to the question

$$22x = 2 \times \frac{22}{7} \times 31.5 \Rightarrow x = 9$$

Smaller side of rectangle = $4x$

$$= 4 \times 9 = 36 \text{ cm}$$

Sol.68.(d)

Let the sides be 4 units and 3 units.

According to the question

$$2(4 \text{ unit} + 3 \text{ unit}) = 56$$

$$1 \text{ unit} = 4 \text{ cm}, 4 \text{ unit} = 16 \text{ cm}, 3 \text{ unit} = 12 \text{ cm}$$

Diagonal of Rectangle = Diameter of circle

$$= \sqrt{AB^2 + BC^2} = \sqrt{16^2 + 12^2} = 20 \text{ cm}$$

$$\text{Radius of the circle} = \frac{20}{2} = 10 \text{ cm}$$

$$\text{Area of circle} = \pi r^2 = \pi 10^2 = 100\pi$$

Sol.69.(d)

$$\text{Curved Surface area of cylinder} = 2\pi rh$$

$$\text{Volume of cylinder} = \pi r^2 h$$

According to the question

$$2\pi rh = 132 \dots (1), \pi r^2 h = 528 \dots (2)$$

$$\text{Dividing eq 2 by eq 1, } r = 8 \text{ cm}$$

$$\text{Put } r = 8 \text{ in eq (1)}$$

$$2\pi rh = 132$$

$$2 \times \frac{22}{7} \times 8 \times h = 132 \Rightarrow h = 2\frac{5}{8} \text{ m}$$

$$\text{Sol.70.(a) Area of Hexagon} = 6 \times \frac{\sqrt{3}}{4} a^2$$

$$\text{Here, } 6 \times \frac{\sqrt{3}}{4} a^2 = 2400\sqrt{3},$$

$$a^2 = 1600, a = 40$$

$$\text{Perimeter of the hexagonal field} = 6a$$

$$6 \times 40 = 240 \text{ metre}$$

$$\text{Required cost of fencing} = 240 \times 18.5$$

$$= \text{Rs. 4440}$$

Sol.71.(c) One of the sides of newly

$$\text{formed cuboid} = 6 \times 2 = 12 \text{ cm}$$

Other two sides will be 2 cm.

$$\text{Surface area of cuboid} = 2(lb + bh + hl)$$

$$= 2(12 \times 2 + 2 \times 2 + 2 \times 12) = 104 \text{ cm}^2$$

$$\text{Sol.72.(d) } 40\% = \frac{2}{5}$$

$$\text{Old : New}$$

$$\text{Radius} \rightarrow 5 : 3$$

$$\text{Volume}(\propto r^3) \rightarrow 125 : 27$$

Let the volumes are 125 unit and 27 unit

$$\text{Required \%} = \frac{125 - 27}{125} \times 100 = 78.4\%$$

Sol.73.(b) Area of triangular field with

$$\text{side } x \text{ m} = \frac{\sqrt{3}}{4} x^2$$

Area of other triangular field

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{Where } s = \frac{a+b+c}{2}$$

$$\Rightarrow \frac{50+70+80}{2} = 100 \text{ m}$$

Area of field =

$$\sqrt{100(100-50)(100-70)(100-80)}$$

$$= \sqrt{100 \times 50 \times 30 \times 20} = 1000\sqrt{3} \text{ m}^2$$

$$\text{According to question, } \frac{\sqrt{3}}{4} x^2 = 1000\sqrt{3}$$

$$x^2 = 4000 \Rightarrow x = 63.25 \text{ m}^2$$

$$\text{Sol.74.(d) Area of Sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$= \frac{150}{360} \times \frac{22}{7} \times 21^2 = 577.5 \text{ cm}^2$$

Sol.75.(d) Area of circle is $\propto r^2$.

$$\Rightarrow \text{Area of three circles will be } 4^2, 5^2, 7^2$$

$$\text{Desired Ratio} = (5^2 - 4^2) : (7^2 - 5^2) \\ = 9 : 24 = 3 : 8$$

Sol.76.(c)

$$\text{Volume of a cylinder} = \pi(R^2 - r^2)h$$

Where R = external radius

r = internal radius

h = height or length of the cylinder

$$\text{Volume} = \frac{22}{7} \times (2.5^2 - 1.5^2) \times 756$$

$$= 22 \times (2.5 + 1.5)(2.5 - 1.5) \times 108$$

$$= 9504 \text{ cm}^3$$

$$\text{Total Weight of pipe} = \text{Volume of pipe} \times \text{density of metal}$$

$$= 9504 \times 7.5 = 71280 \text{ g or } 71.28 \text{ kg}$$

Sol.77.(a)

$$\text{Volume of a cylinder} = \pi(R^2 - r^2)h$$

Where R = external radius

r = internal radius

h = height or length of the cylinder

$$7480 = \frac{22}{7} \times (9^2 - r^2) \times 140$$

$$17 = 81 - r^2 \Rightarrow r^2 = 64 \Rightarrow r = 8 \text{ cm}$$

$$\text{Thickness} = 9 - 8 = 1 \text{ cm}$$

$$\text{Sol.78.(a) Volume of cuboid} = \sqrt{a \times b \times c}$$

Here a, b and c are the area of three faces

$$= \sqrt{32 \times 24 \times 48} = 192 \text{ cm}^3$$

Sol.79.(d) We know that

$$\text{Inradius} = \frac{\text{area}}{\text{semiperimeter}}$$

$$3 = \frac{15}{\text{semiperimeter}} \Rightarrow \text{Semiperimeter} = 5$$

$$\text{Perimeter} = 2 \times 5 = 10 \text{ cm}$$

Sol.80.(c) Required area of sheet = Total surface area of rectangular tank - Area of circular opening

$$= 2 \times (80 + 48 + 60) - \pi \times 1^2$$

$$= 376 - 3.1 = 372.9 \text{ m}^2$$

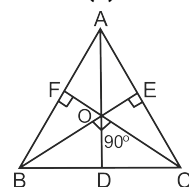
Sol.81.(b) Let side of bigger cube = R

$$R^3 = 13824 \Rightarrow R = \sqrt[3]{13824} = 24$$

Let side of smaller cubes = r

$$13824 = 8 \times r^3 \Rightarrow 1728 = r^3 \Rightarrow 12 = r$$

$$\text{Required ratio} = (6 \times 24^2) : (3 \times 6 \times 12^2) \\ = 4 : 3$$

Sol.82.(b)

$$\triangle OBC, OB = OC = x \Rightarrow BC = 2$$

$$x^2 + x^2 = 2^2 \Rightarrow x^2 = 2 \Rightarrow x = \sqrt{2}$$

$$\text{In } \triangle OBD, BD = 1 \Rightarrow OB = \sqrt{2}$$

$$OD = \sqrt{(\sqrt{2})^2 - 1^2} = 1$$

$$\text{Now, } AO = 2 \times OD \Rightarrow 2 \times 1 = 2$$

$$AD = AO + OD = 3$$

$$\text{Area of } \triangle ABC = \frac{1}{2} \times 3 \times 2 = 3 \text{ cm}^2$$

Sol.83.(b) Volume of cubical box

$$= 30 \times 30 \times 30 = 27000 \text{ cm}^3$$

Required rise in water level

$$= \frac{27000}{60 \times 30} = 15 \text{ cm}$$

Sol.84.(c) Old : New

$$\text{Length} \rightarrow 10 : 13$$

$$\text{Breadth} \rightarrow 5 : 4$$

$$\text{Area} \rightarrow 50 : 52$$

$$52 - 50 = 2 \text{ unit} = 100 \Rightarrow 1 \text{ unit} = 50$$

$$\text{New area} = 52 \times 50 = 2600 \text{ cm}^2$$

Sol.85.(a) Required Cost

$$= \frac{11 \times 6 \times 100}{60} \times 112.50 = 12,375$$

Sol.86.(c) Remaining area

$$= 119 \times 18 - 17 \times 6 = 2040$$

$$\text{Volume of sand} = 17 \times 6 \times 3 = 306$$

Required height

$$= \frac{306 \times 100 \times 100 \times 100}{2040 \times 100 \times 100} = 15 \text{ cm}$$

Sol.87.(d) Total area to be painted

$$= 40 \times 30 + 2(40 \times 2.2 + 30 \times 2.2) = 1508$$

$$\text{Total cost} = 1508 \times 25 = 37700$$

Sol.88.(a) Since, the height of the triangle formed will be half of the diagonal of the square.

Therefore,

$$\text{Height} = \frac{3\sqrt{2}}{2} = \frac{3}{\sqrt{2}} \text{ m}$$

Sol.89.(a) ATQ,

$$\frac{3\sqrt{3}}{2} \times a^2 = 108\sqrt{3} \Rightarrow a = 6\sqrt{2}$$

$$\therefore \text{Perimeter} = 6 \times 6\sqrt{2} = 36\sqrt{2} \text{ cm}$$

Sol.90.(a) Volume = $(12 - 4) \times (12 - 4) \times 2$

$$= 8 \times 8 \times 2 = 128 \text{ cu.cm}$$

Sol.91.(d) Given, in a right angle triangle

Hypotenuse = 25, Base = 24

Then, Height = 7

$$\therefore \text{Area} = \frac{1}{2} \times 24 \times 7 = 84 \text{ cm}^2$$

Sol.92.(a)

$$\text{Longest pole} = \sqrt{16^2 + 8^2 + 11^2} = 21 \text{ m}$$

$$\text{Sol.93.(d) } \frac{\sqrt{3}}{2} a = 10\sqrt{3} \Rightarrow a = 20$$

$$\text{Area} = \frac{\sqrt{3}}{4} a^2$$

$$= \frac{\sqrt{3}}{4} \times 20 \times 20 = 100\sqrt{3} \text{ cm}^2$$

Sol.94.(c)

$$\text{Required time} = \frac{66 \times 35 \times 3}{\frac{22}{7} \times \frac{35}{200} \times \frac{35}{200} \times 8}$$

$$= 9000 \text{ sec} = 2.5 \text{ h}$$

Sol.95.(a)

In the given solid figures. Cuboid has the maximum no. of vertex i.e. 8.

Sol.96.(b)

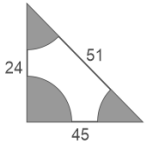
$$\text{Area} = 20 \times (7.5 + 8.6) = 322 \text{ cm}^2$$

Sol.97.(b) Since the chessboard is composed of 64 squares of side 2cm and the border is of thickness 2 cm.

Therefore, the length of each side of chessboard = $2 \times 8 + 4 = 20$

So, the perimeter = $4 \times 20 = 80 \text{ cm}$

Sol.98.(b) Given triangle is a right angled triangle:



$$\text{Area of triangle} = \frac{1}{2} \times 24 \times 45 = 540$$

Area of all sectors of the circle covering

$$\text{the triangle} = \frac{180}{360} \times \frac{22}{7} \times 10.5 \times 10.5$$

$$= 173.25$$

Therefore, Area of triangle excluding the portion covered = $540 - 173.25$

$$= 366.75 \text{ sq. cm}$$

Sol.99.(a) Old : New

$$\text{Radius} \rightarrow 2^2 : 5^2$$

$$\text{Height} \rightarrow \frac{5}{2} : \frac{4}{2}$$

$$\text{Volume} \rightarrow 20 : 100$$

$$\text{Required \%} = \frac{100 - 20}{20} \times 100 = 400\%$$

Sol.100.(c) Given: Surface area of cube

$$(6a^2) = 1728 \text{ cm}^2 \quad \text{So, } a = 12\sqrt{2} \text{ cm}$$

$$\text{Volume of cube } (a^3) = (12\sqrt{2})^3$$

$$= 3456\sqrt{2} \text{ cm}^3$$

Sol.101.(d) Radius = $\frac{8}{\pi}$, Volume = 448

$$\text{Volume of cylinder } (\pi r^2 h) = 448$$

$$\Rightarrow \pi \times \frac{8}{\pi} \times \frac{8}{\pi} \times h = 448$$

$$\Rightarrow h = \frac{448 \times 22}{64 \times 7} \Rightarrow h = 22 \text{ cm}$$

Sol.102.(a)

$$\text{Ratio} \rightarrow \text{Sphere}_1 : \text{Sphere}_2$$

$$\text{Volume} \rightarrow \frac{4}{3}\pi(4)^3 : \frac{4}{3}\pi(5)^3$$

$$64 : 125$$

Sol.103.(c) Volume of toy = Volume of hemisphere + volume of cylinder + volume of cone

$$= \frac{2}{3}\pi r^3 + \pi r^2 h + \frac{1}{3}\pi r^2 h$$

$$= \pi r^2 \left[\frac{2}{3}r + h + \frac{1}{3}h \right]$$

$$= \frac{22}{7} \times 4.2 \times 4.2 \left[\frac{2}{3} \times 4.2 + 7 + \frac{7}{3} \right]$$

$$= 22 \times 0.6 \times 4.2 \times \frac{36.4}{3} = 672.672 \text{ cm}^3$$

Sol.104.(b)

$$\text{Area of regular hexagon} = \frac{3\sqrt{3}}{2} a^2$$

According to the question,

$$\frac{3\sqrt{3}}{2} a^2 = 2048\sqrt{3} \Rightarrow a^2 = \frac{4096}{3}$$

$$a = \sqrt{\frac{4096}{3}} = \frac{64}{\sqrt{3}} = \frac{64}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

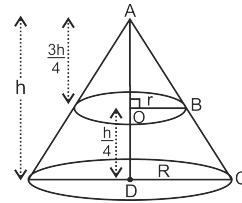
$$= \frac{64\sqrt{3}}{3} \text{ cm}^2$$

Sol.105.(d)

$$\text{CSA of a cone} = \pi R l = \pi R \sqrt{R^2 + h^2}$$

According to question,

$$\text{Height of small cone} = h - \frac{h}{4} = \frac{3h}{4}$$



$\triangle AOB \sim \triangle ADC$ by AAA rule

$$\Rightarrow \frac{r}{R} = \frac{\frac{3h}{4}}{h} \Rightarrow r = \frac{3R}{4}$$

$$\text{CSA of a small cone} = \pi r l_1$$

$$= \pi r \sqrt{r^2 + \left(\frac{3h}{4}\right)^2}$$

$$= \pi \times \frac{3R}{4} \times \sqrt{\left(\frac{3R}{4}\right)^2 + \left(\frac{3h}{4}\right)^2}$$

$$= \pi \times \frac{3R}{4} \times \frac{3}{4} \sqrt{R^2 + h^2}$$

Now,

$$\text{CSA of frustum} = \pi(R+r)l_2$$

$$= \pi \times \left(R + \frac{3R}{4}\right)$$

$$\times \sqrt{\left(R - \frac{3R}{4}\right)^2 + \left(\frac{h}{4}\right)^2}$$

$$= \pi \times \frac{7R}{4} \sqrt{\frac{R^2 + h^2}{16}}$$

$$= \pi \times \frac{7R}{16} \times \sqrt{R^2 + h^2}$$

So,

$$\text{Required ratio} \rightarrow \pi \times \frac{3R}{4} \times \frac{3}{4} \sqrt{R^2 + h^2}$$

$$: \pi \times \frac{7R}{16} \times \sqrt{R^2 + h^2} \text{ or } 9 : 7$$

Sol.106.(c) Volume of cuboid

$$= 5.5 \times 5 \times 3.5 = 96.25 \text{ cm}^3$$

$$\text{Volume of each gold coins} = \pi r^2 h$$

$$= \frac{22}{7} \times (0.875)^2 \times \frac{2}{10} = 0.48125 \text{ cm}^3$$

$$\text{Hence, no. of gold coins} = \frac{96.25}{0.48125}$$

$$= 200$$

Sol.107.(a) Total surface area of the tank = total surface area of cylinder - area of circular cut

$$= 2\pi r(h + R) - \pi r^2$$

$$= 2\pi \times 5(10 + 5) - \pi (0.5)^2$$

$$= 2\pi \times 75 - 0.25\pi = 150\pi - \frac{\pi}{4} = \frac{599\pi}{4} \text{ m}^2$$

Sol.108.(c) TSA of a cylinder = $2\pi r(h + r)$

$$= 2 \times \frac{22}{7} \times 3.5(8 + 3.5)$$

$$= 22 \times 11.5 = 253 \text{ cm}^2$$

According to question,

$$\Rightarrow 5 \text{ cm}^2 = ₹40$$

$$\Rightarrow 253 \text{ cm}^2 = \frac{40}{5} \times 253 = ₹2024$$

Sol.109.(b) Given,

$$\Rightarrow 2\pi r = 264 \text{ cm} \Rightarrow r = \frac{264 \times 7}{44} = 42 \text{ cm}$$

Given Bowl is half -full of beverage

$\therefore$ Volume of half -full beverage

$$\text{hemispherical bowl} = \frac{\frac{2}{3}\pi r^3}{2}$$

Now, radius of hemispherical glasses = 3cm

According to question ,

$\Rightarrow N \times \text{volume of hemispherical glasses} = \text{Volume of half -full beverage hemispherical bowl}$

$$\Rightarrow N = \frac{\frac{2}{3}\pi r^3}{\frac{2}{3}\pi r^3} = \frac{21 \times 42 \times 42}{3 \times 3 \times 3} = 1372$$

Sol.110.(c) Given radius = 14cm

$$\text{Area of circle} = \pi r^2$$

$$= \frac{22}{7} \times 14 \times 14 = 616 \text{ cm}^2$$

Diagonal of square ($\sqrt{2}a$) = diameter

$$\Rightarrow \sqrt{2}a = 28 \Rightarrow a = 14\sqrt{2} \text{ cm}$$

$$\text{Area of Square} = 14\sqrt{2} \times 14\sqrt{2} = 14 \times 14 \times 2 = 392 \text{ cm}^2$$

$$\text{Area of remaining portion} = 616 - 392 = 224 \text{ cm}^2$$

Sol.111.(c) Volume of the hemisphere

$$= \frac{2}{3}\pi r^3 \Rightarrow 486\pi = \frac{2}{3}\pi r^3$$

$$\Rightarrow 243 \times 3 = r^3 \Rightarrow r^3 = 729 \Rightarrow r = 9 \text{ cm}$$

According to question,

$$\Rightarrow \text{Radius } (r) = 9 \times 3 = 27 \text{ cm}$$

$$\text{CSA of a hemisphere} = 2\pi r^2$$

$$= 2 \times \pi \times 27 \times 27 = 1458\pi \text{ cm}^2$$

Sol.112.(c) Given , internal radius of hemispherical bowl = 15cm

Diameter and height of cylindrical shaped bottles = 5cm and 6cm

Number of bottles required to empty the bowl

$$= \frac{\text{Volume of hemispherical bowl}}{\text{volume of cylindrical shape bottles}}$$

$$= \frac{\frac{2}{3}\pi r^3}{\pi r^2 h} = \frac{\frac{2}{3} \times 15 \times 15 \times 15}{\frac{5}{2} \times \frac{5}{2} \times 6} = 60$$

Sol.113.(b) Volume of cone = $\frac{1}{3}\pi r^2 h$

$$= \frac{1}{3}\pi \times (7)^2 \times 18 = 294\pi \text{ cm}^3$$

Volume of hollow sphere

$$= \frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3}\pi[R^3 - (20)^3]$$

According to the question,

$$\frac{2}{3}\pi[R^3 - (20)^3] = 294\pi$$

$$R^3 - (20)^3 = 441$$

$$R^3 = 441 + 8000 \Rightarrow R = \sqrt[3]{8441} \text{ cm}^3$$

Sol.114.(a) Volume of a spherical item =
Volume of a shape

$$\Rightarrow \frac{4}{3}\pi r^3 = \frac{1}{3}\pi r^2 H + \pi r^2 h$$

$$\Rightarrow \frac{4}{3} \times \frac{21}{2} \times \frac{21}{2} \times \frac{21}{2} = \frac{1}{3} \times \frac{21}{2} \times \frac{21}{2}$$

$$\times H + \frac{21}{2} \times \frac{21}{2} \times 7 \Rightarrow \frac{4}{3} \times \frac{21}{2} = \frac{H}{3} + 7$$

$$\Rightarrow 14 = \frac{H}{3} + 7 \Rightarrow H = 21 \text{ cm}$$

Hence, the height of the cone is 21 cm

Sol.115.(b)

Each diagonal of hexagon = 2 × side(a)

$$\text{So, } a = \frac{12}{2} = 6 \text{ cm}$$

$$\text{Now, area of hexagon} = \frac{3\sqrt{3}}{2} a^2$$

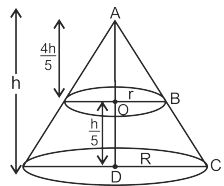
$$= \frac{3\sqrt{3}}{2} \times (6)^2 = 54\sqrt{3} \text{ cm}^2$$

Sol.116.(d)

$$\text{CSA of a cone} = \pi R l = \pi R \sqrt{R^2 + h^2}$$

According to question,

$$\text{Height of small cone} = h - \frac{h}{5} = \frac{4h}{5}$$



$\Delta AOB \sim \Delta ADC$ by AAA rule

$$\Rightarrow \frac{r}{R} = \frac{\frac{4h}{5}}{h}$$

$$\Rightarrow \text{small cone radius } (r) = \frac{4R}{5}$$

TSA of a small cone = CSA of smaller cone + base area of smaller cone

$$\Rightarrow \pi r l_1 + \pi r^2 = \pi \times \frac{4R}{5} \times \left[\frac{4R}{5} + \sqrt{\left(\frac{4R}{5}\right)^2 + \left(\frac{4h}{5}\right)^2} \right] + \pi \left(\frac{4R}{5}\right)^2$$

$$= \pi \times \frac{16R}{25} [R + \sqrt{R^2 + h^2}]$$

$$\text{Now, CSA of frustum} = \pi(R+r)l_2$$

$$= \pi \times \left(R + \frac{4R}{5}\right) \times \sqrt{\left(R - \frac{4R}{5}\right)^2 + \left(\frac{h}{5}\right)^2}$$

$$= \pi \times \frac{9R}{5} \sqrt{\frac{R^2 + h^2}{25}} = \pi \times \frac{9R}{25} \times \sqrt{R^2 + h^2}$$

$$\text{So, Required ratio} = \pi \times \frac{9R}{25} \times \sqrt{R^2 + h^2}$$

$$: \pi \times \frac{16R}{25} \left[R + \sqrt{R^2 + h^2} \right]$$

$$= \pi \times \frac{9R}{25} \times L : \pi \times \frac{16R}{25} [R + L]$$

$$= 9L : 16(R + L)$$

Sol.117.(b) Given, Radius : Height = 8 : 15

$$\text{Slant height} = \sqrt{r^2 + h^2} = \sqrt{8^2 + 15^2}$$

$$= \sqrt{289} = 17 \text{ cm}$$

Sol.118.(d)

Case 1 :- along the length 28 cm

Circumference of the circle = $2\pi r = 28 \text{ cm}$

$$\Rightarrow r = \frac{28}{2\pi} = \frac{14}{\pi} \text{ and height} = 16 \text{ cm}$$

$$\text{Volume of cylinder} = \pi r^2 h = \pi \left(\frac{14}{\pi}\right)^2 \times 16$$

$$= \frac{3136}{\pi} \text{ cm}^3$$

Case 2 :- along the breadth 16 cm

Circumference of the circle = $2\pi R = 16 \text{ cm}$

$$\Rightarrow 16 = 2\pi R \Rightarrow r = \frac{16}{2\pi} = \frac{8}{\pi} \text{ and}$$

height(h) = 28 cm

$$\text{Volume of cylinder} = \pi R^2 h = \pi \left(\frac{8}{\pi}\right)^2 \times 28$$

$$= \frac{1792}{\pi} \text{ cm}^3$$

The ratio of the larger volume to the

$$\text{smaller volume} = \frac{\frac{3136}{\pi}}{\frac{1792}{\pi}} = \frac{3136}{1792} = \frac{7}{4}$$

Hence, Required percentage

$$= \frac{7}{4} \times 100 = 175\%$$

Sol.119.(a)

1 person occupy the space = 5 m²

12 person occupy the space = 12 × 5

$$= 60 \text{ m}^2$$

Then,

$$\Rightarrow \text{Area of the circle} = \pi r^2$$

$$\Rightarrow 60 = \pi r^2 \dots (i)$$

1 person breadth the air = 20 m³

12 person breadth the air

$$= 12 \times 20 = 240 \text{ m}^3$$

Then,

$$\Rightarrow \text{Volume of the cone} = \frac{1}{3}\pi r^2 h$$

$$\Rightarrow 240 = \frac{1}{3} \times 60 \times h \text{ [from eq. (i)]}$$

$$\Rightarrow h = 12 \text{ m}$$

Sol.120.(b) Length of capsule = 15 mm

$$\text{Height of cylindrical part} = 15 - \left(\frac{6}{2}\right)$$

$$= 12 \text{ mm}$$

Now, surface area of capsule = T.S.A of

cylinder + T.S.A of sphere - $2\pi r^2$

$$= (2\pi r^2 + 2\pi r h) + 4\pi r^2 - 2\pi r^2$$

$$= 2\pi r h + 4\pi r^2$$

$$= 2\pi r(h + 2r) = 2 \times \frac{22}{7} \times 3 \times (3 + 12)$$

$$= \frac{12 \times 15 \times 11}{7} \approx 283 \text{ mm}^2$$

Hence, surface area of capsule $\approx 283 \text{ mm}^2$

Sol.121.(b)

Total surface area of hemisphere = $3\pi r^2$

According to the question,

$$3\pi r^2 = 7392$$

$$r^2 = 7392 \times \frac{7}{22} \times \frac{1}{3} = 784$$

$$\text{Radius } (r) = \sqrt{784} = 28$$

$\frac{1}{4}$ th of the hemispherical container is empty

So, total volume of oil container

$$= \frac{3}{4} \times \frac{2}{3}\pi r^3 = \frac{3}{4} \times \frac{2}{3} \times \frac{22}{7} \times (28)^3 \text{ cm}^3$$

Now, oil is to be transferred in small hemispherical bottles

$$\text{Then, volume per bottle} = \frac{2}{3}\pi r^3$$

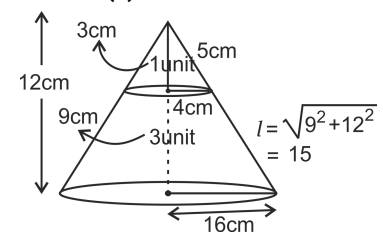
$$= \frac{2}{3} \times \frac{22}{7} \times (6.3)^3 \text{ cm}^3$$

Hence, required no. of bottles

$$= \frac{\frac{3}{4} \times \frac{2}{3} \times \frac{22}{7} \times (28)^3}{\frac{2}{3} \times \frac{22}{7} \times (6.3)^3} = \frac{\frac{3}{4} \times (28)^3}{(6.3)^3}$$

$$= \frac{3 \times 7 \times 28 \times 28}{6.3 \times 6.3 \times 6.3} = \frac{16464}{250} \approx 66$$

Sol.122.(a)



Given, radius (R) = 16 cm, height = 12 cm

and ratio of height of solid = 1 : 3

According to question,

Height of small cone = h_1

$$= \frac{1}{4} \times H = \frac{1}{4} \times 12 = 3 \text{ cm}$$

$$\text{Height of frustum} = h_2 = \frac{1}{4} \times H$$

$$= \frac{3}{4} \times 12 = 9 \text{ cm}$$

The section is parallel to the base, so the ratio of the radii will be equal to the ratio of the heights.

$$\text{radius of small cone } (r) = \frac{h_1}{H} \times R$$

$$= \frac{3}{12} \times 16 = 4 \text{ cm}$$

Slant height of small cone (l_1)

$$= \sqrt{(r_1)^2 + (h_1)^2} = \sqrt{(4)^2 + (3)^2} = 5 \text{ cm}$$

From the question -

$$l_1 : l_2 = 1 : 3 \Rightarrow 5 : l_2 = 1 : 3 \Rightarrow l_2 = 15 \text{ cm}$$

Thus

Slant height of the larger frustum (l_2)

$$= 15 \text{ cm}$$

$$\text{TSA of small cone} = \pi r_1 l_1 + \pi r_1^2$$

$$= \pi(4 \times 5) + \pi(4)^2 = 20\pi + 16\pi = 36\pi$$

Total surface area of the larger frustum

$$= \pi l_2(R + r) + \pi(R^2 + r^2)$$

$$= \pi \times 15 \times (20) + \pi(4^2 + 16^2)$$

$$= 300\pi + 272\pi = 572\pi$$

$$\text{Hence, Required ratio} = \frac{36\pi}{572\pi} = 9 : 143$$

Sol.123.(a)

Radius(r) = 24 cm, height(h) = 7 cm

As, (7, 24, 25) are triplet

Then, slant height (l) = 25 cm

Area of remaining solid = C.S.A. of cylinder + area of base + C.S.A. of cone

$$= 2\pi rh + \pi r^2 + \pi rl$$

$$= \pi r(2h + r + l)$$

$$= \pi \times 24(2 \times 7 + 24 + 25)$$

$$= \pi \times 24 \times 63 = 1512\pi \text{ cm}^2$$

Sol.124.(a)

Let side of one square = x cm

Side of another square = $x + 3$

According to question

$$\Rightarrow x^2 + (x + 3)^2 = 225$$

$$\Rightarrow x^2 + x^2 + 9 + 6x = 225$$

$$\Rightarrow 2x^2 + 6x - 216 = 0 \Rightarrow x^2 + 3x - 108 = 0$$

$$\Rightarrow x^2 + 12x - 9x - 108 = 0$$

$$\Rightarrow x(x + 12) - 9(x + 12) = 0$$

$$\Rightarrow (x - 9)(x + 12) = 0$$

$$x = 9, -12$$

$\therefore$ Sides of two square are = 9 and 12

Hence sum of the perimeter of both the square are = $4 \times 9 + 4 \times 12 = 84$ cm

Sol.125.(b) CSA of a hemisphere = $2\pi r^2$

$$308 = 2 \times \frac{22}{7} \times r^2$$

$$r^2 = \frac{308 \times 7}{2 \times 22} = 49 \Rightarrow r = 7 \text{ cm}$$

$$\text{Volume of the hemisphere} = \frac{2}{3}\pi r^3$$

$$= \frac{2}{3} \times \frac{22}{7} \times 7 \times 7 \times 7 = 718.66 \text{ cm}^3$$

Now,

$$1 \text{ cm}^3 = 8.7 \text{ gm}$$

$$718.66 \text{ cm}^3 = 718.66 \times 8.7 = 6252.342$$

$$\text{gm} \approx 6252.4 \text{ gm}$$

Sol.126.(a) Diagonal of a square who made mid point a square = $\sqrt{2} \times \text{side}$

$$\Rightarrow a = \sqrt{2} \times \text{side} \Rightarrow \text{side} = \frac{a}{\sqrt{2}}$$

$$\text{Area of a square} = \left(\frac{a}{\sqrt{2}}\right)^2 = \frac{a^2}{2}$$

According to question,

$$\text{Remaining area} = a^2 - \frac{a^2}{2}$$

$$= \frac{2a^2 - a^2}{2} = \frac{a^2}{2}$$

Sol.127.(d) External radius (R) = 2.7 cm

$$\text{Internal radius } (r) = 2.7 - \frac{3}{10} = 2.4 \text{ cm}$$

Now, Volume of pipe = $\pi(R^2 - r^2)h$

$$= \frac{22}{7} \times \{(2.7)^2 - (2.4)^2\} \times 420$$

$$= \frac{22}{7} \times \{7.29 - 5.76\} \times 420$$

$$= 22 \times 1.53 \times 60 = 2019.6 \text{ cm}^3$$

Given, $1 \text{ cm}^3 = 25 \text{ gm}$

$$\text{So, } 2019.6 \text{ cm}^3 = 25 \times 2019.6$$

$$= 50490 \text{ gm} = 50.49 \text{ kg}$$

Sol.128.(b) Volume of solid sphere

$$\left(\frac{4}{3}\pi r^3\right) = \text{Volume of cylindrical pipe}$$

$$(\pi r^2 h) \Rightarrow \frac{4}{3} \times 15 \times 15 \times 15 = (6^2 - 4^2)h$$

$$\Rightarrow h = \frac{\frac{4}{3} \times 15 \times 15 \times 15}{20}$$

$$= 225 \text{ cm or } 2.25 \text{ m}$$

Sol.129.(a) According to question,

$$\Rightarrow r + l = h^2 \Rightarrow r + l = 9^2 = 81 \dots\dots(i)$$

$$\Rightarrow r - l = 1 \dots\dots(ii)$$

On solving equation (i) and (ii), we get

$$r = 41 \text{ cm and } l = 40 \text{ cm}$$

TSA of cone = $\pi r(r + l)$

$$= \frac{22}{7} \times 41(41 + 40) = \frac{22 \times 41 \times 81}{7}$$

$$= \frac{73062}{7} = 10437.43 \text{ cm}^2$$

Sol.130.(b) TSA of cylinder = $2\pi r(h + r)$

$$= 2\pi \times 10(30 + 10) = 800\pi \text{ cm}^2$$

According to question,

TSA of larger piece of a cylinder

$$= 800\pi \times \frac{70}{100} = 560\pi \text{ cm}^2$$

$$\Rightarrow 2\pi r(h_1 + r) = 560\pi$$

$$\Rightarrow 2\pi \times 10 \times (h_1 + 10) = 560\pi$$

$$\Rightarrow (h_1 + 10) = 28 \Rightarrow h_1 = 18 \text{ cm}$$

Height(h_2) of smaller cylinder = $30 - 18$

$$= 12 \text{ cm}$$

Now,

Volume of sphere = Volume of smaller piece of cylinder

$$\frac{4}{3}\pi R^3 = \pi r^2 h_2 \Rightarrow \frac{4}{3} \times R^3 = 10 \times 10 \times 12$$

$$R^3 = 900 \Rightarrow R = \sqrt[3]{900} \text{ cm}$$

Sol.131.(a)

For the largest right circular cone Length of cone = height of cube = 6 unit

Diameter of cone = side length of cone = 6 unit

$$\text{Radius of cone} = \frac{6}{2} = 3 \text{ unit}$$

$$\text{Now, volume of cone} = \frac{1}{3}\pi r^2 h$$

$$= \frac{1}{3} \times \pi \times (3)^2 \times 6 = 18\pi \text{ units}^3$$

Sol.132.(b) Internal Volume of the large

$$\text{cylindrical vessel } (V_L) = \pi r_L^2 h_L$$

$$= \pi \times 10 \times 10 \times 14 = 1400\pi \text{ cm}^3$$

Internal Volume of small cylindrical

$$\text{vessel } (V_S) = \pi r_S^2 h_S$$

$$= \pi \times 5 \times 5 \times 7 = 175\pi \text{ cm}^3$$

Unfilled volume of big vessel after transferring 6 times = $1400\pi - 6 \times 175\pi$

$$= 1400\pi - 1050\pi = 350\pi \text{ cm}^3$$

Now,

$$\text{Unfilled volume of big vessel} = 350\pi \text{ cm}^3$$

$$\pi r_3^2 h_3 = 350\pi \text{ cm}^3 \Rightarrow h_3 = \frac{350}{10 \times 10} = 3.5 \text{ cm}$$

Sol.133.(b) Inner radius(r) = $\frac{12}{2} = 6$ cm

Outer radius(R) = $6 + 2 = 8$ cm

Length (h) = 35 cm

T.S.A. of cylindrical tube = C.S.A of inner part + C.S.A of outer part + area of rings

Now,

C.S.A of inner part + C.S.A of outer part :-

$$= 2\pi h(R + r) = 2 \times \frac{22}{7} \times 35 \times (8 + 6)$$

$$= 3080 \text{ cm}^2$$

For area of two ring :-

$$2 \times \pi(R^2 - r^2) = 2 \times \frac{22}{7} \times [(8)^2 - (6)^2]$$

$$= 2 \times \frac{22}{7} \times [64 - 36] = 176 \text{ cm}^2$$

Hence, required area = $3080 + 176$

$$= 3256 \text{ cm}^2$$

Sol.134.(c)

Perimeter of rhombus ($4a$) = 160

Side of rhombus(a) = 40 cm

$$\text{Now, } a^2 = \left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2$$

$$(40)^2 = \left(\frac{d_1}{2}\right)^2 + \left(\frac{50}{2}\right)^2$$

$$1600 = \left(\frac{d_1}{2}\right)^2 + 625 \Rightarrow \left(\frac{d_1}{2}\right)^2 = 975$$

$$d_1 = \sqrt{975 \times 4} = \sqrt{3900} = 10\sqrt{39} \text{ cm}$$

$$\text{Area of rhombus} = \frac{1}{2} \times d_1 \times d_2$$

$$= \frac{1}{2} \times 10\sqrt{39} \times 50 = 250\sqrt{39} \text{ cm}^2$$

Sol.135.(b) According to question,

$$\Rightarrow \sqrt{2}a_1 : \sqrt{2}a_2 = 5 : 6 \Rightarrow a_1 : a_2 = 5 : 6$$

Now,

$$\Rightarrow a_1^2 : a_2^2 = 25 : 36$$

Sol.136.(c)

Given, radius of circle (R) = $\frac{(l+r)}{2}$

According to question,

Total surface area of icecream cone with lid = area of circle

$$\Rightarrow \pi r^2 + \pi r l = \pi r^2$$

$$\Rightarrow \pi r (r + l) = \pi \times \left(\frac{(l+r)}{2}\right)^2$$

$$\Rightarrow 4r^2 + 4rl = l^2 + r^2 + 2lr$$

$$\Rightarrow 4r^2 - r^2 - l^2 + 4rl - 2lr = 0$$

$$\Rightarrow 3r^2 - l^2 + 2rl = 0 \text{ -----(i)}$$

$$\text{Now, (R)} = \frac{(l+r)}{2}$$

$$l = 2R - r \text{ -----(ii)}$$

Putting the value of l in equation (i)

$$3r^2 - (2R - r)^2 + 2r(2R - r) = 0$$

$$3r^2 - 4R^2 - r^2 + 4Rr + 4rR - 2r^2 = 0$$

$$8Rr - 4R^2 = 0$$

$$8Rr = 4R^2$$

$$8r = 4R \Rightarrow R : r = 2 : 1$$

$$\text{Now, } l = 2R - r \Rightarrow l = 2 \times 2 - 1 = 3$$

$$\text{Hence, required ratio} = l : R = 3 : 2$$

Sol.137.(c) Volume of hemisphere

$$= \frac{2}{3} \pi r^3 = \left(\frac{2}{3} \times \pi \times 15 \times 15 \times 15\right) \text{ cm}^3$$

Water displaced out =

$$\left(\frac{2}{5} \times \frac{2}{3} \times \pi \times 15 \times 15 \times 15\right) \text{ cm}^3$$

$$\text{Volume of lead shot} = \frac{4}{3} \pi r^3$$

$$= \left(\frac{4}{3} \pi \times 0.75 \times 0.75 \times 0.75\right) \text{ cm}^3$$

No of lead shots drop in the hemisphere

$$= \frac{\frac{2}{5} \times \frac{2}{3} \times \pi \times 15 \times 15 \times 15}{\frac{4}{3} \pi \times 0.75 \times 0.75 \times 0.75} = 1600$$

Sol.138.(c) Volume of the cone = $\frac{1}{3} \pi r^2 h$

$$= \frac{1}{3} \times \frac{22}{7} \times \frac{13.2}{2} \times \frac{13.2}{2} \times 14$$

$$= 11 \times 4.4 \times 13.2 = 638.88 \text{ cm}^3$$

According to question,

$$\Rightarrow \text{Number of bags} = \frac{638.88}{29.4}$$

$$= 21.73 \approx 22$$

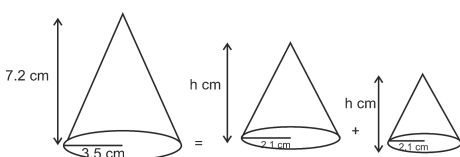
Sol.139.(b) For the largest right circular

cone that cut out from a cube

Diameter of cone = edge of the cube

$$= 8.4 \text{ cm}$$

$$\text{So, Radius of cone} = \frac{8.4}{2} = 4.2 \text{ cm}$$

Sol.140.(b) Volume of cone = $\frac{1}{3} \pi r^2 h$ 

$$\frac{1}{3} \times \pi \times 3.5 \times 3.5 \times 7.2$$

$$= 2 \times \left(\frac{1}{3} \times \pi \times 2.1 \times 2.1 \times h\right)$$

$$3.5 \times 3.5 \times 7.2 = 2 \times (2.1 \times 2.1 \times h)$$

$$h = \frac{3.5 \times 3.5 \times 7.2}{2 \times 2.1 \times 2.1} = 10 \text{ cm}$$

Sol.141.(d) Side = radius ... (given)

$$\frac{\text{Perimeter of square}}{\text{Perimeter of circle}} = \frac{7}{11}$$

$$\frac{\text{area of square}}{\text{area of circle}} = \frac{\text{side}^2}{\pi r^2} = \frac{1}{\frac{22}{7}} = \frac{7}{22}$$

Hence, required ratio = 7 : 22

Sol.142.(b) Area to be painted = external

C.S.A. + Internal C.S.A. + area of ring

$$= 2\pi R^2 + 2\pi r^2 + \pi(R^2 - r^2)$$

$$= 2 \times \frac{22}{7} [(14)^2 + (13.3)^2] + \frac{22}{7}$$

$$[(14)^2 - (13.3)^2]$$

$$= 2 \times \frac{22}{7} [(196 + 176.89) + \frac{22}{7} [196 - 176.89]$$

$$= 2 \times \frac{22}{7} \times 372.89 + \frac{22}{7} \times 19.11$$

$$= 44 \times 53.27 + 22 \times 2.73$$

$$= 2343.88 + 60.06 = 2403.94$$

Cost of painting

$$= 2403.94 \times 3.75 = 9014.775 \approx 9015$$

Sol.143.(b)

$$\frac{\text{curved surface area of cylinder}_1}{\text{curved surface area of cylinder}_2}$$

$$= \frac{2\pi RH}{2\pi rh} = \frac{2 \times 5}{3 \times 4} = \frac{5}{6}$$

Required ratio = 5 : 6

Sol.144.(a) Outer radius (R) = 9 units

Volume of hollow hemispherical steel

$$\text{bowl} = \frac{686}{3} \pi \text{ unit}^3 \Rightarrow \frac{2}{3} \pi r^3 = \frac{686}{3} \pi \text{ unit}^3$$

$$r^3 = 343 \text{ unit}^3$$

Internal radius (r) = 7 unit

$$\text{Steel used} = \frac{2}{3} \pi (R^3 - r^3)$$

$$= \frac{2}{3} \pi (9^3 - 7^3) = \frac{2}{3} \pi (729 - 343)$$

$$= \frac{2}{3} \pi \times 386 = \frac{772}{3} \pi \text{ unit}^3$$

Sol.145.(a) Radius of balloon = 8 feet

$$\text{Volume of spherical balloon} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times 3.14 \times 512$$

Time taken for balloon to empty

$$= \frac{\text{Volume of spherical balloon}}{\text{Rate of leak}}$$

$$= \frac{\frac{4}{3} \times 3.14 \times 512}{24} = \frac{12.56 \times 64}{9}$$

$$= \frac{803.84}{8} = 89.31 \approx 89 \text{ minutes}$$

Sol.146.(a) Volume of hemisphere

$$= N \times \text{volume of cone}$$

$$\frac{2}{3} \pi R^3 = N \times \frac{1}{3} \pi r^2 h$$

$$\frac{2}{3} \times 12^3 = N \times \frac{1}{3} \times 6^2 \times 8$$

$$2 \times 144 \times 12 = N \times 36 \times 8$$

$$N = \frac{144}{12} = 12$$

Sol.147.(a) Let the height of the two

cone be h_1 and h_2 ,

According to the question,

$$\Rightarrow \frac{\frac{1}{3} \pi r_1^2 h_1}{\frac{1}{3} \pi r_2^2 h_2} = \frac{3}{5} \Rightarrow \frac{4 \times h_1}{25 \times h_2} = \frac{3}{5}$$

$$\Rightarrow \frac{h_1}{h_2} = \frac{75}{20} = \frac{15}{4} \Rightarrow h_1 : h_2 = 15 : 4$$

Sol.148.(b) Volume of cylinder = $\pi r^2 h$

$$\Rightarrow \frac{22}{7} \times (8)^2 \times 14 = 2816 \text{ cm}^3$$

Sol.149.(b) Area of sector = $\frac{1}{2} \times l \times r$

$$= \frac{1}{2} \times 11 \times 15.75 = 86.625 \text{ cm}^2$$

Sol.150.(c) Area of isosceles right angle

$$\text{triangle} = \left(\frac{\text{hypotenuse}(H)}{2}\right)^2$$

$$\Rightarrow \left(\frac{H}{2}\right)^2 = 50 \Rightarrow H^2 = 200 \Rightarrow H = 10\sqrt{2} \text{ unit}$$

Sol.151.(d)

$$\text{Perimeter of sector} = 2r + \frac{\theta}{360} \times 2\pi r$$

$$\Rightarrow 2 \times 10 + \frac{30}{360} \times 2 \times 3.14 \times 10$$

$$\Rightarrow 20 + \frac{1}{12} \times 62.8$$

$$\Rightarrow 20 + \frac{1}{12} \times 62.8 = 25.2 \text{ (approx)}$$

Sol.152.(b) Volume of cone = $\frac{1}{3} \pi r^2 h$

$$1232 = \frac{1}{3} \times \frac{22}{7} \times r^2 \times 24$$

$$1232 = \frac{176}{7} r^2 \Rightarrow r^2 = \frac{1232 \times 7}{176}$$

$$= r^2 = 49 \Rightarrow r = 7 \text{ cm}$$

$$\text{Slant height} = \sqrt{7^2 + 24^2} = \sqrt{625} = 25 \text{ cm}$$

TSA of cone = $\pi r(r + l)$

$$= \frac{22}{7} \times 7(7 + 25) = 22 \times 32 = 704 \text{ cm}^2$$

Sol.153.(b) If a cylinder is formed by rolling the paper along its breadth, breadth becomes perimeter of base and length is equivalent to height.

$$\Rightarrow 2\pi r = 22 \Rightarrow r = \frac{7}{2} = 3.5$$

And $h = 50$

Volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 50 = 1925 \text{ cm}^3$$

Sol.154.(d) Length = 495 cm

As $OP = OR$, So with the help of diagram we can say that, $\angle SOQ = 180^\circ$

Sol.176.(c) Height and diameter of a cylinder are 12 cm and 28cm

Total surface area of cylinder $= 2\pi r(r + h)$

$$= 2 \times \frac{22}{7} \times 14 (14 + 12) = 88 \times 26$$

Curved Surface area of cylinder $= 2\pi rh$

$$= 2 \times \frac{22}{7} \times 14 \times 12 = 88 \times 12$$

The ratio of the total surface area to the curved surface area $= 88 \times 26 : 88 \times 12$

$$= 13 : 6$$

Sol.177.(d)

$$\text{Perimeter of the sector} = 2\pi r \times \frac{\theta}{360^\circ} + 2r$$

$$= 2 \times 3.14 \times 14 \times \frac{15^\circ}{360^\circ} + 2 \times 14$$

$$= \frac{3.14 \times 7}{6} + 28 = 3.66 + 28 = 31.66 \text{ cm}$$

Sol.178.(a) Let breadth of the rectangular garden = x m

Length of the rectangular garden

$$= (x + 6) \text{ m}$$

Perimeter of rectangular garden $[2(l + b)]$

$$= 48 \text{ m}$$

$$\Rightarrow 2(x + 6 + x) = 48 \Rightarrow 2x + 6 = 24$$

$$\Rightarrow 2x = 18 \Rightarrow x = 9 \text{ m}$$

Length = 15 m and breadth = 9 m

Area of rectangular garden $= l \times b$

$$= 15 \times 9 = 135 \text{ m}^2$$

Sol.179.(a)

Volume of each cube $= 512 \text{ cm}^3$

$$\therefore \text{Side of cube} = \sqrt[3]{512} = 8 \text{ cm}$$

Now by joining three cube, length of the cuboid $= 8 + 8 + 8 = 24 \text{ cm}$

Breadth = 8 cm and height = 8 cm

$$\therefore \text{Surface area} = 2(lb + bh + hl)$$

$$= 2(24 \times 8 + 8 \times 8 + 8 \times 24)$$

$$= 2(192 + 64 + 192) = 896 \text{ cm}^2$$

Sol.180.(a) Given : radius(r) = 9 cm and

slant height(l) = 49 cm

Curved surface area of a cone $= \pi rl$

$$= \frac{22}{7} \times 9 \times 49 = 1386 \text{ cm}^2$$

Sol.181.(d)

Area of parallelogram = Base $\times$ Height

$$\Rightarrow 26 \times 18 = 468 \text{ cm}^2$$

Sol.182.(d) Base radius = 10.5cm

Curved surface area of cylinder ($2\pi rh$)

$$= 792 \text{ cm}^2$$

$$h = \frac{792 \times 7 \times 10}{2 \times 22 \times 105} = 12 \text{ cm}$$

volume of the cylinder $= \pi r^2 h$

$$= \frac{22}{7} \times 10.5 \times 10.5 \times 12 = 4158 \text{ cm}^3$$

Sol.183.(d) given, $(a + b) = 9$ and $ab = 18$

We know that,

$$(a - b) = \sqrt{(a + b)^2 - 4ab}$$

$$= \sqrt{(9)^2 - 4 \times 18} = \sqrt{81 - 72} = \sqrt{9} = 3$$

then,

$$(a + b) = 9 \dots\dots (i)$$

$$(a - b) = 3 \dots\dots (ii)$$

On solving equation (i) and (ii), we get

$$a = 6 \text{ and } b = 3$$

Now,

$$a^2 - ab + b^2 = 6^2 - 6 \times 3 + 3^2$$

$$= 36 - 18 + 9 = 27$$

Sol.184.(c) Area of sector of a circle $= \pi$

$$r^2 \times \frac{\theta}{360^\circ} \Rightarrow 462 = \frac{22}{7} \times r^2 \times \frac{120^\circ}{360^\circ}$$

$$\Rightarrow r^2 = 462 \times \frac{7}{22} \times 3 = 21 \times 21$$

$$\Rightarrow r = 21 \text{ cm}$$

Sol.185.(c)

Total surface area of cube $= 6a^2$

$$\Rightarrow 1152 = 6a^2 \Rightarrow a^2 = 192 \Rightarrow a = 8\sqrt{3} \text{ cm}$$

Sol.186.(b)

Surface area of four walls $= 2h(l + b)$

$$= 2 \times 5 (15 + 9) = 10 (24) = 240 \text{ m}^2$$

So,

$$\text{The number of cans} = \frac{240}{40} = 6$$

Sol.187.(d) Volume of a right circular

cylindrical box $(\pi R^2 h) = 600 \text{ cm}^3$

$$\Rightarrow \frac{22}{7} \times 30 \times 30 \times h = 600$$

$$\Rightarrow h = \frac{600 \times 7}{22 \times 30 \times 30} = \frac{7}{33} = 0.21 \text{ cm}$$

Sol.188.(d) According to question,

$$\Rightarrow \pi R^2 = \pi r_1^2 + \pi r_2^2$$

$$\Rightarrow R^2 = (8^2 + 15^2) \Rightarrow R = 17 \text{ cm}$$

$$\therefore \text{Perimeter of circle} = 2 \times \pi \times 17 = 34\pi \text{ cm}$$

Sol.189.(b)

Length : breadth : height $= 3 : 2 : 1$

Breadth (2 unit) = 12 cm

Length (3 unit) = 18 cm

Height (1 unit) = 6 cm

$$\text{Volume} = 18 \times 12 \times 6 = 1296 \text{ cm}^3$$

Sol.190.(a) Circumference of base of cylinder = 154 cm

$$2\pi r = 154$$

$$2 \times \frac{22}{7} \times r = 154 \Rightarrow r = \frac{49}{2}$$

Volume of cylindrical vessel $= \pi r^2 h$

$$= \frac{22}{7} \times \frac{49}{2} \times \frac{49}{2} \times \frac{49}{10}$$

$$= 11 \times 7 \times \frac{49}{2} \times \frac{49}{10} = 9243.5 \text{ cm}^3$$

$$\text{In litres} = 9243.5 \times \frac{1}{1000} = 9.244 \text{ litres}$$

Sol.191.(d) $1 \text{ l} = 1000 \text{ cm}^3$

Volume of water level dropped $= \pi r^2 h$

$$616 \times 1000 = \frac{22}{7} \times 14 \times 14 \times h$$

$$h = \frac{616000}{22 \times 14 \times 2} = 1000 \text{ cm} = 10 \text{ m}$$

Sol.192.(b) Circumference of bigger

$$\text{circle} = 2\pi r_1 + 2\pi r_2$$

$$2\pi R = 2\pi(r_1 + r_2)$$

$$R = 12 + 16 = 28 \text{ cm}$$

Radius of bigger circle = 28 cm

then diameter = 56 cm

Sol.193.(c) Volume of earth taken out

$$= \pi r^2 h = \frac{22}{7} \times 14 \times 14 \times 23$$

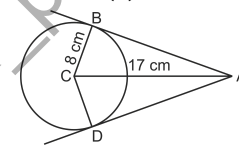
$$= 22 \times 28 \times 23 = 14168 \text{ cm}^3$$

Sol.194.(b) T.S.A. of cone = curved surface area + area of base

$$= 550 + \pi r^2 = 550 + \frac{22}{7} \times 7 \times 7$$

$$= 550 + 154 = 704 \text{ cm}^2$$

Sol.195.(b)



In $\triangle ABC$

$$\angle ABC = 90^\circ \dots (\text{radius} \perp \text{tangent})$$

hence,

$$AB = \sqrt{AC^2 - BC^2} = \sqrt{17^2 - 8^2}$$

$$= \sqrt{289 - 64} = \sqrt{225} = 15 \text{ cm}$$

$$\text{Area of } \triangle ABC = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times AB \times BC = \frac{1}{2} \times 15 \times 8 = 60 \text{ cm}^2$$

Area of quadrilateral ABCD

$$= 2 \times \text{Area of } \triangle ABC = 2 \times 60 = 120 \text{ cm}^2$$

Sol.196.(d) Length of sector = 96π

$$\frac{\theta}{360^\circ} \times 2\pi r = 96\pi \Rightarrow \frac{80^\circ}{360^\circ} \times 2 \times r = 96$$

$$r = \frac{96 \times 36}{16} = 6 \times 36 = 216 \text{ unit}$$

Sol.197.(c)

$$\text{Radius of the sphere} = \frac{14}{2} = 7 \text{ cm}$$

$$\text{Volume of the sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times (7)^3 = 1437 \frac{1}{3} \text{ cm}^3$$

Sol.198.(d) According to the question,

Let radius and height of cylinder be $3x$ and $7x$ respectively,

$$\Rightarrow \frac{\text{CSA of cylinder}}{\text{Volume of cylinder}} = \frac{2\pi rh}{\pi r^2 h}$$

$$\Rightarrow \frac{2}{3} = \frac{2 \times \pi \times 3x \times 7x}{\pi \times 3x \times 3x \times 7x} \Rightarrow x = 1 \text{ unit}$$

So, radius of cylinder $= 3x = 3 \text{ cm}$ and height $= 7x = 7 \text{ cm}$

Now,

Total surface area of cylinder
 $= 2\pi r(r + h) = 2\pi \times 3(3 + 7) = 60\pi \text{ cm}^2$

Sol.199.(b) Ratio of surface area of two spheres
 $= 4\pi r_1^2 : 4\pi r_2^2 = r_1^2 : r_2^2$

$$\text{Given: } \frac{r_1^2}{r_2^2} = \frac{25}{36} \Rightarrow \frac{r_1}{r_2} = \frac{5}{6}$$

Now, ratio of their volumes

$$= \frac{4}{3}\pi r_1^3 : \frac{4}{3}\pi r_2^3 = r_1^3 : r_2^3 = \frac{5^3}{6^3} = \frac{125}{216}$$

Sol.200.(b) Let the length, breadth and height of cuboid be l , b and h respectively,

Let sides of cuboid be $3x$, $4x$ and $5x$ respectively

Then, Surface area of cuboid

$$= 2(lb + bh + hl)$$

$$= 2(12x^2 + 20x^2 + 15x^2) = 2(47x^2) = 94x^2$$

Now, surface area of cube with side 6 cm

$$= 6 \times (6 \text{ unit})^2 = 216 \text{ unit}^2$$

Now, according to the question,

$$94x^2 = 216 \text{ unit}^2$$

$$x^2 = \frac{216}{94} \Rightarrow x = \sqrt{\frac{216}{94}} \approx \frac{14.69}{9.69}$$

$$= 1.51599$$

Volume of cuboid (lbh) = $3x \times 4x \times 5x$

$$= 60x^3 = 60 \times (1.51599)^3 \approx 210 \text{ unit}^3$$

Sol.201.(a) Area of the playground =

$$700 \times \text{CSA of cylinder} = 700 \times 2\pi rh$$

$$= 700 \times 2 \times \frac{22}{7} \times 49 \times 200$$

$$= 4312 \times 10^4 \text{ cm}^2$$

Sol.202.(d)

Curved surface area of the cone = πrl

$$2200 = \frac{22}{7} \times 28 \times l$$

$$(\text{slant height}) l = 25 \text{ cm}$$

Sol.203.(b) Ratio of surface area and

$$\text{volume of cube} = 6a^2 : a^3$$

$$= 6(3)^2 : (3)^3 = 2 : 1$$

Sol.204.(b)

Surface area of sphere = $4\pi r^2$

$$4\pi r^2 = 2464 \Rightarrow 4 \times \frac{22}{7} \times r^2 = 2464$$

$$r^2 = 2464 \times \frac{7}{22} \times \frac{1}{4} = 196$$

$$r = \sqrt{196} = 14 \text{ cm}$$

Now, volume of sphere = $\frac{4}{3}\pi r^3$

$$= \frac{4}{3} \times \frac{22}{7} \times (14)^3 = 11498 \frac{2}{3} \text{ cm}^3$$

Sol.205.(b) By checking options one by one option (b) satisfied

$$\text{i.e. } (12 : 16 : 20) = (3 : 4 : 5)$$

3, 4 and 5 are the pythagorean triplet so, triangle having side 12 cm, 16 cm and 20

cm is right angled triangle.

Sol.206.(d) volume of sphere = $\frac{4}{3}\pi r^3$

$$= \frac{4}{3} \times \frac{22}{7} \times (r)^3 = 38.808 \text{ cm}^3$$

$$(r)^3 = \frac{38.808 \times 3 \times 7}{22 \times 4}$$

$$= \frac{11 \times 8 \times 0.441 \times 3 \times 7}{11 \times 2 \times 4} = 9.261 \text{ cm}^3$$

$$r = \sqrt[3]{9.261} = \sqrt[3]{\frac{9261}{1000}} = \frac{21}{10} = 2.1 \text{ cm}$$

Hence, diameter of spherical ball ($2r$)

$$= 4.2 \text{ cm}$$

Sol.207.(b) Length of arc (l) = $\frac{\theta}{360^\circ} \times 2\pi r$

$$= \frac{45^\circ}{360^\circ} \times 2 \times \pi \times 4$$

$$= \frac{1}{8} \times 2 \times \pi \times 4 = \pi \text{ unit}$$

Now, perimeter of minor sector = $2r + l$

$$= 2 \times 4 + \pi = (8 + \pi) \text{ unit}$$

Sol.208.(c) Curved surface area of

$$\text{hemisphere} = 2\pi r^2 = 2 \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2}$$

$$= 3 \times 11 \times 21 = 693 \text{ cm}^2$$

Sol.209.(a)

Capacity of cylinder = 8316 cm^3

$$\pi r^2 h = 8316 \Rightarrow \frac{22}{7} \times 21 \times 21 \times h = 8316$$

$$h = 8316 \times \frac{7}{22 \times 21 \times 21} = 6 \text{ cm}$$

$$\text{Sol.210.(b)} \quad 72.9\% = \frac{729}{1000}$$

$$\frac{\text{final volume of cube } (a_1^3)}{\text{initial volume of cube } (a^3)} = \frac{729}{1000}$$

$$\frac{a_1}{a} = \frac{9}{10}$$

$$\text{Hence, required \%} = \frac{1}{10} \times 100 = 10\%$$

Sol.211.(d)

Total surface area of hemisphere = $3\pi r^2$

$$\Rightarrow 3 \times 3.14 \times (6.2)^2 = 362.10 \text{ cm}^2$$

$$\text{Sol.212.(b)} \quad \text{Required \%} = 5 + 5 + \frac{5 \times 5}{100}$$

$$= 10 + 0.25 = 10.25\%$$

Sol.213.(d) Required bricks

$$= \frac{(10 \times 100) \times 22.5 \times (6 \times 100)}{25 \times 12 \times 9}$$

$$= 40 \times 2.5 \times 50 = 5000$$

Sol.214.(a) Weight of the empty box

$$= [(60 \times 40 \times 20) - (58 \times 38 \times 18)] \times 50$$

$$= [48000 - 39672] \times 50 = [8328] \times 50$$

$$= 416400 \text{ gm}$$

$$\text{Weight of the empty box} = \frac{416400}{1000}$$

$$= 416.4 \text{ kg}$$

Sol.215.(d)

$$\text{Volume of the hemisphere} = \frac{2}{3}\pi r^3$$

Volume of the hemisphere

$$= \frac{2}{3} \times \frac{22}{7} \times 21 \times 21 \times 21$$

$$\text{Volume of the hemisphere} = 19404 \text{ cm}^3$$

Sol.216.(d) Difference between TSA and CSA of cylinder = $2\pi r(h + r) - 2\pi rh$

$$= 2\pi r[h + r - h] = 2\pi r^2$$

Required difference ($2\pi r^2$)

$$= 2 \times \frac{22}{7} \times 14 \times 14 = 1232 \text{ cm}^2$$

Sol.217.(c) To find the maximum possible area of a triangle with a perimeter of 36 units, we assume it's an equilateral triangle.

Perimeter = $3a = 36$, $a = 12$ units

Area of the equilateral triangle

$$= \frac{\sqrt{3}}{4} \text{ side}^2 = \frac{\sqrt{3}}{4} \times 12 \times 12$$

$$= 36\sqrt{3} \text{ square units}$$

Sol.218.(a) TSA of the sphere = $4\pi r^2$

Required cost

$$= 4 \times \pi \times 5 \times 5 \times \frac{80}{\pi} = ₹ 8000$$

Sol.219.(b)

3, 4, 5 are triplet of right angle triangle so,

Length of the perpendicular

$$= \frac{p \times b}{h} = \frac{3 \times 4}{5} = 2.4 \text{ cm}$$

Sol.220.(c) Volume of the cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times (6x)^2 (7x) = 792$$

$$x^3 = \frac{792 \times 7}{22 \times 36 \times 7} = 1 \text{ so } x = 1$$

So, the radius and height of the cylinder will be 6 and 7

Now, according to the question,

CSA of the cylinder = $2\pi rh$

$$= 2 \times \frac{22}{7} \times 6 \times 7 = 264 \text{ cm}^2$$

Sol.221.(c) Required number of boxes

$$= \frac{2160}{6 \times 10 \times 12} = 3$$

Sol.222.(c) Volume of the cone = $\frac{1}{3}\pi r^2 h$

$$\text{Required ratio} = 6^2 \times 5 : 5^2 \times 3 = 12 : 5$$

Sol.223.(c) Volume of sphere = $\frac{4}{3}\pi r^3$

Ratio - before : after

$$\text{Radius - } 1 : 2$$

$$\text{Volume - } (1)^3 : (2)^3$$

$$= 1 : 8$$

We can clearly see that the volume will be 8 times of the previous volume.

Sol.224.(b) CSA of the cylinder = $2\pi rh$

$$88 = 2 \times \frac{22}{7} \times r \times 14 \Rightarrow r = 1 \text{ cm}$$

$$\begin{aligned} \text{TSA of cylinder} &= 2\pi r(h + r) \\ &= 2 \times \frac{22}{7} \times 1(14 + 1) = \frac{660}{7} \text{ cm}^2 \end{aligned}$$

Sol.225.(c)

$$\begin{aligned} \text{Area of the rectangle} &= \text{length} \times \text{breadth} \\ 3600 &= 45 \times \text{breadth} \\ \text{Breadth} &= 80 \text{ m} \\ \text{Perimeter of the rectangular field} &= 2(\text{length} + \text{breadth}) = 2(45 + 80) \\ &= 250 \text{ m} \end{aligned}$$

Sol.226.(d)

$$\begin{aligned} \text{TSA of cuboid} &= 2(lb + bh + hl) \\ 108 &= 2(6 \times 3 + 3h + 6h) \Rightarrow 54 = 18 + 9h \\ h &= \frac{54 - 18}{9} = 4 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{Sol.227.(b) Area of sector} &= \frac{\theta}{360} \pi r^2 \\ &= \frac{45}{360} \times 3.14 \times (4)^2 = \frac{1}{8} \times 3.14 \times 16 \\ &= 6.28 \text{ cm}^2 \end{aligned}$$

Sol.228.(b)

$$\begin{aligned} \text{Volume of right circular cone} &= \frac{1}{3} \pi r^2 h \\ \text{According to the question,} \\ \text{Volume of right circular cone} &= \frac{1}{3} \times \frac{22}{7} \times 10.5 \times 10.5 \times 24 = 2772 \text{ cm}^3 \end{aligned}$$

Sol.229.(c)

$$\begin{aligned} \text{Effective rate for 2 year} &= x + y + \frac{x \times y}{100} \\ \text{Effective rate for 2 year} &= 12 + 12 + \frac{12 \times 12}{100} = 24 + 1.44 = 25.44 \\ \text{Net change in volume} &= 25.44 + 12 + \frac{25.44 \times 12}{100} \\ &= 37.44 + 3.0528 = 40.49 \% \end{aligned}$$

$$\begin{aligned} \text{Sol.230.(a) Let radius of cylinder} &= r \\ \text{Then according to question,} \\ \text{Height of cylinder} &= 12r \end{aligned}$$

$$\text{And, radius of each spherical ball} = \frac{r}{3}$$

$$\begin{aligned} \text{Volume of cylindrical iron rod} &= x \times \text{volume of each spherical balls} \end{aligned}$$

$$\pi r^2 h = x \times \frac{4}{3} \pi r^3$$

$$\pi r^2 \times 12r = x \times \frac{4}{3} \pi \left(\frac{r}{3}\right)^3$$

$$12r^3 = x \times \frac{4}{3} \times \frac{r^3}{27}$$

$$x = \frac{12 \times 3 \times 27}{4} = 243$$

$$\text{Hence, number of spherical ball (x)} = 243$$

Sol.231.(a)

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$18 = \frac{1}{2} \times 6 \times \text{height} \Rightarrow \text{Height} = 6 \text{ cm}$$

Sol.232.(b) Let the sides of the

$$\text{rectangular box be } 3x, 5x \text{ and } 8x$$

$$\text{TSA of cuboid} = 2(lb + bh + hl)$$

$$632 = 2(15x^2 + 40x^2 + 24x^2)$$

$$\Rightarrow 316 = 79x^2 \Rightarrow x^2 = 4, x = 2$$

$$\text{So the sides of the rectangular box will be } 6, 10 \text{ and } 16$$

$$\text{Volume of cuboid} = lbh$$

$$= 6 \times 10 \times 16 = 960 \text{ cm}^3$$

Sol.233.(d)

$$\begin{aligned} \text{Volume of 3 cubes} &= \text{volume of new cube} \\ \text{Let the sides of new cubes be } n \end{aligned}$$

$$\text{Volume of cube} = a^3$$

$$6^3 + 8^3 + 10^3 = n^3$$

$$216 + 512 + 1000 = n^3$$

$$n^3 = 1728, n = 12 \text{ cm}$$

$$\text{TSA of new cube} = 6a^2$$

$$= 6 \times 12 \times 12 = 864 \text{ cm}^2$$

$$\text{Sol.234.(a) CSA of the cylinder} = 2\pi rh$$

$$352 = 2 \times \frac{22}{7} \times 7 \times h \Rightarrow h = 8 \text{ cm}$$

$$\text{TSA of the cylinder} = 2\pi r(h + r)$$

$$= 2 \times \frac{22}{7} \times 7(8 + 7) = 44 \times 15 = 660 \text{ cm}^2$$

$$\text{Sol.235.(c) } 25\% = \frac{1}{4}$$

$$\begin{aligned} \text{TSA of the sphere} &= 4\pi r^2 \\ \text{Radius} &= 4 : 5 \end{aligned}$$

$$\begin{aligned} \text{TSA of the sphere} &= (4)^2 : (5)^2 \\ &= 16 : 25 \end{aligned}$$

$$\text{increase \%} = \frac{25 - 16}{16} \times 100 = 56.25\%$$

$$\text{Sol.236.(b) CSA of the cylinder} = 2\pi rh$$

$$= 2 \times \frac{22}{7} \times 3 \times 14 = 264 \text{ cm}^2$$

Sol.237.(a)

$$\text{Total surface area of hemisphere} = 3\pi r^2$$

$$= 3 \times \frac{22}{7} \times (14)^2 = 1848 \text{ cm}^2$$

$$\text{Sol.238.(c) Volume of cylinder} = \pi r^2 h$$

$$= \frac{22}{7} \times 2.1 \times 2.1 \times 6.3 = 87.318 \text{ m}^3$$

$$\text{Now, } 1 \text{ m}^3 = 1000 \text{ litre}$$

$$\begin{aligned} \text{Hence, capacity of water tank} &= 87.318 \times 1000 = 87318 \text{ litre} \end{aligned}$$

$$\text{Sol.239.(a) Total surface area of cylinder}$$

$$= 2\pi r(r + h)$$

$$= 2 \times \frac{22}{7} \times 7 \times (7 + 9) = 704 \text{ cm}^2$$

Sol.240.(d)

$$\text{Length of the arc} = \frac{\theta}{360} \times 2\pi r$$

$$5.5 = \frac{30}{360} \times 2 \times \frac{22}{7} \times r$$

$$r = \frac{42 \times 5.5}{22} = 10.5 \text{ cm}$$

Sol.241.(b) TSA of the cylinder

$$= 2\pi r(h + r) = 2 \times \frac{22}{7} \times 15(27 + 15)$$

$$= 2 \times 22 \times 15 \times 6 = 3960 \text{ cm}^2$$

$$\text{Sol.242.(d) Volume of sphere} = \frac{4}{3} \pi r^3$$

$$\text{Volume of 3 small spheres} = \text{volume of 1 big sphere}$$

$$\frac{4}{3} \pi [(3)^3 + (4)^3 + (5)^3] = \frac{4}{3} \pi (R)^3$$

$$27 + 64 + 125 = (R)^3 \Rightarrow 216 = (6)^3 = (R)^3$$

$$R = 6 \text{ cm}$$

$$\text{Sol.243.(d) CSA of the cylinder} = 2\pi rh$$

$$\text{TSA of the cylinder} = 2\pi r(h + r)$$

$$\text{Required ratio} = 2\pi r(10) : 2\pi r(10 + 30)$$

$$= 10 : 40$$

$$= 1 : 4$$

$$\text{Sol.244.(b) Given : the ratio of cuboid dimensions} = 3 : 4 : 5$$

$$\text{Let length(l)} = 3x, \text{ breadth(b)} = 4x \text{ and height(h)} = 5x$$

$$\text{According to question,}$$

$$\text{Volume of cube} = l \times b \times h$$

$$\Rightarrow 3x \times 4x \times 5x = 1620 \Rightarrow 60x^3 = 1620$$

$$\Rightarrow x^3 = 27 \Rightarrow x = 3 \text{ units}$$

$$\text{Total surface area of cuboid}$$

$$= 2(lb + bh + hl) = 2(12x^2 + 20x^2 + 15x^2)$$

$$= 2 \times 47x^2 = 2 \times 47 \times 9 = 846 \text{ m}^2$$

$$\text{Sol.245.(c) Total surface area of cylinder}$$

$$= 2\pi r^2 + 2\pi rh$$

$$\Rightarrow 628 = 2 \times \frac{22}{7} \times 5 \times 5 + 2\pi rh$$

$$\Rightarrow 2\pi rh = 628 - \frac{1100}{7}$$

$$\Rightarrow 2\pi rh = 628 - 157.14$$

$$\Rightarrow 2\pi rh = 470.86 \text{ m}^2 \text{ or } 470.8 \text{ m}^2$$

$$\text{So, curved surface area of a cylinder be } 470.8 \text{ m}^2$$

$$\text{Sol.246.(a) Given : diameter of sphere} = 28 \text{ cm then radius} = 14 \text{ cm}$$

$$\text{Surface area of sphere} = 4\pi r^2$$

$$\Rightarrow \text{Surface area of sphere}$$

$$= 4 \times \frac{22}{7} \times 14 \times 14 = 2464 \text{ cm}^2$$

$$\text{Now,}$$

$$\text{Cost of painting the ball}$$

$$= 2464 \times ₹2 = ₹4928$$

$$\text{Sol.247.(b) } 9 \text{ cm} = \frac{9}{100} \text{ m}$$

$$4 \text{ mm} = \frac{4}{1000} \text{ m}$$

$$\text{Volume of sphere} = \text{volume of cylindrical wire}$$

$$\frac{4}{3} \pi r^3 = \pi r^2 h \Rightarrow \frac{4}{3} \times \frac{9}{100} \times \frac{9}{100} \times \frac{9}{100}$$

$$= \frac{4}{1000} \times \frac{4}{1000} \times h \Rightarrow h = 60.75 \text{ m}$$

Sol.248.(a)

$$\text{Let the height of the triangle} = 3x$$

$$\text{And the base is} = 3x \times \frac{2}{3} = 2x$$

Area of the triangle = $\frac{1}{2} \times \text{base} \times \text{height}$

$$363 = \frac{1}{2} \times 3x \times 2x \Rightarrow 121 = x^2, x = 11$$

Height of the triangle ($3x$) = 33 cm

Sol.249.(b)

Total Surface area of cube = $6a^2$

Ratio - initial : new

Side - a : $3a$

TSA - a^2 : $9a^2$

$$\text{Required \%} = \frac{9a^2 - a^2}{a^2} \times 100 = 800\%$$

Sol.250.(c) Total surface area of cuboid = $2(lb + bh + hl)$

$$2600 = 2(30 \times 20 + 20 \times h + h \times 30)$$

$$\Rightarrow 2600 = 2(600 + 50h)$$

$$\Rightarrow 2600 = 1200 + 100h$$

$$100h = 1400 \Rightarrow \text{height (h)} = 14 \text{ cm}$$

Sol.251.(c) According to question,

$$\Rightarrow \frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle PQR} = \frac{AB^2}{PQ^2}$$

$$\Rightarrow \frac{490}{640} = \frac{21^2}{PQ^2} \Rightarrow \frac{49}{64} = \frac{21^2}{PQ^2}$$

Taking square root both side,

$$\Rightarrow \frac{7}{8} = \frac{21}{PQ} \Rightarrow PQ = 24 \text{ cm}$$

Sol.252.(a) Given:

radius = $\frac{35}{2}$ cm and length = 100 cm

Surface area of a roller

$$= 2 \times \frac{22}{7} \times \frac{35}{2} \times 100 = 11000 \text{ cm}^2$$

1 revolution of roller = surface area of a roller

$$200 \text{ revolution} = 11000 \times 200$$

$$= 2200000 \text{ cm}^2 = 220 \text{ m}^2$$

Hence, Area of the playground = 220 m^2

Sol.253.(c)

Perimeter of rectangle = $2(l + b)$

$$70 = 2(l + b) \Rightarrow l + b = 35 \dots (i)$$

Diagonal of the rectangle = $\sqrt{l^2 + b^2}$

$$25 = \sqrt{l^2 + b^2} \Rightarrow l^2 + b^2 = 625 \dots (ii)$$

Now,

$$l + b = 35$$

(on square both sides)

$$(l + b)^2 = 1225 \Rightarrow l^2 + b^2 + 2lb = 1225$$

$$625 + 2lb = 1225 \quad (\text{from e.q. (i)})$$

$$lb = \frac{1225 - 625}{2} = 300$$

Area of rectangle = lb

Hence, the area of the rectangle is 300 cm^2

$$\text{Sol.254.(a) Volume of sphere} = \frac{4}{3} \pi r^3$$

Let the small number of spheres = n

Volume of big sphere = $n \times \text{volume of small spheres}$

$$\frac{4}{3} \pi R^3 = n \times \frac{4}{3} \pi r^3$$

$$\Rightarrow (8)^3 = n \times (2)^3 \Rightarrow n = 64$$

Sol.255.(b) Surface area of sphere = $4\pi r^2$

According to the question,

$$4\pi(r + 3.5)^2 - 4\pi(r)^2 = 550$$

$$4\pi[(r + 3.5)^2 - r^2] = 550$$

$$[(r + 3.5)^2 - r^2] = 550 \times \frac{7}{22 \times 4}$$

$$(2r + 3.5) \times 3.5 = 43.75 \Rightarrow 2r + 3.5 = 12.5$$

$$2r = 9 \Rightarrow \text{radius (r)} = 4.5 \text{ cm}$$

Sol.256.(a) According to the question,

$$a^2 = \frac{1}{2} \pi r^2 \Rightarrow a^2 = \frac{1}{2} \pi \times (2\sqrt{2})^2 = 4\pi$$

$$a = \sqrt{4\pi} = 2\sqrt{\pi}$$

$$\text{Perimeter of square (4a)} = 8\sqrt{\pi} \text{ cm}$$

Sol.257.(d) Volume of cuboid (lbh)

$$= 36 \text{ cm} \times 22 \text{ cm} \times 12 \text{ cm}$$

Volume of each cube = $6 \times 6 \times 6$

So, required number of cubes

$$= \frac{36 \times 22 \times 12}{6 \times 6 \times 6} = 44$$

Sol.258.(b) Area of isosceles right angle

$$\text{triangle} = \left(\frac{\text{hypotenuse}}{2}\right)^2$$

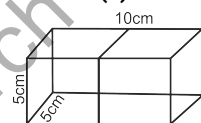
$$16 = \left(\frac{\text{hypotenuse}}{2}\right)^2 \Rightarrow \text{Hypotenuse} = 8 \text{ m}$$

Sol.259.(b) Area of sphere = $4\pi r^2$

$$= 4 \times \frac{22}{7} \times 14 \times 14 = 2464 \text{ cm}^2$$

Cost of painting a ball

$$= 2464 \times 5 = ₹12,320$$

Sol.260.(a)

Length of cuboid (l) = $(5 + 5) \text{ cm} = 10 \text{ cm}$

Breadth of cuboid (b) = 5 cm

Height of cuboid (h) = 5 cm

Surface area of cuboid = $2(lb + bh + hl)$

$$= 2 \times (10 \times 5 + 5 \times 5 + 5 \times 10)$$

$$= 2(50 + 25 + 50) = 250 \text{ cm}^2$$

Sol.261.(a) Let breadth of park be $x \text{ m}$

Length = $(x + 15) \text{ m}$

Perimeter of rectangular park

$$= 2[(x + 15) + (x)] = 4x + 30$$

According to question,

$$4x + 30 = 110 \Rightarrow x = 20$$

Length of rectangular park

$$= x + 15 = 35 \text{ m}$$

Sol.262.(d)

Lateral surface area of cube = $4a^2$

$$4a^2 = 144 \Rightarrow a = 6 \text{ cm}$$

$$\text{Volume of cube (a}^3\text{)} = 6^3 = 216 \text{ cm}^3$$

Sol.263.(d) Surface area of cuboid

$$= 2(lb + bh + hl) = 2(5 + 15 + 27) = 94 \text{ cm}^2$$

Sol.264.(c) Cylindrical tank is closed

So, total surface area cylindrical tank

$$= 2\pi r(r + h)$$

$$= 2 \times \frac{22}{7} \times 1.5 (1.5 + 3) = 42.42 \text{ m}^2$$

Hence, required metal sheet used

$$= 42.42 \text{ m}^2$$

Sol.265.(b)

$$\text{Sides ratio} = \sqrt[3]{1} : \sqrt[3]{125} = 1 : 5$$

Surface area of cube = $6a^2$

$$\text{Required ratio} = 6 \times (1)^2 : 6 \times (5)^2 = 1 : 25$$

Sol.266.(d)

$$\text{Height of the cone} = \sqrt{(10)^2 - (6)^2}$$

$$= \sqrt{100 - 36} = \sqrt{64} = 8 \text{ cm}$$

$$\text{Volume of the cone} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \pi \times 6 \times 6 \times 8 = 96\pi \text{ cm}^3$$

$$\text{Sol.267.(d) Volume of Sphere} = \frac{4}{3} \pi r^3$$

According to the question,

$$n \times \frac{4}{3} \pi (1)^3 = \frac{4}{3} \pi (9)^3$$

Required number of balls (n)

$$= \frac{729}{1} = 729$$

Sol.268.(d)

As we know, 1 litre = 1000 cm^3

Then, 10 litres = 10000 cm^3

Let the height be $15x$

Breadth of cuboidal water purifier

$$= \frac{1}{5} \times 15x = 3x$$

Length of cuboidal water purifier

$$= \frac{1}{3} (3x + 15x) = 6x$$

Now, volume of cuboidal water purifier

$$= 3x \times 6x \times 15x = 270x^3 \Rightarrow 10000 = 270x^3$$

$$\frac{1000}{27} = x^3 \Rightarrow x = \frac{10}{3}$$

So, the height of cuboidal water purifier

$$= 15 \times \frac{10}{3} = 50 \text{ cm}$$

Sol.269.(a)

Total surface area of cube = $6a^2$

$$\text{Required ratio} = 6 \times (2)^2 : 6 \times (4)^2 = 1 : 4$$

Sol.270.(c)

Curved surface area of cone = $\pi r l$

According to the question,

$$\pi \times 4 \times l = 20\pi$$

Slant height (l) = 5 cm

$$\text{Height of cone (h)} = \sqrt{l^2 - r^2}$$

$$= \sqrt{(5)^2 - (4)^2} = \sqrt{9} = 3 \text{ cm}$$

$$\text{Now, volume of cone} = \frac{1}{3} \pi r^2 h$$

$$\Rightarrow \frac{1}{3} \times \pi \times (4)^2 \times 3 = 16\pi \text{ cm}^3$$

Sol.271.(b) Let the radius of outer circle

be R and inner circle be r.

According to the question,

$$2\pi R : 2\pi r = 7 : 6 \Rightarrow R : r = 7 : 6$$

Difference (7 - 6) = 1 unit

1 unit = 45 m

Hence, radius of inner circle (6 unit)

$$= 45 \times 6 = 270 \text{ m}$$

Sol.272.(d)

Let the no. of smaller cube be n

$$\text{Surface area of larger cube} = 6a_1^2 = 54$$

$$a_1^2 = 9 \Rightarrow a_1 = 3 \text{ cm}$$

Surface area of smaller cube

$$= 6a_2^2 = \frac{8}{12} \Rightarrow a_2^2 = \frac{8}{12 \times 6} = \frac{1}{9}$$

$$\Rightarrow a_2 = \frac{1}{3}$$

Volume of larger cube = n × volume of smaller cube

$$a_1^3 = n \times a_2^3 \Rightarrow 3^3 = n \times \frac{1}{3^3}$$

$$27 = \frac{n}{27} \Rightarrow n = 729$$

Sol.273.(d) Length of cuboid = 60 cm

$$\text{Breadth of cuboid} = 60 \times \frac{1}{2} = 30 \text{ cm}$$

$$\text{Height of cuboid} = \frac{1}{3} \times 30 = 10 \text{ cm}$$

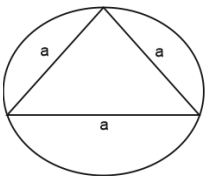
TSA of cardboard box = 2(lb + bh + hl)

$$= 2(60 \times 30 + 30 \times 10 + 10 \times 60)$$

$$= 2(1800 + 300 + 600) = 2 \times 2700$$

$$= 5400 \text{ cm}^2$$

Sol.274.(c)



$$\text{Circumradius (R)} = \frac{a}{\sqrt{3}}, \text{ Radius (r)} = r$$

$$\text{Given that, } \frac{a}{\sqrt{3}} = r \Rightarrow a = \sqrt{3}r \text{ unit}$$

Sol.275.(c)

Area of rhombus = side × height

$$70 = \text{side} \times 5 \Rightarrow \text{Side} = \frac{70}{5} = 14 \text{ cm}$$

So, the perimeter of rhombus

$$= 14 \times 4 = 56 \text{ cm}$$

Sol.276.(c)

$$\text{Median of equilateral triangle} = \frac{\sqrt{3}}{2}a$$

$$\text{Perimeter of equilateral triangle} = 3a$$

$$\text{Required ratio} = 3a : \frac{\sqrt{3}}{2} \times 3a = 2 : \sqrt{3}$$

Sol.277.(c)

$$\text{Inner radius of bowl} = \frac{4}{2} = 2 \text{ cm}$$

$$\text{Outer radius of bowl} = 2 + 0.5 = 2.5 \text{ cm}$$

Required volume of silver used in making

$$\text{the bowl} = \frac{2}{3}\pi(R^3 - r^3)$$

$$= \frac{2}{3} \times 3.14 \times [(2.5)^3 - (2)^3]$$

$$= \frac{2}{3} \times 3.14 \times 7.625 = \frac{47.885}{3}$$

$$= 15.96 \text{ cm}^3$$

Sol.278.(c)

Length = 4x, Breadth = 3x, Height = 2x

$$\text{Volume} = 12288 \Rightarrow l \times b \times h = 12288$$

$$4x \times 3x \times 2x = 12288$$

$$24x^3 = 12288 \Rightarrow x^3 = 512 \Rightarrow x = 8$$

Hence,

$$\text{Length} = 4x = 4 \times 8 = 32 \text{ m}$$

$$\text{Breadth} = 3x = 3 \times 8 = 24 \text{ m}$$

$$\text{Height} = 2x = 2 \times 8 = 16 \text{ m}$$

Total surface area of box = 2(lb + bh + hl)

$$= 2(4x \times 3x + 3x \times 2x + 2x \times 4x)$$

$$= 2(12x^2 + 6x^2 + 8x^2) = 2(26x^2)$$

$$= 52 \times 8^2 = 52 \times 64 = 3328 \text{ m}^2$$

Total cost of polishing = 2 × 3328

$$= 6656 \text{ paise or Rs. 66.56}$$

Sol.279.(c) Let no. of person be x

Perimeter of brim of hemispherical bowl

$$= 2\pi r = 88 \Rightarrow 2 \times \frac{22}{7} \times r = 88 \Rightarrow \frac{44}{7}r = 88$$

$$r = \frac{88 \times 7}{44} = 14 \text{ cm}$$

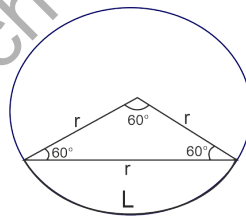
Volume of hemispherical bowl = $\frac{2}{3} \times \pi \times r^3$

Volume of hemispherical glass

$$\frac{2}{3} \times \pi \times 14^3 = x \times \frac{2}{3} \times \pi \times \left(\frac{7}{2}\right)^3$$

$$2744 = \frac{343}{8}x \Rightarrow x = \frac{2744 \times 8}{343} = 64$$

Sol.280.(b)



$$\text{Length of the arc (L)} = \frac{\theta}{360^\circ} \times 2\pi r$$

$$= \frac{60^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 21 = 22 \text{ units}$$

Sol.281.(a) Area of parallelogram

= Base(b) × Height(h)

According to the question,

$$b = 2h \Rightarrow 338 = 2h \times h$$

$$\Rightarrow 338 = 2h^2 \Rightarrow h^2 = 169 \Rightarrow h = 13 \text{ cm}$$

Sol.282.(b) TSA of cone = $\pi R(L + R)$

$$= \frac{\pi r}{4} \left(4l + \frac{r}{4}\right)$$

$$= \frac{\pi r}{4} \times 4 \left(l + \frac{r}{16}\right) = \pi r \left(l + \frac{r}{16}\right)$$

Sol.283.(b) CSA of conical tent = $\pi r l$

Let the required length of cloth be x m

$$\text{Slant height of tent (l)} = \sqrt{r^2 + h^2}$$

$$= \sqrt{(21)^2 + (28)^2} = \sqrt{1225} = 35 \text{ m}$$

Area of cloth = CSA of conical tent

$$5 \times x = \frac{22}{7} \times 21 \times 35 \Rightarrow 5x = 2310$$

$$x = \frac{2310}{5} = 462 \text{ m}$$

Sol.284.(b)

Volume of cube = 4 × volume of cuboid

$$a^3 = 4 \times 32 \times 8 \times 4 = 4096 \Rightarrow a = 16$$

$$\text{Now, } \frac{\text{TSA of cube}}{\text{TSA of cuboid}}$$

$$= \frac{6 \times 16 \times 16}{2(32 \times 8 + 8 \times 4 + 32 \times 4)}$$

$$= \frac{6 \times 16 \times 16}{832} = \frac{24}{13} = 24 : 13$$

Sol.285.(a)

$$\text{Radius of copper sphere} = \frac{12}{2} = 6 \text{ cm}$$

$$\text{Radius of wire} = \frac{4}{2} = 2 \text{ mm} = \frac{2}{10}$$

$$= 0.2 \text{ cm}$$

volume of sphere = volume of wire

$$\frac{4}{3} \pi \times 6^3 = \pi \times 0.2^2 \times h$$

$$288 = 0.04h \Rightarrow h = \frac{288}{0.04} = 7200 \text{ cm}$$

Sol.286.(d) Length of the largest possible rod which can be placed in cubical room = $\sqrt{3}a \Rightarrow 42\sqrt{3} = \sqrt{3}a \Rightarrow a = 42 \text{ m}$

Required TSA of sphere

$$= 4 \times \frac{22}{7} \times \left(\frac{42}{2}\right)^2 = 5544 \text{ m}^2$$

Sol.287.(a) circumference of circle

$$= 2 \times \frac{22}{7} \times r \Rightarrow 88 = \frac{44}{7}r$$

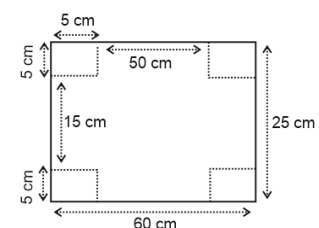
$$r = \frac{88 \times 7}{44} = 14 \text{ cm}$$

$$\text{Area of a minor sector} = \frac{1}{2} \times l \times r$$

(l = minor arc length)

$$= \frac{1}{2} \times 22 \times 14 = 154 \text{ cm}^2$$

Sol.288.(c)



Length of box = 50 cm

Breadth of box = 15 cm

Height of box = 5 cm

$$\text{Volume of box} = l \times b \times h = 50 \times 15 \times 5 = 3750 \text{ cm}^3$$

Sol.289.(b)

Volume of cylindrical shell = $\pi(R^2 - r^2)h$

$$= \frac{22}{7} \times (7^2 - 4^2) \times 42$$

$$= 22 \times (49 - 16) \times 6$$

$$= 22 \times 33 \times 6 = 4356 \text{ cm}^3$$

Sol.290.(d)

$$\text{Volume of laddoo of radius 5 cm} = \frac{4}{3}\pi R^3$$

$$\text{Volume of laddoo of radius 2.5 cm}$$

$$= \frac{4}{3}\pi r^3$$

$$\text{No. of laddoos created} = \frac{\frac{4}{3}\pi R^3}{\frac{4}{3}\pi r^3}$$

$$= \frac{5 \times 5 \times 5}{2.5 \times 2.5 \times 2.5} = 8 \text{ laddoos}$$

Sol.291.(d) T.S.A. of spherical vessel

$$= 4\pi r^2 = 4 \times \frac{22}{7} \times 7 \times 7 = 616 \text{ cm}^2$$

Cost per square centimeter

$$= \frac{\text{total cost}}{\text{T.S.A.}} = \frac{21,560}{616} = 35 \text{ Rs.}$$

Sol.292.(c) $\frac{\text{Volume of cone}}{\text{Volume of cylinder}}$

$$= \frac{\frac{1}{3}\pi r^2 h}{\pi R^2 H} = \frac{\frac{1}{3} \times 5 \times 5 \times 6}{2.5 \times 2.5 \times 3} = \frac{8}{3} = 8 : 3$$

Sol.293.(a) C.S.A. of cylinder = 2200

$$2\pi r h = 2200$$

$$2 \times \frac{22}{7} \times 7 \times h = 2200$$

$$\Rightarrow h = \frac{2200}{44} = 50 \text{ cm}$$

$$\text{T.S.A. of cylinder} = 2\pi r(h + r)$$

$$= 2 \times \frac{22}{7} \times 7(50 + 7) = 44 \times 57$$

$$= 2508 \text{ cm}^2$$

Sol.294.(d) Let R be external radius and r be internal radius of a hollow sphere and radius of a cone be r_1

According to question,

Volume of cone = volume of hollow cylinder

$$\frac{1}{3}\pi r_1^2 h = \frac{4}{3}\pi(R^3 - r^3)$$

$$5^2 \times h = 4(5^3 - 3^3)$$

$$25 \times h = 4 \times 98$$

$$h = \frac{4 \times 98}{25} = 15.68 \text{ cm}$$

Sol.295.(b)

Surface area of spherical box = 528 cm²

$$4\pi r^2 = 528 \text{ cm}^2$$

$$4 \times \frac{22}{7} \times r^2 = 528 \text{ cm}^2$$

$$r^2 = \frac{528 \times 7}{4 \times 22} = 42 \Rightarrow r = \sqrt{42}$$

$$\text{Hence, radius} = \sqrt{42} \text{ cm}$$

Sol.296.(d) Area of trapezium =

$$\frac{1}{2} \times \text{sum of parallel sides} \times \text{height}$$

$$\Rightarrow 1188 = \frac{1}{2} \times \text{sum of parallel sides} \times 18$$

sum of parallel sides = 132 cm

Sol.297.(b) Circumference of circular base of cone = 30π ft.

$$2\pi r = 30\pi \text{ ft.} \Rightarrow r = 15 \text{ ft.}$$

Height (h) of cone = 20 ft.

$$\text{Slant height of cone } (l) = \sqrt{r^2 + h^2}$$

$$\text{Slant height of cone } (l) = \sqrt{225 + 400} = 25 \text{ ft.}$$

$$\text{Area of canvas needed} = \pi \times 15 \times 25 = 375\pi \text{ sq. ft.}$$

Sol.298.(a) Volume of cone = 150 π

$$\frac{1}{3} \times \pi \times r^2 \times 18 = 150 \pi$$

$$r^2 = 25 \Rightarrow r = 5$$

$$\text{Slant height } (l) = \sqrt{h^2 + r^2} = \sqrt{18^2 + 5^2} = \sqrt{324 + 25} = \sqrt{349} = 18.68 \text{ cm}$$

Sol.299.(d) If both playgrounds are similar then their dimension must be in proportion.

$$\text{First playground} = \frac{\text{length}}{\text{breadth}} = \frac{224}{160} = \frac{7}{5}$$

By checking option one by one we find that option (d) does not have the same ratio of length and breadth as the first playground.

Sol.300.(a) Volume of storage tank

$$= \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times 9$$

$$= 21 \times 22 = 462 \text{ cubic ft.}$$

Sol.301.(b)

Ratio → length : breadth : height

$$\frac{2}{2} : \frac{1}{2} : \frac{1}{1}$$

$$\text{Final} \rightarrow 4 : 2 : 1$$

$$\text{Volume } (8 \text{ m}^3) = 1728 \text{ m}^3$$

$$1 \text{ m}^3 = \frac{1728}{8} = 216 \Rightarrow 1 \text{ unit} = 6$$

$$\text{Length } (4 \text{ m}) = 6 \times 4 = 24 \text{ m}$$

Sol.302.(b) Number of candles

$$= \frac{\text{volume of cuboidal solid}}{\text{volume of cylindrical candle}}$$

$$= \frac{70 \times 44 \times 20}{\frac{22}{7} \times 7 \times 7 \times 20} = 20$$

Sol.303.(a) Lateral surface area = πrl

$$13.2 = \frac{22}{7} \times r \times 4.2$$

$$13.2 = 22 \times r \times 0.6 \Rightarrow r = \frac{13.2}{13.2} = 1 \text{ unit}$$

Sol.304.(d) Area of triangle = inradius × semiperimeter

$$= 8 \times \frac{125}{2} = 500 \text{ cm}^2$$

Sol.305.(c) Volume of metal ball

= volume of (27 identical dice + a ball)

$$\frac{4}{3}\pi R^3 = 27 \times \text{side}^3 + \frac{4}{3}\pi r^3$$

$$\frac{4}{3}\pi R^3 - \frac{4}{3}\pi r^3 = 27 \times \text{side}^3$$

$$\frac{4}{3} \times \frac{22}{7} (27 - 1) = 27 \times \text{side}^3$$

$$\frac{4}{3} \times \frac{22}{7} \times 26 = 27 \times \text{side}^3$$

$$\text{side}^3 = \frac{2288}{21 \times 27} \Rightarrow \text{side}^3 = 4.035 \text{ (approx)}$$

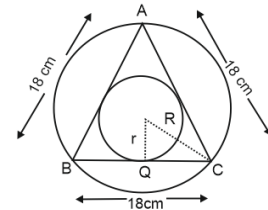
$$\text{Side} = 1.6 \text{ units (approx)}$$

Sol.306.(c) $l : h = 4 : 3$

$$\text{C.S.A. of cone} = 4\sqrt{7}\pi$$

$$\pi r l = 4\sqrt{7}\pi \Rightarrow r \times 4 = 4\sqrt{7} \Rightarrow r = \sqrt{7} \text{ unit}$$

Sol.307.(d)



$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} \times a^2$$

$$= \frac{\sqrt{3}}{4} \times 18 \times 18 = 81\sqrt{3} \text{ cm}^2$$

Relation between area of circumcircle, equilateral triangle and incircle

Circumcircle : equilateral triangle :

$$\text{incircle} = 4\pi : 3\sqrt{3} : \pi$$

Area of circumcircle - area of incircle

$$= 4\pi - \pi = 3\pi \Rightarrow 3\sqrt{3} \text{ unit} = 81\sqrt{3} \text{ cm}^2$$

$$1 \text{ unit} = 27 \text{ cm}^2$$

$$3\pi \text{ unit} = \frac{27}{1} \times 3 \times \frac{22}{7} = 254\frac{4}{7} \text{ cm}^2$$

Sol.308.(d) Area of circle = 256π cm²

$$\pi r^2 = 256\pi \Rightarrow r = 16 \text{ cm}$$

$$\text{Diameter} = 2 \times r = 2 \times 16 = 32 \text{ cm}$$

Diameter is the longest chord,

Hence length of longest chord = 32 cm

Sol.309.(c) Perimeter of one face of cube (4a) = 40 ⇒ a = 10

$$\text{Volume of cube} = a^3 = 10^3 = 1000 \text{ cm}^3$$

$$\text{Sol.310.(c)} \text{ Area of triangle} = \frac{1}{2} \times 15^2 \times$$

$$\sin 30^\circ = \frac{225}{2} \times \frac{1}{2} = \frac{225}{4} = 56.25 \text{ cm}^2$$

$$\text{Sol.311.(a)} \text{ Radius of cone} = \frac{14}{2} = 7 \text{ cm}$$

LSA of cone (πrl) = 550

$$\Rightarrow \frac{22}{7} \times 7 \times l = 550 \Rightarrow 22l = 550$$

$$\Rightarrow l = \frac{550}{22} = 25$$

Height of the cone

$$= \sqrt{25^2 - 7^2} = \sqrt{576} = 24 \text{ cm}$$

Sol.312.(c)

$$\text{Volume of hemisphere} = \frac{2}{3} \pi r^3$$

$$\Rightarrow V = \frac{2}{3} \times \pi \times (9)^3 \Rightarrow V = 486 \pi \text{ cm}^3$$

Sol.313.(c) Given,

Height of cylinder = 9 cm

the base area = $25\pi\text{cm}^2$

$$\Rightarrow \pi r^2 = 25\pi\text{cm}^2 \Rightarrow r = 5\text{ cm}$$

Now,

$$\text{Total Surface Area} = 2\pi rh + 2\pi r^2$$

TSA of cylinder

$$= 2 \times \frac{22}{7} \times 5 \times 9 + 2 \times \frac{22}{7} \times 25$$

$$= 2 \times \frac{22}{7} \times 5 (9 + 5)$$

$$= 2 \times \frac{22}{7} \times 5 \times 14 = 440\text{ cm}^2$$

Sol.314.(b) Let height of the first cone = 3h

And height of the second cone = h

Radius of first cone = r

And radius of the second cone = 2r

Total volume(V) = volume of first cone + volume of second cone

$$\Rightarrow V = \frac{1}{3} \pi r^2 3h + \frac{1}{3} \pi 4r^2 h$$

$$\Rightarrow 100 = \frac{1}{3} \pi (r^2 \cdot 3h + 4r^2 h)$$

$$\Rightarrow 100 = \frac{1}{3} \pi (7r^2 h) \Rightarrow r^2 h = \frac{300}{7\pi}$$

Now,

Difference between the volume of cones

$$= \frac{1}{3} \pi (4r^2 \cdot h - r^2 \cdot 3h)$$

$$\Rightarrow V = \frac{1}{3} \pi (r^2 h) \Rightarrow V = \frac{1}{3} \pi \cdot \frac{300}{7\pi}$$

$$= \frac{100}{7} = 14.28 \approx 14.3\text{ unit}^3$$

Sol.315.(a)

Surface area of cube = 5400 square units

$$\Rightarrow 6a^2 = 5400 \Rightarrow a^2 = 900 \Rightarrow a = 30\text{ unit}$$

Volume of cube = $a^3 = 27000\text{ Unit}^3$

Now, New side = 30 + 5 = 35 unit

New Volume of cube

$$= 35 \times 35 \times 35 = 42875\text{ Unit}^3$$

$$\text{Increment in volume} = \frac{42875 - 27000}{27000} \times$$

$$100 = \frac{15875}{27000} \times 100 = 58.79 \approx 59\%$$

Sol.316.(d) Let radius = r cm

$$\Rightarrow \text{Surface Area of sphere} = 4\pi r^2\text{ cm}^2$$

New radius = r + 2 cm

New surface area of sphere

$$= 4\pi(r + 2)^2\text{ cm}^2$$

According to question,

$$4\pi(r + 2)^2 - 4\pi r^2 = 704$$

$$\Rightarrow 4 \times \frac{22}{7} (r^2 + 4 + 4r - r^2) = 704$$

$$\Rightarrow 4 + 4r = 56 \Rightarrow 4r = 52 \Rightarrow r = 13\text{ cm}$$

Sol.317.(a)

$$10\% = \frac{1}{10}, 20\% = \frac{1}{5}, 40\% = \frac{2}{5}$$

Original New

Length 10 : 9

Breadth 5 : 4

Height 5 : 7

Volume 250 : 252

$$\begin{aligned} \text{Change in volume} &= \frac{252 - 250}{250} \times 100 \\ &= \frac{200}{250} = 0.8\% \text{ increase} \end{aligned}$$

Sol.318.(b) Radius of cone = $\frac{14}{2} = 7\text{ m}$

Volume of cone

$$= \frac{1}{3} \times \frac{22}{7} \times 7^2 \times 12 = 616\text{ m}^3$$

Required time taken to fill the cone

$$= \frac{616}{2} \times 20 = 6160\text{ sec}$$

Sol.319.(b) Volume of a cuboid = $2 \times$

Volume of a cube

$$12 \times 16 \times 18 = 2 \times \text{Volume of a cube}$$

$$3456 = 2 \times \text{Volume of a cube}$$

$$\text{Volume of a cube} = 1728$$

$$a^3 = 1728 \Rightarrow a = 12$$

$$\text{TSA of cube} = 6a^2 = 6 \times 12^2 = 864\text{ cm}^2$$

Sol.320.(c) ATQ,

$$h = 4r$$

$$\frac{\text{CSA of cylinder}}{\text{Base area of cylinder}}$$

$$= \frac{2\pi rh}{\pi r^2} = \frac{2\pi r \times 4r}{\pi r^2} = \frac{8}{1}$$

So, the curved surface area is 8 times its base area.

Sol.321.(a) According to the question,

$$\text{Lateral surface area of cylinder} = 2\pi rh$$

$$3862.2 = 2 \times 3.14 \times r \times 15$$

$$3862.2 = 94.2r \Rightarrow r = 41\text{ cm}$$

$$\text{Volume of cylinder} = \pi r^2 h$$

$$= 3.14 \times 41^2 \times 15 = 79,175.1\text{ cm}^3$$

Sol.322.(a) L = 2B and H = $\frac{1}{2}B$

Volume of cubical shaped cold storage

$$= L \times B \times H$$

$$\Rightarrow 216 = 2B \times B \times \frac{1}{2}B = B^3 \Rightarrow B = 6\text{ m}$$

$$\text{Then, } L = 2 \times 6 = 12\text{ m, } H = \frac{1}{2} \times 6 = 3\text{ m}$$

So, the dimensions are 12 m $\times$ 6 m $\times$ 3 m

Sol.323.(d) Volume of cylinder = $\pi r^2 h$

CSA of cylinder = $2\pi rh$

Let x be multiplied to obtain volume of cylinder

According to the question,

$$2\pi rh \times x = \pi r^2 h \Rightarrow x = \frac{\pi r^2 h}{2\pi rh} = \frac{r}{2}$$

Sol.324.(a) Required no of cubes

$$= \frac{\text{volume of cuboid}}{\text{volume of cube}}$$

$$= \frac{500 \times 200 \times 500}{50 \times 50 \times 50} = 400$$

Sol.325.(c) Perpendicular = $\sqrt{17^2 - 15^2}$

$$= \sqrt{289 - 225} = \sqrt{64} = 8\text{ cm}$$

Area of right angle triangle

$$= \frac{1}{2} \times 8 \times 15 = 60\text{ cm}^2$$

Sol.326.(c)

Let length of rectangle = l meter

Breadth of rectangle = b meter

According to question,

$$\frac{l+6}{b+6} = \frac{4}{3} \Rightarrow 3l - 4b = 6 \text{ ----- (i)}$$

And

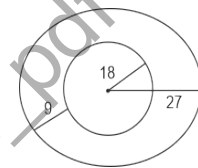
$$\frac{l-6}{b-6} = \frac{2}{1} \Rightarrow l - 2b = -6 \text{ ----- (ii)}$$

On solving equations (i) and (ii) we get,

$$l = 18\text{ meter, } b = 12\text{ meter}$$

Sol.327.(c) Increment in area of circle

$$= 7 + 7 + \frac{7 \times 7}{100} = 14.49\%$$

Sol.328.(b)

Area of the closed figure

$$= \pi(27)^2 - \pi(18)^2$$

$$\Rightarrow \text{Area} = \pi(729 - 324) = 405\pi\text{ cm}^2$$

Sol.329.(c) Required area of hemisphere

$$= 2\pi r^2 + 2\pi r^2 \Rightarrow A = 4\pi r^2$$

$$\Rightarrow A = 4 \times \frac{22}{7} \times 21 \times 21 = 5544$$

Hence, cost of painting

$$= 5544 \times 0.05 = 277.2\text{ Rs.}$$

Sol.330.(b)

$$\text{Area of a regular pentagon} = 5 \times \frac{\sqrt{3}}{4} a^2$$

$$\Rightarrow 3920\sqrt{3} = 5 \times \frac{\sqrt{3}}{4} a^2$$

$$\Rightarrow a^2 = 3136 \Rightarrow a = 56\text{ cm}$$

Sol.331.(a) Given,

Base (b) = 3H, Height (h) = H

Area of parallelogram = 1020 cm^2

$$\Rightarrow b \times h = 1020 \Rightarrow 3H \times H = 1020$$

$$\Rightarrow H^2 = 340 \Rightarrow H = 2\sqrt{85}\text{ cm}$$

Sol.332.(a) According to the question,

Surface area of sphere = Surface area of cube

$$\Rightarrow 4\pi r^2 = 6a^2 \Rightarrow 2\pi r^2 = 3a^2$$

Take square root both side,

$$\Rightarrow \sqrt{2\pi} r = \sqrt{3} a \text{ (i)}$$

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

$$\text{Volume of cube} = a^3$$

Required ratio

$$= \frac{\frac{4}{3} \pi r^3}{a^3} = \frac{\frac{4}{3} \pi r^3}{\left(\frac{\sqrt{2\pi} r}{\sqrt{3}}\right)^3} \text{ [from eq. (i)]}$$

$$= \frac{\sqrt{6}}{\sqrt{\pi}}$$

Sol.333.(d)

Surface area of a sphere = 2464 cm²

$$\Rightarrow 4\pi r^2 = 2464 \Rightarrow 4 \times \frac{22}{7} \times r^2 = 2464$$

$$\Rightarrow r^2 = 196 \Rightarrow r = 14 \text{ cm}$$

Hence,

$$\text{Volume} = \frac{4}{3}\pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times 14 \times 14 \times 14$$

$$= 11498.67 \text{ cm}^3$$

Sol.334.(b)

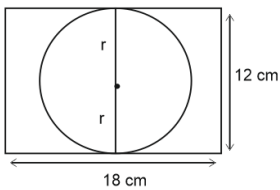
As we know, $(\frac{D_1}{2})^2 + (\frac{D_2}{2})^2 = \text{side}^2$

$$12^2 + (\frac{D_2}{2})^2 = 13^2 \Rightarrow (\frac{D_2}{2})^2 =$$

$$= \sqrt{13^2 - 12^2} = 5 \Rightarrow D_2 = 5 \times 2 = 10 \text{ cm}$$

Now, Area of rhombus = $\frac{1}{2} \times D_1 \times D_2$

$$= \frac{1}{2} \times 24 \times 10 = 120 \text{ cm}^2$$

Sol.335.(d)

Maximum radius (r) = $\frac{12}{2} = 6 \text{ cm}$

$$\text{Area of circle} = \pi r^2 = \pi \times 6^2 = 36\pi \text{ cm}^2$$

Sol.336.(c) LSA of cube = $4a^2$

TSA of cube = $6a^2$

$$\text{ATQ, } 6a^2 - 4a^2 = 2a^2 = 2 \times 11^2 = 242 \text{ cm}^2$$

Sol.337.(d) Volume of cone = $\frac{1}{3}\pi r^2 h$

Volume of cone $\propto r^2 h$

(As $\frac{1}{3}\pi$ is constant)

According to the question,

$$\text{Volume of cone} = \text{Radius}^2 \times \text{Height}$$

$$\text{Original} \rightarrow 100 = 5^2 \times 4$$

$$\text{New} \rightarrow 108 = 6^2 \times 3$$

% Change in volume

$$= \frac{108 - 100}{100} \times 100 = 8\%$$

Sol.338.(b)

Let the width of rectangular garden be x

Then, length = $(x + 8)$ cm

Perimeter of garden = $42 \times 2 = 84$ cm

$$2(x + x + 8) = 84$$

$$2(2x + 8) = 84 \Rightarrow 2x + 8 = 42$$

$$x + 4 = 21 \Rightarrow x = 17$$

So, the area of rectangular garden

$$= 17 \times 25 = 425 \text{ cm}^2$$

Sol.339.(c) Volume of cube = 64 cm^3

$$a^3 = 64 \Rightarrow a = 4$$

$$\text{TSA of cube} = 6a^2 = 6 \times 4^2 = 96 \text{ cm}^2$$

Sol.340.(b)

Before : After

Ratio of radius = 10 : 13

$$\therefore \text{Ratio of volume} = (10)^3 : (13)^3$$

$$= 1000 : 2197$$

% increase in volume

$$= \frac{2197 - 1000}{1000} \times 100 = 119.7\%$$

Sol.341.(b)

Diameter = 21 cm, Radius = $\frac{21}{2}$ cm

$$\text{Volume of hemispherical bowl} = \frac{2}{3}\pi r^3$$

$$= \frac{2}{3} \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times \frac{21}{2}$$

$$= 2425.5 \text{ cm}^3$$

Sol.342.(b) Before : After

$$\text{Radius} \rightarrow 4^2 : 5^2$$

$$\text{Height} \rightarrow 4 : 5$$

$$\text{Volume} \rightarrow 64 : 125$$

$$\% \text{ increase in volume} = \frac{125 - 64}{64} \times 100 = 95.31\%$$

Sol.343.(d) Volume of cube = a^3

$$a^3 = 512 \text{ cubic unit}$$

$\therefore$ side of cube (a) = 8 unit

Now, surface area of cube = $6 \times a^2$

$$= 6 \times 8 \times 8 = 384 \text{ square unit}$$

Sol.344.(d)

Let the total number of packed box = x

$\therefore$ volume of big box = volume of small box $\times x$

$$160 \times 90 \times 60 = 6 \times 5 \times 4 \times x$$

$$\therefore x = 7200$$

Hence no. of packed box = 7200

Sol.345.(c)

Let the length = $6x$, width = $3x$, height = $2x$

ATQ,

$$2(LB + BH + HL) = 1800$$

$$2(6x \times 3x + 3x \times 2x + 2x \times 6x) = 1800$$

$$2 \times 36x^2 = 1800 \Rightarrow x^2 = 25 \Rightarrow x = 5$$

Now, volume of cuboid = $L \times B \times H = 6x \times$

$$= 3x \times 2x = 36x^3 = 36 \times 5^3 = 4500$$

Sol.346.(a)

Let the no of small cubes be n

ATQ,

Volume of solid cube = Volume of new cube formed

$$48 \times 48 \times 48 = n \times 64$$

$$n = \frac{48 \times 48 \times 48}{64} = 1728$$

Sol.347.(b) T.S.A. of cylinder = $2\pi r(h + r)$

$$\frac{18,425}{25} = 2 \times \frac{22}{7} \times r(30 + r)$$

$$\frac{737 \times 7}{11} = 4 \times r(30 + r)$$

$$469 = 4r^2 + 120r$$

$$4r^2 + 120r - 469 = 0$$

$$4r^2 + 134r - 14r - 469 = 0$$

$$2r(2r + 67) - 7(2r + 67) = 0$$

$$(2r + 67)(2r - 7) = 0$$

$$r = -\frac{67}{2} \text{ or } r = \frac{7}{2} \text{ m}$$

Radius cannot be negative

therefore, radius (r) = $\frac{7}{2}$

Volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 30$$

$$\text{Volume of cylinder} = 1155 \text{ m}^3$$

Sol.348.(c) According to the question, T.S.A. of hemisphere - C.S.A. of hemisphere

$$3\pi r^2 - 2\pi r^2 = \pi r^2 = 4\pi \text{ cm}^2$$

Sol.349.(a) According to the question,

C.S.A of cone = $\pi r l$

$$\text{Required cost} = \frac{22}{7} \times 21 \times 25 \times 1.5$$

$$= 2475 \text{ Rs}$$

Sol.350.(b)

Let the number of hemisphere = n

Volume of sphere = $n \times$ volume of hemisphere

$$\frac{4}{3}\pi \times 12 \times 12 \times 12 = n \times \frac{2}{3}\pi \times 4 \times 4 \times 4$$

$$n = 54$$

Sol.351.(d) Given, Perimeter of rhombus

= 52 cm and diagonal (D_1) = 10 cm

Then, side of the rhombus = 13 cm

$$\text{Now, } D_1^2 + D_2^2 = 4\text{side}^2$$

$$D_2^2 = 676 - 100 \Rightarrow D_2 = 24 \text{ cm}$$

Now, area of rhombus

$$= \frac{1}{2} \times D_1 \times D_2 = \frac{1}{2} \times 10 \times 24 = 120 \text{ cm}^2$$

Sol.352.(c) side of square base = b

slant height = s

height of pyramid = h

$$\text{base area of pyramid } (b)^2 = \frac{3}{8} \times 1536$$

$$= 576 \text{ m}^2$$

side of square base (b) = 24 m

T.S.A. of square based pyramid

$$= (b)^2 + 2 \times b \times s$$

$$1536 = 576 + 2 \times 24 \times s$$

$$\text{Slant height } (s) = \frac{1536 - 576}{48}$$

$$= \frac{960}{48} = 20 \text{ m}$$

$$\text{Height of pyramid } (h) = \sqrt{s^2 - (\frac{b}{2})^2}$$

$$= \sqrt{400 - 144} = \sqrt{256} = 16 \text{ m}$$

$$\text{Volume of pyramid} = (b)^2 \times \frac{h}{3}$$

$$= 24 \times 24 \times \frac{16}{3} = 3072 \text{ m}^3$$

Sol.353.(a)

$$\text{Volume of sphere} = \frac{4}{3} \times \pi \times (\text{radius})^3$$

when diameter is reduced to half then new radius will also reduced to half of the former sphere

Therefore,

Volume of new sphere

$$= \frac{4}{3} \times \pi \times \left(\frac{\text{radius}}{2}\right)^3$$

$$= \frac{1}{8} \times \frac{4}{3} \times \pi \times (\text{radius})^3$$

= reduced by $\frac{1}{8}$ of the former volume

Sol.354.(c) According to the question,

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

Volume of 1st cone : volume of 2nd cone

$$\frac{1}{3} \pi \times 5^2 \times 7 : \frac{1}{3} \pi \times \left(\frac{21}{2}\right)^2 \times 5$$

$$25 \times 7 : \frac{441}{4} \times 5 \Rightarrow 20 : 63$$

Sol.355.(d)

$$\text{Diagonal of rectangle} = \sqrt{4^2 + 3^2} = 5$$

According to the question,

Diagonal of rectangle = diameter of hemisphere = 5

So, radius = 2.5 cm

Now, total surface area of hemisphere

$$= 3\pi \times 2.5 \times 2.5 = \frac{75\pi}{4}$$

Sol.356.(d) Let the height and radius of the right circular cone = 12x and 5x

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$314 = \frac{1}{3} \times 3.14 \times 25x^2 \times 12x$$

$$x^3 = \frac{314 \times 100 \times 3}{314 \times 25 \times 12} = 1 \Rightarrow x = 1$$

$$\text{Hence, Slant height} = \sqrt{(12x)^2 + (5x)^2}$$

$$= \sqrt{144 + 25} = 13 \text{ cm}$$

Sol.357.(d)

$$\text{Given, diagonal of square} = 24\sqrt{2}$$

Then, side of the square = 24 cm

Lateral surface area of square pyramid

= 2 × side × slant height

$$= 2 \times 24 \times l = 624 \Rightarrow l = 13 \text{ cm}$$

$$\text{Now, height of pyramid} = \sqrt{13^2 - \left(\frac{24}{2}\right)^2}$$

$$= 5 \text{ cm}$$

So, volume of pyramid

$$= \frac{1}{3} \times \text{side}^2 \times \text{height}$$

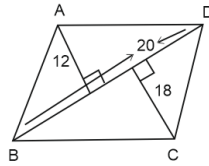
$$= \frac{1}{3} \times 24^2 \times 5 = 960 \text{ cm}^3$$

Sol.358.(d) According to the question,
Volume of cylinder = Volume of sphere

$$\pi r^2 h = \frac{4}{3} \pi r^3 \Rightarrow 12 \times 12 \times h$$

$$= \frac{4}{3} \times 9 \times 9 \times 9 \Rightarrow h = \frac{4 \times 9 \times 9 \times 9}{3 \times 12 \times 12}$$

$$h = 6.75 \text{ cm}$$

Sol.359.(d)

According to the question,
Area of quadrilateral ABCD

$$= \frac{1}{2} \times 20 \times 12 + \frac{1}{2} \times 20 \times 18 = 300 \text{ m}^2$$

Sol.360.(c)

Ratio - ball P : ball Q

Diameter - 8 : 2

Radius - 4 : 1

Required ratio = $4^3 : 1^3 = 64 : 1$

Sol.361.(c) Given, Slant height (l) = 29 cm

Height (h) = 20 cm

$$\text{Radius (r)} = \sqrt{(29)^2 - (20)^2}$$

$$= \sqrt{841 - 400} = \sqrt{441} = 21 \text{ cm}$$

According to question,

$$\frac{\text{T.S.A. of cone}}{\text{Volume of cone}} = \frac{\pi r(l+r)}{\frac{1}{3} \pi r^2 h} = \frac{(l+r)}{\frac{1}{3} r h}$$

$$\frac{\text{T.S.A. of cone}}{\text{Volume of cone}} = \frac{(29+21)}{\frac{1}{3} \times 21 \times 20}$$

$$= \frac{50}{140} = 5 : 14$$

Sol.362.(b) Given, Radius (R) = 3 cm

Radius of each hemispherical ball after recasting (r) = 1.5 cm

$$\text{Number of recasted balls} = \frac{\frac{4}{3} \pi R^3}{\frac{4}{3} \pi r^3}$$

$$= \frac{2 \times (3)^3}{(1.5)^3} = 2 \times \left(\frac{3}{1.5}\right)^3 = 2 \times 8 = 16$$

Sol.363.(c) According to the question,

Volume of hollow cylinder = $\pi(R^2 - r^2)h$

$$= \frac{22}{7} \times (10+7)(10-7) \times 49 = 7854 \text{ cm}^3$$

Sol.364.(b)

Volume of sphere = volume of wire

$$\frac{4}{3} \pi r^3 = \pi r^2 h$$

$$\frac{4}{3} \times 3 \times 3 \times 3 = 0.2 \times 0.2 \times h$$

$$h = 900 \text{ cm or } 9 \text{ m}$$

Sol.365.(c) Number of small cubes n

$$= \frac{8 \times 6 \times 6}{0.5 \times 0.5 \times 0.5} = 2304$$

Sol.366.(c) Let radius and height of the cone = 3x and 4x

$$\text{Slant height (l)} = \sqrt{(3x)^2 + (4x)^2}$$

$$= \sqrt{25x^2} = 5x$$

According to the question,

$$\text{Volume of the cone} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times 3.14 \times 9x^2 \times 4x = 301.44$$

$$= 12x^3 \times 3.14 = 301.44$$

$$= x^3 = \frac{301.44}{12 \times 3.14} = 8 \Rightarrow x = 2$$

$$\text{Slant height (5x)} = 5 \times 2 = 10 \text{ cm}$$

Sol.367.(c) According to the question,

$$1 \text{ liter} = 1000 \text{ cm}^3$$

$$50 \text{ liter} = 50,000 \text{ cm}^3$$

$$50 \text{ liter} = 50,000 \times \frac{1}{1000000} = 0.05 \text{ m}^3$$

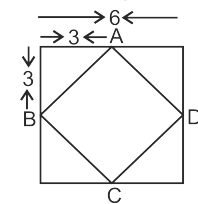
$$\text{Required time} = \frac{120}{0.05}$$

$$= 2400 \text{ min. or } 40 \text{ hours}$$

Sol.368.(c) according to the question,

Area of the square = side^2

$$36 = \text{side}^2, \text{side} = 6$$



$$\text{Length of AB} = \sqrt{3^2 + 3^2} = \sqrt{18}$$

$$\text{Area of square ABCD} = (\sqrt{18})^2 = 18 \text{ square units}$$

Sol.369.(a) Both the median and the altitude are the same in equilateral triangle.

(height) Median of equilateral triangle

$$= \frac{\sqrt{3}}{2} a = \frac{\sqrt{3}}{2} a = 6\sqrt{3} \Rightarrow a = 12 \text{ cm}$$

$$\text{So, area of equilateral triangle} = \frac{\sqrt{3}}{4} a^2$$

$$= \frac{\sqrt{3}}{4} \times 12 \times 12 = 36\sqrt{3} \text{ cm}^2$$

Sol.370.(b) Volume of cuboid When areas of three adjacent faces of a cuboid are given = $\sqrt{pqr} \text{ unit}^3$ **Sol.371.(b)**

Total surface area of cylinder = $2\pi r(h+r)$

$$2\pi r(h+r) = 64\pi$$

$$2\pi r \times 2r = 64\pi \text{ (given } r = h)$$

$$4\pi r^2 = 64\pi \Rightarrow r = 4, (h = 4)$$

$$\text{Now, } R = 2, H = 8$$

Total surface area of cylinder

$$= 2\pi \times 2(8+2) = 40\pi$$

Sol.372.(c) According to the question,
Volume of hollow metal sphere = volume of solid cone

$$\frac{4}{3} \times \pi \times (6^3 - 5^3) = \frac{1}{3} \times \pi \times (5.2)^2 \times h$$

$$h = \frac{4 \times 91}{5.2 \times 5.2} \approx 13.5$$

Sol.373.(a) Area of rhombus

$$= \frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 7.5 \times 5.6 = 21 \text{ cm}^2$$

Sol.374.(a) According to the question,
Volume of hollow sphere = volume of cone

$$\frac{4}{3} \pi (R^3 - r^3) = \frac{1}{3} \pi r^2 h$$

$$4(3^3 - 2^3) = 4^2 h \Rightarrow 4 \times 19 = 16h$$

$$(\text{height}) h = 4 \frac{3}{4} \text{ cm}$$

Sol.375.(d)

$$\text{Total area of ground} (\pi r^2) = 15 \times 5 = 75 \text{ m}^2$$

$$\text{Volume of the cone} = \frac{1}{3} \pi r^2 h$$

According to the question,

$$\frac{1}{3} \pi r^2 h = 30 \times 15 \Rightarrow \frac{1}{3} \times 75 \times h = 30 \times 15$$

$$\text{Height of cone (h)} = 18 \text{ m}$$

Sol.376.(a)

Volume of cuboid = Volume of cone

$$11 \times 8 \times 6 = \frac{1}{3} \times \frac{22}{7} \times r^2 \times 14$$

$$r^2 = 36 \Rightarrow r = 6 \text{ cm}$$

Sol.377.(b)

Let length, breadth and height of cuboid be $6x$, $4x$ and $3x$ respectively

Total surface area of the cuboid

$$= 2(lb + bh + hl)$$

$$2(6x \times 4x + 4x \times 3x + 3x \times 6x) = 972$$

$$108x^2 = 972, x = 3$$

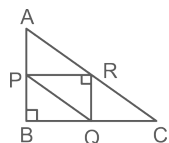
then, length, breadth and height = 18 cm, 12 cm and 9 cm

Diagonal of the cuboid

$$= \sqrt{(18)^2 + (12)^2 + (9)^2}$$

$$= \sqrt{549} = 3\sqrt{61} \text{ cm}$$

Sol.378.(a) (5, 12, 13) is a triplate so the triangle will be a right angle triangle.



triangle made by the mid point of the sides ABC triangle will be also right angle triangle

And the area of ΔPRQ will be $\frac{1}{4}$ of the area of ΔABC .

$$\text{So the area of } \Delta PRQ = \frac{1}{4} \times \frac{1}{2} \times 5 \times 12$$

$$= 7 \frac{1}{2} \text{ cm}^2$$

Sol.379.(d) C.S.A of cone = $\pi r l$

$$\text{T.S.A of cone} = \pi r(l + r)$$

$$\text{slant height (l)} = \sqrt{h^2 + r^2}$$

$$l = \sqrt{20^2 + 15^2} = 25 \text{ cm}$$

$$\text{C.S.A of cone} = \pi \times 15 \times 25 = 375\pi \text{ cm}^2$$

$$\text{T.S.A of cone} = \pi \times 15(25 + 15) = 600\pi \text{ cm}^2$$

Sol.380.(c) Surface area of spherical water tank = $4\pi r^2$

$$= 4 \times \frac{22}{7} \times 3.5 \times 3.5 \Rightarrow 88 \times 5 \times 3.5$$

$$= 44 \times 3.5 = 154 \text{ m}^2$$

$$\text{Required cost} = 154 \times 13 = 2002 \text{ Rs.}$$

Sol.381.(a)

Pythagoras triplets $\therefore (20, 21, 29)$

Radius of cone = 20 cm

$$\text{Volume of cone} = \frac{1}{3} \times \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times 20 \times 20 \times 21$$

$$= 400 \times 22 = 8800 \text{ cm}^3$$

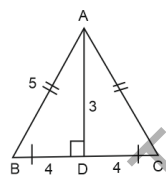
Sol.382.(d)

Perimeter of parallelogram = $2(l + b)$

$$= 48 \text{ cm} \Rightarrow b = 24 - 8 = 16 \text{ cm}$$

$$\text{Area} = \text{base} \times \text{height} = 16 \times 6 = 96 \text{ cm}^2$$

Sol.383.(a)



Pythagoras triplets : (3, 4, 5)

$$AD = 3 \text{ cm}$$

$$\text{Area of triangle} = \frac{1}{2} \times 8 \times 3 = 12 \text{ cm}^2$$

Sol.384.(a) Radius of sphere = 3.5 cm

$$\text{Volume} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times 3.5 \times 3.5 \times 3.5$$

$$= \frac{4}{3} \times 22 \times 0.5 \times 3.5 \times 3.5$$

$$= \frac{44}{3} \times 12.25 = 179.67 \text{ cm}^3$$

Sol.385.(d) 0.2 cm refers to 1 km

1 cm refers to 5 km

1 cm² refers to 25 km²

$$\text{Now, } 22500 \text{ km}^2 = \frac{1}{25} \times 22500 = 900 \text{ cm}^2$$

Sol.386.(d)

$$\text{Total volume} = (15)^3 + (18)^3 + (25)^3$$

Side of newly formed cube

$$= \sqrt[3]{(15)^3 + (18)^3 + (25)^3}$$

$$= \sqrt[3]{3375 + 5832 + 15625}$$

$$= \sqrt[3]{24832} = \sqrt[3]{64 \times 388} = 4\sqrt[3]{388}$$

Sol.387.(d)

$$\frac{4}{3} \times \pi \times (9)^3 = x \times \frac{4}{3} \times \pi \times (3)^3$$

$$\Rightarrow x = 27 \text{ laddoos}$$

Sol.388.(a)

$$\text{Side of square} = \frac{172}{4} = 43 \text{ cm}$$

$$\text{Area of triangle} = \frac{43 \times 43}{2} = 924.5 \text{ cm}^2$$

Sol.389.(d) 1 m³ = 1000 litre

$$\text{Volume of cuboid} = 12 \times 9 \times 7.5$$

$$= 810 \text{ m}^3 = 810000 \text{ litre}$$

Sol.390.(a)

$$\text{Third side} = 20 - (5 + 6) = 9 \text{ meter}$$

$$\text{Semiperimeter} = 10$$

$$\text{Area} = \sqrt{10 \times (10 - 5)(10 - 6)(10 - 9)}$$

$$= \sqrt{10 \times (5) \times (4) \times (1)} = 10\sqrt{2} \text{ m}^2$$

$$\text{Sol.391.(d) Length of arc} = \frac{\theta}{360^\circ} \times 2\pi r$$

$$= 20.6 - 2 \times 7.4 = 5.8 \text{ cm}$$

$$\text{Area of sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$= \left(\frac{\theta}{360^\circ} \times 2\pi r \right) \times \frac{r}{2} = 5.8 \times \frac{7.4}{2}$$

$$= 21.46 \text{ cm}^2$$

Sol.392.(b)

$$\text{Length of arc} = 15.5 - (2 \times 5) = 5.5 \text{ cm}$$

$$\text{Area of sector} = \text{length of arc} \times \frac{\text{radius}}{2}$$

$$= 5.5 \times \frac{5}{2} = 13.75 \text{ cm}^2$$

Sol.393.(a)

$$\text{Side of the square} = \sqrt{30.25} = 5.5 \text{ cm}$$

$$\text{Circumference of circle} = 4 \times 5.5 = 22 \text{ cm}$$

$$\text{Radius of circle} = 22 \times \frac{7}{2 \times 22} = 3.5 \text{ cm}$$

$$\text{Area of circle} = \frac{22}{7} \times 3.5 \times 3.5 = 38.5 \text{ cm}^2$$

Sol.394.(a) Inner radius of hemispherical bowl = 4 cm

$$\text{So, outer radius} = (4 + 1) = 5 \text{ cm}$$

Volume of hemispherical bowl

$$= \frac{2}{3} \times \pi \times (5^3 - 4^3) = \frac{122}{3} \pi = 40 \frac{2}{3} \pi$$

Sol.395.(a) Volume of cylinder = $\pi r^2 h$

According to the question,

$$\pi \times 5^2 \times 52 = \pi \times r^2 \times 100 \Rightarrow r^2 = 13$$

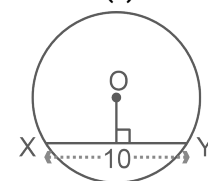
$$\text{Radius of the wire (r)} = \sqrt{13}$$

Sol.396.(c)

Volume of the cylindrical flask = $\pi r^2 h$

$$= \frac{22}{7} \times 6 \times 6 \times 21 = 2376 \text{ cm}^3$$

Sol.397.(a)



Area of the circle = πr^2

$$616 = \frac{22}{7} \times r^2 \Rightarrow r = 14 \text{ cm}$$

$$\text{Required distance} = \sqrt{14^2 - 5^2} = \sqrt{171} \text{ cm}$$

Sol.398.(b) Area of the triangles

$$= \frac{1}{2} \times \text{base} \times \text{height}$$

According to the question,

$$\frac{5}{3} = \frac{5 \times \text{base}_1}{7 \times \text{base}_2} \Rightarrow \text{Required ratio} = 7 : 3$$

Sol.399.(b)

$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} a^2$$

According to question,

Net % increase in area

$$= 34 + 34 + \frac{34 \times 34}{100} = 79.56 \%$$

Sol.400.(a) Angle Subtended by arc

$$(\text{in radian}) = \frac{\text{Length of arc}}{\text{Radius}}$$

According to question,

$$\frac{\pi}{10} = \frac{23.1}{\text{Radius}} \\ \Rightarrow \text{radius} = \frac{231 \times 7}{22} = 73.5 \text{ cm}$$

$$\text{Required area of Circle} = \pi r^2$$

$$= \frac{22}{7} \times 73.5 \times 73.5 = 16978.5 \text{ cm}^2$$

Sol.401.(a)

$$\text{Perimeter of one face of cube} = 4a$$

According to the question,

$$4a = 40 \Rightarrow a = 10 \text{ cm}$$

$$\text{Volume of cube} (a^3) = 10^3 = 1000 \text{ cm}^3$$

Sol.402.(d) According to the question,

$$\text{Volume of the ball} = \frac{4}{3} \pi r^3$$

$$1 \text{ big ball volume} = 3 \text{ small balls volume}$$

$$\frac{4}{3} \pi \times \left(\frac{3}{2}\right)^3 = \frac{4}{3} \pi \left\{ \left(\frac{3}{4}\right)^3 + (1)^3 + (r_3)^3 \right\}$$

$$\left(\frac{27}{8}\right) = \left\{ \frac{27}{64} + 1 + (r_3)^3 \right\}$$

$$(r_3)^3 = \frac{27}{8} - \frac{91}{64} = \frac{125}{64} \Rightarrow r_3 = \frac{5}{4} = 1.25$$

$$\text{Diameter of the 3rd ball} = 1.25 \times 2 = 2.5 \text{ cm}$$

Sol.403.(a)

$$\text{Area of the sector} = \left(\frac{\theta}{360}\right) \pi r^2$$

$$32\pi = \left(\frac{\theta}{360}\right) \pi \times 12 \times 12$$

$$\theta = \frac{32 \times 360}{12 \times 12} = 80^\circ$$

$$\text{Length of the arc} = 2\pi r \left(\frac{\theta}{360}\right)$$

$$= 2\pi \times 12 \left(\frac{80}{360}\right) = \frac{16}{3} \pi \text{ cm}$$

Sol.404.(a) Area of isosceles triangle

$$= \frac{b}{4} \times \sqrt{4a^2 - b^2}$$

where, b is an unequal side and a is equal

sides.

$$30\sqrt{11} \text{ cm}^2 = \frac{5x}{4} \times \sqrt{4 \times 9x^2 - 25x^2}$$

$$30\sqrt{11} \text{ cm}^2 = \frac{5 \times x^2}{4} \times \sqrt{11}$$

$$30 \text{ cm}^2 = \frac{5 \times x^2}{4} \Rightarrow x = \sqrt{24} = 2\sqrt{6}$$

$$\text{Length of unequal side} (5x) = 10\sqrt{6} \text{ cm}$$

Sol.405.(b)

$$\text{Volume of conical tent} = \frac{1}{3} \pi r^2 h$$

$$\text{Volume} = \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times 24$$

$$= 22 \times 7 \times 8 = 22 \times 56 = 1232 \text{ m}^3$$

$$\text{Let, n number of full bags of rice emptied in space} = 1232 \text{ m}^3 = n \times 2 \text{ m}^3$$

$$n = \frac{1232}{2} = 616 \text{ bags}$$

$$\text{Sol.406.(d)} \text{ Surface area of sphere} = 4\pi r^2$$

According to the question,

$$4\pi r^2 = 9\pi \Rightarrow r = \frac{3}{2}$$

$$\text{diameter} (2r) = 3 \text{ cm}$$

Sol.407.(b)

$$\text{Curved surface area of cone} = \pi r l$$

$$\text{Area of base} = \pi r^2$$

$$\text{According to question, } \pi r l = 2 \times \pi r^2$$

$$= \pi r \times 6\sqrt{3} = 2 \times \pi r^2 \Rightarrow r = 3\sqrt{3} \text{ cm}$$

By using pythagoras theorem,

$$\text{Height of cone} = \sqrt{(6\sqrt{3})^2 - (3\sqrt{3})^2} = 9 \text{ cm}$$

Sol.408.(b)

$$2(L + B) = 68 \Rightarrow (L + B) = 34 \dots \text{e.q. (1)}$$

$$(L - B) = 2 \dots \text{e.q. (2)}$$

On solving e.q. (1) and e.q. (2), we get the length of the rectangle (L) = 18

Sol.409.(a)

$$\text{Initial Ratio} = \text{Length} : \text{Breadth} : \text{Height}$$

$$= 1 : 2 : 3$$

$$\text{Initial volume} = 1 \times 2 \times 3 = 6 \text{ unit}^3$$

$$\text{Final Ratio} = \text{Length} : \text{Breadth} : \text{Height}$$

$$= 1 \times 300\% : 2 \times 400\% : 3 \times 400\%$$

$$= 3 : 8 : 12$$

$$\text{Final volume} = 3 \times 8 \times 12 = 288 \text{ unit}^3$$

$$\text{Required increase in volume} = \frac{288 - 6}{6}$$

$$= 47 \text{ times}$$

Sol.410.(d) According to the question,

$$l \times b \times h = 2 \times a^3$$

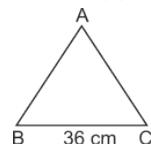
$$= 8 \times 8 \times 16 = 2 \times a^3$$

$$a^3 = 512 \Rightarrow a = 8 \text{ cm}$$

$$\text{Now, total surface area of the cube} = 6a^2$$

$$= 6 \times 8 \times 8 = 384 \text{ m}^2$$

Sol.411.(d)



Now, semi-perimeter of the triangle

$$= \frac{100}{2} = 50 \text{ cm}$$

$$\text{Sol.412.(b)} \text{ Area of sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$112 = \frac{\theta}{360^\circ} \times \frac{22}{7} \times 28 \times 28$$

$$\theta = \frac{360}{22} \text{ (in degree)}$$

$$\theta = \frac{360}{22} \times \frac{22}{7 \times 180} \text{ (in radian)}$$

By Using formula :-

$$\theta (\text{in radian}) = \frac{\text{length of arc}}{\text{radius}}$$

$$\text{Length of arc} = \frac{360}{22} \times \frac{22}{7 \times 180} \times 28$$

$$= 8 \text{ cm}$$

Sol.413.(c)

$$\text{Volume of wall} = 14 \times 0.125 \times 5 = 8.75 \text{ m}^3$$

$$\text{Required no. of bricks} = \frac{8.75}{0.0014}$$

$$= 6250 \text{ bricks}$$

Sol.414.(d)

$$\text{Inner radius of the bowl} = 5 \text{ cm}$$

$$\text{Outer radius of the bowl} = 5.25 \text{ cm}$$

$$\text{Required Volume of the hemisphere}$$

$$= \frac{2}{3} \times \pi (r_2^3 - r_1^3)$$

$$= \frac{2}{3} \times \frac{22}{7} \times \{(5.25)^3 - (5)^3\}$$

$$= \frac{2}{3} \times \frac{22}{7} \times 19.70 = 41.28 \text{ cm}^3$$

Sol.415.(a) Let the inner radius of pipe be 'r' and outer radius of pipe be 'R'.

$$\text{Volume of pipe} = \pi r^2 h \Rightarrow \frac{88}{7} = \frac{22}{7} \times r^2 \times 1$$

$$\text{Inner radius} = 2 \text{ m and outer radius} = 4 \text{ m}$$

Therefore, Required quantity of material

$$= \pi R^2 h - \pi r^2 h$$

$$= \frac{22}{7} \times 4 \times 4 \times 1 - \frac{88}{7} = 37.7 \text{ m}^3$$

Sol.416.(c) Let the length and width of rectangle be $3x$ and $2x$.

According to question,

$$3x \times 2x = 150 \Rightarrow x = 5$$

$$\text{Require perimeter} = 2(\text{length} + \text{width})$$

$$= 2[(3 \times 5) + (2 \times 5)] = 50 \text{ cm}$$

Sol.417.(d) Circumradius of equilateral

$$\text{triangle} = \frac{a}{\sqrt{3}} = \frac{12}{\sqrt{3}} = 4\sqrt{3} \text{ cm}$$

Short trick :-

$$\text{Ratio} \Rightarrow \text{side} : \text{circumradius} = 6 : 2\sqrt{3}$$

Here,

$$6 \text{ units} = 12 \text{ cm} \Rightarrow 2\sqrt{3} \text{ units} = 4\sqrt{3} \text{ cm}$$

$$\text{Sol.418.(c)} \text{ Volume of Sphere} = \frac{4}{3} \pi r^3$$

Let inner radius be r and outer radius be R.

According to question,

$$\text{Required material} = \frac{4}{3} \pi R^3 - \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \pi (3^3 - 2^3) = \frac{76}{3} \pi$$

Sol.419.(c)

Equating the volumes of the cubes :-

$$12 \times 12 \times 12 = 6 \times 6 \times 6 + 8 \times 8 \times 8 + x^3$$

$$1728 = 216 + 512 + x^3$$

$$x^3 = 1728 - 728 = 1000 \Rightarrow x = 10 \text{ cm}$$

Sol.420.(c)

Pythagoras triplets :- 150, 80, 170

Length of its diagonal = 170 cm

Perimeter = $2(150 + 80) = 460$

$$\text{Now, } \frac{460}{170} = 2.7 \text{ times}$$

Sol.421.(b)

Required area = $2(bh + hl) + lb$

$$= 2(20 \times 15 + 15 \times 10) + 10 \times 20 = 1100 \text{ m}^2$$

So, cost of whitewashing the walls of the inside of the hall and ceiling =

$$1100 \times 10.20 = ₹11,220$$

Sol.422.(c) Total surface area of cuboid = Area of square

$$2(lb + bh + hl) = \text{side}^2$$

$$2(4 \times 6 + 6 \times 8 + 8 \times 4) = \text{side}^2$$

$$2 \times 104 = \text{side}^2$$

$$\text{side} = \sqrt{208} = 14.42 \text{ units}$$

Sol.423.(c) Given, radius = 35 cm

Surface area of sphere = $4\pi r^2$

$$= 4 \times \frac{22}{7} \times 35 \times 35 = 15400 \text{ cm}^2$$

Sol.424.(a) $1 \text{ m}^3 = 1000 \text{ liters}$

Volume of the tank = lbh

$$= 50 \times 25 \times 10 = 12500 \text{ m}^3$$

the capacity of the petrol tank

$$= 12500 \times 1000 = 1,25,00,000 \text{ liters.}$$

Sol.425.(a) $l = r\theta$

$$5r = r\theta \Rightarrow \theta = 5^\circ = 5 \times \frac{180^\circ}{\pi}$$

$$\text{Area of sector AOB} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$= \frac{5 \times 180^\circ}{360^\circ \times \pi} \times \pi \times r^2 = 2.5r^2$$

Sol.426.(a) CSA of hemispherical bowl inside as well as outside = $4\pi r^2$

$$= 4 \times \frac{22}{7} \times 7^2 = 616 \text{ cm}^2$$

Cost of painting the hemispherical bowl

$$= 616 \times \frac{40}{20} = ₹1232$$

Sol.427.(a)

$$\text{Perimeter of arc} = \frac{\theta}{360^\circ} \times 2\pi r + 2r$$

$$= \frac{45}{360^\circ} \times 8 \times \frac{22}{7} + 8$$

$$= \frac{22 + 56}{7} = \frac{78}{7} \text{ cm}$$

Sol.428.(c) Volume of cube = $(\text{Side})^3$

$$\text{Side} = \sqrt[3]{12167} = 23 \text{ cm}$$

$$\text{Surface area of cube} = 6 \times (\text{Side})^2$$

$$= 6 \times 529 = 3174 \text{ cm}^2$$

Sol.429.(a) Diagonal of cube = $\text{side}\sqrt{3}$

$$10 = \text{side}\sqrt{3} \Rightarrow \text{Side} = \frac{10}{\sqrt{3}} = 5.77 \text{ m}$$

Sol.430.(c) Volume of cylinder = area of base $\times$ height

$$\text{Height} = \frac{1650 \times 7}{550} = 21 \text{ m}$$

Sol.431.(d)

$$\text{Diagonal of cuboid} = \sqrt{l^2 + b^2 + h^2}$$

$$= \sqrt{144 + 100 + 64} = \sqrt{308} = 17.54 \text{ m}$$

Sol.432.(c) Equating the volumes :-

Volume of hollow hemisphere = Volume of cylinder

$$\frac{2}{3} \pi (R^3 - r^3) = \pi \times 7 \times 7 \times H$$

$$\frac{2}{3} (5^3 - 3^3) = 7 \times 7 \times H$$

$$H = \frac{2 \times (125 - 27)}{7 \times 7 \times 3} = \frac{4}{3} = 1.33 \text{ cm}$$

Sol.433.(a)

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{Height}$$

As we know, the base and height of an isosceles right angle triangle are equal.

According to question,

$$200 = \frac{1}{2} \times a \times a \Rightarrow a = 20 \text{ cm}$$

By using pythagoras theorem,

$$\text{Hypotenuse} = 20\sqrt{2} \text{ cm}$$

Perimeter of isosceles right-angle

$$\text{triangle} = 20\sqrt{2} + 40 = 68.3 \text{ cm}$$

Sol.434.(b) Surface area of sphere = $4\pi r^2$

According to question,

$$616 = 4 \times \frac{22}{7} \times r^2 \Rightarrow r^2 = 49 \Rightarrow r = 7 \text{ cm}$$

Now, total surface area of hemisphere

$$= 3\pi r^2 = 3 \times \frac{22}{7} \times 7 \times 7 = 462 \text{ cm}^2$$

Sol.435.(b)

$$\frac{\pi r^2 h}{2\pi r h} = \frac{1848}{528} \Rightarrow r = \frac{2 \times 7}{2} = 7 \text{ m}$$

$$\text{Now, } 2\pi r h = 528 \text{ m}^2$$

$$h = \frac{528 \times 7}{2 \times 22 \times 7} = 12 \text{ m}$$

Required ratio = 7 : 12

Sol.436.(a) Total surface area of Cuboid

$$= 2(lb + bh + lh)$$

$$= 2\{(48 \times 27) + (27 \times 30) + (30 \times 48)\}$$

$$= 2\{1296 + 810 + 1440\} = 7092 \text{ sq. inch}$$

$$= \frac{7092}{12 \times 12} = 49.25 \text{ sq. foot}$$

Sol.437.(c) Let, the breadth of rectangle be x metre, then length be $3x$ metre.

$$\text{Area} = x \times 3x = 3x^2$$

$$3x^2 = 2700 \text{ m}^2 \Rightarrow \text{breadth } (x) = 30 \text{ metre}$$

Sol.438.(a) Area of square $(\text{side}^2) = 5929$

Side = 77

Perimeter of square = perimeter of circle

$$4 \times 77 = 2\pi r \Rightarrow 2 \times \frac{22}{7} \times r = 308 \Rightarrow r = 49$$

Area of the circle = (πr^2)

$$= \frac{22}{7} \times 49 \times 49 = 7546 \text{ cm}^2$$

Sol.439.(d)

Perimeter of the semi-circle = $\pi r + 2r$

$$3.6 = r\left(\frac{22}{7} + 2\right)$$

$$3.6 = r\left(\frac{36}{7}\right) \Rightarrow r = \frac{7}{10}$$

Area of semi-circle = $\frac{1}{2} \pi r^2$

$$\frac{1}{2} \times \frac{22}{7} \times \frac{7}{10} \times \frac{7}{10} = 0.77 \text{ m}^2$$

Sol.440.(d) Let, inner radius and outer radius be r_2 and r_1 respectively.

According to the question,

$$\frac{2\pi r_1}{2\pi r_2} = \frac{17}{15} \Rightarrow \frac{r_1}{r_2} = \frac{17}{15} \Rightarrow r_1 = \frac{17}{15} r_2$$

$$\text{Now, } r_1 - r_2 = 40 \Rightarrow \frac{17}{15} r_2 - r_2 = 40$$

$$r_2 = 300 \text{ and } r_1 = 340$$

Radius of inner circle (r_2) = 300 m

Sol.441.(c) Area of the square = (side^2)

$$529 = (\text{side}^2) \Rightarrow \text{Side} = 23$$

Diagonal of the square

$$= \text{side}\sqrt{2} = 23\sqrt{2} \text{ cm}$$

Sol.442.(b) Let length of cuboid be l , breadth be b and height be h

According to the question,

Breadth of cuboid = 3cm

$$l = 6b = 6 \times 3 = 18 \text{ cm}$$

$$\text{And, } h = 4b = 4 \times 3 = 12 \text{ cm}$$

Hence,

total surface area of cuboid

$$= 2(lb + bh + hl)$$

$$= 2(18 \times 3 + 3 \times 12 + 12 \times 18) = 612 \text{ cm}^2$$

Sol.443.(b) Ratio of the radius of cylinder and cone = 3 : 1

Volume of cylinder = $\pi r^2 h$

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

According to the question,

$$\pi(3r)^2 h : \frac{1}{3} \pi(r)^2 h = 9 : \frac{1}{3} = 27 : 1$$

Sol.444.(c) Area of scalene triangle

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{Semiperimeter } (s) = \frac{11 + 11 + 4}{2} = 13$$

$$\sqrt{13(13-11)(13-11)(13-4)}$$

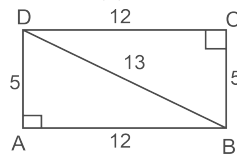
$$= \sqrt{13 \times 2 \times 2 \times 9} = 6\sqrt{13} \text{ unit}$$

According to the question,

$$6\sqrt{13} \text{ unit} = 2\sqrt{117} \text{ cm}^2 \Rightarrow 1 \text{ unit} = 1$$

Required side = 11 cm

Sol.445.(a) Triplate = 5, 12, 13



So, the area of the parallelogram

$$= 2 \left(\frac{1}{2} \times 12 \times 5 \right) = 60 \text{ cm}^2$$

Sol.446.(a) Volume of cuboid = $l \times b \times h$

Volume of cube = $side^3$

Volume of cuboid = volume of cube

$$8 \times 27 \times 64 = side^3$$

$$Side = 2 \times 3 \times 4 = 24 \text{ cm}$$

$$TSA \text{ of cuboid} = 2(lb + bh + hl)$$

$$TSA \text{ of cube} = 6side^2$$

Required ratio

$$= 2(8 \times 27 + 27 \times 64 + 64 \times 8) : 6 \times 24^2$$

$$= 2(216 + 1728 + 512) : 6 \times 576$$

$$= 2 \times 2456 : 6 \times 576 = 307 : 216$$

Sol.447.(b)

$$\text{Volume of hemisphere} = \frac{2}{3} \pi r^3$$

$$= \frac{2}{3} \times \frac{22}{7} \times 4.2 \times 4.2 \times 4.2 = 155.232 \text{ m}^3$$

Now,

$$\frac{1}{21} \text{th of volume} = \frac{155.232}{21} = 7.3 \text{ m}^3$$

Sol.448.(b) After joining two cubes each

of edge 9 cm to form cuboid

Length of cuboid(l) = 18cm,

width (b) = 9cm and height (h) = 9 cm

Then, Total surface area of cuboid

$$= 2(lb + bh + hl)$$

$$= 2(18 \times 9 + 9 \times 9 + 9 \times 18) = 810 \text{ cm}^2$$

Sol.449.(b) CSA of cone = $\pi r l$

$$\text{Slant height of the cone } (l) = \sqrt{r^2 + h^2}$$

$$l = \sqrt{8^2 + 15^2} = \sqrt{64 + 225} = 17$$

$$\text{Required cost} = \frac{\pi \times 8 \times 17 \times 54}{\pi} = ₹ 7344$$

Sol.450.(a) For largest possible sphere,
diameter = side of cube

$$\text{Radius of sphere} = \frac{6}{2} = 3 \text{ cm}$$

$$\text{Volume of the sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times 3 \times 3 \times 3 = \frac{792}{7} \text{ cm}^3$$

Sol.451.(c)

Formula used :- area of rhombus

$$= \frac{1}{2} \times \text{first diagonal} \times \text{second diagonal}$$

According to the question,

$$\text{Area of rhombus} = \frac{1}{2} \times 16 \times 20 = 160 \text{ cm}^2$$

Sol.452.(b) Volume of cube = $side^3$

Volume of 2 small cube = volume of 1 big cube

$$7^3 + 11^3 = \text{volume of 1 big cube}$$

$$\text{volume of 1 big cube} = 343 + 1331$$

$$= 1674 \text{ cm}^3$$

Sol.453.(a)

$$\text{Area of rectangle} = 58 \times 30 = 1740 \text{ m}^2$$

$$\text{Required cost} = 1740 \times 20 = ₹ 34800$$

Sol.454.(c) Volume of the cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times \frac{126}{10} \times \frac{126}{10} \times 400 \text{ cm}^3$$

$$= 199584 \text{ cm}^3 \Rightarrow 1 \text{ cm}^3 = 2 \text{ grams}$$

$$199584 \text{ cm}^3 = \frac{2}{1} \times 199584 \times \frac{1}{1000}$$

$$= 399.168 \text{ kg}$$

Sol.455.(c) According to the question,

$$\text{Base} = 56 \times \frac{1}{4} = 14 \text{ cm}$$

$$\text{Area of the triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 14 \times 56 = 392 \text{ cm}^2$$

Sol.456.(d)

Outer radius of spherical shell (R) = 13cm

Inner radius (r) = $13 - 3 = 10 \text{ cm}$

$$\text{Volume of the spherical shell} = \frac{4}{3} \pi (R^3 - r^3)$$

$$\text{Required volume} = \frac{4}{3} \times \frac{22}{7} \times (13^3 - 10^3)$$

$$= \frac{4}{3} \times \frac{22}{7} \times 1197 = 5016 \text{ cm}^3$$

Sol.457.(c) According to question,

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times (65)^3 = 1150809.52$$

$$= 1.15 \times 10^6 \text{ cm}^3$$

Sol.458.(a)

$$\frac{\text{C.S.A of cone}}{\text{base area of cone}} = \frac{\pi r l}{\pi r^2} = \frac{l}{r}$$

According to the question,

$$\frac{l}{r} = \frac{3}{1}, \frac{l}{\frac{1}{2}} = \frac{3}{1} \Rightarrow l = \frac{3}{2}$$

$$\text{T.S.A of cone} = \pi r (l + r)$$

$$= \pi \times \frac{1}{2} \left(\frac{3}{2} + \frac{1}{2} \right) = \pi \text{ cm}^2$$

Sol.459.(c) Let radius of sphere = R

$$\text{According to the question, } R = \frac{3}{4} r$$

Ratio of volumes

$$= \frac{4}{3} \times \pi \times \left(\frac{3}{4} r \right)^3 : \frac{2}{3} \times \pi \times (r)^3$$

$$\Rightarrow 2 \times \frac{27}{64} : 1 = 27 : 32$$

Sol.460.(b) According to the question,
 $720 \times h = 28800$

$$\text{Depth of the water } (h) = \frac{28800}{720}$$

= 40 meters.

Sol.461.(c)

From the given sides of triangle

$$= (6, 8, 10)$$

$$6^2 + 8^2 = 10^2 \Rightarrow 100 = 100$$

So, the given triangle is a right angled triangle.

Therefore, Area of triangle

$$= \frac{1}{2} \times \text{base} \times \text{height} = \frac{6 \times 8}{2} = 24 \text{ cm}^2$$

Sol.462.(b) According to the question,
circumference of the base of the cylinder
 $= 2\pi r$

$$22 = 2 \times \frac{22}{7} \times r \Rightarrow r = 3.5 \text{ cm}$$

$$\text{Height} = 3.5 \times 4 = 14 \text{ cm}$$

$$\text{C.S.A of cylinder} = 2\pi r h$$

$$2 \times \frac{22}{7} \times 3.5 \times 14 = 308 \text{ cm}^2$$

Sol.463.(d) Volume of Cylinder = $\pi r^2 h$
According to question,

$$\text{Required ratio} = \frac{\pi r_1^2 h_1}{\pi r_2^2 h_2}$$

$$= \frac{\pi \times \frac{6}{2} \times \frac{6}{2} \times 7}{\pi \times \frac{7}{2} \times \frac{7}{2} \times 9} = \frac{4}{7}$$

Sol.464.(d) According to the question,

$$2\pi r h = \frac{2}{5} \times 2\pi r (r + h)$$

$$3h = 2r \Rightarrow \frac{r}{h} = \frac{3}{2}$$

Hence, ratio of diameter to height

$$= 3 \times 2 : 2 = 3 : 1$$

Sol.465.(c)

circumferences of circles = $2\pi r$

Let the radius of larger circle be R .

According to question,

$$\left(2 \times \frac{22}{7} \times R \right) - \left(2 \times \frac{22}{7} \times 1.4 \right) = 4.4$$

$$\left(2 \times \frac{22}{7} \times R \right) = 13.2 \Rightarrow R = 2.1 \text{ cm}$$

Sol.466.(a) According to the question,

$$9 \times \pi r^2 = 3 \times side^2 \Rightarrow \frac{r^2}{side^2} = \frac{1}{3 \times \pi}$$

$$\frac{r}{side} = \sqrt{\frac{1}{3 \times \pi}} = \frac{1}{\sqrt{3\pi}}$$

$$\text{Diameter of the circle} = 2r = 2 \times 1 = 2$$

$$\text{Diagonal of the square} = \sqrt{2} \text{ side}$$

$$= \sqrt{2} \times \sqrt{3\pi}$$

$$\text{Required ratio} = 2 : \sqrt{2} \times \sqrt{3\pi} = \sqrt{2} : \sqrt{3\pi}$$

Sol.467.(b)

$$\text{Volume of hemispherical bowl} = \frac{2}{3} \pi r^3$$

$$\frac{2}{3} \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times \frac{21}{2} = 2425.5 \text{ m}^3$$

$$1 \text{ m}^3 = 1000 \text{ liter}$$

$$2425.5 \text{ m}^3 = 2425500 \text{ liter}$$

Sol.468.(c) Total Surface area of Cylinder

$$= 2\pi r(r + h) = 2 \times \frac{22}{7} \times 28(28 + 24)$$

$$= 9152$$

Total Surface area of bisected cylinder

$$= 2\{2\pi r(r + \frac{h}{2})\}$$

$$= 2 \times 2 \times \frac{22}{7} \times 28(28 + \frac{24}{2}) = 14080$$

$$\text{Required}\% = \frac{14080 - 9152}{9152} \times 100 = 53.85\%$$

Sol.469.(c)

Curved surface area of Cylinder = $2\pi rh$

$$330 = 2 \times \frac{22}{7} \times r \times 15$$

$$r = \frac{330 \times 7}{2 \times 15 \times 22} = 3.5 \text{ cm}$$

Required Diameter = 7 cm

Sol.470.(c) According to question,

Length, Breadth and Height of room

$$= 640 \text{ cm}, 480 \text{ cm and } 320 \text{ cm}$$

$$\text{HCF}(640, 480, 320) = 160 \text{ cm}$$

So, Required length of tape = 1 m 60 cm

Sol.471.(c) According to the question,

$$\text{Volume of hemisphere} = \frac{2}{3}\pi r^3$$

$$\text{Volume of cone} = \frac{1}{3}\pi r^2 h$$

$$N \text{ number of cone} = \frac{\text{Volume of hemisphere}}{\text{volume of cone}}$$

$$= \frac{\frac{2}{3}\pi r^3}{\frac{1}{3}\pi r^2 h} \Rightarrow \frac{2 \times 8 \times 8 \times 8}{4 \times 4 \times 2} = 32$$

Sol.472.(d) According to the question ,

$$\frac{2\pi R}{2\pi r} = \frac{11}{7} \Rightarrow \frac{R}{r} = \frac{11}{7}$$

Now, width of path (11 - 7) unit

$$= 20 \text{ meter} \Rightarrow 1 \text{ unit} = 5 \text{ meter}$$

Radius of inner circle (r) = 7 × 5 = 35 meter

$$\text{Sol.473.(d)} \quad S = \frac{a+b+c}{2} \text{ and Area of}$$

$$\text{triangle} = \sqrt{S(S-a)(S-b)(S-c)}$$

According to question,

$$S = \frac{7+4+5}{2} = 8$$

$$\text{Area of triangle} = \sqrt{8(8-7)(8-4)(8-5)}$$

$$= \sqrt{8 \times 1 \times 4 \times 3} = \sqrt{96} = 4\sqrt{6} \text{ cm}^2$$

$$\text{Sol.474.(d)} \quad \text{Area of square} = \frac{(\text{diagonal})^2}{2}$$

$$= \frac{(18.4)^2}{2} = \frac{338.56}{2} = 169.28 \text{ cm}^2$$

$$\text{Sol.475.(d)} \quad \text{Volume of cylinder} = \pi r^2 h$$

$$\text{Volume of cone} = \frac{1}{3}\pi r^2 h$$

Let the Radius of cylinder = x

And the radius of cone = 2x

According to the question,

$$\frac{1}{3}\pi r^2 h : \pi r^2 h = \frac{1}{3} \times 2x \times 2x : x \times x$$

Required ratio = 4 : 3

Sol.476.(c) According to the question,

$$l - b = 40$$

Now, perimeter of rectangle $2(l + b) = 96$

$$l + b = 48$$

Then $l = 44$ and $b = 4$

$$\text{Area of rectangle} = 44 \times 4 = 176 \text{ m}^2$$

Sol.477.(b)

$$\text{Area of Trapezium} = \frac{(a+b)}{2} \times h$$

According to question,

Area of Trapezium

$$= \frac{7.58 + 6.42}{2} \times 6.50 = 45.5 \text{ square yard}$$

$$\text{Required price} = 45.5 \times 2000 = \text{Rs. } 91000$$

Sol.478.(a) According to question,

$$\pi r^2 H = \pi \left(\frac{88r}{100}\right)^2 h$$

$$r \times r \times H = \frac{22r}{25} \times \frac{22r}{25} \times h \Rightarrow \frac{H}{h} = \frac{484}{625}$$

$$\text{Required \%} = \frac{(625 - 484)}{484} \times 100 = 29.13\%$$

Sol.479.(c) According to the question,

$$\text{Height} = 7 \times 3 = 21 \text{ cm}$$

$$\text{CSA of cylinder} = 2\pi rh$$

$$= 2 \times \frac{22}{7} \times 7 \times 21 = 924 \text{ cm}^2$$

Sol.480.(a) Semiperimeter of triangle (s)

$$= \frac{3x + 3x + 4x}{2} = 5x$$

Now, area of triangle

$$= \sqrt{5x(5x - 3x)(5x - 3x)(5x - 4x)}$$

$$= \sqrt{5x(2x)(2x)(x)} = 2\sqrt{5} x^2$$

According to the question,

$$2\sqrt{5} x^2 = 32\sqrt{5} \Rightarrow x = 4 \text{ cm}$$

$$\text{Length of equal sides } (3x) = 12 \text{ cm}$$

Sol.481.(d) Length = 4cm

ATQ

Breadth = 16cm , Height = 8cm

Lateral surface area of cuboid

$$= 2(\text{length} + \text{breadth})\text{height}$$

$$= 2(4 + 16) \times 8 = 2(20) \times 8 = 320 \text{ cm}^2$$

Sol.482.(b)

Let length of longer Diagonal be 10 unit

Then, the length of the smaller diagonal is 7 units.

$$\text{Area of Rhombus} = \frac{\text{product of Diagonals}}{2}$$

$$= \frac{7 \times 10}{2} = 35$$

square of the length of larger diagonal

$$= 100$$

According to question,

$$\text{Required Ratio} = \frac{35}{100} = 7 : 20$$

Sol.483.(b) According to question

initial : final

$$\text{Radius} \rightarrow (25)^2 : (21)^2$$

$$\text{Height} \rightarrow h_1 : h_2$$

$$\text{Volume} \rightarrow 625 h_1 : 441 h_2$$

According to question ,

volume remains same

Volume is directly proportional to height-

Increase in height:-

$$\frac{625 - 441}{441} \times 100 = 41.72\%$$

Sol.484.(a) According to the question,

$$\text{Height of the cuboid} = 16 \times \frac{1}{4} = 4 \text{ cm}$$

CSA of cuboid

$$= 2(\text{length} + \text{breadth}) \times \text{height}$$

$$= 2(16) \times 4 = 128 \text{ cm}^2$$

Sol.485.(a) According to question,

Length, breadth and height of Cuboid

= 10 cm , 5 cm and 20 cm respectively

lateral surface area of the cuboid

$$= 2(\text{length} + \text{breadth}) \times \text{height}$$

$$= 2(10 + 5) \times 20 = 600 \text{ cm}^2$$

Sol.486.(b) Perimeter of rectangle

$$= 2(\text{length} + \text{breadth})$$

$$56 = 2(l + b) \Rightarrow (l + b) = 28 \text{ m}$$

$$\text{Area of rectangle} = l \times b \Rightarrow l \times b = 192$$

$$(l + b) = 28 \text{ m (squaring both sides)}$$

$$l^2 + b^2 + 2 \times 192 = 784 \Rightarrow l^2 + b^2 = 400$$

$$\text{Diagonal of the rectangle} = \sqrt{l^2 + b^2}$$

$$\text{Diagonal of the rectangle} = \sqrt{400}$$

$$= 20 \text{ meter}$$

Sol.487.(b) According to the question,

$$\text{Area of circle} = \pi r^2 \Rightarrow \pi \times r^2 = 1560$$

total surface area of the hemisphere

$$= 3\pi r^2 \Rightarrow 3 \times 1560 = 4680 \text{ m}^2$$

Sol.488.(c) According to the question,

$$\text{Height of the cuboid} = 40 \times \frac{1}{5} = 8$$

Lateral Surface area of cuboid

$$= 2(l + b)h \Rightarrow 2(40) \times 8 = 640 \text{ cm}^2$$

Sol.489.(d)

$$\text{Volume of the sphere} = \frac{4}{3}\pi r^3$$

After decreasing the radius of sphere by

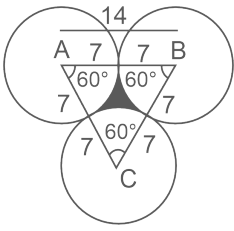
10% then volume of the sphere

$$= \frac{4}{3}\pi \left(\frac{9}{10}r\right)^3 = \frac{4}{3}\pi \left(\frac{729}{1000}r^3\right)$$

Therefore, decrement in the volume

$$= \frac{r^3 - \left(\frac{729}{1000}r^3\right)}{r^3} \times 100$$

$$= \frac{271r^3}{1000r^3} \times 100 \Rightarrow 27.1 \%$$

Sol.490.(a)

Side of the equilateral triangle $(2r) = 14\text{ cm}$
So, area of the equilateral triangle

$$= \frac{\sqrt{3}}{4} \times 14 \times 14 = 49\sqrt{3} \text{ cm}^2$$

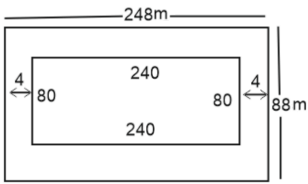
Now, the area of the each sector

$$= 3 \times \frac{60^\circ}{360^\circ} \times \pi r^2$$

$$= 3 \times \frac{1}{6} \times \frac{22}{7} \times 7 \times 7 = 77 \text{ cm}^2$$

Therefore, area of the portion enclosed by three circle = (area of equilateral triangle - area of all three sector)

$$\Rightarrow (49\sqrt{3} - 77) \text{ cm}^2$$

Sol.491.(d)

$$\begin{aligned} \text{Area of the path} &= (248 \times 88 - 240 \times 80) \\ &= (21824 - 19200) \text{ m}^2 = 2624 \text{ m}^2 \end{aligned}$$

Sol.492.(d) According to the question,

$$l + b + h = 20\text{ cm}$$

Total Surface area of the cuboid

$$= 2(lb + bh + hl) = 256 \text{ cm}^2$$

$$\text{Now, } (l + b + h)^2 = (l^2 + b^2 + h^2) + 2(lb + bh + hl)$$

$$\Rightarrow (l^2 + b^2 + h^2) = 400 - 256$$

$$= 144 \text{ cm}^2$$

Maximum Length of stick that can be placed inside the cuboid box

$$= \sqrt{(l^2 + b^2 + h^2)} = \sqrt{144} = 12 \text{ cm}$$

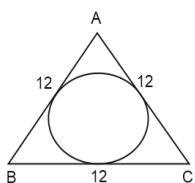
Sol.493.(b) Let side of third triangle (z) = a

According to the question,

$$\text{Area of the triangle} \rightarrow X + Y = Z$$

$$\frac{\sqrt{3}}{4} \times (6)^2 + \frac{\sqrt{3}}{4} \times (8)^2 = \frac{\sqrt{3}}{4} \times (a)^2$$

$$(a)^2 = (6)^2 + (8)^2 \Rightarrow a = 10 \text{ cm}$$

Sol.494.(d)

$$\text{Inradius} = \frac{\text{side}}{2\sqrt{3}}$$

Side of the equilateral triangle

$$= \text{inradius} \times 2\sqrt{3}$$

$$\text{inradius} = \frac{12}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 2\sqrt{3}$$

$$\begin{aligned} \text{Area of the equilateral triangle} &= \frac{\sqrt{3}}{4} \times a^2 \\ &= \frac{\sqrt{3}}{4} \times 144 = 36 \times 1.732 = 62.35 \text{ sq. cm} \end{aligned}$$

$$\text{Area of the incircle} = \pi r^2$$

$$= \frac{22}{7} \times (2\sqrt{3})^2 = 37.71 \text{ sq. cm}$$

$$\begin{aligned} \text{Therefore, area of the triangle which is not in the incircle (shaded region)} \\ &= (62.35 - 37.71) = 24.64 \text{ sq. cm} \end{aligned}$$

Sol.495.(a) Area of the rhombus = $\frac{1}{2}$

$$\times \text{first diagonal} \times \text{second diagonal}$$

$$= \frac{1}{2} \times 8 \times 13 = 52 \text{ cm}^2$$

Sol.496.(c)

$$\text{Volume of the sphere} = \frac{4}{3} \pi (R)^3$$

$$\text{Volume of the hemisphere} = \frac{2}{3} \pi (r)^3$$

According to the question,

$$\text{Required ratio} \rightarrow \frac{4}{3} \pi (R)^3 : \frac{2}{3} \pi (r)^3$$

$$\frac{4}{3} \pi (3r)^3 : \frac{2}{3} \pi (r)^3 \Rightarrow 54 : 1$$

Sol.497.(c) Let, the radius of the 1st cylinder be 'R' and radius of the smaller cylinder be 'r'.

$$\text{Ratio} \rightarrow \text{1st cylinder} : \text{2nd cylinder}$$

$$\text{Volume} \rightarrow 32 : 9$$

$$\text{Height} \rightarrow 8 : 9$$

According to the question,

$$\text{Area of base of 2nd Cylinder } (\pi r^2)$$

$$= 616 \text{ cm}^2 \Rightarrow r = \sqrt{616 \times \frac{7}{22}} = 14 \text{ cm.}$$

Now,

$$\frac{\text{Volume of 1st Cylinder } (\pi R^2 H)}{\text{Volume of 2nd Cylinder } (\pi r^2 h)} = \frac{32}{9}$$

$$\frac{\pi r^2 (8)}{\pi \times 196 \times (9)} = \frac{32}{9}$$

$$r^2 = 784 \Rightarrow r = 28 \text{ cm}$$

Sol.498.(a) $\frac{2(l+b)}{2} = 30 \text{ cm}$

$$(l+b) = 30 \text{ cm}$$

According to question;

$$l = (6+x) \text{ and } b = x$$

$$(6+x) + x = 30 \text{ cm} \Rightarrow x = 12$$

$$l = (6+12) = 18 \text{ cm and } b = 12 \text{ cm}$$

$$\text{Area of rectangle} = l \times b = 18 \times 12 = 216 \text{ cm}^2$$

Alternate method :

Formula : Perimeter of rectangle

$$= 2(l+b) \Rightarrow \frac{2(l+b)}{2} = 30 \text{ cm}$$

$$(l+b) = 30 \text{ cm} \dots \text{eq(1)}$$

$$(l-b) = 6 \text{ cm} \dots \text{eq(2)}$$

Solving the eq(1) and eq(2), we get,

$$l = 18 \text{ cm, and } b = 12 \text{ cm,}$$

$$\begin{aligned} \text{Area of rectangle} &= l \times b = 18 \times 12 \\ &= 216 \text{ cm}^2 \end{aligned}$$

Sol.499.(a) $(a \times b \times c)^2 = 1024 \text{ units}$
 $a \times b \times c = 32$

$$\text{Area of its circumcircle} = 16\pi$$

$$\pi R^2 = 16\pi \Rightarrow R = 4$$

$$\text{circumradius}(R) = \frac{abc}{4 \times \text{area of } \Delta}$$

$$4 = \frac{32}{4 \times \text{area of } \Delta}$$

$$\text{Area of given triangle} = 2 \text{ sq. units}$$

Sol.500.(d)

Let the length of the rectangle = $5x$

Breadth = $3x$

$$\text{Area of rectangle} = \text{length} \times \text{breadth}$$

$$60 = 5x \times 3x \Rightarrow x \rightarrow 2$$

Now,

$$\text{Length} = 5x \rightarrow 10\text{ m, Breadth} = 3x \rightarrow 6\text{ m}$$

$$\text{Difference between length and breadth}$$

$$\Rightarrow 10 - 6 \rightarrow 4 \text{ m}$$

Sol.501.(a) Total surface area of cylinder

$$2\pi r(h+r) = 2 \times \frac{22}{7} \times 7(5+7) = 528 \text{ cm}^2$$

When Cylinder is bisected from its height

$$\text{, then Increased area} \rightarrow 2 \times (\pi r^2)$$

$$= 2 \times \frac{22}{7} \times 7 \times 7 = 308 \text{ cm}^2$$

Required % increase

$$= \frac{308}{528} \times 100 \approx 58.33\%$$

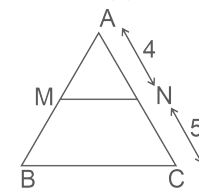
Alternate method :

After cutting the cylinder in two identical cylinder the total increase in the surface area will be $2\pi r^2$

% increase in the surface area

$$= \frac{2\pi r^2}{2\pi r(h+r)} \times 100 \Rightarrow \frac{r}{(r+h)} \times 100$$

$$= \frac{7}{(5+7)} \times 100 = 58.33\%$$

Sol.502.(c)

$$\text{Area of } \Delta AMN = (4)^2 \Rightarrow 16 \text{ units}$$

$$\text{Area of } \Delta ABC = (9)^2 \Rightarrow 81 \text{ units}$$

$$\text{Area of quadrilateral MBCN} = \text{Area of } \Delta ABC - \text{Area of } \Delta AMN$$

$$81 - 16 \Rightarrow 65 \text{ units}$$

$$65 \text{ units} = 130 \text{ cm}^2$$

$$\text{Area of } \Delta AMN (16 \text{ units}) = 32 \text{ cm}^2$$

Sol.503.(a)

$$\text{Circumference of circle} = 2\pi r (\text{Small circle})$$

$$2\pi r = 264 \text{ m} \Rightarrow r = 42 \text{ m}$$

$$(\text{Big circle}) \Rightarrow 2\pi R = 308 \text{ m, } R = 49 \text{ m}$$

$$\text{Area of circle} = \pi r^2$$

Difference between bigger and smaller

circle's area = $\pi(49^2 - 42^2) \rightarrow 2002 \text{ m}^2$

Sol.504.(c) Ratio of their heights $\rightarrow 2 : 3$

Ratio of their radius $\rightarrow 6 : 5$

$$\frac{\text{Volume}_1}{\text{Volume}_2} = \frac{\pi \times 6 \times 6 \times 2}{\pi \times 5 \times 5 \times 3} = \frac{24}{25}$$

Sol.505.(b) Length + breadth = 4 cm

Length = 3 cm, Breadth = 1 cm

$$\text{Height} = \frac{4}{2} = 2 \text{ cm}$$

$$\begin{aligned} \text{T.S.A cuboid} &= 2(\text{length} \times \text{breadth} + \text{breadth} \times \text{height} + \text{height} \times \text{length}) \\ &= 2(3 \times 1 + 1 \times 2 + 2 \times 3) = 22 \text{ cm}^2 \end{aligned}$$

Sol.506.(c) Area of square = side^2

$$1156 = \text{side}^2, \text{side} = 34 \text{ cm}$$

So the length of rectangle = 34×2

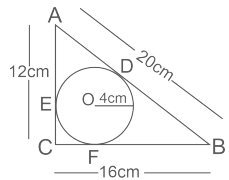
$$= 68 \text{ cm}$$

Breadth of rectangle = $34 - 14 = 20 \text{ cm}$

Ratio of length and breadth of rectangle

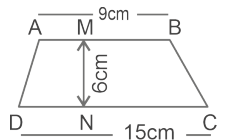
$$= 68 : 20 \Rightarrow 17 : 5$$

Sol.507.(c)



$$\text{Area of } \triangle ABC = \frac{16 \times 12}{2} = 96 \text{ cm}^2$$

Sol.508.(b)



$$\text{Area of Trapezium} = \frac{1}{2} \times (AB + DC) \times MN$$

$$= \frac{1}{2} \times (9 + 15) \times 6 = 72 \text{ cm}^2$$

Total cost of copper plating

$$= 72 \times 3 = \text{Rs. } 216$$

Sol.509.(b) According to the question,

Semiperimeter of the $\triangle ABC$,

$$s = \frac{22}{2} = 11 \text{ cm}$$

Then, Area of triangle $\triangle ABC$

$$\Rightarrow \sqrt{11(7)(3)(1)} = \sqrt{231} \text{ cm}^2$$

$\therefore \triangle ABC \sim \triangle PQR$

$$\text{Therefore, } \frac{\text{ar}(\triangle PQR)}{\text{ar}(\triangle ABC)} = \left(\frac{QR}{BC}\right)^2$$

$$= \frac{\text{ar}(\triangle PQR)}{\sqrt{231}} = \frac{256}{64}$$

$$= \text{ar}(\triangle PQR) = 4\sqrt{231} \text{ cm}^2$$

Sol.510.(c) Length of each side of the

cube = 21 cm

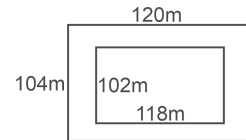
$$\text{Radius of sphere} = \frac{21}{2} \text{ cm}$$

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times \frac{21}{2}$$

$$= 4851 \text{ cm}^3$$

Sol.511.(d)



Required perimeter for marking

$$= 2(l + b) = 2(102 + 118) = 440 \text{ metre}$$

$$\text{Total price} = 440 \times 2.5 = 1100 \text{ Rs.}$$

Sol.512.(c)

$$\text{TSA of Pyramid} = \text{Area of Base} + \left(\frac{1}{2} \times \text{Perimeter of Base} \times \text{Slant height}\right)$$

$$\text{Slant Height } (l) = \sqrt{r^2 + h^2}$$

$$l = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ cm}$$

$$\text{TSA of Pyramid} = 64 + \left(\frac{1}{2} \times 32 \times 5\right)$$

$$= 144 \text{ cm}^2.$$

Sol.513.(c) Longest diagonal in a room

$$= \sqrt{l^2 + b^2 + h^2}$$

According to Question,

$$\text{Breadth} = \frac{\text{Length}}{2} \text{ and}$$

$$\text{Breadth} = 2 \times \text{height}$$

$$\begin{aligned} \text{Now, Diagonal} &= \sqrt{(2b)^2 + (b)^2 + \left(\frac{b}{2}\right)^2} \\ &= \sqrt{\frac{16b^2 + 4b^2 + b^2}{4}} = \frac{\sqrt{21} b}{2} \end{aligned}$$

$$\text{Sol.514.(d)} \text{ LSA} = \frac{1}{2} \times \text{Perimeter of Base} \times \text{Slant height}$$

$$\text{TSA of pyramid} = \text{LSA} + \text{Area of Base}$$

$$(\text{Edge})^2 = (\text{Slant height})^2 + \left(\frac{\text{Side}}{2}\right)^2$$

$$(24)^2 = (\text{Slant height})^2 + \left(\frac{8}{2}\right)^2$$

$$\Rightarrow \text{Slant height} = 4\sqrt{35} \text{ cm}$$

$$\text{LSA} = 48\sqrt{35} \text{ cm}$$

$$\text{TSA} = 16\sqrt{3} + 48\sqrt{35} \text{ cm}^2$$

Sol.515.(c) Let the parallel sides of trapezium be a and b cm.

$$\text{Area of Trapezium} = \frac{1}{2} \times \text{Sum of two parallel sides} \times \text{height}$$

$$1380 = \frac{1}{2} \times (a + b) \times 60$$

$$a + b = 46 \text{ and } a - b = 12$$

On Solving these equations, we get

$$a = 29 \text{ cm and } b = 17 \text{ cm.}$$

Sol.516.(b) Volume of rectangular leads

$$= n \times \text{volume of spherical lead}$$

$$L \times B \times H = n \times \frac{4}{3} \times \pi \times r^3$$

$$88 \times 63 \times 42 = n \times \frac{4}{3} \times \frac{22}{7} \times 4.2^3$$

$$n = \frac{63 \times 10 \times 21}{4.2 \times 4.2} = 750$$

Sol.517.(d) If any 3D figure cast into another 3D figure always equate their Volume.

Volume of sphere = Volume of Cone

$$\frac{4}{3} \pi r_1^3 = \frac{1}{3} \pi r_2^2 h$$

$$\frac{4}{3} \times (13)^3 = \frac{1}{3} \times \left(\frac{13}{2}\right)^2 h$$

$$h = \frac{4 \times 13 \times 13 \times 13 \times 3 \times 2 \times 2}{3 \times 13 \times 13} = 208 \text{ cm}$$

Sol.518.(a)

Statement 1 :- volume of cuboid

$$= \text{Length} \times \text{Breadth} \times \text{height}$$

$$= 6 \times 7 \times 8 = 336 \text{ cm}^3$$

Statement 2 :- volume of cube

$$= (\text{Side})^3 = (7)^3 = 343 \text{ cm}^3$$

Clearly, the cube has more volume

Sol.519.(a) Curved surface area of hemisphere = $2\pi r^2$

$$\text{C.S.A.} = 2 \times \frac{22}{7} \times 42 \times 42 = 11088 \text{ cm}^2$$

$$\text{Cost of tin painting} = 11088 \text{ cm}^2 \times \frac{21}{100}$$

$$= 2328.48 \text{ Rs.}$$

Sol.520.(a)

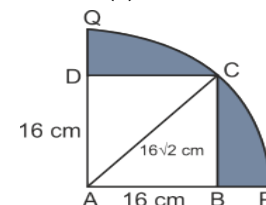
Height of the cylinder = 3.5 cm

Then radius of the cylinder = 17.5 cm

Volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times (17.5)^2 \times 3.5 = 3368.75 \text{ cm}^3$$

Sol.521.(d)



$$\text{Area of square} = (\text{Side})^2 = 16^2 = 256 \text{ cm}^2$$

$$\text{Diagonal of square} = \sqrt{2} \times \text{Side} = 16\sqrt{2} \text{ cm}$$

Here, diagonal of square is the radius for the quadrant,

$$\text{So, area of the quadrant} = \frac{1}{4} \times \pi r^2$$

$$= \frac{1}{4} \times 3.14 \times 16\sqrt{2} \times 16\sqrt{2} = 401.92 \text{ cm}^2$$

Now, Area of shaded portion

$$= \text{Area of quadrant} - \text{area of square}$$

$$= 401.92 \text{ cm}^2 - 256 \text{ cm}^2 = 145.92 \text{ cm}^2$$

$$\text{Sol.522.(b)} \text{ Volume of sphere} = \frac{4}{3} \times \pi r^3$$

$$\text{Now, } 24416.64 = \frac{4}{3} \times 3.14 \times r^3$$

$$r^3 = \frac{7776 \times 3}{4} = 5832 \Rightarrow r = 18 \text{ cm}$$

Now, surface area of sphere = $4\pi r^2$

$$= 4 \times 3.14 \times 18 \times 18 = 4069.44 \text{ cm}^2$$

Sol.523.(d) Height = 2.1 cm

According to question, radius = 6.3 cm

Volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 6.3 \times 6.3 \times 2.1 = 261.95 \text{ cm}^3$$

Sol.524.(d) If any solid object cast into another solid object always equates their Volume.

Volume of sphere = Volume of cylinder

$$\frac{4}{3} \pi r_1^3 = \pi r_2^2 h$$

$$\frac{4}{3} \times \frac{22}{7} \times 9^3 = \frac{22}{7} \times 2^2 \times h$$

$$h = \frac{729}{3} = 243 \text{ cm}$$

Sol.525.(a) Side of given cube = 5.6 cm.

For largest sphere to be inscribed in the cube,

diameter of sphere should be equal to edge length of cube,

$$\Rightarrow 2 \times \text{radius} = 5.6 \text{ cm} \Rightarrow \text{radius (r)} = 2.8$$

Volume of the sphere = $\frac{4}{3} \pi r^3$

$$= \frac{4}{3} \pi (2.8)^3 = 91.98 \text{ cm}^3$$

Sol.526.(a) Volume of hollow sphere

$$= \frac{4}{3} \pi (R^3 - r^3) = \frac{4}{3} \pi (4^3 - 1^3) = 84 \pi \text{ cm}^3$$

Sol.527.(b) Diagonal of cube = $\text{side} \sqrt{3}$

Side of cube = 6 cm.

Three cube of equal volume are joined end to end, then a cuboid is formed, having length = 18 cm, breadth = 6 cm and height = 6 cm

Now, Surface area of cuboid

$$= 2(lb + bh + hl)$$

Surface area of cuboid

$$= 2(108 + 36 + 108) = 504 \text{ cm}^2$$

Sol.528.(b) When a 3D figure is cast into another 3D figure, then their volume is equal. According to the Question,

$$\frac{\text{C.S.A. of cylinder}}{\text{T.S.A. of cylinder}} = \frac{2}{5}$$

$$\frac{2\pi rh}{2\pi r(h+r)} = \frac{2}{5}$$

$$\frac{h}{(h+r)} = \frac{2}{5} \Rightarrow r = 3x$$

Let height is 2x then radius be 3x

Again, Volume of solid hemisphere

$$= \text{volume of cylinder} = \frac{2}{3} \pi R^3 = \pi r^2 h$$

$$\frac{2}{3} \times 21 \times 21 \times 21 = 9x^2 \times 2x$$

$$x^3 = \frac{2 \times 7 \times 21 \times 21}{9 \times 2} \Rightarrow x = 7,$$

then radius = 3x = 3 × 7 = 21 cm

Sol.529.(c) Area of sector = $\frac{\theta}{360^\circ} \times \pi r^2$

$$\text{Area} = \frac{90^\circ}{360^\circ} \times 3.14 \times 10^2 = 78.5 \text{ m}^2$$

Sol.530.(c) Circumference of circle = $2\pi r$

Ratio of circumference of circle is directly proportional to ratio of radius of circle. $R : r = 5 : 4$

Width of path

= (outer radius - inner radius) = 1 unit

According to question, 1 unit = 50 m

Then, Inner radius (r) = 200 m

Sol.531.(d)Diagonal of the first square = $8\sqrt{2}$

$$\text{Side of the first square} = \frac{8\sqrt{2}}{\sqrt{2}} = 8 \text{ cm}$$

Area of the square = $8 \times 8 = 64 \text{ cm}^2$ Area of the new square = $64 \times 3 = 192 \text{ cm}^2$ Side of the new square = $\sqrt{192} = 8\sqrt{3} \text{ cm}$

Diagonal of the new square

$$= 8\sqrt{3} \times \sqrt{2} = 8\sqrt{6} \text{ cm}$$

Sol.532.(b) Given, inner radius = 6 cm,

thickness = 0.37 cm

Outer radius = $6 + 0.37 = 6.37 \text{ cm}$ Outside curved surface area = $2\pi R^2$

$$= 2 \times \frac{22}{7} \times 6.37 \times 6.37 = 255.0548 \text{ cm}^2$$

Sol.533.(d) Given, diameter of the rod

= 4 cm, radius = 2 cm and height = 12 cm

Volume of cylinder = $\pi R^2 h$

$$= 2 \times 2 \times 12 \times \pi = 48\pi$$

Now, new length = 12m = 1200 cm

$$\pi r^2 \times 1200 = 48\pi$$

$$r = \sqrt{\frac{48}{1200}} = 0.2 \text{ cm}$$

So, the thickness of the wire is 0.4 cm

Sol.534.(c) Volume of cone

$$= \frac{1}{3} \pi \times \left(\frac{42}{2}\right)^2 \times 42 = 6174\pi$$

Volume of sphere

$$= \frac{4}{3} \pi \times (24)^3 = 18,432\pi$$

The amount of wood wasted

$$= 18,432\pi - 6174\pi = 12,258\pi$$

Required % of wood wasted

$$= \frac{12,258\pi}{18,432\pi} \times 100 = 66.50\%$$

Sol.535.(a) diameter the largest sphere that can be taken out from cube will be 2.8cm

$$\text{Volume of sphere} = \frac{4}{3} \times \frac{22}{7} \times \left(\frac{2.8}{2}\right)^3$$

$$= \frac{34.496}{3} \approx 11.50 \text{ cm}^3$$

Sol.536.(c) CSA of cone = $\pi r l$

$$594 = \frac{22}{7} \times 21 \times l$$

$$594 = 66 \times l \Rightarrow l = \frac{594}{66} = 9 \text{ cm}$$

Sol.537.(a)

Let number of solid sphere be 'n'

$$n \times \frac{4}{3} \pi \times \left(\frac{6}{2}\right)^3 = \pi \times \left(\frac{4}{2}\right)^2 \times 90$$

$$n \times \frac{4}{3} \pi \times 27 = \pi \times 4 \times 90$$

$$n \times 36\pi = 360\pi \Rightarrow n = \frac{360}{36} = 10$$

Sol.538.(b) Volume of a frustum

$$= \frac{1}{3} \times \pi \times h(R^2 + r^2 + R \times r)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 7(5^2 + 3^2 + 5 \times 3)$$

$$= \frac{22}{3} (25 + 9 + 15)$$

$$= \frac{22 \times 49}{3} = \frac{1078}{3} = 359.3 \text{ cm}^3$$

Sol.539.(d)

Let a be the edge of the cubical room.

ATQ, Area of the floor = $a^2 = 64 \Rightarrow a = 8 \text{ m}$

So,

length of the longest rod = $\sqrt{3} \times 8 = 8\sqrt{3} \text{ m}$ **Sol.540.(a)**

Let the height of water raised be h

ATQ,

(Volume of conical = (Volume of cylindrical vessel)

$$\frac{1}{3} \pi \times 20 \times 20 \times 27 = \pi \times 15 \times 15 \times h$$

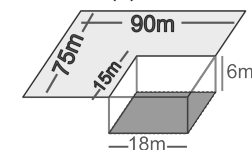
$$3600 = 225h \Rightarrow h = \frac{3600}{225} = 16 \text{ cm}$$

Sol.541.(d) CSA of cylinder = $2\pi RH$ Volume of cylinder = $\pi R^2 H$

ATQ,

$$2 \times 2\pi RH = \pi R^2 H, \Rightarrow R = 4$$

So, radius of cylinder = 4 unit

Sol.542.(b)

Let the rise in the level of the field be h

The area of field = $90 \times 75 = 6750 \text{ m}^2$ Area of pit = $18 \times 15 = 270 \text{ m}^2$

Area of the remaining part of the field

$$= 6750 - 270 = 6480 \text{ m}^2$$

ATQ, Volume of earth taken out = Volume of earth spread over remaining part of field $270 \times 6 = 6480 \times h \Rightarrow 1620 = 6480 h$

$$h = \frac{1620}{6480} = \frac{1}{4} \text{ m} = 25 \text{ cm}$$

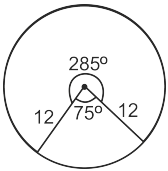
Sol.543.(d) Area of sector OAB

$$= \frac{\text{length of arc} \times \text{radius}}{2} = \frac{2r \times r}{2} = r^2$$

Sol.544.(a) TSA of hemisphere = $3\pi r^2$

$$= 3\pi \times \frac{25}{\pi} = 75 \text{ unit}^2$$

Sol.545.(b)



Perimeter of major sector

$$= \frac{285^\circ}{360^\circ} \times 2\pi \times 12 + 12 \times 2 = (19\pi + 24)m$$

Sol.546.(b)

Let the no of sphere formed be N

Volume of frustum = N × Volume of sphere

$$\frac{1}{3} \pi (R^2 + r^2 + Rr)h_1 = N \times \frac{4}{3} \pi \times R_1^3$$

$$(25^2 + 20^2 + 500)18 = 4N$$

$$(625 + 400 + 500)18 = 4N$$

$$1525 \times 18 = 4N \Rightarrow N = \frac{1525 \times 18}{4} = 6862$$

Sol.547.(c) C.S.A of hemisphere = $2\pi R^2$

$$75\% \text{ of CSA of hemisphere} = \frac{3}{4} \times 2 \times$$

$$\frac{22}{7} \times 21 \times 21 = 2079 \approx 2080 \text{ cm}^2$$

Sol.548.(a) TSA of hemisphere = $3\pi r^2$,

$$462 = 3 \times \frac{22}{7} \times r^2$$

$$r^2 = \frac{462 \times 7}{66} = 49 \Rightarrow r = 7 \text{ cm}$$

Volume of hemisphere = Volume of cone

$$\frac{2}{3} \pi \times 7^3 = \frac{1}{3} \pi \times 7^2 \times h$$

$$686 = 49h \Rightarrow h = \frac{686}{49} = 14 \text{ cm}$$

Sol.549.(b) Let the radius of each of the smaller semicircle be r

Then, the diameter of larger semicircle = 6r

The area of larger semicircle

$$= \frac{\pi}{2} \times \left(\frac{6r}{2}\right)^2 = 4.5\pi r^2$$

The area of unshaded region

$$= 3 \times \frac{1}{2} \pi \times r^2 = 1.5\pi r^2$$

$$\text{The required ratio} = 1.5\pi r^2 : \pi r^2 (4.5 - 1.5)$$

$$= 1.5\pi r^2 : 3\pi r^2 = 1 : 2$$

Sol.550.(b) Length of the longest pole

$$= \sqrt{60^2 + 30^2 + 20^2}$$

$$= \sqrt{3600 + 900 + 400} = \sqrt{4900} = 70 \text{ feet}$$

Sol.551.(c) Let the breadth of wall be x

Then, height of wall = 6x

And length of wall = 3x

ATQ,

$$3x \times x \times 6x = 23958 \Rightarrow 18x^3 = 23958$$

$$x^3 = \frac{23958}{18} = 1331 \Rightarrow x = 11 \text{ cm}$$

Sol.552.(c)

$$\text{CSA of cylinder} = 2\pi rh = 2200 \text{ ---- (1)}$$

Circumference at the base of cylinder = 2

$$\pi r = 110 \text{ ---- (2)}$$

Dividing eqn(1) and (2) we have :

$$h = \frac{2200}{110} = 20 \text{ cm}$$

Sol.553.(d)

Let the breadth of a cold storage be x

Then, length of the cold storage = 5x

The area of four walls = $2(l+b) \times h$

$$720 = 2(5x + x) \times 8$$

$$45 = 6x \Rightarrow x = \frac{45}{6} = 7.5 \text{ m}$$

Now,

$$2lb = 2 \times 5x \times x = 10x^2 = 10 \times 7.5^2 = 562.5$$

TSA of cold storage = $2(l + b) \times h + 2lb$

$$= 720 + 562.5 = 1282.5 \text{ m}^2$$

Sol.554.(a) ATQ, r = 6 cm, h = 8 cm

Slant height of cone

$$= \sqrt{6^2 + 8^2} = \sqrt{100} = 10 \text{ cm}$$

Total CSA of cones

$$= 3 \times \pi \times 6 \times 10 = 180\pi \text{ cm}^2$$

Sol.555.(c) ATQ,

$$\frac{2\pi R_1}{2\pi R_2} = \frac{5}{3} \Rightarrow \frac{R_1}{R_2} = \frac{5}{3}$$

$$\text{So, } \frac{\text{volume of cone 1}}{\text{volume of cone 2}} = \frac{\frac{1}{3} \pi R_1^2 h_1}{\frac{1}{3} \pi R_2^2 h_2}$$

$$= \frac{5^2 \times 1}{3^2 \times 5} = \frac{25}{45} = \frac{5}{9}$$

Sol.556.(a) The area of sector

$$= \frac{\text{length of arc} \times \text{radius}}{2} \Rightarrow 128 \times 2 = 64r$$

$$\Rightarrow 256 = 64r \Rightarrow r = \frac{256}{64} = 4 \text{ cm}$$

Sol.557.(d)

Volume of cone-shaped storage

$$= \frac{1}{3} \times 3.14 \times 7^2 \times 9 = 461.58 \text{ cubic feet}$$

Sol.558.(d) TSA of hall to be painted

$$= 2(l + b) \times h + l \times b$$

$$= 2(10 + 5) \times 6 + 10 \times 5 = 2 \times 15 \times 6 + 50$$

$$= 180 + 50 = 230 \text{ m}^2$$

Sol.559.(d) Volume of hemispherical

bowl = volume of cylindrical bottles

$$\frac{2}{3} \pi R^3 = n \times \pi r^2 h$$

Required number of bottles

$$= \frac{2 \times 6 \times 6 \times 6}{3 \times 1 \times 1 \times 4} = 36 \text{ bottles}$$

Sol.560.(c)

Given, s - a = 18, s - b = 17 and s - c = 25

$$S = \frac{a + b + c}{2} \Rightarrow a + b + c = 2s$$

Now, s - a + s - b + s - c = 60

$$3s - (a + b + c) = 60$$

$$3s - 2s = 60 \Rightarrow s = 60$$

So, area of the triangle

$$= \sqrt{s(s - a)(s - b)(s - c)}$$

$$= \sqrt{60 \times 18 \times 17 \times 25} = 30\sqrt{510} \text{ cm}^2$$

Sol.561.(c)

$$\text{CSA} + \text{TSA of cylinder} = 2068 \text{ cm}^2$$

$$2\pi rh + (2\pi rh + 2\pi r^2) = 2068$$

$$2\pi r(2h + r) = 2068$$

$$2 \times \frac{22}{7} \times 7(2h + 7) = 2068$$

$$2h + 7 = \frac{2068}{44} = 47$$

$$2h = 40 \Rightarrow h = 20 \text{ cm}$$

So, the volume of cylinder

$$= \frac{22}{7} \times 7^2 \times 20 = 3080 \text{ cm}^3$$

Sol.562.(d) Slant height l = 20 cm

According to the question,

$$\text{Area of base of cone} = 616 \text{ cm}^2$$

$$\pi r^2 = 616 \Rightarrow \frac{22}{7} \times r^2 = 616$$

$$r^2 = 196 \Rightarrow r = 14 \text{ cm}$$

So, CSA of cone = $\pi r l$

$$= \frac{22}{7} \times 14 \times 20 = 880 \text{ cm}^2$$

Sol.563.(d) Given that, r = 14 cm

According to the question,

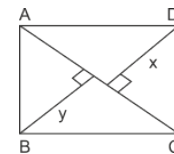
$$\text{Height } h = r + 6 = 14 + 6 = 20 \text{ cm}$$

So, volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 14^2 \times 20 = 12320 \text{ cm}^3$$

Sol.564.(c) Let the length of the perpendicular drawn from point D to AC and point B to AC be x and y respectively.

According to the question ,



So, area of quadrilateral

$$= \text{Area of } \triangle ABC + \text{Area of } \triangle ADC$$

$$= \frac{1}{2} \times AC \times y + \frac{1}{2} \times AC \times x$$

$$= \frac{1}{2} \times AC \times (x + y)$$

$$= \frac{1}{2} \times 24 \times (10) = 120 \text{ cm}^2$$

Sol.565.(c) Let the radius of the inner and outer circle be r and R respectively.

According to the question,

Difference between the areas of two concentric circles = 264 cm^2

$$\pi R^2 - \pi r^2 = 264 \Rightarrow \pi(R^2 - r^2) = 264$$

$$\frac{22}{7} \times (R^2 - r^2) = 264$$

$$R^2 - r^2 = 84 \text{ cm}^2$$

So, option (c) is the correct answer.

Sol.566.(b) According to the question,

$$\text{Volume of cone} = 400 \text{ cm}^3$$

Volume of cone with radius r and height

$$h = \frac{1}{3} \pi r^2 h = 400$$

$$\frac{1}{3} \times \frac{22}{7} \times r^2 h = 400 \Rightarrow r^2 h = \frac{4200}{11}$$

Volume of cone with radius $2r$ and height

$$h = \frac{1}{3} \pi (2r)^2 h = \frac{4}{3} \pi r^2 h$$

$$= \frac{4}{3} \times \frac{22}{7} \times \frac{4200}{11} = 1600 \text{ cm}^3$$

Sol.567.(d)

Given, radius of the cone = 4 cm,

C.S.A of cone = $\pi r l$

$$\pi r l = 20\pi \Rightarrow l = \frac{20}{4} = 5 \text{ cm}$$

$$\text{Now, height of the cone} = \sqrt{(l)^2 - (r)^2}$$

$$= \sqrt{25 - 16} = 3 \text{ cm}$$

$$\text{Volume of the conical vessel} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \pi \times 4 \times 4 \times 3 = 16\pi \text{ cm}^3$$

$$\text{Sol.568.(c) Area of rhombus} = \frac{1}{2} \times d_1 \times d_2$$

$$= \frac{1}{2} \times 6 \times 8 = 24 \text{ cm}^2$$

Sol.569.(c) Volume of a cube = a^3

Where a = side of a cube

$$\frac{\text{Side of smaller cube}}{\text{Side of larger cube}} =$$

$$\sqrt[3]{\frac{\text{Volume of smaller cube}}{\text{volume of larger cube}}}$$

$$\frac{\text{Side of smaller cube}}{\text{Side of larger cube}} = \sqrt[3]{\frac{8}{27}} = \frac{2}{3}$$

Then, side of smaller cube = $2x$

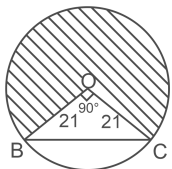
and side of larger cube = $3x$

According to question,

$$27x^3 - 8x^3 = 513 \text{ cm}^3 \Rightarrow 19x^3 = 513$$

$$x^3 = 27 \Rightarrow x = 3$$

So, Side of smaller cube = $2 \times 3 = 6 \text{ cm}$

Sol.570.(c)

Area of minor sector(OBC)

$$= \frac{90}{360} \times \frac{22}{7} \times 21 \times 21 = 346.5 \text{ cm}^2$$

Area of $\triangle BOC$

$$= \frac{1}{2} \times 21 \times 21 \times \sin 90^\circ = 220.5 \text{ cm}^2$$

Area of minor segment

$$= 346.5 - 220.5 = 126 \text{ cm}^2$$

Now, area of major segment = area of circle - area of minor segment

$$= \frac{22}{7} \times 21^2 - 126 = 1386 - 126 = 1260 \text{ cm}^2$$

Short Trick:- Area of major segment :- area of major sector + area of triangle

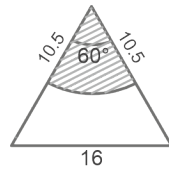
$$\text{Required area} :- \frac{270}{360} \times \frac{22}{7} \times 21 \times 21 +$$

$$\frac{1}{2} \times 21 \times 21 \times \sin 90^\circ$$

$$\text{Required area} :- \frac{21}{2} (99 + 21)$$

$$= 60 \times 21 = 1260 \text{ cm}^2$$

Sol.571.(c) As it is given that plot is in the shape of an equilateral triangle then, the grazed area is part of circle of radius 10.5m. so, angle is 60°



Hence, required area

$$= \frac{60^\circ}{360^\circ} \times \frac{22}{7} \times 10.5 \times 10.5$$

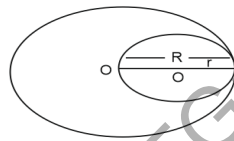
$$= \frac{2425.5}{42} = 57.75 \text{ m}^2$$

Sol.572.(d) Let the radius and height of the cone be r and h respectively.

ATQ,

$$\frac{\text{New volume}}{\text{Original volume}} = \frac{\frac{1}{3} \pi \times 4r^2 \times 2h}{\frac{1}{3} \pi \times r^2 \times h} = 8$$

So, the volume of the cone becomes 8 times its previous volume.

Sol.573.(a)

Area of the smaller circle = 192

$$\pi r^2 = 192 \Rightarrow \pi \times \left(\frac{R}{2}\right)^2 = 192$$

$$\pi R^2 = 192 \times 4 = 768$$

So, area of the bigger circle = 768 cm^2

Sol.574.(d) Let the length of the rectangle be L cm and the side of the square be a cm.

$$\text{Area of square} = 400 \text{ cm}^2$$

$$a^2 = 400 \Rightarrow a = 20 \text{ cm}$$

ATQ, Perimeter of square = perimeter of rectangle

$$4a = 2(L + 12) \Rightarrow 4 \times 20 = 2(L + 12)$$

$$40 = L + 12 \Rightarrow L = 40 - 12 = 28$$

So, the area of rectangle

$$= 28 \times 12 = 336 \text{ cm}^2$$

Sol.575.(d) ATQ,

TSA of Sphere = TSA of cube

$$4\pi r^2 = 6s^2 \Rightarrow 4 \times \frac{22}{7} \times r^2 = 6s^2$$

$$\frac{r^2}{s^2} = \frac{6 \times 7}{4 \times 22} = \frac{42}{88} = \frac{21}{44}$$

So, we can see that $r < s$

Sol.576.(c) The ratio of length, breadth and height of the wall = $3 : 1 : 6$,

Let the length, breadth and height of the

wall be $3x$, x and $6x$ respectively

ATQ,

The volume of wall = 23958

$$6x \times 3x \times x = 23958$$

$$18x^3 = 23958 \Rightarrow x^3 = \frac{23958}{18} = 1331$$

$$x = \sqrt[3]{1331} = 11 \text{ cm}$$

Sol.577.(b)

Diameter of a wire = 1 mm = 0.1 cm

$$\text{Radius of a wire} = \frac{0.1}{2} = 0.05 \text{ cm}$$

Let the no of gold chain required be n

ATQ,

Volume of wire $\times n$ = Volume of gold

$$\frac{22}{7} \times 0.05 \times 0.05 \times 56 \times n = 110$$

$$n = \frac{110 \times 7}{22 \times 0.05 \times 0.05 \times 56} = 250$$

Sol.578.(d) The outer dimensions of a closed metal box is 15 cm, 18 cm, 8 cm

Thickness of the metal = 1.5 cm

So, the inner dimension of the metal box-

$$= 15 - 1.5 \times 2 = 15 - 3 = 12 \text{ cm},$$

$$= 18 - 1.5 \times 2 = 18 - 3 = 15 \text{ cm},$$

$$= 8 - 1.5 \times 2 = 8 - 3 = 5 \text{ cm}.$$

So, the required volume of the metal

$$= 15 \times 18 \times 8 - (12 \times 15 \times 5)$$

$$= 2160 - 900 = 1260 \text{ cm}^3$$

Sol.579.(b)

Perimeter of an equilateral triangle = $3a$

Length of equilateral triangle = a

ATQ,

$$3a - a = 120 \Rightarrow 2a = 120 \Rightarrow a = 60 \text{ cm}$$

So, the area of equilateral triangle -

$$= \frac{\sqrt{3}}{4} \times 60 \times 60 = 900\sqrt{3} \text{ cm}^2$$

Sol.580.(b) It is given that,

$$L + B = 37 - (1), \quad L - B = 13 - (2)$$

Adding eqn. (1) and (2), we have :

$$L = \frac{37 + 13}{2} = \frac{50}{2} = 25 \text{ cm}, \quad B = 12 \text{ cm}$$

ATQ, Volume of cuboid = $25 \times 12 \times H$

$$3300 = 300H \Rightarrow H = \frac{3300}{300} = 11 \text{ cm}$$

Sol.581.(a) Area of the rectangle -

$$= 40 \times 35 = 1400 \text{ cm}^2$$

We can cut the biggest square by taking

side of breadth of rectangle

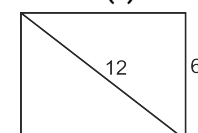
Breadth of rectangle = 35cm

Then area of the square -

$$= 35 \times 35 = 1225 \text{ cm}^2$$

Area of remaining part-

$$= 1400 - 1225 = 175 \text{ cm}^2$$

Sol.582.(c) In rectangle

$$\text{Another side} = \sqrt{12^2 - 6^2} = \sqrt{144 - 36}$$



$$= \sqrt{108} = 6\sqrt{3}$$

$$\text{Area of rectangle} = l \times b = 6\sqrt{3} \times 6 = 36\sqrt{3}$$

Sol.583.(c) Sphere is melted and recast into pyramid

Then the volume of sphere = volume of pyramid

$$\frac{4}{3}\pi r^3 = \frac{1}{3} \times (\text{area of base}) \times \text{height}$$

Let side of pyramid having square base = a

$$4 \times \frac{22}{7} \times 21^3 = a^2 \times 66$$

$$\{\text{area of square} = (\text{side})^2\}$$

$$a^2 = 4 \times \frac{22}{66 \times 7} \times 21 \times 21 \times 21$$

$$a^2 = 4 \times 21 \times 21$$

$$\text{Therefore, } a = \sqrt{21 \times 21 \times 4} = 42 \text{ cm}$$

Sol.584.(a) According to the question, The total surface area of cube having 8 cm side = area of a square + 128 cm²

Let side of square is, X

$$6 \times 8 \times 8 = X^2 + 128 \Rightarrow X^2 + 128 = 384$$

$$X^2 = 256 \Rightarrow X = \sqrt{256} \Rightarrow 16 \text{ cm}$$

Sol.585.(d) According to the question, height of cylinder = 4 m

$$\text{Radius of cone and cylinder}(r) = \frac{10.5}{2} \text{ m}$$

And slant height of cone (l) = 10 m

Canvas used for all tents = 3861 cm²

Let number of tents = N,

$$= N \times 3861 = N \times [\pi r(l + 2h)] = 3861$$

$$= N \times \left[\frac{22}{7} \times \frac{10.5}{2} (10 + 8) \right] = 3861$$

$$= N \times \frac{22}{7} \times \frac{10.5}{2} \times 18 = 3861 \Rightarrow N = 13$$

Sol.586.(a)

Area of a rectangle = l × b = 480

Perimeter of rectangle = 2(l + b) = 92

$$l + b = 46 = 30 + 16$$

$$l \times b = 480 = 30 \times 16$$

So clearly length is 30 cm and breadth is 16 cm

Now diagonal of a rectangle

$$= \sqrt{l^2 + b^2} = \sqrt{30^2 + 16^2}$$

$$= \sqrt{1156} = 34 \text{ cm}$$

Sol.587.(a) According to question,

Circumference of circle = perimeter of square,

$$\Rightarrow 2\pi r = 4a \Rightarrow r = \frac{2a}{\pi}$$

$$\text{Area of circle} = \pi r^2$$

$$= \pi \times \left(\frac{2a}{\pi} \right)^2 = \frac{4a^2}{\pi} = \frac{14}{11} \times a^2$$

$$\text{Area of square} = a^2$$

Hence, we can see that the area of a circle is more than the area of a square.

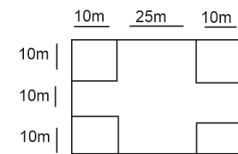
Sol.588.(d) If length is 2x then breadth is x

Perimeter of rectangle = 2(l + b) = 54

$$2(2x + x) = 54 \Rightarrow x = 9$$

$$\text{Length} = 2x = 2 \times 9 = 18 \text{ cm}$$

Sol.589.(d)



When the 4 square shaped cuts are made in rectangular sheet then

Length = 25m, Breadth = 10m,

Height = 10m

$$\text{Volume} = l \times b \times h = 25 \text{ m} \times 10 \text{ m} \times 10 \text{ m}$$

$$= 2500 \text{ m}^3$$

$$\text{Sol.590.(a)} \text{ Side of the cube} = \frac{50\sqrt{3}}{\sqrt{3}} = 50$$

$$\text{Total surface area of cube} = 6 \text{ side}^2$$

$$= 6 \times 50 \times 50 = 15000 \text{ cm}^2$$

Sol.591.(c) Let the smaller diagonal be x

$$\text{and larger diagonal be} = \frac{150}{100} x \Rightarrow \frac{3}{2} x$$

$$432 = \frac{1}{2} \times \frac{3}{2} x^2$$

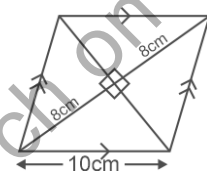
$$x = \sqrt{\frac{432 \times 4}{3}} = 24 \text{ cm}$$

So, d₁ = 24 cm and d₂ = 36 cm

Side of the rhombus

$$= \sqrt{12^2 + 18^2} = 6\sqrt{13}$$

Sol.592.(d)

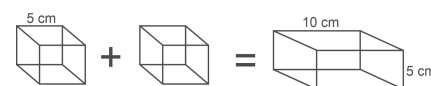


$$D_2 = 2 \times 6 = 12 \text{ cm}$$

$$\text{Area of rhombus} = \frac{1}{2} \times 16 \times 12 = 96 \text{ cm}^2$$

Sol.593.(b) T.S.A of cube (6side²) = 150

side = 5 cm



Now,

$$\text{T.S.A of cuboid} = 2(10 \times 5 + 5 \times 5 + 10 \times 5)$$

$$= 2(50 + 25 + 50) = 250 \text{ cm}^2$$

Sol.594.(b) According to the question,

Radius of circle = 84 cm

Circumference of circle = perimeter of an equilateral triangle.

$$2\pi r = 3 \times a \quad (\text{let side of triangle} = a)$$

$$2 \times \frac{22}{7} \times 84 = 3 \times a$$

$$3 \times a = 2 \times 22 \times 12$$

$$a = 2 \times 22 \times 4 = 176 \text{ cm}$$

Sol.595.(a) Height of a cylinder = 10 cm

According to the question,

$$\text{radius} = 10 - 3 = 7 \text{ cm}$$

$$\text{So, volume of cylinder} = \pi r^2 h$$

$$\frac{22}{7} \times 7 \times 7 \times 10 \Rightarrow 22 \times 7 \times 10 = 1540 \text{ cm}^3$$

Sol.596.(d) Area of large field = area of three smaller field

$$\Rightarrow \pi R^2 = \pi 8^2 + \pi 9^2 + \pi 12^2$$

$$\Rightarrow \pi R^2 = \pi(8^2 + 9^2 + 12^2)$$

$$\Rightarrow R^2 = 64 + 81 + 144$$

$$\Rightarrow \text{Radius (R)} = \sqrt{289} \Rightarrow 17 \text{ m}$$

Sol.597.(c)

Radius of common base r = 15 cm

$$\text{Volume of toy} = \frac{2}{3} \pi r^3 + \frac{1}{3} \pi r^2 h$$

$$\Rightarrow 2850 \pi = \frac{1}{3} \pi r^2 (2r + h)$$

$$\Rightarrow \frac{1}{3} \times 15 \times 15 \times (30 + h) = 2850$$

$$\Rightarrow (30 + h) = 38 \Rightarrow h = 8$$

Hence, by Pythagoras

$$L(\text{Slant Height}) = \sqrt{15^2 + 8^2} = 17 \text{ cm}$$

Therefore,

$$\text{Surface area of toy} = 2\pi r^2 + \pi r l$$

$$\Rightarrow \pi r(2r + l) \Rightarrow \pi \times 15 \times (30 + 17) = 705\pi \text{ cm}^2$$

Sol.598.(c) According to the question,

Side of the large cube = 12 cm

Radius of small spheres = 3 cm

$$\text{So} \Rightarrow 12 \times 12 \times 12$$

$$= N \times \frac{4}{3} \pi r^3 + (N + 2) 4 \times 4 \times 6.5$$

$$= N \times \frac{4}{3} \pi r^3 + (N + 2) 4 \times 4 \times 6.5 = 1728$$

$$= N \times \frac{4}{3} \times \frac{22}{7} \times 3 \times 3 \times 3 + 104N + 208$$

$$= 1728$$

$$N \left[\frac{792}{7} + 104 \right] = 1728 - 208$$

$$N \left[\frac{792 + 728}{7} \right] = 1520$$

$$N \left[\frac{1520}{7} \right] = 1520 \Rightarrow N = 7$$

Sol.599.(c) Area of trapezium

$$= \frac{1}{2} (68 \times 75) = 2550 \text{ cm}^2$$

$$\text{Area of square} = \frac{2550 \times 17}{6} = 7225 \text{ cm}^2$$

$$\text{Side of square} = \sqrt{7225} = 85 \text{ cm}$$

$$\text{Diagonal of square} = \sqrt{2} \times 85 = 119.85 \text{ cm}$$

Sol.600.(d) D = 2R = 7 cm

Hence, R = 3.5 cm

$$\text{Total surface area} = 3\pi R^2$$

$$3 \times \frac{22}{7} \times 3.5 \times 3.5 = 115.5 \text{ cm}^2$$

Sol.601.(c) Let the length and width of the rectangle be 3x, 2x

$$\text{Area of rectangle} = 6x^2$$

According to question,

$$\text{Now, } \left(\frac{125}{100} \times 3x \right) \times 2x - 6x^2 = 24$$

$$\frac{15}{2}x^2 - 6x^2 = 24 \Rightarrow x = \sqrt{\frac{24 \times 2}{3}} = 4$$

Width of the rectangle = 8 m

Sol.602.(c) ratio of an external angle and an internal angle = 1 : 5

As per question,

$$x + 5x = 180 \Rightarrow x = 30$$

$$\text{Number of sides} = \frac{360}{30} = 12$$

Sol.603.(c) Area of a square = 1296

$$a^2 = 1296 \Rightarrow a = 36$$

$$\text{Perimeter of square} = 4a = 4 \times 36 = 144$$

$$\text{Radius} = 36 \times \frac{7}{6} = 42$$

Circumference of circle -

$$= 2 \times \frac{22}{7} \times 42 = 264$$

$$\text{Required ratio} = 144 : 264 = 6 : 11$$

Sol.604.(b) According to question,

$$\frac{2}{3} = \frac{4 \times 3 \times H_1}{5 \times 4 \times H_2} \Rightarrow H_1 : H_2 = 10 : 9$$

Sol.605.(d)

Let breadth of the rectangle = 4x

Radius of the circle = 5x

Side of the square = 25x

$$\text{Area} = 625 \Rightarrow (25x)^2 = 625 \Rightarrow x = 1$$

So,

$$\text{breadth of the rectangle} = 4x = 4$$

$$\text{Length} = 20$$

$$= \text{Area of rectangle} = 4 \times 20 = 80 \text{ cm}^2$$

Sol.606.(a)

Let the diameter of circle = 7x

Base of triangle = 4x

$$\text{Area of the triangle} = \frac{1}{2} \times b \times h$$

$$= 56 = \frac{1}{2} \times 4x \times 14 \Rightarrow x = 2$$

$$\text{Diameter} = 7x = 14 \text{ cm} \Rightarrow \text{Radius} = 7 \text{ cm}$$

Circumference of circle -

$$= 2 \times \frac{22}{7} \times 7 = 44 \text{ cm or } 0.44 \text{ m}$$

Sol.607.(c)

Given length of side of square = p

$$\text{So, length of rectangle} = \frac{p}{2}$$

$$\text{Breadth of rectangle} = \frac{p}{6}$$

Now, required area = area of square -

$$\text{area of rectangle} = p^2 - \frac{p^2}{12} = \frac{11p^2}{12}$$

Sol.608.(d) Desired area

$$= \frac{22}{7} \times [(9)^2 - (5)^2] = 176 \text{ m}^2$$

Sol.609.(d)

Since, Pillar is cylindrical in shape.

So, to reach the required answer we have to find the volume of the cylinder.

$$\text{Volume of cylinder (pillar)} = \pi r^2 h$$

$$= \frac{22}{7} \times 11 \times 11 \times 21 = 7986 \text{ m}^3$$

Sol.610.(a) Diagonal of cube = diameter of sphere = $14\sqrt{3}$ cm

$$\text{Radius of sphere} = \frac{14\sqrt{3}}{2} = 7\sqrt{3} \text{ cm}$$

$$\text{Volume of Sphere} = \frac{4}{3} \times \frac{22}{7} \times 7\sqrt{3} \times$$

$$7\sqrt{3} \times 7\sqrt{3} = 4312\sqrt{3} \text{ cm}^3$$

Sol.611.(c) $1 \text{ m}^3 = 1000 \text{ litre}$

$$\text{Volume of tank} = 1.5 \times 3 \times 3.5 \times 1000 = 15750 \text{ litre.}$$

Sol.612.(a) Edge of cube = 10 cm

Height of the largest cone = Edge of the cone = 10 cm

$$\text{Radius of the largest cone} = \frac{10}{2} = 5 \text{ cm}$$

Volume of the largest cone

$$= \frac{1}{3} \times \pi \times 5 \times 5 \times 10 = \frac{250\pi}{3} \text{ cm}^3$$

Sol.613.(c) Volume of 15 spherical ball

$$= 15 \times \frac{4}{3} \times \pi \times 5^3 = 2500\pi \text{ cm}^3$$

Since 35% of the mass wasted,

So the volume of sphere

$$= 2500\pi \times \frac{65}{100} = 1625\pi \text{ cm}^3$$

Let the radius of new formed sphere = r

Volume of big Sphere = volume of new

$$\text{formed sphere} \Rightarrow 1625\pi = \frac{4}{3} \times \pi \times r^3$$

$$r^3 = \frac{1625 \times 3}{4} = \frac{4875}{4}$$

$$r = \sqrt[3]{1218.75} = 10.68 \text{ cm}$$

Sol.614.(c)

$$\text{Volume of sphere} = r^3 \left(\frac{4}{3}\pi \text{ is constant} \right)$$

$$\text{Original Volume} \quad 343 = 7^3$$

$$\text{New Volume} \quad 2744 = 14^3$$

$$\text{Required ratio} = 343 : 2744 = 1 : 8$$

Sol.615.(a)

Volume of cylinder = Volume of cone

$$\pi \times 15^2 \times 12 = \frac{1}{3} \times \pi \times 24^2 \times H$$

$$H = \frac{225 \times 36}{576} = \frac{8100}{576} = 14.0625 \text{ cm}$$

Sol.616.(c) According to the question,

Percentage contribution of hemisphere in total volume of toy = $100 - (50 + 20)$

$$= 100 - 70 = 30\%$$

$$\text{So, required ratio} = 20 : 50 : 30 = 2 : 5 : 3$$

Sol.617.(c) Circumference of the base of cylinder = $2\pi r$

$$\Rightarrow 4\pi = 2\pi r \Rightarrow r = 2 \text{ cm}$$

$$\text{CSA of cylinder} = 2\pi rh,$$

$$12\pi = 2 \times \pi \times 2 \times h \Rightarrow h = \frac{12}{4} = 3 \text{ cm}$$

$$\text{Volume of cylinder} = \pi \times 2 \times 2 \times 3 = 12\pi \text{ cm}^3$$

Sol.618.(b) $L + B + H = 23 \text{ cm}$ (Given)

Length of the diagonal of cuboid

$$= \sqrt{L^2 + B^2 + H^2}$$

$$\Rightarrow 5\sqrt{7} = \sqrt{L^2 + B^2 + H^2}$$

$$\Rightarrow 175 = L^2 + B^2 + H^2$$

$$\Rightarrow (L + B + H)^2 = L^2 + B^2 + H^2 + 2(LB + B$$

$$H + HL)$$

$$\Rightarrow 23^2 = 175 + 2(LB + BH + HL)$$

$$\Rightarrow 529 = 175 + 2(LB + BH + HL)$$

$$\Rightarrow 354 = 2(LB + BH + HL)$$

$$\text{So, TSA of cuboid} = 354 \text{ cm}^2$$

Sol.619.(c)

CSA of cone-shaped temple = πrl

$$= \frac{22}{7} \times 11 \times 10 = \frac{2420}{7} \text{ m}^2$$

Total cost for painting its CSA

$$= 7 \times \frac{2420}{7} = 2420 \text{ Rs.}$$

Sol.620.(b)

Let the base of the triangle = 24cm

$$\text{Area of } \Delta = \frac{1}{2} \times 24 \times \text{height}$$

$$96 = 12 \times \text{height} \Rightarrow H = \frac{96}{12} = 8 \text{ cm}$$

Sol.621.(a) According to the question,

$$\text{Length} = \text{Breadth} + 2(\text{Breadth})$$

$$= 8 + 2(8) = 8 + 16 = 24 \text{ cm}$$

$$\text{Breadth} = 2(\text{Height})$$

$$8 = 2(\text{Height}), \text{ Height} = 4 \text{ cm}$$

$$\text{Volume of cuboid} = \text{Length} \times \text{Breadth} \times$$

$$\text{Height} = 24 \times 8 \times 4 = 768 \text{ cm}^3$$

Sol.622.(b)

$$\text{Diagonal of cuboid} = \sqrt{l^2 + b^2 + h^2}$$

$$= \sqrt{20^2 + 5^2 + 4^2} = \sqrt{441} = 21 \text{ m}$$

Sol.623.(a) Volume of Hemisphere

$$= \frac{2}{3} \times \frac{22}{7} \times 7 \times 7 \times 7$$

$$= \frac{2156}{3} = 718 \frac{2}{3} \text{ cm}^3$$

Sol.624.(d) Length of each side of hexagon = 10 cm

Area of Hexagon = 6 × area of equilateral triangle

$$\text{Area of Hexagon} = 6 \times \frac{\sqrt{3}}{4} \times 10 \times 10$$

$$= \frac{600\sqrt{3}}{4} = 150\sqrt{3} \text{ cm}^2$$

Sol.625.(a)

$$\text{Area of square} = 9m^2 + 12m + 4$$

$$= (3m)^2 + 2 \times 3m \times 2 + 2^2 \Rightarrow (3m + 2)^2$$

$$\text{Side of square} = \sqrt{(3m + 2)^2} = 3m + 2$$

Sol.626.(b) Area of square = $4x^2 - 12x + 9$

$$= (2x)^2 - 2 \times 2x \times 3 + 3^2 = (2x - 3)^2$$

$$\text{Side of the square} = \sqrt{(2x - 3)^2} = 2x - 3$$

Sol.627.(a) $L \times B = 729 \text{ cm}^2$, $B \times H = 529 \text{ cm}^2$, $H \times L = 289 \text{ cm}^2$
 $L^2 \times B^2 \times H^2 = 729 \times 529 \times 289$
 $L \times B \times H = \sqrt{729 \times 529 \times 289}$
 $= 27 \times 23 \times 17 = 10557 \text{ cm}^3$

Sol.628.(c) TSA of cone = $\pi r (r + l)$
 $= \frac{22}{7} \times \frac{10}{2} \left(\frac{10}{2} + 16 \right)$
 $= \frac{22}{7} \times 5 \times 21 = 330 \text{ cm}^2$

Sol.629.(b)

Diagonal of rhombus(D_1) = 8cm

Area of rhombus = $\frac{1}{2} \times 8 \times D_2$

$48 = 4 \times D_2 \Rightarrow D_2 = \frac{48}{4} = 12 \text{ cm}$

Now, side of rhombus

$= \sqrt{\left(\frac{8}{2}\right)^2 + \left(\frac{12}{2}\right)^2} = \sqrt{16 + 36}$
 $= \sqrt{52} = 2\sqrt{13} \text{ cm}$

Sol.630.(c) TSA of cylinder = $2\pi r (r + h)$
 $= 2\pi r \left(r + \frac{r}{2} \right) = 2\pi r \times \frac{3r}{2} = 3\pi r^2$

Sol.631.(b) Circumference of the base of hemispherical bowl = 132 cm

$2 \times \frac{22}{7} \times r = 132 \Rightarrow \frac{44}{7} \times r = 132$

$r = \frac{132 \times 7}{44} = 21 \text{ cm}$

The capacity of the bowl

$= \frac{2}{3} \times \frac{22}{7} \times 21 \times 21 \times 21 = 19404 \text{ cm}^3$

Sol.632.(a)

Radius of cylinder = $\frac{14}{2} = 7 \text{ cm}$

Diameter of sphere = height of cylinder
 ... (Since cylinder is circumscribing a sphere)

So, volume of cylinder

$= \frac{22}{7} \times 7^2 \times 14 = 2156 \text{ cm}^3$

Sol.633.(c) Surface area of spherical volleyball = 100π

$4\pi r^2 = 100\pi \Rightarrow r^2 = 25 \Rightarrow r = 5$

Volume of air

$= \frac{4}{3} \times \pi \times 5 \times 5 \times 5 = \frac{500}{3}\pi$

Sol.634.(d) Length of diagonal of cuboid

$= \sqrt{13^2 + 7^2 + 3^2} = \sqrt{169 + 49 + 9}$
 $= \sqrt{227} \text{ m}$

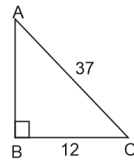
Sol.635.(d) Area of playground = CSA of roller $\times$ Number of complete revolutions

$= 2\pi r h \times 235$

$= 2 \times \frac{22}{7} \times 14 \times 20 \times 235$

$= 413600 = 4136 \times 10^2 \text{ cm}^2$

Sol.636.(d) According to the question,

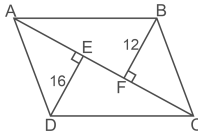


$AB^2 = \sqrt{AC^2 - BC^2} = \sqrt{37^2 - 12^2}$
 $= \sqrt{1369 - 144} = \sqrt{1225} = 35 \text{ cm}$

Area of triangle = $\frac{1}{2} \times AB \times BC$

$= \frac{1}{2} \times 35 \times 12 = 210 \text{ cm}^2$

Sol.637.(d)



Area of ABCD

= area of ADC + area of ABC

$336 = \frac{1}{2} \times AC \times DE + \frac{1}{2} \times AC \times BF$

$336 = \frac{1}{2} \times AC (DE + BF)$

$672 = AC(16 + 12)$

$672 = AC \times 28 \Rightarrow AC = 24 \text{ cm}$

Sol.638.(d)

Let the number of spherical balls be n.

According to the question,

Volume of n spherical balls

= Volume of spherical clay ball

$n \times \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3 \Rightarrow n \times r^3 = R^3$

$n \times 5^3 = 15^3 \Rightarrow n = 3^3 = 27$

Sol.639.(c) Total surface area = $2\pi r(h + r)$

$60\pi = 2 \times \pi \times 3(h + 3)$

$10 = h + 3 \Rightarrow h = 10 - 3 = 7 \text{ cm}$

Sol.640.(d) Let,

the rise of water level = a cm

According to the question,

Volume of cube = Length of rectangle $\times$

Breadth of rectangle $\times$ a

$= 8 \times 8 \times 8 = 16 \times 8 \times a \Rightarrow a = 4 \text{ cm}$

Sol.641.(d)

Let, the rise of water level = a

According to the question,

Volume of cube = Length of rectangle $\times$

Breadth of rectangle $\times$ a

$8 \times 8 \times 8 = 16 \times 8 \times a \Rightarrow h = 4 \text{ cm}$

Sol.642.(c) Required difference = Volume of sphere - Volume of cone

$= \frac{4}{3} \pi r^3 - \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi r^2 (4r - h)$

$= \frac{1}{3} \times \frac{22}{7} \times 7^2 \times (4 \times 7 - 10)$

$= \frac{1}{3} \times 154 \times (18) = 924 \text{ cm}^3$

Sol.643.(b) Length of square

$= \frac{\text{Diagonal}}{\sqrt{2}} = \frac{(a + b)}{\sqrt{2}}$

So, Area of square = $\left[\frac{(a + b)}{\sqrt{2}} \right]^2$

$= \frac{a^2 + b^2 + 2ab}{2} = \frac{1}{2}(a^2 + b^2) + ab$

Sol.644.(b) According to the question,

$y = 14x$

Volume of sphere = $14 \times$ surface area of sphere

$\frac{4}{3} \pi r^3 = 14 \times 4\pi r^2$

$\frac{1}{3} \times r = 14 \Rightarrow r = 42 \text{ cm}$

Sol.645.(c) Curved surface area of the hemisphere = $2\pi r^2$

Base area of the hemisphere = πr^2

Effective area increase on each face of the cube = $2\pi r^2 - \pi r^2 = \pi r^2$

Area of each face of the cube

$= (10)^2 + \pi r^2 = 100 + \frac{22}{7} \times 2^2 = 112 \frac{4}{7} \text{ cm}^2$

Total surface area of the cube

$= 6 \times 112 \frac{4}{7} = 675 \frac{3}{7} \text{ cm}^2$

Sol.646.(a) Here, length (l) = 20 cm,

breadth (b) = 10 cm and height (h) = 8 cm

Area of a cardboard needed

$= 2(lb + bh + hl)$

$= 2(20 \times 10 + 10 \times 8 + 8 \times 20)$

$= 2(200 + 80 + 160) = 2(440) \Rightarrow 880 \text{ cm}^2$

Sol.647.(b) Volume of cube = volume of sphere + remaining part volume

$a^3 = \frac{4}{3} \pi r^3 + 163 \frac{1}{3}$

$7^3 = \frac{4}{3} \times \frac{22}{7} \times r^3 + \frac{490}{3}$

$343 - \frac{490}{3} = \frac{4}{3} \times \frac{22}{7} \times r^3$

$\frac{539}{3} = \frac{4}{3} \times \frac{22}{7} \times r^3$

$r^3 = \frac{7 \times 7 \times 7}{2 \times 2 \times 2} \Rightarrow r = \frac{7}{2} \Rightarrow d = 2r = 7 \text{ cm}$

Sol.648.(b) When we melt and convert any solid then volume of both the solids remains same. $R = 3 \text{ cm} =$ radius of sphere

$r = 0.1 \text{ mm} = \frac{1}{100} \text{ cm} =$ radius of wire

h = length of wire

$\frac{4}{3} \pi R^3 = \pi r^2 h$

$= \frac{4}{3} \pi \times 3 \times 3 \times 3 = \pi \frac{1}{10000} h$

$h = 360000 \text{ cm}$

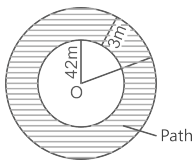
Sol.649.(c) Take inner radius = r

Outer radius = $R = (r + 3)$

According to the question

$2\pi r = 264 \text{ m} \Rightarrow r = \frac{264 \times 7}{22 \times 2} = 42 \text{ m}$

$R = (r + 3) = 42 + 3 = 45 \text{ m}$

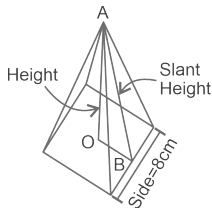


Area of path = $\pi(R^2 - r^2)$

$$\frac{22}{7}(R + r)(R - r)$$

$$= \frac{22}{7}(87\text{m} \times 3\text{m}) = 820.28 \approx 820 \text{ m}^2$$

Sol.650.(c)



Total surface area of square based pyramid = Surface area of base + 4 × surface area of wall

$$8^2 + 4 \times \frac{1}{2} \times 8 \times AB = 208 \text{ cm}^2$$

$$AB = \text{Slant height} = \frac{208 - 64}{16} = 9 \text{ cm}$$

Sol.651.(b) Upper radius = $\frac{21}{2} = 10.5$

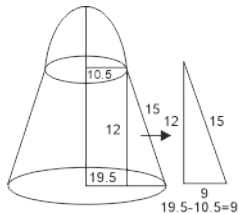
$$\text{Lower radius} = \frac{39}{2} = 19.5$$

Height of frustum = 22.5 - radius of hemisphere

$$\text{Height of frustum} = 22.5 - 10.5 = 12$$

Now by pythagoras theorem

$$\text{Slant height (l)} = \sqrt{12^2 + 9^2} = 15$$



Total surface area = surface area of hemisphere + surface area of frustum

$$\text{Total surface area} = 2\pi r^2 + \pi(R + r)l$$

$$= 2 \times \frac{22}{7} \times 10.5 \times 10.5 + \frac{22}{7} \times (19.5 +$$

$$10.5) \times 15 = \frac{14751}{7} = 2107 \frac{2}{7} \text{ cm}^2$$

Sol.652.(a) Let, side of each cube so formed be a.

Volume of cuboid = number of cubes formed $\times a^3$

$$\text{Volume of cuboid} = 8 \times a^3$$

$$12 \times 54 \times 72 = 8 \times a^3$$

$$a^3 = 5832 \Rightarrow a = 18 \text{ cm}$$

Lateral surface area of 2 such cube

$$= 2 \times (4 \times 18^2) = 8 \times 324 = 2592 \text{ cm}^2$$

Sol.653.(b) Ratio of old radius to new radius = 3.5 : 4.9 = 5 : 7.

Old : New

Radius 5 : 7

$$\text{Volume } 5^3 : 7^3$$

$$125 : 343$$

Required percentage increase

$$= \frac{343 - 125}{125} \times 100 = 174.4\%$$

Sol.654.(c) Let the height of the cylindrical tank be "h"

$$\text{Curved surface area} = 2\pi \times 4 \times h = 8\pi h$$

3 times of sum of areas of two circular

$$\text{faces} = 6\pi r^2 = 6\pi \times 4 \times 4 = 96\pi$$

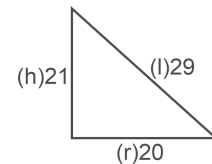
According to the question,

$$96\pi = 2 \times 8\pi h \Rightarrow h = 6$$

Volume of cylindrical tank

$$= \pi r^2 h = \pi \times 4 \times 4 \times 6 = 96\pi.$$

Sol.655.(c)



As height, radius and slant height of a cone form a triplet, we have height 21x and radius = 20x

$$\frac{1}{3} \pi (20x)^2 \times 21x = 4838.4\pi$$

$$x^3 = 1.728 \Rightarrow x = 1.2$$

$$\text{radius} = 20x = 20 \times 1.2 = 24 \text{ cm}$$

Sol.656.(b) According to the question,

$$(l \times b) \times (b \times h) \times (h \times l)$$

$$= 216 \times 96 \times 144$$

$$(lbh)^2 = 216 \times 96 \times 144$$

$$lbh = \sqrt{216 \times 96 \times 144} = 1728 \text{ cm}^3$$

Let the edge of each cube = x

So,

volume of cuboid = 8 (volume of cube)

$$lbh = 8(x)^3$$

$$1728 = 8(x)^3 \Rightarrow (x)^3 = 216 \Rightarrow x = 6 \text{ cm}$$

Lateral surface area of three cubes

$$= 3 \times 4(x)^2 = 3 \times 4(6)^2 = 432 \text{ cm}^2$$

Sol.657.(d)

Let inner radius = r and outer radius = R

According to the question,

$$R - r = 2 \Rightarrow R - 3 = 2 \Rightarrow \text{So, } R = 5 \text{ m}$$

Volume of the soil taken out = Volume of embankment

$$\pi r^2 h = \pi(R^2 - r^2)H \Rightarrow r^2 h = (R^2 - r^2)H$$

$$(3)^2 \times 6 = (5^2 - 3^2) \times H$$

$$54 = 16 \times H \Rightarrow H = \frac{54}{16} = 3 \frac{3}{8} \text{ m}$$

Sol.658.(c)

Let "n" small spheres have been formed.

$$\frac{4}{3} \pi (4)^3 = n \times \frac{4}{3} \pi (2)^3 \Rightarrow n = \frac{64}{8} = 8.$$

Surface area of original sphere

$$= 4\pi \times 4^2 = 64\pi$$

Surface area of 8 small spheres

$$= 8 \times 4\pi \times 2^2 = 128\pi$$

Required ratio = 1 : 2.

Sol.659.(c) Volume of the sheet

= volume of cylinder with outer radius - volume of cylinder with inner radius.

$$\text{Inner radius} = 4 \text{ cm} - 0.5 \text{ cm} = 3.5 \text{ cm}$$

$$h = 2 \text{ m} = 200 \text{ cm}.$$

$$\text{Required volume} = \pi \times 200(4^2 - 3.5^2)$$

$$= \pi \times 200(16 - 12.25)$$

$$= \pi \times 200 \times 3.75 = 750\pi$$

Sol.660.(c) Lateral surface area of prism

= Perimeter of base $\times$ Height

$$480 = 40 \times \text{Height} \Rightarrow \text{Height} = 12 \text{ cm}$$

Volume of prism = Area of base $\times$ height

$$= \left(\frac{1}{2} \times 8 \times 15\right) \times 12 = 720 \text{ cm}^3$$

Sol.661.(d) According to the question,

$$\text{CSA of cone} = 2310 \text{ cm}^2$$

$$\frac{22}{7} \times 21 \times l = 2310 \Rightarrow l = 35 \text{ cm}$$

$$\text{So, } h = \sqrt{l^2 - r^2} = \sqrt{35^2 - 21^2}$$

$$= \sqrt{1225 - 441} = \sqrt{784} = 28 \text{ cm}$$

Let new height = H and new radius = R

$$\text{So, } H = 28 \times \frac{50}{100} = 14 \text{ cm and } R$$

$$= 21 \times \frac{200}{100} = 42 \text{ cm}$$

$$\text{Capacity of cone} = \frac{1}{3} \pi R^2 H$$

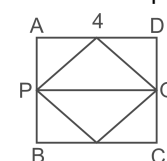
$$= \frac{1}{3} \times \frac{22}{7} \times (42)^2 \times 14 = 25872 \text{ cm}^3$$

So, capacity (in litres)

$$= \frac{25872}{1000} = 25.872 \approx 25.9 \text{ litres}$$

Sol.662.(b) Let the side of square = 4

$$\text{So area of square } S = 4^2 = 16$$



For square formed by joining mid point, diagonal (PQ) = 4

So, area of square which formed

$$= \frac{1}{2} \times (\text{diagonal})^2 = \frac{1}{2} \times (4)^2 = 8$$

$$\text{So, } \frac{A}{S} = \frac{8}{16} = \frac{1}{2} = 0.5$$

Sol.663.(c)

Radius of spherical droplet (in cm) = R

$$= \frac{2}{10 \times 2} = \frac{1}{10} \text{ cm}$$

Radius of cone = Height of cone = r

Volume of 128000 droplets = Volume of cone

$$128000 \times \frac{4}{3} \pi R^3 = \frac{1}{3} \pi r^2 h$$

$$128000 \times 4 \left(\frac{1}{10}\right)^3 = r^2 \times r$$

$$r^3 = 512 \Rightarrow r = 8 \text{ cm}$$

Sol.664.(b) TSA of cylinder = 880 cm^2

$$2\pi r(h + r) = 880$$

$$2 \times \frac{22}{7} \times r(13 + r) = 880$$

$$13r + r^2 = 140 \Rightarrow r^2 + 13r - 140 = 0$$

$$r^2 + 20r - 7r - 140 = 0$$

$$r(r + 20) - 7(r + 20) = 0 \Rightarrow r = 7 \text{ cm}$$

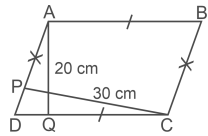
According to question,

Volume of cylinder = $11k$

$$\pi r^2 h = 11k$$

$$= \frac{22}{7} \times (7)^2 \times 13 = 11k \Rightarrow k = 182$$

Sol.665.(b) According to question,



Area of parallelogram = Base $\times$ Altitude

For base AD,

$$300 = AD \times CP \Rightarrow 300 = AD \times 30$$

$$\Rightarrow AD = 10 \text{ cm},$$

$$BC = AD = 10 \text{ cm}$$

Similarly, for base CD,

$$300 = CD \times AQ \Rightarrow 300 = CD \times 20$$

$$\Rightarrow CD = 15 \text{ cm}, AB = CD = 15 \text{ cm}$$

So, perimeter = $AB + BC + CD + AD$

$$= 10 + 15 + 10 + 15 = 50 \text{ cm}$$

Sol.666.(a) Volume of frustum

$$= \frac{1}{3} \pi h (r_1^2 + r_2^2 + r_1 r_2)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 12 \times (20^2 + 13^2 + 20 \times 13)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 12 \times (400 + 169 + 260)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 12 \times 829 \approx 10422 \text{ cm}^3$$

$$= 10422 \text{ cm}^3 \Rightarrow 10.4 \text{ liters}$$

Sol.667.(a)

Let each side of the square be "a"

Side including the road

$$= a + 2 \times 4.5 = a + 9$$

According to the question,

$$(a + 9)^2 - a^2 = 252$$

$$a^2 + 18a + 81 - a^2 = 252$$

$$18a = 171 \Rightarrow a = \frac{171}{18} = 9.5 \text{ m}$$

Sol.668.(b)

Volume of the cube = base area of the cylinder $\times$ (raised height of water)

$$12 \times 12 \times 12 = \frac{22}{7} \times 20 \times 20 \times h$$

$$h = \frac{12 \times 12 \times 12 \times 7}{22 \times 20 \times 20} \approx 1.4 \text{ cm}$$

Sol.669.(c) For box prepared after

cutting squares from rectangular sheet,

$$\text{Length} = 48 - 2(3.5) = 48 - 7 = 41 \text{ cm}$$

$$\text{Breadth} = 27 - 2(3.5) = 27 - 7 = 20 \text{ cm}$$

Height = 3.5 cm

$$\text{Volume of the box} = 41 \times 20 \times 3.5$$

$$= 2870 \text{ cm}^3$$

Sol.670.(b) Here the thickness = 1 cm,

so for outer surface : Length

$$= 30 + 2(1) = 32 \text{ cm},$$

$$\text{Breadth} = 18 + 2(1) = 20 \text{ cm},$$

$$\text{Height} = 23 + 2(1) = 25 \text{ cm}$$

Space occupied by rectangular box

$$= \text{Length} \times \text{Breadth} \times \text{Height}$$

$$= 32 \times 20 \times 25 = 16000 \text{ cm}^3$$

Sol.671.(a) According to question,

$$\frac{90}{100} \times [\text{Volume of cylinder}] = \text{Volume}$$

$$\text{of sphere} = \frac{90}{100} \times [\pi R^2 h] = \frac{4}{3} \pi r^3$$

$$= \frac{9}{10} \times [3^2 \times 120] = \frac{4}{3} r^3$$

$$= \frac{9}{10} \times [1080] = \frac{4}{3} r^3$$

$$= r^3 = 729 \Rightarrow r = 9 \text{ cm}$$

Sol.672.(b) Difference between areas of

two concentric circles = 88 cm^2

$$\pi R^2 - \pi r^2 = 88 \Rightarrow \pi(R^2 - r^2) = 88$$

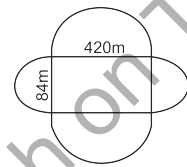
$$\frac{22}{7} (R^2 - 6^2) = 88$$

$$R^2 - 36 = 28 \Rightarrow R^2 = 64$$

$$\text{Area of largest circle} = \pi R^2$$

$$= \frac{22}{7} \times 64 = 201.14 \approx 201 \text{ cm}^2$$

Sol.673.(c)



Total area(x) = Area of rectangular plot +
Area of two semi-circular of diameter 84
m + Area of two semi-circular of
diameter 420 m

$$= 420 \times 84 + 2\left(\frac{\pi r^2}{2}\right) + 2\left(\frac{\pi R^2}{2}\right)$$

$$= 35280 + \pi(r^2 + R^2)$$

$$= 35280 + \frac{22}{7} \left[\left(\frac{84}{2}\right)^2 + \left(\frac{420}{2}\right)^2 \right]$$

$$= 35280 + \frac{22}{7} [(42)^2 + (210)^2]$$

$$= 35280 + \frac{22}{7} [1764 + 44100]$$

$$= 35280 + \frac{22}{7} [45864]$$

$$= 35280 + 144144 = 179424 \text{ m}^2$$

$$1 \text{ hectare} = 10000 \text{ m}^2$$

so, Area of plot = x

$$= \frac{179424}{10000} = 17.9 \text{ hectares}$$

Sol.674.(b)

Let the side of cubical block be x.

Volume of cubical box

$$= x^3 = 10^3 = 1000 \text{ cm}^3$$

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times (4.2)^3 = 310.464 \text{ cm}^3$$

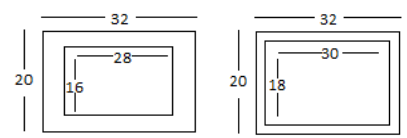
So, wood wasted

$$= 1000 - 310.464 = 689.536 \text{ cm}^3$$

Sol.675.(a)

(Figure 1)

(figure 2)



When margin of 2 cm is left, area
available for typing (figure 1)

$$= (32 - 4) \times (20 - 4) = 28 \times 16 = 448 \text{ cm}^2$$

When margin of 1 cm is left, area
available for typing (figure 2)

$$= (32 - 2) \times (20 - 2) = 30 \times 18 = 540 \text{ cm}^2$$

So percentage increase in typing area

$$= \frac{540 - 448}{448} \times 100 = \frac{92}{448} \times 100$$

$$= 20.54\%$$

Sol.676.(a) $1 \text{ m}^3 = 1000 \text{ litres}$, so 126

$$\text{kilolitres} = 126 \times 1000 = 126000 \text{ litres}$$

which is equivalent to 126 m^3 .

$$\text{Width of reservoir} = \frac{126}{8 \times 3.5} = 4.5 \text{ m}$$

Sol.677.(c) Area of four walls = Area of
floor and ceiling

$$2h(l + b) = 2(l \times b)$$

$$2h(18 + 12) = 2(18 \times 12)$$

$$60h = 432 \Rightarrow h = 7.2 \text{ m}$$

Volume of hall = $l \times b \times h$

$$= 18 \times 12 \times 7.2 = 1555.2 \text{ m}^3$$

Sol.678.(d)

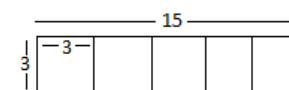
Let the edge of each cube be x.

Volume of each cube so formed

$$= 27 \text{ cm}^3 \Rightarrow x^3 = 27 \Rightarrow x = 3$$

So length of cuboid = $3 \times 5 = 15 \text{ cm}$,

breadth = 3 cm, and height = 3 cm



So, TSA of cuboid = $2(lb + bh + hl)$

$$= 2(15 \times 3 + 3 \times 3 + 3 \times 15)$$

$$= 2(45 + 9 + 45) = 2(99) = 198 \text{ cm}^2$$

Sol.679.(d) For hollow cylindrical shell:

Outer radius $R = b$, Inner radius $P = b - a$
, and $R + P = b + (b - a) = 2b - a$

According to the question,

Volume of sphere = Volume of hollow
cylindrical shell

$$\frac{4}{3} \pi r^3 = \pi(R^2 - P^2)H$$

$$\frac{4}{3} r^3 = (R^2 - P^2)H$$

$$\frac{4}{3} r^3 = (R - P)(R + P)H$$

$$\frac{4}{3}r^3 = (a) \times (2b - a) \times H$$

$$H = \frac{\frac{4}{3}r^3}{(a) \times (2b - a)} = \frac{4r^3}{3a(2b - a)}$$

Sol.680.(c)

Volume of cone = Volume of hemisphere

$$\frac{1}{3}\pi r^2 h = \frac{2}{3}\pi r^3, h = 2r$$

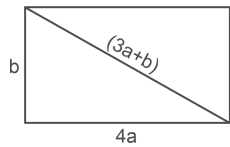
$$\text{So, slant height of cone} = \sqrt{r^2 + h^2}$$

$$= \sqrt{r^2 + (2r)^2} = \sqrt{5r^2} = \sqrt{5}r = 14\sqrt{5} \text{ cm}$$

$$\text{CSA of cone} = \pi r l$$

$$= \frac{22}{7} \times 14 \times 14\sqrt{5} = 616\sqrt{5} \text{ cm}^2$$

Sol.681.(c) Let the larger side of the rectangle be "4a", and other side be "b"
According to the question, length of diagonal = 4a + b - a = 3a + b



Applying pythagoras theorem we get,

$$b^2 + (4a)^2 = (3a + b)^2$$

$$b^2 + 16a^2 = 9a^2 + b^2 + 6ab$$

$$7a^2 = 6ab \Rightarrow \frac{a}{b} = \frac{6}{7},$$

ratio of larger side : smaller side

$$= 4a : b \Rightarrow 4 \times 6 : 7 = 24 : 7$$

Sol.682.(c) Area of the square = 6400 m²

$$\text{Side of the square} = \sqrt{6400} = 80 \text{ m}$$

So, radius of circle = 80 m

$$\text{Length of rectangle} = \frac{3}{5} \times 80 = 48 \text{ m}$$

Area of rectangle = length × breadth

$$= 48 \times 20 = 960 \text{ m}^2$$

Sol.683.(d) Volume of 3 cylinder to be filled with grain

$$= 3 \times \text{volume of each cylinder}$$

$$= 3 \times (\pi r^2 h) = 3 \times \left(\frac{22}{7} \times (20)^2 \times 70\right)$$

$$= 3 \times (88000) = 264000 \text{ cm}^3$$

3000 cm³ corresponds to 1 kg of grain,

$$\text{so } 264000 \text{ cm}^3 = \frac{264000}{3000} = 88 \text{ kg}$$

$$\text{Grain left in the bag} = 100 - 88 = 12 \text{ kg}$$

Sol.684.(d) Surface area of original balloon : Surface area of inflated balloon

$$= 4\pi r^2 : 4\pi R^2 = r^2 : R^2 = (6)^2 : (10)^2$$

$$= 36 : 100 = 9 : 25$$

Sol.685.(d) Let the original radius r = 8,

$$\text{so new radius } R = \frac{r}{2} = \frac{8}{2} = 4$$

Let the height of original cylinder h = 10
and new height = H

Volume of original cylinder = Volume of new cylinder

$$\pi r^2 h = \pi R^2 H \Rightarrow r^2 h = R^2 H$$

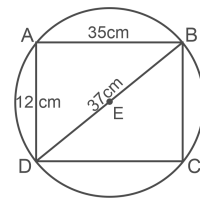
$$8^2 \times 10 = 4^2 \times H$$

$$H = 40 = 4(10) = 4H, \text{ so value of } k = 4$$

$$\text{Sol.686.(d)} \text{ BD} = \sqrt{AB^2 + AD^2}$$

$$= \sqrt{35^2 + 12^2} = \sqrt{1225 + 144}$$

$$= \sqrt{1369} = 37 \text{ cm}$$

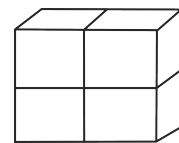


Diagonal of rectangle = Diameter of circle = BD = 37 cm

Circumference of circle

$$= 2\pi r = 2 \times \frac{22}{7} \times \frac{37}{2} = \frac{814}{7} = 116\frac{2}{7} \text{ cm}$$

Sol.687.(b) Clearly, the following cuboidal figure with dimensions 24 × 24 × 12 will be formed.



Total surface area of a cuboid

$$= 2(lb + bh + lh)$$

$$= 2(24 \times 24 + 24 \times 12 + 24 \times 12)$$

$$= 2(576 + 288 + 288)$$

$$= 2 \times 1152 = 2304 \text{ cm}^2$$

Sol.688.(c)

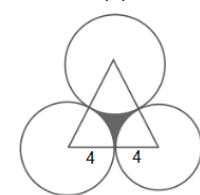
For maximum volume of the cone we must have the height and diameter of the cone equal to the side of the cube.

The side of the cube be 14 units, then the height of the cone is 14 units and its radius is 7 units.

Required ratio

$$= 14 \times 14 \times 14 : \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times 14$$

$$= 42 : 11$$

Sol.689.(b)

Required area = Area of equilateral triangle - Area of 3 sectors with angle 60°

Area of equilateral triangle

$$= \frac{\sqrt{3}}{4} a^2 = \frac{\sqrt{3}}{4} \times 8 \times 8 = 16\sqrt{3} \text{ cm}^2$$

Area of three sectors with central angle 60° and radius 4 cm

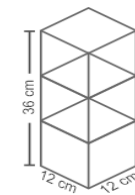
$$= 3 \times \frac{60^\circ}{360^\circ} \times \pi \times 4 \times 4 = 8\pi.$$

$$\text{Required area} = 16\sqrt{3} - 8\pi = 8(2\sqrt{3} - \pi)$$

Sol.690.(d)

$$\text{Side of each cube} = \sqrt[3]{1728} = 12 \text{ cm}$$

By putting three cubes one above the other, the resulting figure will be cuboid with dimension of 36 cm × 12 cm × 12 cm



Total surface area of the cuboid

$$= 2(LB + BH + HL)$$

$$= 2(12 \times 12 + 12 \times 36 + 12 \times 36)$$

$$= 2 \times 1008 = 2016 \text{ cm}^2$$

Sol.691.(c) Area of shaded region

$$= 80 \times 50 = 4000 \text{ m}^2$$

So, area of unshaded region

$$= \frac{35}{100} \times 4000 = 1400 \text{ m}^2$$

$$\text{Total area} = (80 + x) \times (50 + x)$$

$$4000 + 1400 = 4000 + 130x + x^2$$

$$x^2 + 130x - 1400 = 0$$

$$x^2 + 140x - 10x - 1400 = 0$$

$$x(x + 140) - 10(x + 140) = 0$$

$$x \neq -140, \text{ so } x = 10$$

Sol.692.(a) Here, (r + h) = 46 cm,

Total surface area of cylinder

$$= 2\pi r(r + h) \Rightarrow 2\pi r(r + h) = 6072$$

$$2 \times \frac{22}{7} \times r \times 46 = 6072$$

$$r = \frac{6072 \times 7}{2 \times 22 \times 46}, r = 21 \text{ cm}$$

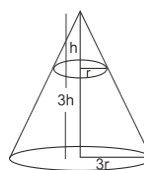
$$h = 46 - 21 = 25 \text{ cm.}$$

$$\text{Volume} = \pi r^2 h = \pi \times 21 \times 21 \times 25 = 11025\pi$$

Sol.693.(c) The respective ratio of height and radius of the two cones will be 1 : 3.

Let radius and height of smaller cone be "r" and "h"

Radius and height of whole cone become 3r and 3h



$$= \frac{\text{volume of smaller cone}}{\text{volume of whole cone}}$$

$$= \frac{\frac{1}{3}\pi r^2 h}{\frac{1}{3}\pi (3r)^2 3h} = \frac{1}{3^2 \times 3} = \frac{1}{27}$$

Sol.694.(c) Let the radius of the outer circumference be 'R' and the radius of the inner circumference be 'r'.

$$\text{Now, } 2\pi R = 748 \text{ m.}$$

$$2 \times \frac{22}{7} \times R = 748.$$

$$R = \frac{748 \times 7}{22 \times 2} \Rightarrow R = 119 \text{ m.}$$

Again, $2\pi r = 396 \text{ m.}$

$$2 \times \frac{22}{7} \times r = 396 \Rightarrow r = \frac{396 \times 7}{22 \times 2} = 63 \text{ m.}$$

$$\text{Width of the track} = (R - r) \\ = (119 - 63) = 56 \text{ m.}$$

Sol.695.(b)

Area of the floor = $l \times b = 18 \times 12 = 216$

Area of four walls

$$= 2 \times (l + b) \times h \Rightarrow 2 \times (18 + 12) \times h$$

$$\text{Given, } 2 \times (30) \times h = 216$$

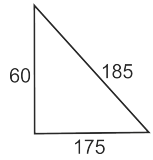
$$\text{So, } h = 3.6$$

Now volume of a cuboidal shape

$$= l \times b \times h = 12 \times 18 \times 3.6 = 777.6 \text{ cm}^3$$

Sol.696.(d)

Sides of the triangular field are triplets.



Area of Δ = Area of rectangular park

$$\frac{1}{2} \times 60 \times 175 = 21x \times 10x$$

$$x^2 = 25 \Rightarrow x = 5$$

Perimeter of rectangular park

$$= 2 [21x + 10x] = 62x = 62(5) = 310 \text{ m}$$

$$\text{Sol.697.(b)} \quad r = \frac{56}{2} = 28 \text{ cm}$$

Length of circular wire

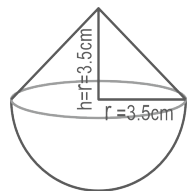
$$= 2\pi r = 2 \times \frac{22}{7} \times 28 = 176 \text{ cm}$$

Length of circular wire = Perimeter of rectangle

$$176 = 2(7x + 4x) \Rightarrow 88 = 11x \Rightarrow x = 8$$

$$\text{Area of rectangle} = 7x \times 4x = 28x^2$$

$$= 28(8^2) = 28(64) = 1792 \text{ cm}^2$$

Sol.698.(c)

Height of cone = radius of base, so $h = r$

Volume of metal used = Volume of cone + volume of hemisphere

$$= \frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3$$

$$= \frac{1}{3} \pi r^3 + \frac{2}{3} \pi r^3$$

$$= \frac{1}{3} \pi r^3 (1 + 2) = \frac{1}{3} \pi r^3 (3)$$

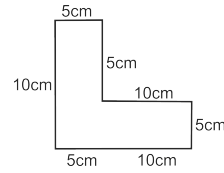
$$= \frac{22}{7} \times \left(\frac{7}{2}\right)^3 = \frac{539}{4} = 134\frac{3}{4} \text{ cm}^3$$

Sol.699.(d) Volume of frustum

$$= \frac{1}{3} \pi h [r_1^2 + r_2^2 + (r_1 \times r_2)]$$

$$= \frac{1}{3} \times \frac{22}{7} \times 24(20^2 + 10^2 + 20 \times 10)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 24 \times 700 = 17600 \text{ cm}^3$$

Sol.700.(c) The given figure is

Total perimeter

$$= (5 + 10 + 5 + 10 + 5 + 10 + 5) = 50 \text{ cm.}$$

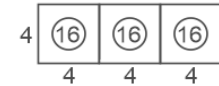
Sol.701.(d) Let the radius of the hemisphere be "r" and the side of the triangle be "a".

$$\text{Now, given } 3\pi r^2 = \frac{\sqrt{3}}{4} a^2$$

$$= \frac{a^2}{r^2} = \frac{3\pi \times 4}{\sqrt{3}} \Rightarrow \frac{a}{r} = (4\pi\sqrt{3})^{0.5}$$

Sol.702.(c)

Each side of the square = $\sqrt{16} = 4 \text{ cm}$



Clearly, the breadth of the rectangle is 4 cm while the length of the rectangle is $3 \times 4 = 12 \text{ cm}$

Perimeter of the rectangle

$$= 2(l + b) = 2(4 + 12) = 32 \text{ cm.}$$

Sol.703.(d) Area of sector making angle

$$\theta \text{ at the centre} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$= \frac{\theta}{360^\circ} \times \pi (2)^2 = \frac{4\pi}{5} \Rightarrow \theta = \frac{360^\circ}{5} = 72^\circ$$

Sol.704.(a)

Radius = 4.2m, Height = 1.75m

As radius and height is in the ratio of 12 : 5 So slant height = 13 unit

$$\text{Slant height } l = \frac{1.75}{5} \times 13 = 4.55 \text{ m}$$

Curved surface area of the canvas

$$= \pi r l = \frac{22}{7} \times 4.2 \times 4.55$$

$$= 13.2 \times 4.55 = 60.06 \text{ m}^2$$

Sol.705.(c)

$$\text{Side of the square} = \sqrt{1764} = 42 \text{ m}$$

$$\text{Breadth} = \frac{1}{3} \text{ of } 42 = 14 \text{ m}$$

$$\text{Length} = 28 \text{ m, Area} = 28 \times 14$$

$$\text{Cost of levelling} = 28 \times 14 \times 15 = ₹ 5880$$

Sol.706.(c) Let width of the wall = x

$$\text{Height of the wall} = 10x$$

$$\text{Length of the wall} = 80x$$

Volume of the wall

$$= 800 x^3 = 51.2 \text{ m}^3 \Rightarrow x = 0.4 \text{ m}$$

$$\text{Length of the wall} = 32 \text{ m}$$

$$\text{Height of the wall} = 4 \text{ m}$$

$$\text{Area of the one side of the wall} = 128 \text{ m}^2$$

$$\text{Cost of painting} = 128 \times 100 = ₹ 12800$$

Sol.707.(b)

circumference of the circle = $2\pi r = 44 \text{ cm}$

$$\text{Side of the square} = a = \frac{44}{4} = 11 \text{ cm}$$

$$\text{Ratio of area} = \pi r^2 : a^2$$

$$= \frac{22}{7} \times 7 \times 7 : 11 \times 11 = 14 : 11$$

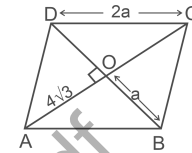
Sol.708.(c)

Area of the ring = $\pi(R^2 - r^2) = 3080$

$$(R^2 - r^2) = 3080 \times \frac{7}{22} = 980$$

$$(R + r)(R - r) = 980$$

$$(R + r) \times 10 = 980 \Rightarrow R + r = 98$$

Sol.709.(d)

Let side of the rhombus = $2a$

Other diagonal = $2a$; $OB = OD = a \text{ cm}$

In ΔOAB ; $AB = 2a$, $OB = a$, $AO = 4\sqrt{3}$

$$(2a)^2 = a^2 + (4\sqrt{3})^2$$

$$\Rightarrow 4a^2 = a^2 + 48 \Rightarrow a = 4$$

Other diagonal = $2a = 8$

Area of the rhombus

$$= \frac{1}{2} \times 8 \times 8\sqrt{3} = 32\sqrt{3}$$

Short trick : Let the other diagonal = a

So, side = a In a rhombus,

$$4a^2 = d_1^2 + d_2^2 \Rightarrow 4a^2 = (8\sqrt{3})^2 + a^2$$

$$a^2 = 64, a = 8$$

Area of the rhombus =

$$\frac{1}{2} \times 8 \times 8\sqrt{3} = 32\sqrt{3} \text{ cm}^2$$

Sol.710.(a) As we know that 15, 20 and 25 are pythagoras triplet so third side is $25\sqrt{3}$

So, total perimeter

$$= 15\sqrt{3} + 20\sqrt{3} + 25\sqrt{3} = 60\sqrt{3}$$

Sol.711.(a) Let the equal side of the isosceles triangle = $4x$ and its base = $5x$.

$$4x + 4x + 5x = 91 \Rightarrow x = 7 \text{ cm}$$

Length of its base = 35 cm

Sol.712.(a) Let the sides of a triangle are $3x$, $4x$ and $5x$ respectively.

$$5x - 3x = 3.6 \Rightarrow x = 1.8 \text{ cm}$$

$$\text{Area of triangle} = \frac{1}{2} \times 3x \times 4x$$

$$= 6 \times 1.8 \times 1.8 = 19.44 \text{ cm}^2$$

Sol.713.(b) Diameter = 28 cm

Radius = 14 cm

Perimeter of circular pizza

$$= 2\pi r = 88 \text{ cm}$$

Perimeter of circular part of one - quarter

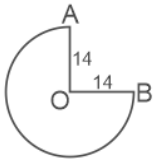
$$= \frac{1}{4} \times 88 \text{ cm} = 22 \text{ cm}$$

Perimeter of the circular part of remaining pizza = $(88 - 22) = 66$ cm

Perimeter of remaining pizza

$$= \text{Perimeter of the circular part} + 2 \times r \\ = 66 + 28 = 94 \text{ cm}$$

Alternate method :



Perimeter of remaining pizza = length of arc AB + AO + OB

$$= \frac{3}{4} \times 2 \times \frac{22}{7} \times 14 + 14 + 14 = 94 \text{ cm}$$

Sol.714.(b) After cutting,

$$\text{Length of Box} = 32 - (3 + 3) = 26$$

$$\text{Breadth of Box} = 18 - (3 + 3) = 12$$

$$\text{Height} = 3$$

$$\text{Volume of the box} = 26 \times 12 \times 3 = 936 \text{ cm}^3$$

Sol.715.(d)

Volume of n sphere = volume of cube

$$\frac{4}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{7}{2} \times n \\ = 77 \times 77 \times 77 \Rightarrow n = 2541$$

Sol.716.(a) Diagonal of the square = $7\sqrt{2}$

Side of the square = 7 cm

Perimeter = $4a = 28$ = base of a triangle

Area of another square = 169

Side of another square = height of the triangle = 13

Required area of triangle

$$= \frac{1}{2} \times 28 \times 13 = 182 \text{ cm}^2$$

Sol.717.(b)

Let the area and breadth of the rectangle are $9x$ and x respectively.

$$\text{Area} = l \times b \Rightarrow 9x = l \times x \Rightarrow l = 9$$

$$\text{Perimeter of rectangle} = 2(l + b)$$

$$86 = 2(l + b) \Rightarrow l + b = 43$$

$$= 9 + b = 43 \Rightarrow b = 34 \text{ cm}$$

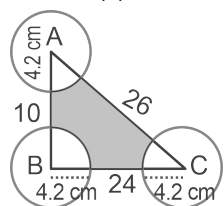
Sol.718.(d) If side of a cube is = a

then its diagonal = $a\sqrt{3}$ and total surface area = $6a^2$

$$(\sqrt{3}a)^2 = 2175 \text{ then } 3a^2 = 2175,$$

$$a^2 = 725 \text{ and } 6a^2 = 6 \times 725 = 4350 \text{ cm}^2$$

Sol.719.(d)



Since we know that $(10, 24, 26)$ is a pythagorean triplet so we can easily find out the area of the triangle to be

$$= \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 24 \times 10 = 120 \text{ cm}^2$$

So the area of the triangle will be equal to 120. At each vertices a circle of radius 4.2 cm is drawn so the total area of circle will be = $\frac{1}{2} \times \text{area of a circle}$ (as the sum of angles of a triangle is equal to 180°)

$$\text{Area of circle} = \frac{1}{2} \times \pi \times 4.2 \times 4.2$$

$$= 27.72$$

$$\text{Area of triangle excluding the portion covered by circle} = 120 - 27.72 = 92.28 \text{ cm}^2$$

Sol.720.(d)

$$\text{Volume of cube} = a^3 = 9^3 = 729 \text{ cm}^3$$

$$\text{Volume of cuboid} = 5 \times 13 \times 31 = 2015$$

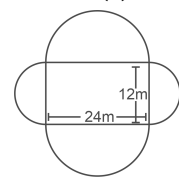
$$\text{Total volume} = 2015 + 729 = 2744 \text{ cm}^3$$

$$\text{Side of resultant cube} = 14 \text{ cm}$$

$$\text{Surface area of resultant cube}$$

$$= 6a^2 = 6 \times 14^2 = 1176 \text{ cm}^2$$

Sol.721.(a)



Total area of rectangular lawn

$$= 24 \times 12 + \pi R^2 + \pi r^2 \text{ (Where } R = \text{radius of bigger semicircle and } r = \text{radius of smaller semicircle)}$$

$$\text{Area} = 288 + 3.14 \times (12^2 + 6^2)$$

$$= 288 + 565.2 = 853.2 \text{ m}^2$$

$$\text{Cost} = 853.2 \times 100 = 85320$$

Sol.722.(b)

$$\text{Volume of cube} = a^3 = 9^3 = 729 \text{ cm}^3$$

$$\text{Volume of cuboid} = 5 \times 13 \times 31 = 2015$$

$$\text{Total volume} = 2015 + 729 = 2744 \text{ cm}^3$$

$$\text{Side of a new cube} = 14 \text{ cm}$$

$$\text{Surface area of a new cube}$$

$$= 6a^2 = 6 \times 14^2 = 1176 \text{ cm}^2$$

$$\text{Cost} = 1176 \times 10 = 11,760$$

$$\text{Sol.723.(d) Area of a hexagon} = \frac{3\sqrt{3}}{2} a^2$$

$$= 1944\sqrt{3} = \frac{3\sqrt{3}}{2} a^2 \Rightarrow a = 36$$

$$\text{Perimeter} = 36 \times 6 = 216 \text{ m}$$

$$\text{Cost} = 216 \times 11.50 = \text{Rs. } 2,484$$

Sol.724.(a) Diagonal of square = $6\sqrt{2}$ cm

Side of square = 6 cm

perimeter of square = 24 cm

Base of triangle = 24 cm

Height of triangle = $\sqrt{144} = 12$ (area of square = side²)

$$\text{Area of triangle} = \frac{1}{2} \times 12 \times 24 = 144 \text{ cm}^2$$

Sol.725.(d) Perimeter of square = 40 m

$$\text{Perimeter of rectangle} = 2 \times 40 = 80 \text{ m}$$

Let length of rectangle = $3x$

Breadth of rectangle = $2x$

Total perimeter of rectangle

$$= 2(3x + 2x) \Rightarrow 10x = 80 \Rightarrow x = 8$$

length of rectangle = 24 m

Breadth of rectangle = 16 m

$$\text{area of rectangle} = 24 \times 16 = 384 \text{ m}^2$$